



Analytical Tools Interface
for Landscape Assessments



Analytical Tools Interface for Landscape Assessments (ATtILA) User Guide

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Overview

ATtILA for ArcGIS Pro is an Esri ArcGIS toolbox that allows users to easily calculate many common landscape metrics. GIS expertise is not required, but some experience with ArcGIS is recommended. Three metric groups (toolsets) are currently included in ATtILA: Landscape Characteristics, People in the Landscape, and Riparian Characteristics, along with a Utilities section. ATtILA for ArcGIS Pro is written using the Python programming language and is designed to accommodate spatial data from a variety of sources.

Download the latest version here: <https://github.com/USEPA/ATtILA2/releases> or as a zipped file at <https://www.epa.gov/enviroatlas/attila-toolbox>.

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Credits

Python code for ATtILA was written by Don Ebert (EPA), Michael Jackson (former EPA), Baohong Ji (former Woolpert), and Torrin Hultgren (GDIT), with special assistance from Ellen D'Amico (EPA), Doug Browning (former EPA Student Services Contractor), Allison Killea (former Student Services Contractor), and Jacob Tafrate (ORAU Student Services Contractor).

David Gottlieb (former EPA Student Services Contractor) wrote the code for the Land Cover Coding Editor.

The original ATtILA ArcView extension tool was written in Avenue code by Don Ebert (EPA) and Tim Wade (former EPA) with assistance from Dennis Yankee (Tennessee Valley Authority, Public Power Institute) who wrote code for the PCTIA_RD metric.

The following programmers were not directly involved with the project, but gave permission for their code to be incorporated into the original ATtILA and we greatly appreciate their generosity:

- Laine, Jarko. 1998. Intersec Script, Novo.
- Eichenlaub, Bill. 1998. Profiler Script (Version 1.0), National Park Service.
- O'Malley, Kevin. 1999. Two Theme Analyst Extension.
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- Cosmas, Tom. 1999. Table.RenameField Script, New Jersey Department of Environmental Protection.
- Schultz, Ron. 2003. Bearing extension.
- Jenness, Jeff. 2005. Distance/Azimuth Tools (Version 1.4b), Jenness Enterprises.

Tim Wade (former EPA), Deborah Chaloud (former EPA), Megan Culler (GDIT), Steven Jett (Woolpert), Barbara Rosenbaum (Woolpert), Caroline Erickson (former EPA) and Bob Ohman (former EPA Senior Environmental Employment Program) wrote the help files for the many versions of ATtILA.

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License

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Installing ATtILA

This section provides basic information on installing ATtILA using the downloadable zip file available at <https://www.epa.gov/enviroatlas/attila-toolbox>. For more detailed instructions on installing and using custom toolboxes in ArcGIS Pro, please see, [‘Connect to a toolbox’](#), in ArcGIS Pro’s Online Help. New users should read through this wiki at least once to familiarize themselves with potential pitfalls associated with spatial data, ArcGIS limitations, and ATtILA processes.

Requirements

ATtILA requires ArcGIS Pro 3.0 or later and the Spatial Analyst extension. ATtILA has been tested on ArcGIS Pro through version 3.3.

In ArcGIS Pro 3.4, several ATtILA for ArcGIS Pro tools have failed when spaces are in the toolbox’s file path. All tools operate as expected during testing when no spaces are found.

The Spatial Analyst or 3D Analyst extensions are required.

Installation

1. Download the ATtILA distribution zip file to a safe location, such as a Downloads directory.
2. Extract the contents of the ATtILA zip file to a location on the file system where you commonly store personal Toolboxes. After extraction, ensure the ATtILA v2.tbx file and the ToolboxSource folder are at the same directory level. ATtILA’s directory structure is illustrated below.

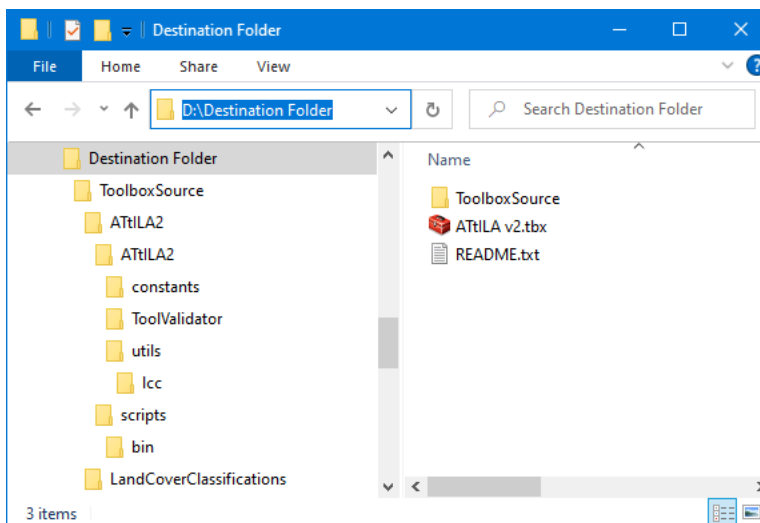


Figure 1. ATtILA directory structure

3. Start ArcGIS Pro. In the Catalog View or Catalog Pane, right click on Toolboxes, and click on Add Toolbox.

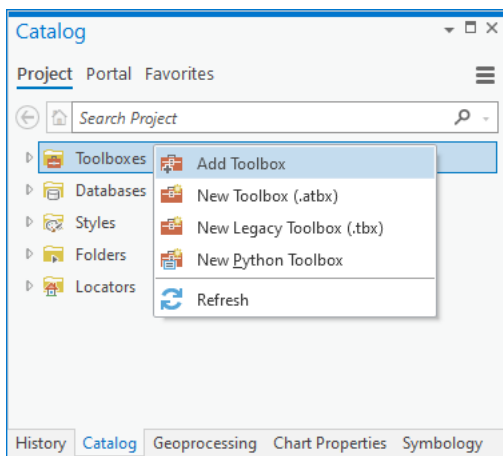


Figure 2. Adding general toolbox

4. Navigate to ATtILA v{#}.tbx, select it and click OK.

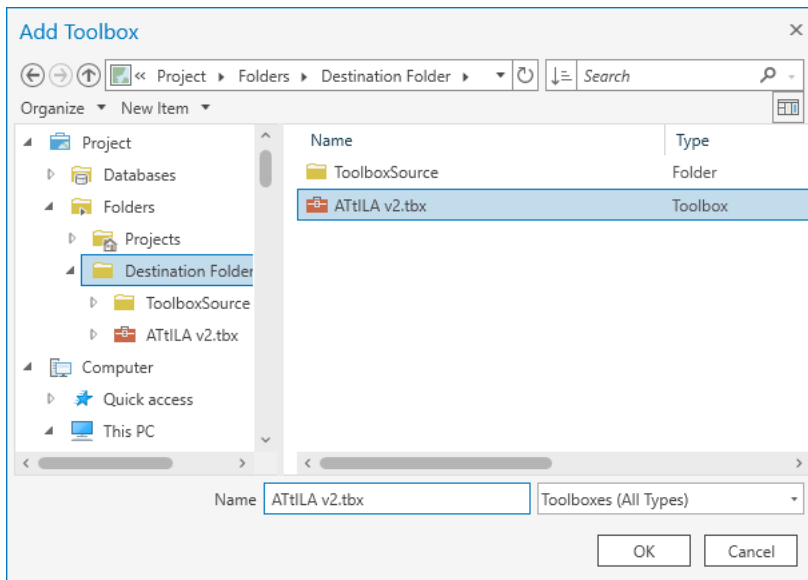


Figure 3. Adding ATtILA Toolbox

5. Check to see if the toolbox installed correctly, i.e., open Toolbox to see tools contained within.

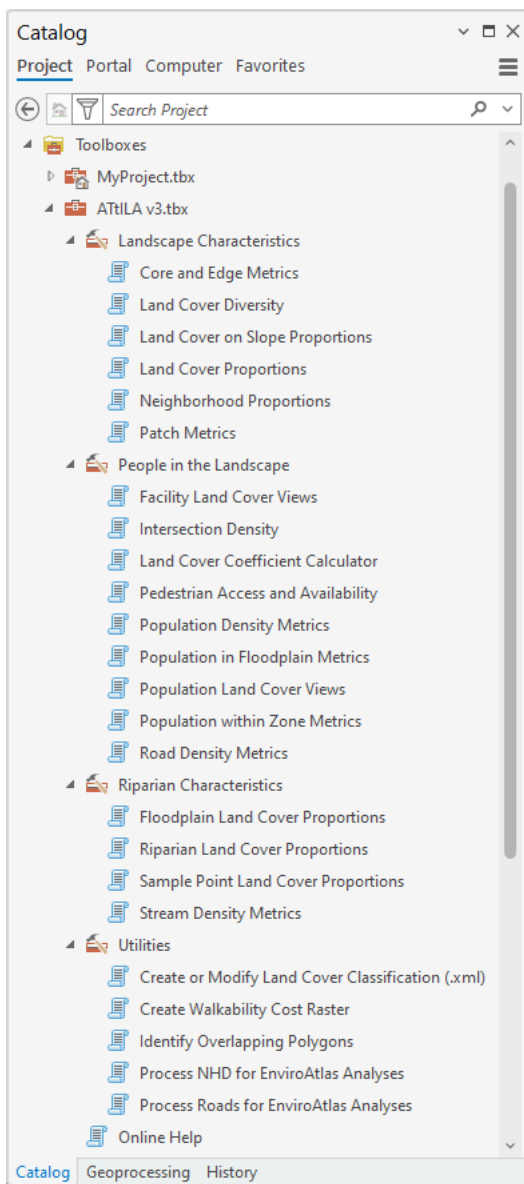


Figure 4. ATtILA Toolbox with expanded toolsets view

NOTE: To make the ATtILA toolbox available to all new projects, right-click on the toolbox and select 'Add to New Projects'. This will add a link to ATtILA toolbox in project favorites. For more details on accessing ATtILA this way, see 'Project favorites', in ArcGIS Pro's Online Help.

ATtILA Metrics

ATtILA metrics are measurements of a component or components within the landscape, derived from combinations or overlays of data, which are used to characterize the composition and spatial configuration of those components in the landscape. A seminal work on metrics, which may be considered the genesis of ATtILA, can be found in “*An Ecological Assessment of the United States Mid-Atlantic Region: A Landscape Atlas*” (Jones et al,1997). Many of the metrics originally conceived and developed for the atlas are included in ATtILA for ArcGIS.

The following toolsets are included in the ATtILA toolbox:

- **Landscape Characteristics**
- **People in the Landscape**
- **Riparian Characteristics**
- **Utilities**

Each toolset is comprised of several tools for calculating metrics. Input data varies depending on the tool being used, and the dialog box for each tool guides the user, step by step, through the acceptable input data types and formats. Help is available from within each tool’s dialog box by clicking the **ATtILA’s wiki** hyperlink in the tool’s Description. Additionally, each required dataset is described in detail in the tool-specific sections of this [manual](#).

ATtILA tools are extremely flexible and accept a wide range of data as input. Users are responsible for choosing data appropriate for their application or assessment. Spatial datasets are available from many sources, with a broad range of scales and spatial and thematic accuracy. There are numerous online clearinghouses that distribute many types of spatial data. Good places to start include Esri’s [Living Atlas](#) Web site, the Federal Geographic Data Committee’s (FGDC) [GeoPlatform](#) or the United States Geological Survey’s (USGS) [National Map Downloader](#). EPA’s spatial data repository may be accessed through the [Environmental Dataset Gateway](#) website. Datasets that cover the conterminous United States are available for download free of charge from these and other Internet sites. They are suitable for regional to national scale assessments. For local studies, higher resolution data is recommended, if available.

ATtILA tools are not restricted to use of particular datasets, but can be run with any input dataset that matches the required dataset type as defined in the tool-specific sections of this manual. Again, ***it is the responsibility of the user to select datasets of the appropriate type, format, and resolution for the metric analysis being performed.***

Landscape Characteristics

Natural land cover and anthropogenic land use have a profound effect on landscapes and environmental stressors. For example, cutting down forests reduces uptakes of nitrogen and other nutrients, disrupts habitats for various fauna, and exposes soils to increased runoff. Erosion may be further accelerated if the exposed area is on a slope. The proportions of various land cover/use types within a watershed can provide information about the expected condition of surface waters and aquatic habitats within that watershed.

The Landscape Characteristics toolset consists of the following metric tools:

- **Core and Edge Metrics** Calculates the edge-to-area ratio within each reporting unit for selected land cover classes and creates an output table. Edge width is provided by the user.
- **Land Cover Diversity** Calculates land cover diversity within each reporting unit and creates an output table. Metrics include Shannon-Weiner index, standardized Shannon-Weiner index, Simpson's index, and simple diversity.
- **Land Cover on Slope Proportions** Calculates percentages of land cover types on slopes greater than a specified threshold value within reporting unit polygons and creates an output table.
- **Land Cover Proportions** Calculates percentages of selected land cover types within reporting unit polygons and creates an output table. Optionally, per capita area of selected land cover types within each reporting unit may also be reported (square meters of land cover class per person).
- **Neighborhood Proportions** Calculates the area percentage of selected land cover classes within a given neighborhood. The neighborhood is square in shape. Its size is provided by the user. An optional secondary raster can be created which reclassifies those values into percentage zones.
- **Patch Metrics** Calculates the number of patches (NUM), largest patch size (LRG), average patch size (AVG), median patch size (MDN), the number of patches per square kilometer (DENS), and proportion of largest patch area to total patch area (PLGP) within each reporting unit for selected land cover classes and creates an output table. Minimum patch size and maximum patch separation are provided by the user. Optionally, Mean Distance to Closest Patch (MDCP) within each reporting unit for selected land cover classes may also be calculated.

Tool Core and Edge Metrics

Available with Spatial Analyst license.

Summary

Calculates the edge-to-area ratio within each reporting unit for selected land cover classes and creates an output table. Edge width is provided by the user.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- **NOTE:** *This tool is resource-intensive in terms of both processing time and disk space. Processing time varies depending on the extent and resolution of the input **Land cover grid**, the number of reporting units, and the number of classes selected for processing. To minimize processing time, consider creating a new **Reporting unit feature** layer that does not include unwanted reporting units. Also select only the land cover classes in the **Report metrics for these classes** parameter that are important for analysis. This tool creates multiple interim rasters during processing which may consume considerable disk space in the Scratch Workspace. If the Scratch Workspace resides on a disk with limited disk space, the tool may fail during processing. To prevent this, consider changing the Scratch Workspace in Environments > Workspace > Scratch Workspace to a location with sufficient available disk space, or free up disk space on the disk in which the current Scratch Workspace resides to accommodate the interim rasters.*
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.

- As the **Reporting unit feature** is a vector dataset, ArcGIS will perform a vector to raster conversion during processing.
 - To better control the vector to raster conversion, the tool defaults the **Snap raster** and the **Processing cell size** to that of the **Land cover grid**. These may be changed from within the tool.
 - If the extent of the **Reporting unit feature** is smaller than that of the **Land cover grid**, the user may wish to check the **Reduce land cover grid to smallest recommended size** option. This creates a temporary land cover raster with the processing extent equal to that of the **Reporting unit feature** to avoid unnecessary processing.
- If the **Reporting unit feature** has overlapping features, the zonal analysis will be performed for each individual feature.
- If a feature in the **Reporting unit feature** zone data does not overlap any cell centers in the secondary class rasters (e.g., **Land cover grid**), those zones will not get converted to the internal zone raster. As a result, those zones will not be represented in the **Output table**.
- The **Land cover classification scheme** must correspond to the **Land cover grid**.
 - Schemes for common land cover datasets are included with ATtILA. Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD 2001 ALL vs. NLCD 2001 LAND). Schemes designated as “ALL” include all land cover classes in reporting unit area calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded. More information about each of the classification schemes supplied with ATtILA may be found in [Supplied Land Cover Classification Schemas](#).
 - In addition to the common land cover schemes, the tool permits a user-defined land cover classification scheme to be used by specifying a **Land cover classification file**. Refer to [Land Cover Classification](#) for more information on how ATtILA may be customized for user-specific needs.
 - **NOTE:** *When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. For example, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. See [Values Element in ATtILA’s LCC XML Document](#) and [Values Window](#) in the [Land Cover Classification Editor](#) for more information on excluded values.*
- **Edge width** must be an integer value measuring the number of grid cells. For example, if the **Land cover grid** is at 30-meter cell resolution, an **Edge width** of five (5) would result in a width of $(30 \times 5) = 150$ meters.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - **NOTE:** *For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.

- When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.
- Field names in the **Output table** follow this naming scheme:
 - **[class caemField]** or **[class Id]_E2A[edge width]** - The edge-to-area ratio for the land cover class within the reporting unit.
 - **[class caemField]** is used if a caemField attribute is provided in the class's element in the selected **Land cover classification file**.
 - (e.g. **FOREST_EdgeRatio** for <class Id="for" Name="Forest Land Cover" filter="" caemField="FOREST_EdgeRatio">)
 - **[class Id]_E2A[edge width]** - is used if a caemField attribute is not provided.
 - (e.g. **for_E2A5** for <class Id="for" Name="Forest Land Cover" filter=""> and a five-cell edge width is used)
 - **NOTE:** Output field names can be altered by editing the **Land cover classification file** with the *Land Cover Classification Editor* (Refer to *Classes Element* in *ATtILA's LCC XML Document* for general information, or the individual **Land cover classification file** for details).
 - **NOTE:** The output field name for each class is shown as the second item next to the class's check box in **Report metrics for these classes**.
- To describe the amounts of core and edge within each reporting unit for each selected land cover class, two ancillary fields are provided in the **Output table**. These field names follow the same naming scheme regarding the caemField class attribute in the **Land cover classification file** as described above. The ancillary field names in this **Output table** are as follows:
 - **[class caemField]_COR[edge width]** or **[class Id]_COR[edge width]** - The percent of the reporting unit comprised of core cells for the land cover class.
 - **[class caemField]_EDG[edge width]** or **[class Id]_EDG[edge width]** - The percent of the reporting unit comprised of edge cells for the land cover class.
- In **Additional Options**, the user may elect to 'Add Quality Assurance Fields', 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Add Quality Assurance Fields
 - Choosing **QAFIELDS** adds a number of fields to the **Output table** to facilitate quality assurance checking.
 - **CAEM_OVER** - The percent overlap between the **Reporting unit feature** layer and the **Land cover grid** for each reporting unit.
 - **CAEM_TOTA** - The total raster zonal area for each reporting unit (zone) after the **Reporting unit feature** layer has been rasterized.
 - **CAEM_EFFA** - The effective raster area of each reporting unit (zone) after excluded classes have been removed.

- **CAEM_EXCA** - The raster area of the excluded classes within each reporting unit (zone) based on exclusions in the **Land cover classification scheme**.
- Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **caem_Raster[class name][edge width]_#** (raster) - Rasters created for each class selected in the **Report metrics for these classes** parameter of the tool with reclassified values corresponding to the following scheme: 1 = Excluded, 2 = Other, 3 = Edge, 4 = Core.
 - **caem_TabArea[class name][edge width]_#** (table) - A table of areas corresponding to the values in the **caem_Raster[class name][edge width]_#** within each reporting unit.
 - ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
 - ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*
 - Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - ***NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.*
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:

- When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
- When the Current Workspace is a folder, the log file will be located in that folder.
- **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
- **NOTE:** ATtILA will report the name and location of the log file in the Tool Details' Messages section.

Syntax

CAEM (Reporting_unit_feature, Reporting_unit_ID_field, Land_cover_grid, Land_cover_classification_scheme, Land_cover_classification_file, Report_metrics_for_these_classes, Edge_width, Output_table, {Processing_cell_size}, {Snap_raster}, Select_options, {Reduce_land_cover_grid_to_smallest_recommended_size})

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Land_cover_grid	The raster dataset representing land cover classes upon which core and edge area metrics will be derived within each Reporting unit feature. The class input must be an integer raster layer.	Raster Layer
Land_cover_classification_scheme	The land cover classification schemes included in ATtILA and a User Defined option. The default schemes correspond to common input land cover datasets. Two schemes are supplied for each dataset included in ATtILA:	String

Parameter	Explanation	Data Type
Land_cover_classification_file	The full pathname to the user-defined .xml file for custom or non-standard land cover classification schemes. Pathname is automatically filled when a default scheme is selected.	File
Report_metrics_for_these_classes	A list of the land cover classes and metric combinations for processing. Check the box to calculate metrics for each land cover class and/or combination class desired within the reporting units.	Multiple Value
Edge_width	The width of the edge, measured in number of grid cells. The input must be an integer value.	Long
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved within a file geodatabase.	Table
Processing_cell_size (Optional)	The Processing cell size for the zonal operation. The default Processing cell size is the cell size of the input raster land cover data. Optionally, the user may select a different cell size.	Analysis cell size
Snap_raster (Optional)	The raster that the cell alignment of the Land cover grid and rasterized Reporting unit feature layer will be matched to during processing. The default Snap raster is the Land cover grid.	Raster Layer
Select_options	Tool options available to provide additional	Multiple Value

Parameter	Explanation	Data Type
	information:	
Reduce_land_cover_grid_to_smallest_recommended_size (Optional)	Specifies whether the Land cover grid extent will be clipped to that of the Reporting unit feature layer during processing.	Boolean

Code Samples

Core and Edge Metrics example 1 (Python window)

This example returns a table with the following fields: the proportion of the reporting unit zone comprised of forest edge, the proportion of the reporting unit zone comprised of forest core, and the ratio of forest edge to total forest area in the reporting unit zone. Edge width was defined as three grid cells. Calculations are then repeated for the All Natural Land Cover class and written to the output table. The options to retain the intermediate calculation products, to record the process steps into a log file, and to generate the quality assurance fields have also been selected.

NOTE: *ATtILA tools that utilize the **Report metrics for these classes** parameter will clear all class selections when the tool is opened. This results in setting any variable set for that parameter to **None** when calling the tool from a script. Because of this, the standard scripting syntax (e.g. `arcpy.ATtILA.LCP(variable 1, variable 2, variable 3, etc.)` will fail. To overcome this idiosyncrasy, use the scripting example provided below. When using this technique, all variables are passed in as strings.*

NOTE: *By running scripts which launch an ATtILA tool via 'metric.run' within ArcGIS Pro's Python Window, all geoprocessing functions run by that tool can be viewed in the History Pane.*

```
import arcpy
from arcpy.sa import *

from ATtILA2 import metric

toolPath = "D:\\Destination Folder\\ToolboxSource\\ATtILA2\\scripts\\CoreAndEdgeMetrics_ArcGIS.py"
Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Land_cover_grid = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Land_cover_classification_scheme = "NLCD LAND"
Land_cover_classification_file = "D:\\Destination Folder\\ToolboxSource\\LandCoverClassifications\\NLCD
LAND.xml"
Report_metrics_for_these_classes = "for - [for_E2A] Forest'; 'NI - [NI_E2A] All Natural Land Use"
Edge_width = "3"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_CAEM"
Processing_cell_size = "30"
Snap_raster = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Select_options = "'LOGFILE - Record Process Steps Taken During Metric Calculation'; 'QAFIELDS - Add
Quality Assurance Fields'; 'INTERMEDIATES - Retain Intermediate Layers Generated During Metric
Calculation'"
Reduce_land_cover_grid_to_smallest_recommended_size = "true"

metric.runCoreAndEdgeMetrics(
    toolPath,
    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Land_cover_grid,
```

```

Land_cover_classification_scheme,
Land_cover_classification_file,
Report_metrics_for_these_classes,
Edge_width,
Output_table,
Processing_cell_size,
Snap_raster,
Select_options,
Reduce_land_cover_grid_to_smallest_recommended_size
)

```

Core and Edge Metrics example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```

Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Constructing the ATtILA metric output table: Watersheds_CAEM
|YYYY-MM-DD hh:mm:ss| Reducing input Land cover grid to smallest recommended size
|YYYY-MM-DD hh:mm:ss| Reduction complete
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Generating core and edge grid for Class: FOR
|YYYY-MM-DD hh:mm:ss| Step 1 of 4: Reclassing land cover grid to Class = 3, Other = 2, and Excluded = 1
|YYYY-MM-DD hh:mm:ss| Step 2 of 4: Setting Class areas to Null
|YYYY-MM-DD hh:mm:ss| Step 3 of 4: Finding distance from Other
|YYYY-MM-DD hh:mm:ss| Step 4 of 4: Delimiting Class areas to Edge = 3 and Core = 4
|YYYY-MM-DD hh:mm:ss| Class FOR core and edge grid complete. Intermediate: caem_RasterFOR3_0
|YYYY-MM-DD hh:mm:ss| Generating a zonal tabulate area table
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Generating core and edge grid for Class: NI
|YYYY-MM-DD hh:mm:ss| Step 1 of 4: Reclassing land cover grid to Class = 3, Other = 2, and Excluded = 1
|YYYY-MM-DD hh:mm:ss| Step 2 of 4: Setting Class areas to Null
|YYYY-MM-DD hh:mm:ss| Step 3 of 4: Finding distance from Other
|YYYY-MM-DD hh:mm:ss| Step 4 of 4: Delimiting Class areas to Edge = 3 and Core = 4
|YYYY-MM-DD hh:mm:ss| Class NI core and edge grid complete. Intermediate: caem_RasterNI3_0
|YYYY-MM-DD hh:mm:ss| Generating a zonal tabulate area table
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)

```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Land Cover Diversity

Available with Spatial Analyst license.

Summary

Calculates land cover diversity within each reporting unit and creates an output table. Metrics include Shannon-Weiner index, standardized Shannon-Weiner index, Simpson's index, and simple diversity.

Diversity metrics quantify landscape composition. These metrics are influenced by two components - richness, the number of different patch types present, and evenness, the distribution of area among the patch types. The four diversity metrics calculated in this tool are:

- Shannon-Weiner index (H) - a measure of the diversity of land cover types throughout the reporting unit. The index value increases with the number of land cover types in the reporting unit.

$$H = - \sum_{i=1}^m P_i \times \ln P_i$$

Figure 5. LCD_Formula_H

- Standardized Shannon-Weiner index (H') - standardizes the Shannon-Weiner index to account for the variety of distinct land cover classes present in the reporting unit.

$$H' = (- \sum_{i=1}^m P_i \times \ln P_i) / \ln m$$

Figure 6. LCD_Formula_HPrime

- Simpson's index (C) - a measure of the evenness of the distribution of land cover classes throughout the reporting unit. Simpson's index is most sensitive to the presence of common land cover types. Simpson's index values range from 0 to 1, with 1 representing perfect evenness of all land cover types.

$$C = 1 - \sum_{i=1}^m P_i^2$$

Figure 7. LCD_Formula_C

- Simple diversity (S) - a count of the number of distinct land cover classes present in the reporting unit.

$$S = m$$

Figure 8. LCD_Formula_S

- For all of the above,
 - m = the number of different land cover types in the reporting unit
 - P_i = the proportion of the reporting unit comprised of land cover type i

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.

- **NOTE:** For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.
- **NOTE:** It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.
- **NOTE:** Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.
- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.
- As the **Reporting unit feature** is a vector dataset, ArcGIS will perform a vector to raster conversion during processing.
 - To better control the vector to raster conversion, the tool defaults the **Snap raster** and the **Processing cell size** to that of the **Land cover grid**. These may be changed from within the tool.
 - If a large size disparity exists between the extents of the **Reporting unit feature** and the **Land cover grid**, the user may wish to set the Extent in Environment Settings > Processing Extent to the smaller of the two to avoid unnecessary processing.
- If the **Reporting unit feature** has overlapping features, the zonal analysis will be performed for each individual feature.
- If a feature in the **Reporting unit feature** zone data does not overlap any cell centers in the secondary class rasters (e.g., **Land cover grid**), those zones will not get converted to the internal zone raster. As a result, those zones will not be represented in the **Output table**.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - **NOTE:** For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.

- Field names in the **Output table** are as follows:
 - H - The Shannon-Weiner index metric for the reporting unit.
 - H_Prime - The Standardized Shannon-Weiner index metric for the reporting unit.
 - C - The Simpson's index metric for the reporting unit.
 - S - The simple diversity metric for the reporting unit.
- In **Additional Options**, the user may elect to 'Add Quality Assurance Fields', 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Add Quality Assurance Fields
 - Choosing **QAFIELDS** adds a number of fields to the **Output table** to facilitate quality assurance checking.
 - **LCD_OVER** - The percent overlap between the **Reporting unit feature** layer and the **Land cover grid** for each reporting unit.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **lcd_TabArea#** (table) - A table of areas corresponding to the values in the Land cover raster within each reporting unit.
 - ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
 - ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*
 - Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named "attilaScratchWorkspace" is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.

- **NOTE:** *ATtILA will report the location of the saved intermediate products in the Tool Details' Messages section.*
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** *In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.*
 - **NOTE:** *ATtILA will report the name and location of the log file in the Tool Details' Messages section.*

Syntax

LCD (Reporting_unit_feature, Reporting_unit_ID_field, Land_cover_grid, Output_table, {Processing_cell_size}, {Snap_raster}, Select_options)

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Land_cover_grid	The raster dataset representing land cover classes upon which diversity metrics will be derived within each Reporting unit feature. The class input must be an integer raster layer.	Raster Layer
Output_table	The output reporting unit metrics table to be created.	Table

Parameter	Explanation	Data Type
	It is recommended that the Output table be saved within a file geodatabase.	
Processing_cell_size (Optional)	The Processing cell size for the zonal operation. The default Processing cell size is the cell size of the input raster land cover data. Optionally, the user may select a different cell size.	Analysis Cell Size
Snap_raster (Optional)	The raster that the cell alignment of the Land cover grid and rasterized Reporting unit feature layer will be matched to during processing. The default Snap raster is the Land cover grid.	Raster Layer
Select_options	Tool options available to provide additional information:	Multiple Value

Code Samples

Land Cover Diversity example 1 (Python window)

This example returns a table with the Shannon-Weiner index, standardized Shannon-Weiner index, Simpson's index, and simple diversity index of land cover categories within each reporting unit. The options to retain the intermediate calculation products, to record the process steps into a log file, and to generate the quality assurance fields have also been selected.

```
import arcpy

Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Land_cover_grid = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_LCD"
Processing_cell_size = "30"
Snap_raster = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Select_options = "'LOGFILE - Record Process Steps Taken During Metric Calculation';'QAFIELDS - Add Quality Assurance Fields';'INTERMEDIATES - Retain Intermediate Layers Generated During Metric Calculation'"

Reporting_unit_feature,
Reporting_unit_ID_field,
Land_cover_grid,
Output_table,
Processing_cell_size,
```

```
Snap_raster,  
Select_options  
)
```

Land Cover Diversity example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM  
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt  
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables  
Intermediates are stored in this directory: workspace path  
|YYYY-MM-DD hh:mm:ss| Constructing the ATtILA metric output table: Watersheds_LCD  
|YYYY-MM-DD hh:mm:ss| Generating a zonal tabulate area table  
|YYYY-MM-DD hh:mm:ss| Processing the tabulate area table and computing metric values  
Summarizing the ATtILA metric output table to log file  
Summary complete  
Log file closed  
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
```

Tags

There are no tags for this item.

Credits

Shannon, C. E. (1948). A mathematical theory of communication. *The Bell System Technical Journal*, 27, 379-423. Simpson, E. H. (1949). Measurement of diversity. *Nature*, 163, 4148, 688-688.

Use Limitations

There are no access and use limitations for this item.

Tool Land Cover on Slope Proportions

Available with Spatial Analyst license.

Summary

Calculates percentages of land cover types on slopes greater than a specified threshold value within reporting unit polygons and creates an output table.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.
- As the **Reporting unit feature** is a vector dataset, ArcGIS will perform a vector to raster conversion during processing.
 - To better control the vector to raster conversion, the tool defaults the **Snap raster** and the **Processing cell size** to that of the **Land cover grid**. These may be changed from within the tool.
 - If the extent of the **Reporting unit feature** is smaller than that of the **Land cover grid**, the user may wish to check the **Reduce land cover grid to smallest recommended size** option. This creates a temporary land cover raster with the processing extent equal to that of the **Reporting unit feature** to avoid unnecessary processing.

- If the **Reporting unit feature** has overlapping features, the zonal analysis will be performed for each individual feature.
- If a feature in the **Reporting unit feature** zone data does not overlap any cell centers in the secondary class rasters (e.g., **Land cover grid**), those zones will not get converted to the internal zone raster. As a result, those zones will not be represented in the **Output table**.
- The **Land cover classification scheme** must correspond to the **Land cover grid**.
 - Schemes for common land cover datasets are included with ATtILA. Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD 2001 ALL vs. NLCD 2001 LAND). Schemes designated as “ALL” include all land cover classes in reporting unit area calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded. More information about each of the classification schemes supplied with ATtILA may be found in [Supplied Land Cover Classification Schemas](#).
 - In addition to the common land cover schemes, the tool permits a user-defined land cover classification scheme to be used by specifying a **Land cover classification file**. Refer to [Land Cover Classification](#) for more information on how ATtILA may be customized for user-specific needs.
 - ***NOTE:** When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. For example, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. See [Values Element in ATtILA’s LCC XML Document](#) and [Values Window in the Land Cover Classification Editor](#) for more information on excluded values.*
- The **Slope grid** may contain slope values measured in either percent or degrees. Enter the **Slope threshold** value in the same units of measure as the **Slope grid** to analyze land cover proportions on slopes equal to or greater than the threshold value.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - ***NOTE:** For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.
- Field names in the **Output table** follow this naming scheme:
 - **[class lcospField] or [class Id]_SL[slope threshold]** - The percent of the reporting unit occupied by the land cover class on slopes that are equal to or exceed the **Slope threshold**.
 - **[class lcospField]** is used if an lcospField attribute is provided in the class’s element in the selected **Land cover classification file**.
 - (e.g., **AgOnSlopes** for <class Id=“agr” Name=“Agriculture” filter=“” lcospField=“AgOnSlopes”>)

- **[class Id]_SL[slope threshold]** - is used if an lcospField attribute is not provided.
 - (e.g., **agr_SL3** for <class Id="agr" Name="Agriculture" filter=""> and a 3 percent slope threshold is used)
- **NOTE:** *Output field names can be altered by editing the **Land cover classification file** with the Land Cover Classification Editor (Refer to Classes Element in ATtILA's LCC XML Document for general information, or the individual **Land cover classification file** for details).*
- **NOTE:** *The output field name for each class is shown as the second item next to the class's check box in **Report metrics for these classes**.*
- In **Additional Options**, the user may elect to 'Add Quality Assurance Fields', 'Add Area Fields', 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Add Area Fields
 - Choosing **AREAFIELDS** adds fields to record the raster area measurement of the selected land cover classes occurring on slopes greater than or equal to the **Slope threshold** within each reporting unit. The field naming conventions is dependent on whether an lcospField is provided in the **Land cover classification file**:
 - **[class lcospField]_A** or **[class Id]SL[slope threshold]_A** - The area of the rasterized reporting unit occupied by the land cover class on slopes that are equal to or exceed the **Slope threshold**.
 - Add Quality Assurance Fields
 - Choosing **QAFIELDS** adds a number of fields to the **Output table** to facilitate quality assurance checking.
 - **LCOSP_OVER** - The percent overlap between the **Reporting unit feature** layer and the **Land cover grid** for each reporting unit.
 - **LCOSP_TOTA** - The total raster zonal area for each reporting unit (zone) after the **Reporting unit feature** layer has been rasterized.
 - **LCOSP_EFFA** - The effective raster area of each reporting unit (zone) after excluded classes have been removed.
 - **LCOSP_EXCA** - The raster area of the excluded classes within each reporting unit (zone) based on exclusions in the **Land cover classification scheme**.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **lcosp_Raster_SL[slope threshold]_#** (raster) - A land cover raster resulting from the overlay between all **Land cover grid** cells and **Slope grid** cells that do not meet the **Slope threshold**. Land cover cells in overlay areas are reclassified with a new value that is outside of the range of valid class values as recognized by the **Land cover**

classification file. The purpose of this intermediate raster product is to exclude land cover cells that are in low slope areas as defined by the **Slope threshold** from metric calculations. The tool uses this raster to calculate the proportion of each reporting unit that is covered by each valid land cover type on slopes that meet or exceed the **Slope threshold**. The processes involved in creating this raster are as follows:

1. Examine the values in both the Land cover grid and the Land cover classification file to determine which has the highest maximum value.
2. Calculate a new value by adding one (1) to the highest maximum value. For example, if the high value in the input grid is 95 and the high value in the classification file is 99, the calculated value for cells meeting the **Slope threshold** is $(99+1) = 100$ in the new intermediate raster.
3. Create a new raster that assigns this new value to all cells that are below the **Slope threshold**. Retain all other land cover cell values.
4. Name the intermediate raster “lcosp_Raster_SL[slope threshold]_#” where “[slope threshold]” is the input **Slope threshold** and “#” is a sequential number added when the raster is saved.

- **lcospTabArea#** (table) - A table of areas corresponding to the values in the intermediate raster within each reporting unit. Areas with reclassified values that do not meet the **Slope threshold** are included.
- **NOTE:** *To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
- **NOTE:** *The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*

- Intermediate products are saved to one of the following locations:

- When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
- When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
- **NOTE:** *ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.*

- Record Process Steps Taken During Metric Calculation

- Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.

- Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
- Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
 - **NOTE:** ATtILA will report the name and location of the log file in the Tool Details' Messages section.

Syntax

LCOSP (Reporting_unit_feature, Reporting_unit_ID_field, Land_cover_grid, Land_cover_classification_scheme, Land_cover_classification_file, Report_metrics_for_these_classes, Slope_grid, Slope_threshold, Output_table, {Processing_cell_size}, {Snap_raster}, Select_options, {Reduce_land_cover_grid_to_smallest_recommended_size})

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Land_cover_grid	The raster dataset representing land cover classes upon which core and edge area metrics will be derived within each Reporting unit feature. The class input must be an integer raster layer.	Raster Layer
Land_cover_classification_scheme	The land cover classification schemes included in ATtILA and a User Defined option. The default schemes correspond to common input land cover datasets. Two schemes are supplied	String

Parameter	Explanation	Data Type
	for each dataset included in ATtILA:	
Land_cover_classification_file	The full pathname to the user-defined .xml file for custom or non-standard land cover classification schemes. Pathname is automatically filled when a default scheme is selected.	File
Report_metrics_for_these_classes	A list of the land cover classes and metric combinations for processing. Check the box to calculate metrics for each land cover class and/or combination class desired within the reporting units.	Multiple Value
Slope_grid	The raster dataset representing slope. The value units may be in either percent or degrees.	Raster layer
Slope_threshold	The slope value at or above which land cover proportion metrics are to be derived. Threshold value must be an integer value zero or above and in the same units of measure as the Slope grid	Long
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved within a file geodatabase.	Table
Processing_cell_size (Optional)	The Processing cell size for the zonal operation. The default Processing cell size is the cell size of the input raster land cover data. Optionally, the user may select a different cell size.	Analysis cell size

Parameter	Explanation	Data Type
Snap_raster (Optional)	The raster that the cell alignment of the Land cover grid and rasterized Reporting unit feature layer will be matched to during processing. The default Snap raster is the Land cover grid.	Raster Layer
Select_options	Tool options available to provide additional information:	Multiple Value
Reduce_land_cover_grid_to_smallest_recommended_size (Optional)	Specifies whether the Land cover grid extent will be clipped to that of the Reporting unit feature layer during processing.	Boolean

Code Samples

Land Cover on Slope Proportions example 1 (Python window)

The example outputs a table with the percent of the reporting unit occupied by agriculture and pasture land cover classes on slopes that are equal to or exceed the Slope threshold of 6 percent. The option to record the process steps into a log file has also been selected.

NOTE: ATtILA tools that utilize the **Report metrics for these classes** parameter will clear all class selections when the tool is opened. This results in setting any variable set for that parameter to **None** when calling the tool from a script. Because of this, the standard scripting syntax (e.g. `arcpy.ATtILA.LCP(variable 1, variable 2, variable 3, etc.)`) will fail. To overcome this idiosyncrasy, use the scripting example provided below. When using this technique, all variables are passed in as strings.

NOTE: By running scripts which launch an ATtILA tool via 'metric.run' within ArcGIS Pro's Python Window, all geoprocessing functions run by that tool can be viewed in the History Pane.

```
import arcpy
from arcpy.sa import *

from ATtILA2 import metric

toolPath = "D:\\Destination
Folder\\ToolboxSource\\ATtILA2\\scripts\\LandCoverOnSlopeProportions_ArcGIS.py"
Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Land_cover_grid = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Land_cover_classification_scheme = "NLCD LAND"
Land_cover_classification_file = "D:\\Destination Folder\\ToolboxSource\\LandCoverClassifications\\NLCD
LAND.xml"
Report_metrics_for_these_classes = "agr - [agr_SL] Agriculture';'agrc - [agrc_SL] Cultivated Crop"
Slope_grid = "D:\\pyexamples\\data.gdb\\Slope"
Slope_threshold = "6"
```

```

Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_LCOSP"
Processing_cell_size = "30"
Snap_raster = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Select_options = "'LOGFILE - Record Process Steps Taken During Metric Calculation'"
Reduce_land_cover_grid_to_smallest_recommended_size = ""

metric.runLandCoverOnSlopeProportions(
    toolPath,
    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Land_cover_grid,
    Land_cover_classification_scheme,
    Land_cover_classification_file,
    Report_metrics_for_these_classes,
    Slope_grid,
    Slope_threshold,
    Output_table,
    Processing_cell_size,
    Snap_raster,
    Select_options,
    Reduce_land_cover_grid_to_smallest_recommended_size
)

```

Land Cover on Slope Proportions 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```

Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
|YYYY-MM-DD hh:mm:ss| Generating land cover above slope threshold grid
|YYYY-MM-DD hh:mm:ss| Inserting EXCLUDED values into areas below slope threshold
|YYYY-MM-DD hh:mm:ss| Constructing the ATtILA metric output table: Watersheds_LCOSP
|YYYY-MM-DD hh:mm:ss| Generating a zonal tabulate area table
|YYYY-MM-DD hh:mm:ss| Processing the tabulate area table and computing metric values
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)

```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Land Cover Proportions

Available with Spatial Analyst license.

Summary

Calculates percentages of selected land cover types within reporting unit polygons and creates an output table.

Optionally, per capita area of selected land cover types within each reporting unit may also be reported (square meters of land cover class per person).

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.
- As the **Reporting unit feature** is a vector dataset, ArcGIS will perform a vector to raster conversion during processing.
 - To better control the vector to raster conversion, the tool defaults the **Snap raster** and the **Processing cell size** to that of the **Land cover grid**. These may be changed from within the tool.

- If a large size disparity exists between the extents of the **Reporting unit feature** and the **Land cover grid**, the user may wish to set the Extent in Environment Settings > Processing Extent to the smaller of the two to avoid unnecessary processing.
- If the **Reporting unit feature** has overlapping features, the zonal analysis will be performed for each individual feature.
- If a feature in the **Reporting unit feature** zone data does not overlap any cell centers in the secondary class rasters (e.g., **Land cover grid**), those zones will not get converted to the internal zone raster. As a result, those zones will not be represented in the **Output table**.
- The **Land cover classification scheme** must correspond to the **Land cover grid**.
 - Schemes for common land cover datasets are included with ATtILA. Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD 2001 ALL vs. NLCD 2001 LAND). Schemes designated as “ALL” include all land cover classes in reporting unit area calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded. More information about each of the classification schemes supplied with ATtILA may be found in Supplied Land Cover Classification Schemas.
 - In addition to the common land cover schemes, the tool permits a user-defined land cover classification scheme to be used by specifying a **Land cover classification file**. Refer to Land Cover Classification for more information on how ATtILA may be customized for user-specific needs.
 - ***NOTE:** When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. For example, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. See Values Element in ATtILA’s LCC XML Document and Values Window in the Land Cover Classification Editor for more information on excluded values.*
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - ***NOTE:** For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.
- When the optional **PER CAPITA** (square meters of land cover type per individual) selection is checked, the tool allows the user to select a **Population raster or polygon feature** and a corresponding **Population field**.
 - The **Population raster or polygon feature** will allow you to choose either a raster, or a polygon feature.
 - When choosing a raster, this tool assumes the grid cell value to be the population count for the grid cell.

- When choosing a polygon feature, this tool assumes that population is distributed evenly throughout each feature polygon. The tool apportions population by area weighting. For example, if 50% of a population polygon is within a reporting unit, the tool will assign 50% of the value in the polygon's **Population field** to that reporting unit. Caution should be exercised when population polygons do not have even population distributions as this could result in an overweighting or underweighting of population when the tool performs the apportionment. Generally, greater accuracy will be achieved if the population polygons are smaller than the smallest **Reporting unit feature** polygons.
- ***NOTE:** The **Population field** dropdown is only available when the **Population raster or polygon feature** is a polygon feature.*
- ***NOTE:** The **Population field** will default to 'Value' when the **Population raster or polygon feature** is an integer grid.*
- ***NOTE:** The **Population field** will be empty when the **Population raster or polygon feature** is a non-integer grid.*
- Field names in the **Output table** follow this naming scheme:
 - **[class lcpField]** or **p[class Id]** - The percent of the non-excluded reporting unit occupied by the land cover class.
 - **[class lcpField]** is used if an lcpField attribute is provided in the class's element in the selected **Land cover classification file**.
 - (e.g., **NINDEX** for <class Id="NI" Name="All Natural Land Use" filter="" lcpField="NINDEX">)
 - **p[class Id]** - is used if an lcpField attribute is not provided.
 - (e.g., **pNI** for <class Id="NI" Name="All Natural Land Use" filter="">)
 - When the **PER CAPITA** option is selected:
 - **[class lcpField]_PC** or **p[class Id]_PC** - The area (m²) per person of the land cover class by reporting unit.
 - ***NOTE:** Output field names can be altered by editing the **Land cover classification file** with the Land Cover Classification Editor (Refer to Classes Element in ATtILA's LCC XML Document for general information, or the individual **Land cover classification file** for details).*
 - ***NOTE:** The output field name for each class is shown as the second item next to the class's check box in **Report metrics for these classes**.*
- In **Additional Options**, the user may elect to 'Add Quality Assurance Fields', 'Add Area Fields', 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Add Area Fields
 - Choosing **AREAFIELDS** adds fields to record the raster area measurement of the selected land cover classes within each reporting unit. The field naming conventions is dependent on whether an lcpField is provided in the **Land cover classification file**:

- **[class lcpField]_A** or **p[class Id]_A** - The area of the rasterized reporting unit occupied by the land use class.
- Add Quality Assurance Fields
 - Choosing **QAFIELDS** adds a number of fields to the **Output table** to facilitate quality assurance checking.
 - **LCP_OVER** - The percent overlap between the **Reporting unit feature** layer and the **Land cover grid** for each reporting unit.
 - **LCP_TOTA** - The total raster zonal area for each reporting unit (zone) after the **Reporting unit feature** layer has been rasterized.
 - **LCP_EFFA** - The effective raster area of each reporting unit (zone) after excluded classes have been removed.
 - **LCP_EXCA** - The raster area of the excluded classes within each reporting unit (zone) based on exclusions in the **Land cover classification scheme**.
- Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **lcp_TabArea#** (table) - A table of areas corresponding to the values in the **Land cover grid** within each reporting unit. The name of the intermediate table has the prefix “lcpTabArea” followed by a number (e.g. lcpTabArea0, lcpTabArea1, lcpTabArea2, etc.). The number suffix is added when the table is saved in order to give each successive table a unique name when the tool is executed more than once and results saved to the same output location.
 - **lcp_[Population polygon feature]#** (vector) - A copy of the **Population raster or polygon feature** with all but its the **Population field** removed. A tmpClass field is added to the attribute table and populated with a value of 1 to facilitate reporting unit population calculations. This feature class is only produced when the optional **PER CAPITA** selection is checked and the **Population raster or polygon feature** is a polygon feature.
 - **lcp_populationCnt#** (table) - A table with the population count within each reporting unit. It is produced only when the optional **PER CAPITA** selection is checked.
 - If the **Population raster or polygon feature** is a polygon feature, the table is generated by performing a Tabulate Intersection operation using the **Reporting unit feature** as the Input Zone Feature, the **Reporting unit ID field** as the Zone Field, and the **lcp_[Population polygon feature]#** as the Input Class Features.
 - If the **Population raster or polygon feature** is a raster, the table is generated by performing a Zonal Statistics as Table operation using the **Reporting unit feature** as the Input raster or feature zone data, the **Reporting unit ID field** as the Zone field, and the **Population raster or polygon feature** as the Input value raster. The statistics type is “Sum” and the Ignore NoData in calculations is checked.

- **NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the `CreateScratchName` ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.
- **NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.
- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - **NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
 - **NOTE:** ATtILA will report the name and location of the log file in the Tool Details’ Messages section.

Syntax

LCP (Reporting_unit_feature, Reporting_unit_ID_field, Land_cover_grid, Land_cover_classification_scheme, Land_cover_classification_file, Report_metrics_for_these_classes, Output_table, {Processing_cell_size}, {Snap_raster}, Select_options)

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Land_cover_grid	The raster dataset representing land cover classes upon which core and edge area metrics will be derived within each Reporting unit feature. The class input must be an integer raster layer.	Raster Layer
Land_cover_classification_scheme	The land cover classification schemes included in ATtILA and a User Defined option. The default schemes correspond to common input land cover datasets. Two schemes are supplied for each dataset included in ATtILA:	String
Land_cover_classification_file	The full pathname to the user-defined .xml file for custom or non-standard land cover classification schemes. Pathname is automatically filled when a default scheme is selected.	File
Report_metrics_for_these_classes	A list of the land cover classes and metric combinations for processing. Check the box to calculate metrics for each land cover class and/or combination class desired within the reporting units.	Multiple Value

Parameter	Explanation	Data Type
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved within a file geodatabase.	Table
PER_CAPITA (Optional)	Specifies whether the per capita value for each land cover class and/or combination class desired within the reporting units will be added to the output table.	Boolean
Population_raster_or_polygon_feature	The raster or vector polygon dataset that contains population data.	Feature Layer or Raster Layer
Population_field (Optional)	The field in the Population raster or polygon feature that contains population data. The behavior of this parameter is dependent on the input for the Population or polygon feature:	Field
Processing_cell_size (Optional)	The Processing cell size for the zonal operation. The default Processing cell size is the cell size of the input raster land cover data. Optionally, the user may select a different cell size.	Analysis cell size
Snap_raster (Optional)	The raster that the cell alignment of the Land cover grid and rasterized Reporting unit feature layer will be matched to during processing. The default Snap raster is the Land cover grid.	Raster Layer
Select_options	Tool options available to provide additional information:	Multiple Value

Code Samples

Land Cover Proportions example 1 (Python window)

This example returns a table with the proportion of the reporting unit zone comprised of agriculture and all natural land cover classes. The options to calculate the square meters of each land cover class per capita, to retain the intermediate calculation products, to record the process steps into a log file, and to generate the quality assurance fields have also been selected.

NOTE: *ATtILA tools that utilize the **Report metrics for these classes** parameter will clear all class selections when the tool is opened. This results in setting any variable set for that parameter to **None** when calling the tool from a script. Because of this, the standard scripting syntax (e.g. `arcpy.ATtILA.LCP(variable 1, variable 2, variable 3, etc.)` will fail. To overcome this idiosyncrasy, use the scripting example provided below. When using this technique, all variables are passed in as strings.*

NOTE: *By running scripts which launch an ATtILA tool via 'metric.run' within ArcGIS Pro's Python Window, all geoprocessing functions run by that tool can be viewed in the History Pane.*

```
import arcpy
from arcpy.sa import *

from ATtILA2 import metric

toolPath = "D:\\Destination Folder\\ToolboxSource\\ATtILA2\\scripts\\LandCoverProportions_ArcGIS.py"
Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Land_cover_grid = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Land_cover_classification_scheme = "NLCD LAND"
Land_cover_classification_file = "D:\\Destination Folder\\ToolboxSource\\LandCoverClassifications\\NLCD
LAND.xml"
Report_metrics_for_these_classes = "agr - [pagr] Agriculture';'NI - [NINDEX] All Natural Land Use"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_LCP"
PER_CAPITA = "true"
Population_raster_or_polygon_feature = "D:\\pyexamples\\data.gdb\\Census2010"
Population_field = "TOTALPOP"
Processing_cell_size = "30"
Snap_raster = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric
Calculation';'LOGFILE - Record Process Steps Taken During Metric Calculation';'QAFIELDS - Add
Quality Assurance Fields'"

metric.runLandCoverProportions(
    toolPath,
    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Land_cover_grid,
    Land_cover_classification_scheme,
    Land_cover_classification_file,
    Report_metrics_for_these_classes,
    Output_table,
    PER_CAPITA,
    Population_raster_or_polygon_feature,
    Population_field,
    Processing_cell_size,
    Snap_raster,
    Select_options
)
```

Land Cover Proportions example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Constructing the ATtILA metric output table: Watersheds_LCP
|YYYY-MM-DD hh:mm:ss| Generating a zonal tabulate area table
|YYYY-MM-DD hh:mm:ss| Calculating population within each reporting unit
|YYYY-MM-DD hh:mm:ss| Processing the tabulate area table and computing metric values
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Neighborhood Proportions

Available with Spatial Analyst license.

Summary

Calculates the area percentage of selected land cover classes within a given neighborhood. The neighborhood is square in shape. Its size is provided by the user.

An optional secondary raster can be created which reclassifies those values into percentage zones.

Usage

- **NOTE:** *It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Land cover classification scheme** must correspond to the **Land cover grid**.
 - Schemes for common land cover datasets are included with ATtILA. Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD 2001 ALL vs. NLCD 2001 LAND). Schemes designated as “ALL” include all land cover classes in reporting unit area calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded. More information about each of the classification schemes supplied with ATtILA may be found in [Supplied Land Cover Classification Schemas](#).
 - In addition to the common land cover schemes, the tool permits a user-defined land cover classification scheme to be used by specifying a **Land cover classification file**. Refer to [Land Cover Classification](#) for more information on how ATtILA may be customized for user-specific needs.
 - **NOTE:** *When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. For example, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. See [Values Element in ATtILA’s LCC XML Document](#) and [Values Window in the Land Cover Classification Editor](#) for more information on excluded values.*

- When the optional **Burn in areas of excluded values** selection is checked, the tool allows the user to select a **Burn in value** and a **Minimum patch size for burn in** to identify areas of excluded values in the output raster. Grid values are marked as excluded in the **Land cover classification scheme**.
 - ***NOTE:** If no excluded values are indicated in the **Land cover classification scheme**, the burn in operation will not be performed and ATtILA will alert the user with a warning message in the tool's View Details box and the Geoprocessing > History window.*
 - **Burn in value** is restricted to a value not possible in the output raster. It can be any integer value outside the range of 0 - 100.
- When the optional **Create zone raster** selection is checked, the tool allows the user to select **Zone proportion bins**. This will produce a second output raster which is a reclassification of the output neighborhood proportions grid categorized into the **Zone proportion bins**. For example, if a **Zone proportion bins** of 20 is selected, the neighborhood proximity raster would be reclassified into bins of 0 - 20, 21 - 40, 41 - 60, 61 - 80, and 81 - 100. The respective attribute values for the optional zone raster would be 20, 40, 60, 80, and 100 and any **Burn in value** if provided. Possible zone bins to select are 5, 10, 20, 25, and 50.
- When the optional **Overwrite existing outputs** selection is checked, ATtILA will set the geoprocessing option to 'Allow geoprocessing tools to overwrite existing datasets' to TRUE and output rasters will be named using preset ATtILA defaults (i.e., output rasters will terminate with ' _Prox' and ' _ProxZone'). ATtILA will not use the CreateScratchName ArcPy function to generate unique output filename by appending a number to the end of the filename. The geoprocessing option to 'Allow geoprocessing tools to overwrite existing datasets' will be reset to its original setting upon tool completion.
- In **Output workspace**, select the geodatabase where the standard and optional tool outputs are to be stored.
 - Standard outputs:
 - **[class Id]_[neighborhood width]_Prox#** (raster) - Rasters created for each class selected in the **Report metrics for these classes** parameter of the tool with values representing the proportion of the **Neighborhood width x Neighborhood width** rectangle around each grid cell that is [class ID]. The values can range from 0 percent to 100 percent plus the **Burn in value** if the **Burn in areas of excluded values** option was selected.
 - **[class Id]_[neighborhood width]_Zone#** (raster) - A copy of the **[class Id]_[neighborhood width]_Prox#** raster reclassified into the specified **Zone proportion bins**. If the **Burn in areas of excluded values** option was selected, the **Burn in value** is also included. This raster is only saved if the **Create zone raster** option is selected.
 - ***NOTE:** To ensure unique filenames, output filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the tool product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
- In **Additional Options**, the user may elect to 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Retain Intermediate Layers Generated During Metric Calculation

- Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - ****np_[minimum patch size]_BurnIn#**** (raster) - A reclassified version of the **Land cover grid** where patches of excluded areas (e.g., water bodies) equal to or greater than the **Minimum patch size for burn in** are coded with the **Burn in value**. All other areas are coded as 0. This raster is only saved if the **Burn in areas of excluded values** option is selected.
 - The processes involved in creating this raster are as follows:
 1. Reclass the **Land cover grid** so that cells representing excluded values = 1, and all other land cover cells = 0.
 2. Perform a RegionGroup operation to calculate the size of excluded area patches with the reclassified raster as the **in_raster**, eight as the **number_neighbors**, **WITHIN** as the **zone_connectivity**, and **add_link** set to **TRUE**.
 3. Use a Con statement on the RegionGroup output to set any patches with a count greater than or equal to the **Minimum patch size for burn in** and a **LINK** value equal to one to the **Burn in value** and everywhere else set to zero.
 - **[class Id]_[neighborhood width]_Cnt#** (raster) - Rasters created for each class selected in the **Report metrics for these classes** parameter of the tool with values representing the count of [class ID] cells within a **Neighborhood width x Neighborhood width** rectangle around each grid cell. The values can range from 0 to the square of the **Neighborhood width**.
 - The processes involved in creating this raster are as follows:
 1. Reclass the **Land cover grid** so that cells representing the selected class = 1, and all other land cover cells = 0.
 2. Perform a focal statistics sum on the reclassified raster using a rectangular neighborhood of **Neighborhood width x Neighborhood width**.
- **NOTE:** *To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
- **NOTE:** *The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*

- **NOTE:** The ending number for the output filenames is assigned only when the geoprocessing option 'Allow geoprocessing tools to overwrite existing datasets' setting is not selected and the tool option, **Overwrite existing outputs**, is also unselected.
- Intermediate products are saved in the **Output workspace**.
 - **NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details' Messages section.
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
 - **NOTE:** ATtILA will report the name and location of the log file in the Tool Details' Messages section.

Syntax

NP (Land_cover_grid, Land_cover_classification_scheme, Land_cover_classification_file, Generate_output_for_these_classes, Neighborhood_width, {Burn_in_areas_of_excluded_values}, Burn_in_value, Minimum_patch_size_for_burn_in, {Create_zone_raster}, Zone_proportion_bins, {Overwrite_existing_outputs}, Output_workspace, {Select_options})

Parameter	Explanation	Data Type
Land_cover_grid	The raster dataset representing land cover classes upon which core and edge area metrics will be derived within each Reporting unit feature.	Raster Layer
	The class input must be an integer raster layer.	
Land_cover_classification_scheme	The land cover classification schemes included in ATtILA and a User Defined option.	String

Parameter	Explanation	Data Type
	The default schemes correspond to common input land cover datasets. Two schemes are supplied for each dataset included in ATtILA:	
Land_cover_classification_file	The full pathname to the user-defined .xml file for custom or non-standard land cover classification schemes. Pathname is automatically filled when a default scheme is selected.	File
Generate_output_for_these_classes	A list of the land cover classes and metric combinations for processing. Check the box to generate proximity percentage zone rasters for each land cover class and/or combination class desired.	Multiple Value
Neighborhood_width	The width of the neighborhood, measured in number of grid cells. The neighborhood is square in shape. The input must be an integer value.	Long
Burn_in_areas_of_excluded_values (Optional)	Specifies whether areas where excluded values occur in the land cover grid will be burned into the output raster.	Boolean
Burn_in_value	Value to identify areas where excluded values occur in the land cover grid. The default is -99999. Values between 0 and 100 are not permitted. The input must be an integer value.	Long

Parameter	Explanation	Data Type
Minimum_patch_size_for_burn_in	Minimum number of connected land cover grid cells with excluded values to be considered a patch. Cells are considered connected if they share a side or are diagonally adjacent. The input must be an integer value.	Long
Create_zone_raster (Optional)	Create secondary output raster that reclassifies the neighborhood proportions raster into percentage zones.	Boolean
Zone_proportion_bins	The size of the percentage bins to define output zones. Neighborhood analyses will be reclassified into these zones. For example, if a Zone percentage bin of 20 is selected, the output raster will contain a minimum of five values: 20, 40, 60, 80, and 100. If the burn in option is selected, the supplied burn in value will also be included.	Long
Overwrite_existing_outputs (Optional)	Controls whether this tool can overwrite existing files when run.	Boolean
Output_workspace	The path to the geodatabase.	Workspace
Select_options (Optional)	Tool options available to provide additional information:	Multiple Value

Code Samples

Neighborhood Proportions example 1 (Python window)

This example returns a raster with the area percentage of the ‘All Natural Land Use’ land cover class within a nine cell x nine cell neighborhood around each grid cell. Patches of excluded values designated in the ‘NLCD LAND’ land cover classification file which were larger than or equal to ten cells in size were given a value of -99999 in the output raster. The options to retain the intermediate calculation products, and to record the process steps into a log file were also selected.

NOTE: *ATtILA tools that utilize the **Report metrics for these classes** parameter will clear all class selections when the tool is opened. This results in setting any variable set for that parameter to **None** when calling the tool from a script. Because of this, the standard scripting syntax (e.g. `arcpy.ATtILA.LCP(variable 1, variable 2, variable 3, etc.)` will fail. To overcome this idiosyncrasy, use the scripting example provided below. When using this technique, all variables are passed in as strings.*

NOTE: *By running scripts which launch an ATtILA tool via ‘metric.run’ within ArcGIS Pro’s Python Window, all geoprocessing functions run by that tool can be viewed in the History Pane.*

```
import arcpy
from arcpy.sa import *

from ATtILA2 import metric

toolPath = "D:\\Destination Folder\\ToolboxSource\\ATtILA2\\scripts\\NeighborhoodProportions_ArcGIS.py"
Land_cover_grid = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Land_cover_classification_scheme = "NLCD LAND"
Land_cover_classification_file = "D:\\Destination Folder\\ToolboxSource\\LandCoverClassifications\\NLCD
LAND.xml"
Report_metrics_for_these_classes = "NI - [NI_Prox] All Natural Land Use"
Neighborhood_width = "9"
Burn_in_areas_of_excluded_values = "true"
Burn_in_value = "-99999"
Minimum_visible_patch_size = "10"
Create_zone_raster = "false"
Zone_proportion_bins = ""
Overwrite_existing_outputs = "false"
Output_workspace = "D:\\pyexamples\\results.gdb"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric
Calculation';'LOGFILE - Record Process Steps Taken During Metric Calculation'"

metric.runNeighborhoodProportions(
    toolPath,
    Land_cover_grid,
    Land_cover_classification_scheme,
    Land_cover_classification_file,
    Report_metrics_for_these_classes,
    Neighborhood_width,
    Burn_in_areas_of_excluded_values,
    Burn_in_value,
    Minimum_visible_patch_size,
    Create_zone_raster,
    Zone_proportion_bins,
    Overwrite_existing_outputs,
    Output_workspace,
    Select_options
)
```

Neighborhood Proportions example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool’s processing steps.


```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Processing BURN IN areas
|YYYY-MM-DD hh:mm:ss| Reclassing excluded values in land cover to 1. All other values = 0
|YYYY-MM-DD hh:mm:ss| Calculating size of excluded area patches
|YYYY-MM-DD hh:mm:ss| Assigning -99999 to excluded area patches >= 10 cells in size
|YYYY-MM-DD hh:mm:ss| Saving intermediate grid: np_10_BurnIn_0
|YYYY-MM-DD hh:mm:ss| Save intermediate grid complete: np_10_BurnIn_0
|YYYY-MM-DD hh:mm:ss| Processing neighborhood proportions grid for NI
|YYYY-MM-DD hh:mm:ss| Reclassifying selected NI land cover class to 1. All other values = 0
|YYYY-MM-DD hh:mm:ss| Performing focal SUM on reclassified raster using 9 x 9 cell neighborhood
|YYYY-MM-DD hh:mm:ss| Calculating the proportion of land cover class within 9 x 9 cell neighborhood
|YYYY-MM-DD hh:mm:ss| Burning excluded areas into proportions grid
|YYYY-MM-DD hh:mm:ss| Saving proportions grid: NI_9_Prox_0
|YYYY-MM-DD hh:mm:ss| Save proportions grid complete: NI_9_Prox_0
|YYYY-MM-DD hh:mm:ss| Saving intermediate grid: NI_9_Cnt_0
|YYYY-MM-DD hh:mm:ss| Save intermediate grid complete: NI_9_Cnt_0
Adding NI_9_Prox_0 to Map view
Log file closed
Succeeded at Day, Month dd, yyyy hh:mm:ss AM/PM (Elapsed Time: ##.## units)
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use

There are no access and use limitations for this item.

Tool Patch Metrics

Available with Spatial Analyst license.

Summary

Calculates the number of patches (NUM), largest patch size (LRG), average patch size (AVG), median patch size (MDN), the number of patches per square kilometer (DENS), and proportion of largest patch area to total patch area (PLGP) within each reporting unit for selected land cover classes and creates an output table. Minimum patch size and maximum patch separation are provided by the user.

Optionally, Mean Distance to Closest Patch (MDCP) within each reporting unit for selected land cover classes may also be calculated.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- **NOTE:** *This tool is resource-intensive in terms of both processing time and disk space. Processing time varies depending on the extent and resolution of the input **Land cover grid**, the number of reporting units, and the number of classes selected for processing. To minimize processing time, consider creating a new **Reporting unit feature** layer that does not include unwanted reporting units. Also select only the land cover classes in the **Report metrics for these classes** parameter that are important for analysis. This tool creates multiple interim rasters and other data layers during processing which may consume considerable disk space in the Scratch Workspace. If the Scratch Workspace resides on a disk with limited disk space, the tool may fail during processing. To prevent this, consider changing the Scratch Workspace in Environments > Workspace > Scratch Workspace to a location with sufficient available disk space, or free up disk space on the disk in which the current Scratch Workspace resides to accommodate the interim datasets.*
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Reporting unit feature** is a zone dataset.

- A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
- When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.
- As the **Reporting unit feature** is a vector dataset, ArcGIS will perform a vector to raster conversion during processing.
 - To better control the vector to raster conversion, the tool defaults the **Snap raster** and the **Processing cell size** to that of the **Land cover grid**. These may be changed from within the tool.
 - If the extent of the **Reporting unit feature** is smaller than that of the **Land cover grid**, the user may wish to check the **Reduce land cover grid to smallest recommended size** option. This creates a temporary land cover raster with the processing extent equal to that of the **Reporting unit feature** to avoid unnecessary processing.
- If the **Reporting unit feature** has overlapping features, the zonal analysis will be performed for each individual feature.
- If a feature in the **Reporting unit feature** zone data does not overlap any cell centers in the secondary class rasters (e.g., **Land cover grid**), those zones will not get converted to the internal zone raster. As a result, those zones will not be represented in the **Output table**.
- The **Land cover classification scheme** must correspond to the **Land cover grid**.
 - Schemes for common land cover datasets are included with ATtILA. Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD 2001 ALL vs. NLCD 2001 LAND). Schemes designated as “ALL” include all land cover classes in reporting unit area calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded. More information about each of the classification schemes supplied with ATtILA may be found in Supplied Land Cover Classification Schemas.
 - In addition to the common land cover schemes, the tool permits a user-defined land cover classification scheme to be used by specifying a **Land cover classification file**. Refer to Land Cover Classification for more information on how ATtILA may be customized for user-specific needs.
 - **NOTE:** *When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. For example, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. See Values Element in ATtILA’s LCC XML Document and Values Window in the Land Cover Classification Editor for more information on excluded values.*
- **Minimum patch size** sets the lower threshold of connected grid cells necessary to be considered a distinct patch. An eight-cell neighborhood is used to determine connectivity.
- The **Maximum separation** is the maximum distance in raster cells for disjunct patches to be considered continuous. The default is zero, requiring all cells in a patch to be continuous.

- **NOTE:** Due to the nature of the algorithm, the number entered is half the amount used in processing. For example, entering a 1 will result in patches 2 cells apart to be considered continuous.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - **NOTE:** For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.
- Field names in the **Output table** follow this naming scheme:
 - **[class Id]_PLGP** - proportion of largest patch area to total patch area for the selected class within the reporting unit (e.g. “for_PLGP” is the name of the field for the proportion of largest forest patch area to all forest area within the reporting unit for the NLCD “Forest” class).
 - **[class Id]_NUM** - The number of distinct patches for the selected class within the reporting unit (e.g. “nat_NUM” is the name of the field for the patch number metric for the NLCD “All natural land use” class).
 - **[class Id]_LRG** - The size in squared map units of the largest class patch within the reporting unit (e.g. “agr_LRG” is the name of the field for the largest patch metric for the NLCD “All agriculture” class).
 - **[class Id]_AVG** - The average size in squared map units of all of the selected class patches within the reporting unit (e.g. “shb_AVG” is the name of the field for the average patch metric for the NLCD “Shrubland” class).
 - **[class Id]_MDN** - The median size in squared map units of all of the selected class patches within the reporting unit (e.g. “for_MDN” is the name of the field for the median patch metric for the NLCD “Forest” class).
 - **[class Id]_DENS** - The density of the selected class patches within the reporting unit given as the number of patches per square kilometer (e.g. “wtl_DENS” is the name of the field for the patch density metric for the NLCD “All wetland” class).
 - **NOTE:** Output field names can be altered by editing the **Land cover classification file** with the Land Cover Classification Editor (Refer to Classes Element in ATtILA's LCC XML Document for general information, or the individual **Land cover classification file** for details).
 - **NOTE:** The output field name for each class is shown as the second item next to the class's check box in **Report metrics for these classes**.
 - **NOTE:** Only the **[class Id]_PLGP** field name is presented in the **Report metrics for these classes**.
 - When the **MDCP** option is selected:
 - **[class Id]_MDCP** - measures the shortest edge-to-edge distance between distinct patches and calculates the mean value in the reporting unit.

- ***NOTE:** The [class Id]_MDCP field will populated with a value of -9999 when no patches of [class Id] are located in the reporting unit.*
- In **Additional Options**, the user may elect to ‘Add Quality Assurance Fields’, ‘Retain Intermediate Layers Generated During Metric Processing’, and/or ‘Record Process Steps Taken During Metric Calculation’.
 - Add Quality Assurance Fields
 - Choosing **QAFIELDS** adds a number of fields to the **Output table** to facilitate quality assurance checking.
 - **PM_OVER** - The percent overlap between the **Reporting unit feature** layer and the **Land cover grid** for each reporting unit.
 - **PM_TOTA** - The total raster zonal area for each reporting unit (zone) after the **Reporting unit feature** layer has been rasterized.
 - **PM_EFFA** - The effective raster area of each reporting unit (zone) after excluded classes have been removed.
 - **PM_EXCA** - The raster area of the excluded classes within each reporting unit (zone) based on exclusions in the **Land cover classification scheme**.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **pm_[class Id]_PatchRast#** (raster) - Rasters created for each class selected in the **Report metrics for these classes** parameter of the tool with values corresponding to the following scheme: -9999 = Excluded, 0 = Other, and 1 thru n = a numbered patch.
 - The processes involved in creating this raster are as follows:
 1. Reclass the input land cover grid so that cells representing the selected class = 3, cells with values marked as excluded = -9999, and all other land cover cells = 0.
 2. If a **Maximum separation** value is NOT provided: Assign unique values to each distinct clump of class cells by use of the regiongroup command using the eight neighbor rule to determine connectivity.
 3. If a **Maximum separation** value is provided: Create a single value class raster by setting to NULL cells in the reclassified raster with values less than 3. Use a euclidean distance operation on the class patch/NULL raster to buffer the classes out by the supplied maximum separation distance. Assign unique values to each distinct clump of class cells by use of the regiongroup command using the eight neighbor rule to determine connectivity. Trim the buffered regiongroup patches to the original boundaries of the patch clumps by use of the con statement and the reclass grid (if it's a patch in the reclass grid (i.e., its value = 3) assign it the regiongroup number, else leave the other values (-9999, 0) alone.

4. If a **Minimum patch size** value is provided: Eliminate all patches below the **Minimum patch size** (i.e., any regiongroup patch with COUNT field entry less than the **Minimum patch size**, set its VALUE field entry to 0).
5. Add the excluded class areas back to the raster if any are indicated.

- **pm_[class Id]_PatchPoly#** (vector) - A polygon feature class of all patches of [class Id] derived from the **pm_[class Id]_PatchRast#** using the Raster to Polygon conversion operation. The Simplify polygons option was unchecked to force the polygon edges to conform to the cell boundaries of the input raster. The gridcode field contains the unique identifier for each distinct patch.
- **pm_[class Id]_PatchPoly_Diss#** (vector) - A polygon feature class of all patches of [class Id] after performing a Dissolve on the **pm_[class Id]_PatchPoly#** feature class using gridcode as the dissolve field. Gridcode is the unique identifier for each distinct patch.
 - ***NOTE:** Due to the complex pattern of some patches, the dissolve function may not create a single multipart polygon for each distinct patch. A quick check to see if duplicate values are found in the gridcode field by using either the Frequency tool or by the field Summarize... option.*
- **pm_[class Id]_PatchCentroids#** (vector) - A point feature class depicting the cell centroids of cells delimiting patches in the **pm_[class Id]_PatchRast#** raster with the unique patch identifier found in the grid_code field.
 - ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
 - ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*
- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - ***NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.*
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.

- Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
- Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
 - **NOTE:** ATtILA will report the name and location of the log file in the Tool Details' Messages section.

Syntax

PM (Reporting_unit_feature, Reporting_unit_ID_field, Land_cover_grid, Land_cover_classification_scheme, Land_cover_classification_file, Report_metrics_for_these_classes, Minimum_Patch_Size, Maximum_Separation, Output_table, {MDCP}, {Processing_cell_size}, {Snap_raster}, Select_options, {Reduce_land_cover_grid_to_smallest_recommended_size})

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Land_cover_grid	The raster dataset representing land cover classes upon which core and edge area metrics will be derived within each Reporting unit feature. The class input must be an integer raster layer.	Raster Layer
Land_cover_classification_scheme	The land cover classification schemes included in ATtILA and a User Defined option. The default schemes correspond to common input land cover datasets. Two schemes are supplied	String

Parameter	Explanation	Data Type
	for each dataset included in ATtILA:	
Land_cover_classification_file	The full pathname to the user-defined .xml file for custom or non-standard land cover classification schemes. Pathname is automatically filled when a default scheme is selected.	File
Report_metrics_for_these_classes	A list of the land cover classes and metric combinations for processing. Check the box to calculate metrics for each land cover class and/or combination class desired within the reporting units.	Multiple Value
Minimum_patch_size	The minimum patch size, measured in number of grid cells, necessary to be considered a distinct patch. Connectivity is defined within the immediate eight-cell neighborhood. Cells of the same class to the right, left, above, below, or diagonal to each other are considered as part of the same patch. The input must be an integer value.	Long
Maximum_separation	The maximum distance separating non-adjointing cells before those cells are considered a distinct patch. The value entered is in number of cells. The processing cell size determines the actual linear distance. Non-adjointing cells are buffered by the maximum distance. If the buffers	Long

Parameter	Explanation	Data Type
	touch, the non-adjointing cells are considered to be one patch. The input must be an integer value.	
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved within a file geodatabase.	Table
MDCP (Optional)	Specifies whether the mean distance to closest patch (MDCP) metric will be included in the output table.	Boolean
Processing_cell_size (Optional)	The Processing cell size for the zonal operation. The default Processing cell size is the cell size of the input raster land cover data. Optionally, the user may select a different cell size.	Analysis cell size
Snap_raster (Optional)	The raster that the cell alignment of the Land cover grid and rasterized Reporting unit feature layer will be matched to during processing. The default Snap raster is the Land cover grid.	Raster Layer
Select_options	Tool options available to provide additional information:	Multiple Value
Reduce_land_cover_grid_to_smallest_recommended_size (Optional)	Specifies whether the Land cover grid extent will be clipped to that of the Reporting unit feature layer during processing.	Boolean

Code Samples

Patch Metrics example 1 (Python window)

This example creates a table with the number of patches (NUM), largest patch size (LRG), average patch size (AVG), the number of patches per square kilometer (DENS), and proportion of largest patch area to total patch area (PLGP) within each reporting unit for the forest land cover class as defined in the 'NLCD LAND' land cover classification file. Patches needed to be clumps of four or more connected cells, but adjacent patches would be considered connected if they were within the maximum patch separation distance. The options to retain the intermediate calculation products, to record the process steps into a log file, and to generate the quality assurance fields have also been selected.

NOTE: *ATtILA tools that utilize the **Report metrics for these classes** parameter will clear all class selections when the tool is opened. This results in setting any variable set for that parameter to **None** when calling the tool from a script. Because of this, the standard scripting syntax (e.g. `arcpy.ATtILA.LCP(variable 1, variable 2, variable 3, etc.)` will fail. To overcome this idiosyncrasy, use the scripting example provided below. When using this technique, all variables are passed in as strings.*

NOTE: *By running scripts which launch an ATtILA tool via 'metric.run' within ArcGIS Pro's Python Window, all geoprocessing functions run by that tool can be viewed in the History Pane.*

```
import arcpy
from arcpy.sa import *

from ATtILA2 import metric

toolPath = "D:\\Destination Folder\\ToolboxSource\\ATtILA2\\scripts\\PatchMetrics_ArcGIS.py"
Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Land_cover_grid = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Land_cover_classification_scheme = "NLCD LAND"
Land_cover_classification_file = "D:\\Destination Folder\\ToolboxSource\\LandCoverClassifications\\NLCD
LAND.xml"
Report_metrics_for_these_classes = "for - [for_PLGP] Forest"
Minimum_patch_size = "4"
Maximum_separation = "2"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_PM"
MDCP = "true"
Processing_cell_size = "30"
Snap_raster = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric
Calculation';'LOGFILE - Record Process Steps Taken During Metric Calculation';'QAFIELDS - Add
Quality Assurance Fields'"
Reduce_land_cover_grid_to_smallest_recommended_size = ""

metric.runPatchMetrics(
    toolPath,
    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Land_cover_grid,
    Land_cover_classification_scheme,
    Land_cover_classification_file,
    Report_metrics_for_these_classes,
    Minimum_patch_size,
    Maximum_separation,
    Output_table,
    MDCP,
    Processing_cell_size,
    Snap_raster,
    Select_options,
    Reduce_land_cover_grid_to_smallest_recommended_size
)
```

Patch Metrics example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up initial environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Constructing the ATtILA metric output table: Watersheds_PM
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Creating Patch Grid for Class:for
|YYYY-MM-DD hh:mm:ss| Reclassing land cover to Class:for = 3, Other = 0, and Excluded = -9999...
|YYYY-MM-DD hh:mm:ss| Connecting clusters of Class:for within maximum separation distance...
|YYYY-MM-DD hh:mm:ss| Assigning unique numbers to each unconnected cluster of Class:for...
|YYYY-MM-DD hh:mm:ss| Eliminating clusters below minimum patch size...
|YYYY-MM-DD hh:mm:ss| Patch Grid Completed for Class:for
|YYYY-MM-DD hh:mm:ss| Calculating Patch Numbers by Reporting Unit for Class:for
1 out of 4 loops, 25.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
2 out of 4 loops, 50.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
3 out of 4 loops, 75.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
4 out of 4 loops, 100.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
|YYYY-MM-DD hh:mm:ss| Patch analysis has been run for Class:for
|YYYY-MM-DD hh:mm:ss| Calculating MDCP for Class:for
|YYYY-MM-DD hh:mm:ss| Converting raster patches to polygons...
|YYYY-MM-DD hh:mm:ss| Dissolving patch polygons by value field...
|YYYY-MM-DD hh:mm:ss| Converting raster patch cells to centroid points...
|YYYY-MM-DD hh:mm:ss| Analyzing MDCP by reporting unit...
1 out of 4 loops, 25.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
2 out of 4 loops, 50.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
3 out of 4 loops, 75.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
4 out of 4 loops, 100.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
|YYYY-MM-DD hh:mm:ss| MDCP analysis has been run for Class:for
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

People in the Landscape

Human land uses and associated changes to land cover are one source of environmental stressors. For example, agriculture usually produces higher levels of nutrients like nitrogen and phosphorus than natural land cover. Excess export of nitrogen and phosphorus to streams leads to eutrophication of streams, lakes, bays, and other water bodies. Most of these nutrients come from non-point sources like animal waste from livestock operations and fertilizer applications on agricultural fields, golf courses, and residential lawns. Urban land uses increase the amount of impervious surfaces (roads, rooftops, parking lots, etc.), which increases the volume and force of precipitation runoff. Increased runoff leads to higher risk of flooding and also carries larger amounts of pollutants and sediment into surface waters.

Population density and change can be used to assess the potential magnitude of human impacts on the environment. Population growth can contribute to changes to land cover and landscape patterns, such as forest fragmentation, loss of interior forest, and increased impervious surface, which can put pressures on ecosystems. Looking at population change can be helpful in identifying reporting units experiencing high rates of growth, making them vulnerable to adverse impacts, or to analyze the history of land use changes in a region.

Roads are another type of human-related stressor. Density of roads may be a surrogate for population density. Roads near streams or other water bodies increase the potential for pollution from runoff. Bridges or other stream crossing may alter the flow hydrology of streams as well as provide access points for dumping.

The People in the Landscape toolset includes the following metric tools:

- **Facility Land Cover Views** counts the number of facilities in each reporting unit polygon that have less than a certain percentage of selected land cover classes within a given distance and creates an output table. The percentage threshold value and buffer distance are provided by the user.
- **Intersection Density** estimates the intersection density of roads within a given search radius for any pixel in the output raster. Intersections are defined as any point where three or more roads meet and density is calculated using kernel density, where closer intersections are weighted higher than further intersections.
- **Land Cover Coefficient Calculator** encompasses the nutrient (nitrogen and phosphorus) loading and the impervious surface estimate metrics contained in an earlier release of ATtILA. For nitrogen (N), phosphorus (P), and impervious surface, coefficients for various land cover classes are applied (multiplied) to a grid of land cover. Coefficient values based on literature reviews are included as default values for the default land cover classification schemes, but the user is free to alter these values to suit his/her individual assessment needs. The user may also use a land cover classification other than the ones supplied with this release of ATtILA, which then requires the user to enter their own custom coefficients. The land cover coefficient calculator may also be used for other custom purposes beyond land cover. The only required input datasets are a grid with discrete values and a polygon theme of reporting units. Custom coefficient values are also required if the default values are not to be used.
- **Pedestrian Access and Availability** estimates the availability of parkland per person based on accessibility of parkland. Accessibility is determined using walkable routes along a cost surface raster and a user-specified maximum travel distance. The output is a raster.

- **Population Density Metrics** calculates the population count and density by reporting unit; it can also be used to calculate population change. Population density is reported as number of people per km². Population change is reported as percentage change in population by reporting unit. The population change tool is quite versatile, like many of the metrics in ATtILA, and can be used beyond its stated purpose. For example, instead of selecting population counts from different time periods, the user could select fields containing counts of male and female members of a population at a given time. The POPCHG metric could then be used to show the percentage difference between the number of males and females for the given reporting unit. Another possible application would be to use ATtILA to calculate the total area of a land cover class in a reporting unit for two different time periods. Using these two themes as inputs into the POPCHG metric, the user could then calculate the percentage change in the land cover type over the given time period.
- **Population in Floodplain Metrics** calculates percentage of total population living in floodplain areas for each reporting unit polygon and creates an output table. Counts for the total population in the reporting unit and within the floodplain area are also reported.
- **Population Land Cover Views** estimates the percentage of the population in each reporting unit that have potential views of selected land cover classes within a given distance from their location and creates an output table. A land cover class is visible if it is found within the neighborhood defined by the View radius and in a group of cells larger than or equal to the Minimum visible patch size. The Minimum visible patch size and View radius are provided by the user. Additional fields for total population count within the reporting unit, the total population count in the reporting unit with potential views, the total population count in the reporting unit without potential views, and the percentage of the total population in the reporting unit without potential views are also output.
- **Population within Zone Metrics** calculates percentage of total population living within or near a zone for each reporting unit polygon and creates an output table. Counts for the total population in the reporting unit and within the zone are also reported.
- **Road Density Metrics** includes road density, roads crossing streams, and roads near streams metrics. The road density metrics require only an input reporting unit polygon theme and a line theme of roads, although any other line theme could be substituted. To calculate roads near streams and roads crossing streams in addition to road density, a line theme of streams is also required as an input. Again, any line theme which makes sense for the analysis could be substituted. For example, this metric could be used to assess the density of power lines or to determine the frequency of railroad tracks crossing roads.

Tool Facility Land Cover Views

Available with Spatial Analyst license.

Summary

Counts the number of facilities in each reporting unit polygon that have less than or equal to a threshold percentage of selected land cover classes within a given distance and creates an output table. The percentage threshold value and buffer distance are provided by the user.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- ***NOTE:** For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- ***NOTE:** It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.*
- ***NOTE:** Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.
- If a feature in the **Reporting unit feature** zone data does not contain any features from the **Facility feature**, those zones will not be represented in the **Output table**.
- The tool buffers a copy of the points in the **Facility feature** and uses Tabulate Area to calculate the percentage of the buffer area that is a selected land cover class. Tabulate Area performs a vector to raster conversion of the buffer area to use as zones for its operation. Results may vary depending on the settings of the Raster Analysis Cell Size and Snap Raster Environment settings

- To better control the vector to raster conversion, the tool defaults the Snap raster and the Processing cell size to that of the **Land cover grid**. These may be changed from within the tool.
- If a large size disparity exists between the extents of the **Reporting unit feature** and the **Land cover grid**, the user may wish to set the Extent in Environment Settings > Processing Extent to the smaller of the two to avoid unnecessary processing.
- **WARNING:** - *If the buffer zone for a facility does not overlap any cell centers in the **Land cover grid** (e.g., the **View radius** is smaller than the **Processing cell size** or a facility exists beyond the extent of the **Land cover grid**), that facility will not be included in any metric calculations. As a result, values recorded for any reporting unit where facilities were dropped will not be accurately represented in the **Output table**. ATtILA will alert the user when the calculated number of facilities above and below the **View threshold** do not add up to the total number of facilities located within a reporting unit. If intermediate products are saved (see **Additional options** below), the table, **flcy_ViewLCP_RUID[view radius]_#**, can be used to identify the specific facilities with missing land cover information.*
- The **Land cover classification scheme** must correspond to the Land cover grid.
 - Schemes for common land cover datasets are included with ATtILA. Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD 2001 ALL vs. NLCD 2001 LAND). Schemes designated as “ALL” include all land cover classes in reporting unit area calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded. More information about each of the classification schemes supplied with ATtILA may be found in Supplied Land Cover Classification Schemas.
 - In addition to the common land cover schemes, the tool permits a user-defined land cover classification scheme to be used by specifying a **Land cover classification file**. Refer to Land Cover Classification for more information on how ATtILA may be customized for user-specific needs.
 - **NOTE:** *When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. For example, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. See Values Element in ATtILA’s LCC XML Document and Values Window in the Land Cover Classification Editor for more information on excluded values.*
- **Facility feature** must be a vector point dataset. Multipoint features are not accepted. The **View radius** specifies the distance around the point features for which buffer zones are created.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - **NOTE:** *For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.

- Field names in the **Output table** follow this naming scheme:
 - **[class flcvField] or [class Id]_Low[view threshold]** - The number of facilities in the reporting unit with percentages of the selected land cover class below the **View threshold** value in facility's buffer area.
 - **[class flcvField]** is used if a flcvField attribute is provided in the class's element in the selected **Land cover classification file**.
 - (e.g. **LowForestViews** for <class Id="for" Name="Forest Land Cover" filter="" flcvField="LowForestViews">)
 - **[class Id]_Low[view threshold]** - is used if a flcvField attribute is not provided.
 - (e.g. **for_Low10** for <class Id="for" Name="Forest Land Cover" filter=""> and a 10 meter view threshold is used)
 - **NOTE:** *Output field names can be altered by editing the **Land cover classification file** with the Land Cover Classification Editor (Refer to Classes Element in ATtILA's LCC XML Document for general information, or the individual **Land cover classification file** for details).*
 - **NOTE:** *The output field name for each class is shown as the second item next to the class's check box in **Report metrics for these classes**.*
- **FAC_Count** will be included in the **Output table** as an ancillary field. It reports the total number of facilities in the reporting unit.
- An additional ancillary field is provided in the **Output table** to report the number of facilities in the reporting unit with percentages of the selected land cover class at or above the **View threshold**. The field name follows a similar naming scheme regarding the flcvField class attribute in the **Land cover classification file** as described above. The ancillary field name in this **Output table** is as follows:
 - **[class flcvField]_High[view threshold] or [class Id]_High[view threshold]** - The number of facilities in the reporting unit with percentages of the selected land cover class at or above the **View threshold**.
 - **NOTE:** *The calculated number of facilities above and below the **View threshold** should add up to the total number of facilities located within a reporting unit.*
- In **Additional Options**, the user may elect to 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **flcv_Facility[view radius]_#** (vector) - Copy of the **Facility feature** with all but the OBJECTID and Shape fields removed.

- **flcv_FacilityRUID[view radius]_#** (vector) - Copy of the **flcv_Facility[view radius]_#** feature with the overlying **Reporting unit feature**'s ID attached via Identity. A facility point that occurs in overlapping **Reporting Unit feature** areas will produce a separate output feature for each of the **Reporting unit features** it occurs in.
- **flcv_ViewBuffer[view radius]_#** (vector) - **flcv_FacilityRUID[view radius]_#** feature buffered to the **View radius**. The attribute table will contain fields for the **Reporting unit feature** ID, the buffer distance (**View radius**), and a field named, 'ORIG_FID'. The ORIG_FID is used as a unique identifier for future operations to tag each facility/reporting unit combination.
- **flcv_TabArea#** (table) - Tabulate Area output table using **flcv_ViewBuffer[view radius]_#**'s ORIG_FID field as the zone identifier and the **Land cover grid** as the raster to be summarized.
- **flcv_ViewLCP[view radius]_#** (table) - Table containing the percentage of each facility's view buffer area composed of selected land cover classes. Additional fields in the attribute table include: FLCV_TOTA, FLCV_EFFA, FLCV_EXCA, and two separate fields for each selected land cover class named, [class Id]_Below and [class Id]_Above.
 - **FLCV_TOTA** is the sum of the **Land cover grid** cell areas within the **View radius**.
 - **FLCV_EFFA** is the sum of the **Land cover grid** cell areas within the **View radius** that are not marked as "Excluded" in the **Land cover classification scheme XML** file.
 - **FLCV_EXCA** is the sum of the **Land cover grid** cell areas within the **View radius** that are marked as "Excluded" in the **Land cover classification scheme XML** file.
 - **[class Id]_Below** is a binary field where land cover class percentage below **View threshold** = 1, and land cover class percentage at or above **View threshold** = 0.
 - **[class Id]_Above** is a binary field where land cover class percentage below **View threshold** = 0, and land cover class percentage at or above **View threshold** = 1.
- **flcv_ViewLCP_RUID[view radius]_#** (table) - The **flcv_ViewLCP[view radius]_#** table with the **Reporting unit feature ID** field attached. The ID was obtained by a join with the **flcv_FacilityRUID[view radius]_#** table and the common 'ORIG_FID' field.
- **flcv_ViewStats[view radius]_#** (table) - Summary Statistics output table where each [class Id]_Below field and [class Id]_Above field in the **flcv_ViewLCP_RUID[view radius]_#** table is summed by **Reporting unit feature ID**.
- **NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.

- **NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.
- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - **NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
 - **NOTE:** ATtILA will report the name and location of the log file in the Tool Details’ Messages section.

Syntax

FLCV (Reporting_unit_feature, Reporting_unit_ID_field, Land_cover_grid, Land_cover_classification_scheme, Land_cover_classification_file, Report_metrics_for_these_classes, Facility_feature, View_radius, View_threshold, Output_table, {Processing_cell_size}, {Snap_raster}, {Select_options})

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer

Parameter	Explanation	Data Type
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Land_cover_grid	The raster dataset representing land cover classes to be summarized within each Reporting unit feature. The grid input must be an integer raster layer.	Raster Layer
Land_cover_classification_scheme	The land cover classification schemes included in ATtILA and a User Defined option. The default schemes correspond to common input land cover datasets. Two schemes are supplied for each dataset included in ATtILA:	String
Land_cover_classification_file	The full pathname to the user-defined .xml file for custom or non-standard land cover classification schemes. Pathname is automatically filled when a default scheme is selected.	File
Report_metrics_for_these_classes	A list of the land cover classes and metric combinations for processing. Check the box to calculate metrics for each land cover class and/or combination class desired within the reporting units.	Multiple Value
Facility_feature	The vector point dataset that provides the basis for the buffer zones.	Feature Layer

Parameter	Explanation	Data Type
	Land cover proportions are calculated for the area within the buffer zones.	
View_radius	The distance around the Facility point features which comprises the buffer zones. The value must be an integer. If the distance linear units are not specified or are entered as Unknown, the linear unit of the input features' spatial reference is used.	Linear Unit
View_threshold	Threshold number used to determine if the percentage of a selected land cover class within the View radius is identified as low. Low percentage values include the View threshold.	Double
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved within a file geodatabase.	Table
Processing_cell_size (Optional)	The Processing cell size for the zonal operation. The default Processing cell size is the cell size of the input raster land cover data. Optionally, the user may select a different cell size.	Analysis Cell Size
Snap_raster (Optional)	The raster that the cell alignment of the Land cover grid and rasterized Reporting unit feature layer will be matched to during processing. The default Snap raster is the Land cover grid.	Raster Layer
Select_options (Optional)	Tool options available to provide additional	Multiple Value

Parameter	Explanation	Data Type
information:		

Code Samples

Facility Land Cover Views example 1 (Python window)

This example returns a table with number of daycares in the reporting unit and the number of daycares in the reporting unit with less than 10 percent coverage of natural land cover within 250 meters of the facility. The options to retain the intermediate calculation products, and to record the process steps into a log file have also been selected.

NOTE: *ATtILA tools that utilize the **Report metrics for these classes** parameter will clear all class selections when the tool is opened. This results in setting any variable set for that parameter to **None** when calling the tool from a script. Because of this, the standard scripting syntax (e.g. `arcpy.ATtILA.LCP(variable 1, variable 2, variable 3, etc.)` will fail. To overcome this idiosyncrasy, use the scripting example provided below. When using this technique, all variables are passed in as strings.*

NOTE: *By running scripts which launch an ATtILA tool via 'metric.run' within ArcGIS Pro's Python Window, all geoprocessing functions run by that tool can be viewed in the History Pane.*

```
import arcpy
from arcpy.sa import *

from ATtILA2 import metric

toolPath = "D:\\Destination Folder\\ToolboxSource\\ATtILA2\\scripts\\FacilityLandCoverViews_ArcGIS.py"
Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Land_cover_grid = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Land_cover_classification_scheme = "NLCD LAND"
Land_cover_classification_file = "D:\\Destination Folder\\ToolboxSource\\LandCoverClassifications\\NLCD
LAND.xml"
Report_metrics_for_these_classes = "NI - [NI_Low] All Natural Land Use"
Facility_feature = "D:\\pyexamples\\data.gdb\\Daycares"
View_radius = "250 Meters"
View_threshold = "10"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_FLCV"
Processing_cell_size = "30"
Snap_raster = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric
Calculation'; 'LOGFILE - Record Process Steps Taken During Metric Calculation'"

metric.runFacilityLandCoverViews(
    toolPath,
    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Land_cover_grid,
    Land_cover_classification_scheme,
    Land_cover_classification_file,
    Report_metrics_for_these_classes,
    Facility_feature,
    View_radius,
    View_threshold,
    Output_table,
    Processing_cell_size,
    Snap_raster,
```

```
Select_options
)
```

Facility Land Cover Views example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Creating a copy of the Facility feature. Intermediate: flcv_Facility250_0
|YYYY-MM-DD hh:mm:ss| Deleting unnecessary fields from flcv_Facility250_0
|YYYY-MM-DD hh:mm:ss| Assigning reporting unit ID to flcv_Facility250_0. Intermediate:
flcv_FacilityRUID250_0
|YYYY-MM-DD hh:mm:ss| Buffering flcv_FacilityRUID250_0 to 250 Meters. Intermediate: flcv_ViewBuffer250_0
|YYYY-MM-DD hh:mm:ss| Constructing facility buffer land cover proportions table. Intermediate:
flcv_ViewLCP250_0
|YYYY-MM-DD hh:mm:ss| Constructing the ATtILA metric output table: Watersheds_FLCV
|YYYY-MM-DD hh:mm:ss| Generating a zonal tabulate area table for view buffer areas
|YYYY-MM-DD hh:mm:ss| Completed zonal tabulate area table. Intermediate: flcv_TabArea0
|YYYY-MM-DD hh:mm:ss| Processing the tabulate area table and computing metric values
|YYYY-MM-DD hh:mm:ss| Finding facilities with views below threshold limit for selected class(es)...
|YYYY-MM-DD hh:mm:ss| Summarizing facilities with low views by Reporting Unit...
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Intersection Density

Available with Spatial Analyst license.

Summary

Estimates the intersection density of roads within a given search radius for any pixel in the output raster. Intersections are defined as any point where three or more roads meet and density is calculated using kernel density, where closer intersections are weighted higher than further intersections.

The kernel density operation uses the Planar option for calculations. For accurate results, the analysis should to be performed at a local scale with a projection that accurately maintains the correct distance and area.

Optionally, divided roads (i.e., roads of the same classification that run parallel for a significant distance) in the input feature class can be merged into single center lines before determining the number and location of intersections.

Usage

- This tool is designed to use a vector line dataset that represent roads, but any vector line dataset may be used. Caution should be exercised when interpreting results, particularly when using line datasets that do not represent roads.
- **NOTE:** *To ensure desired results, the **Road feature** must be in a projected coordinate system with linear units of measure.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- If no specific geoprocessing environment extent is set, ATtILA will temporarily set it to match the **Road feature**. It will be reset when the tool finishes.
- When the optional **Merge divided roads** selection is checked, the tool allows the user to select a **Merge field** and a **Merge distance** to merge road segments that are close to and run parallel to each other into one center line. Merging close, parallel road segments may be desirable if the user wishes to count instances of a road crossing a feature such as a divided highway or boulevard as a single intersection instead of multiple ones. For more information about Merge Divided Roads, search on “How Merge Divided Roads works” in the ArcGIS help documentation.
 - **Merge field** is the field in the **Road feature** attribute table that contains road classification values. Only roads with the same classification value are available to be merged. A road segment with a classification of 0 in the **Merge field** is locked and will not be merged regardless of the presence of any neighboring, parallel segments.
 - If no **Merge field** is supplied, all parallel roads within the specified **Merge distance** will be merged.

- **Merge distance** is the maximum separation distance for two parallel road segments of the same classification to be merged.
- ***NOTE:** If the input feature class for **Road feature** is the [prefix]_RdsIntDens feature class produced by ATtILA's Process Roads for EnviroAtlas Analyses utility, the road features have already been merged.*
- The parameters **Output Coordinate System**, **Density raster cell size**, **Density raster search radius**, and **Density raster area units** are used to control the Spatial Analyst Kernel Density function for generating the final **Output raster**. For more information about the Kernel Density operation, search on “How Kernel Density works” in the ArcGIS help documentation.
 - **NOTE:** *The default values for **Density raster cell size** (10) and **Density raster search radius** (750) are the values used in EnviroAtlas's community-level 'Estimated Intersection Density of Walkable Roads' (Fact Sheet).*
- Final output is written to the location specified in the **Output raster** parameter. The **Output raster** may be saved in a file folder or a geodatabase.
 - **NOTE:** *For most consistent results, It is highly recommended that tool output be saved to a file geodatabase.*
 - When saving in a File Geodatabase, no extension is permitted for the **Output raster** name.
 - When saving in a file folder, the following extensions are allowed for the **Output raster** name:
 - .img for ERDAS IMAGINE
 - .tif for TIFF
 - No extension for Esri Grid
- In **Additional Options**, the user may elect to ‘Retain Intermediate Layers Generated During Metric Processing’, and/or ‘Record Process Steps Taken During Metric Calculation’.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **id_[Road_feature]_Prj_#** - A copy of the **Road feature** projected into the **Output Coordinate System**. This feature class is only produced if the coordinate system for the **Road feature** is different than the **Output Coordinate System**.
 - **id_[Road_feature]_SglPrt_#** - A copy of the **Road feature** or **id_[Road_feature]_Prj_#** feature class with any multipart features separated into singlepart features. This feature class is only produced when the optional **Merge divided roads** selection is checked.
 - **id_[Road_feature]_Merged_#** - A copy of **id_[Road_feature]_SglPrt_#** where matched pairs of divided road lanes within the **Merge distance** are combined into single-line road features. If a **Merge field** was supplied, only parallel, proximate roads of equal classification were merged. Road features with a value of 0 (zero) in the

Merge field are locked, and were not merged. If the **Merge field** parameter was left blank, all parallel roads within the specified distance were merged. This feature class is only produced when the optional **Merge divided roads** selection is checked.

- **id_[Road_feature]_UnSplit_#** - A feature class where input line features with coincident endpoints are aggregated into a single unsplit line. Depending on the tool options selected, the input line source is one of the following: **Road feature**, **id_[Road_feature]_Prj_#**, or **id_[Road_feature]_Merged_#**.
- **id_[Road_feature]_Intersections_#** - A point feature class showing locations of all road intersections. This feature class is produced by performing an Intersect operation on **id_[Road_feature]_UnSplit_#** with itself, and then deleting all identical points based on the “Shape” field.
- ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
- ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*
- Intermediate products are saved to one of the following locations:
 - When the **Output raster** is saved in a File Geodatabase, intermediate products are placed in the same file geodatabase.
 - When the **Output raster** is saved in a file folder, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output raster**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - ***NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.*
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.

- **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
- **NOTE:** ATtILA will report the name and location of the log file in the Tool Details' Messages section.

Syntax

ID (Road_feature, {Merge_divided_roads}, {Merge_field}, Merge_distance, Output_Coordinate_System, Density_raster_cell_size, Density_raster_search_radius, Density_raster_area_units, Output_raster, {Select_options})

Parameter	Explanation	Data Type
Road_feature	The vector line dataset representing roads or other linear features to be measured.	Feature Layer
Merge_divided_roads (Optional)	Specifies whether to merge matched pairs of divided roads into single-line road features before calculating intersection points. Matched pairs of roads are merged if they are the same road class, trend generally parallel to one another, and are within the merge distance apart.	Boolean
Merge_field (Optional)	The field that contains road classification information. Only parallel, proximate roads of equal classification will be merged. A value of 0 (zero) locks a feature to prevent it from participating in merging. This parameter may be left blank. If blank, all parallel roads within the specified distance will be merged.	
Merge_distance	The minimum distance apart, in the specified units, for equal-class, relatively parallel road features to be merged. This distance must be greater than zero.	Linear Unit
Output_Coordinate_System	The coordinate system used for the output data.	Coordinate System

Parameter	Explanation	Data Type
	Its linear units are used to define the distance units in Merge distance, Density raster cell size, and Density raster search radius parameters.	
Density_raster_cell_size	The cell size of the output raster that will be created. Units use the linear unit of the Output Coordinate System.	Double
Density_raster_search_radius	The search radius within which to calculate density. Units are based on the linear unit of the projection of the output spatial reference. For example, if the units are meters—to include all features within a one-mile neighborhood—set the search radius equal to 1609.344 (1 mile = 1609.344 meters). The default search radius is computed specifically for the input dataset using a spatial variant of Silverman’s Rule of Thumb (Silverman, 1986) that is robust enough for spatial outliers (points that are far away from the rest of the points). See the usage tips for a description of the algorithm.	Double
Density_raster_area_units	The area units of the output density values.	String
Output_raster	The output intersection density raster. It is always a floating point raster.	Raster Dataset
Select_options (Optional)	Tool options available to provide additional	Multiple Value

Parameter	Explanation	Data Type
information:		

Code Samples

Intersection Density example 1 (Python window)

This example returns a raster with an estimate of the number of road intersections per square kilometer for every cell in the output raster. The option to merge roads of the same class in close proximity to each other was not selected. The options to retain the intermediate calculation products, and to record the process steps into a log file have also been selected.

```
import arcpy
from arcpy.sa import *

Road_feature = "D:\\pyexamples\\data.gdb\\Roads"
Merge_divided_roads = ""
Merge_field = ""
Merge_distance = ""
Output_Coordinate_System =
    "PROJCS['Albers_Conical_Equal_Area',GEOGCS['GCS_North_American_1983',DATUM['D_North_American_1983',SPHEROID['GF
Density_raster_cell_size = "10"
Density_raster_search_radius = "750"
Density_raster_area_units = "SQUARE KILOMETERS"
Output_raster = "D:\\pyexamples\\results.gdb\\Road_Intersection_Hotspots"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric
Calculation';'LOGFILE - Record Process Steps Taken During Metric Calculation'"

    Road_feature,
    Merge_divided_roads,
    Merge_field,
    Merge_distance,
    Output_Coordinate_System,
    Density_raster_cell_size,
    Density_raster_search_radius,
    Density_raster_area_units,
    Output_raster,
    Select_options
)
```

Intersection Density example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Using Roads's extent for geoprocessing steps.
|YYYY-MM-DD hh:mm:ss| Unsplitting Roads. Intermediate: id_Roads_UnSplit_0
|YYYY-MM-DD hh:mm:ss| Finding intersections. Intermediate: id_Roads_Intersections_0.
|YYYY-MM-DD hh:mm:ss| Deleting identical intersections...
|YYYY-MM-DD hh:mm:ss| Performing kernel density: Result saved as Road_Intersection_Hotspots.
Adding Road_Intersection_Hotspots to Map view
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Land Cover Coefficient Calculator

Available with Spatial Analyst license.

Summary

Calculates metrics based on coefficient values (multipliers) associated with specific land cover classes within each reporting unit polygon and creates an output table.

Within the reporting unit polygons, the area of each land cover class present is multiplied by its supplied coefficient. The products for each land cover class are summed to produce a value for each coefficient type.

In this version of ATtILA, this tool produces three metrics: percent impervious area, nitrogen loading (kg/ha/yr), and phosphorus loading (kg/ha/yr).

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- **NOTE:** *To ensure desired results, the **Land cover grid** must be in a projected coordinate system with linear units of measure.*
- Three coefficients for which metrics are calculated are included in this tool: IMPERVIOUS, NITROGEN, and PHOSPHORUS. Coefficient values are stored in the **Land cover classification file**. For more information on how coefficients are coded into the **Land cover classification file**, please refer to the [Coefficients Element](#) and [Values Element](#) sections of [ATtILA's LCC XML Document](#). To view specific coefficient values associated with supplied land cover schemes, refer to the particular scheme in [Supplied Land Cover Classification Schemas](#). To view and optionally edit coefficient values in a **Land cover classification file**, use either the [Create or Modify Land Cover Classification](#) tool or a text editor.

- IMPERVIOUS coefficient - Represents an estimate of the percent impervious area for each land cover class. For example, many land cover classification schemes contain multiple “developed” land cover classes based on the intensity of development. Generally, high-intensity developed areas (e.g. city centers) contain a greater proportion of impervious surfaces per unit area than low-intensity developed areas (e.g. rural residential). Hence, a high-intensity developed land cover class should be assigned a higher coefficient value (perhaps 0.9 for 90% impervious) than that of a medium- or low-intensity developed land cover class (the coefficient values of which perhaps range from 0.7 to 0.25 for 70% to 25% impervious). Generally, percent impervious area in land cover classes other than developed is very low or zero.
- NITROGEN coefficient - Represents an estimate of the amount of nitrogen output (stream loading) in kilograms per hectare per year for each land cover class. Nitrogen loading is generally found to be highest in land cover classes that are primarily human-use oriented (agricultural and developed land); hence ATtILA assigns a higher nitrogen loading coefficient value to these classes as compared to natural land cover classes. Considering only the various natural land cover classes, nitrogen loading is generally found to be highest in classes with high vegetative biomass per unit area (forest) and ATtILA assigns a higher nitrogen loading coefficient value than those of other natural land cover classes with lower biomass per unit area.
- PHOSPHORUS coefficient - Represents an estimate of the amount of phosphorus output (stream loading) in kilograms per hectare per year for each land cover class. Phosphorus loading follows a similar pattern as nitrogen loading—it is found to be highest in human-use oriented land cover classes to which ATtILA assigns higher phosphorus loading coefficient values versus natural land cover classes. Again, as with nitrogen loading, within the various natural land cover classes ATtILA assigns the highest phosphorus loading coefficient values to classes with high vegetative biomass per unit area (forest), with coefficient values decreasing with diminishing biomass per unit area.
- Reporting unit metrics are based on the coefficient values as coded into the **Land cover classification file** for each land cover class. These metrics are calculated in the following manner:
 - IMPERVIOUS - The percent impervious metric uses a percent area method to calculate values. For each land cover class, the area within each reporting unit is multiplied by the impervious coefficient associated with that class, resulting in a product for each class. Products for all land cover classes are summed to produce the percent impervious area metric for each reporting unit.
 - **NOTE:** *The percent area method expects coefficient values to be expressed as decimal values between 1 and 0 (e.g., 0.9 for 90%).*
 - **NOTE:** *The output calculated value is supplied as a percentage.*
 - NITROGEN - The nitrogen loading metric uses a per-unit-area method to calculate values. For each land cover class, the area within each reporting unit is converted to hectares then multiplied by the nitrogen coefficient associated with that class, resulting in a product for each class. Products for all land cover classes are summed, then divided by the total number of hectares within the reporting unit to provide an average nitrogen loading value across each reporting unit in kilograms per hectare per year.
 - PHOSPHORUS - The phosphorus loading metric uses a per-unit-area method to calculate values. For each land cover class, the area within each reporting unit is converted to hectares then multiplied by the phosphorus coefficient associated with that class, resulting in a product

for each class. Products for all land cover classes are summed, then divided by the total number of hectares within the reporting unit to provide an average phosphorus loading value across each reporting unit in kilograms per hectare per year.

- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.
- As the **Reporting unit feature** is a vector dataset, ArcGIS will perform a vector to raster conversion during processing.
 - To better control the vector to raster conversion, the tool defaults the **Snap raster** and the **Processing cell size** to that of the **Land cover grid**. These may be changed from within the tool.
 - If a large size disparity exists between the extents of the **Reporting unit feature** and the **Land cover grid**, the user may wish to set the Extent in Environment Settings > Processing Extent to the smaller of the two to avoid unnecessary processing.
- If the **Reporting unit feature** has overlapping features, the zonal analysis will be performed for each individual feature.
- If a feature in the **Reporting unit feature** zone data does not overlap any cell centers in the secondary class rasters (e.g., **Land cover grid**), those zones will not get converted to the internal zone raster. As a result, those zones will not be represented in the **Output table**.
- The **Land cover classification scheme** must correspond to the **Land cover grid**.
 - Schemes for common land cover datasets are included with ATtILA. Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD 2001 ALL vs. NLCD 2001 LAND). Schemes designated as “ALL” include all land cover classes in reporting unit area calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded. More information about each of the classification schemes supplied with ATtILA may be found in [Supplied Land Cover Classification Schemas](#).
 - In addition to the common land cover schemes, the tool permits a user-defined land cover classification scheme to be used by specifying a **Land cover classification file**. Refer to [Land Cover Classification](#) for more information on how ATtILA may be customized for user-specific needs.
 - **NOTE:** *When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. For example, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. See [Values Element in ATtILA’s LCC XML Document](#) and [Values Window in the Land Cover Classification Editor](#) for more information on excluded values.*

- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - ***NOTE:** For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.
- Field names in the **Output table** are as follows:
 - PCTIA - The percent impervious area metric for the reporting unit.
 - N_Load - The average annual nitrogen load per hectare for the reporting unit (kg/ha/yr).
 - P_Load - The average annual phosphorus load per hectare for the reporting unit (kg/ha/yr).
 - ***NOTE:** Field names are obtained from the fieldName attribute in each coefficient element in the specified **Land cover classification file** (See the Coefficients Element in ATtILA's LCC XML Document). Output field names can be changed by editing the **Land cover classification file** with the Land Cover Classification Editor.*
 - ***NOTE:** The output field name for each coefficient is shown as the second item next to the coefficient's check box in **Report metrics for these coefficients**.*
- In **Additional Options**, the user may elect to 'Add Quality Assurance Fields', 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Add Quality Assurance Fields
 - Choosing **QAFIELDS** adds a number of fields to the **Output table** to facilitate quality assurance checking.
 - **LCCC_OVER** - The percent overlap between the **Reporting unit feature** layer and the **Land cover grid** for each reporting unit.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **lccc_TabArea#** (table) - A table of areas corresponding to the values in the Land cover raster within each reporting unit.
 - ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*

- **NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.
- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - **NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
 - **NOTE:** ATtILA will report the name and location of the log file in the Tool Details’ Messages section.

Syntax

LCCC (Reporting_unit_feature, Reporting_unit_ID_field, Land_cover_grid, Land_cover_classification_scheme, Land_cover_classification_file, Report_metrics_for_these_coefficients, Output_table, {Processing_cell_size}, {Snap_raster}, Select_options)

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that	Field

Parameter	Explanation	Data Type
	contains the unique ID for each reporting unit. It may be an integer or a string data type.	
Land_cover_grid	The raster dataset representing land cover classes upon which core and edge area metrics will be derived within each Reporting unit feature. The class input must be an integer raster layer.	Raster Layer
Land_cover_classification_scheme	The land cover classification schemes included in ATtILA and a User Defined option. The default schemes correspond to common input land cover datasets. Two schemes are supplied for each dataset included in ATtILA:	String
Land_cover_classification_file	The full pathname to the user-defined .xml file for custom or non-standard land cover classification schemes. Pathname is automatically filled when a default scheme is selected.	File
Report_metrics_for_these_classes	A list of the land cover classes and metric combinations for processing. Check the box to calculate metrics for each land cover class and/or combination class desired within the reporting units.	Multiple Value
Output_table	The output reporting unit metrics table to be created. It is recommended that the	Table

Parameter	Explanation	Data Type
	Output table be saved within a file geodatabase.	
Processing_cell_size (Optional)	The Processing cell size for the zonal operation. The default Processing cell size is the cell size of the input raster land cover data. Optionally, the user may select a different cell size.	Analysis cell size
Snap_raster (Optional)	The raster that the cell alignment of the Land cover grid and rasterized Reporting unit feature layer will be matched to during processing. The default Snap raster is the Land cover grid.	Raster Layer
Select_options	Tool options available to provide additional information:	Multiple Value

Code Samples

Land Cover Coefficient Calculator example 1 (Python window)

This example returns a table with an estimate of the amount of nitrogen output (stream loading) in kilograms per hectare per year and an estimate of the percent impervious area for each reporting unit zone. The options to retain the intermediate calculation products, to record the process steps into a log file, and to generate the quality assurance fields have also been selected.

NOTE: ATtILA tools that utilize the **Report metrics for these classes** parameter will clear all class selections when the tool is opened. This results in setting any variable set for that parameter to **None** when calling the tool from a script. Because of this, the standard scripting syntax (e.g. `arcpy.ATtILA.LCP(variable 1, variable 2, variable 3, etc.)`) will fail. To overcome this idiosyncrasy, use the scripting example provided below. When using this technique, all variables are passed in as strings.

NOTE: By running scripts which launch an ATtILA tool via 'metric.run' within ArcGIS Pro's Python Window, all geoprocessing functions run by that tool can be viewed in the History Pane.

```
import arcpy
from arcpy.sa import *

from ATtILA2 import metric

toolPath = "D:\\Destination
            Folder\\ToolboxSource\\ATtILA2\\scripts\\LandCoverCoefficientCalculator_ArcGIS.py"
Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
```

```

Land_cover_grid = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Land_cover_classification_scheme = "NLCD ALL"
Land_cover_classification_file = "D:\\Destination Folder\\ToolboxSource\\LandCoverClassifications\\NLCD
ALL.xml"
Report_metrics_for_these_classes = "IMPERVIOUS - [PCTIA] Percent Cover Total Impervious Area';'NITROGEN
- [N_Load] Estimated Nitrogen Loading Based on Land Cover';'PHOSPHORUS - [P_Load] Estimated
Phosphorus Loading Based on Land Cover"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_LCCC"
Processing_cell_size = "30"
Snap_raster = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric
Calculation';'LOGFILE - Record Process Steps Taken During Metric Calculation';'QAFIELDS - Add
Quality Assurance Fields'"

metric.runLandCoverCoefficientCalculator(
    toolPath,
    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Land_cover_grid,
    Land_cover_classification_scheme,
    Land_cover_classification_file,
    Report_metrics_for_these_classes,
    Output_table,
    Processing_cell_size,
    Snap_raster,
    Select_options
)

```

Land Cover Coefficient Calculator example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```

Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Generating a zonal tabulate area table
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)

```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Pedestrian Access And Availability

Available with Spatial Analyst license.

Summary

Estimates the availability of parkland per person based on accessibility of parkland. Accessibility is determined using walkable routes along a cost surface raster and a user-specified maximum travel distance. The output is a raster.

Optionally, park polygons which abut or overlap each other can be dissolved into a single unit if no physical barrier to movement is known to exist before determining accessibility and availability.

Usage

- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- This tool processes all parks in the **Park feature** regardless of selections set. The ability to limit calculations to only selected park polygons is not supported in this release.
- Park features are converted internally to a raster before performing the analysis. The resolution of the raster is controlled with the **Processing cell size** parameter. Setting the **Snap raster** will control the alignment of the raster cell boundaries.
- The internal rasterization process uses the cell center method. This means that park features that do not overlap the center of any intermediate rasterized cells will not be represented in the output calculations.
- This tool performs the following steps for each individual park:
 - Select park by its id value and create a feature layer
 - Create a buffer around the park extending 5% beyond the maximum travel distance
 - Create Cost Distance raster to the Maximum travel distance with the buffer area as the processing extent
 - Designate the accessible area by setting all cell values to 1 for any cell in the Cost Distance raster with a value ≥ 0
 - Expand the accessible area if indicated by the Expand area served parameter
 - Determine Population within accessible area
 - if Population parameter is a raster:
 1. Set the geoprocessing snap raster and processing cell size environments to match the population raster
 2. Zonal Statistics As Table with the SUM option is used to calculate the surrounding population
 - if Population parameter is a polygon feature:
 1. the accessible area raster is converted to a polygon

2. Tabulate Intersection is used with this polygon and the Population polygon feature to get an area-weighted estimate of the surrounding population
 - Determine Availability (Park area in square meters divided by population in the accessible area)
- The **Population raster or polygon feature** will allow you to choose either a raster, or a polygon feature.
 - When choosing a raster, this tool assumes the grid cell value to be the population count for the grid cell.
 - When choosing a polygon feature, this tool assumes that population is distributed evenly throughout each feature polygon. The tool apportions population by area weighting. For example, if 50% of a population polygon is within a reporting unit, the tool will assign 50% of the value in the polygon's **Population field** to that reporting unit. Caution should be exercised when population polygons do not have even population distributions as this could result in an overweighting or underweighting of population when the tool performs the apportionment. Generally, greater accuracy will be achieved if the population polygons are smaller than the smallest **Reporting unit feature** polygons.
 - ***NOTE:** The **Population field** dropdown is only available when the **Population raster or polygon feature** is a polygon feature.*
 - ***NOTE:** The **Population field** will default to 'Value' when the **Population raster or polygon feature** is an integer grid.*
 - ***NOTE:** The **Population field** will be empty when the **Population raster or polygon feature** is a non-integer grid.*
 - Final output is written to the location specified in the **Output raster** parameter. The **Output raster** may be saved in a file folder or a geodatabase.
 - ***NOTE:** For most consistent results, It is highly recommended that tool output be saved to a file geodatabase.*
 - When saving in a File Geodatabase, no extension is permitted for the **Output raster** name.
 - When saving in a file folder, the following extensions are allowed for the **Output raster** name:
 - .img for ERDAS IMAGE
 - .tif for TIFF
 - No extension for Esri Grid
 - In **Additional Options**, the user may elect to 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:

- **paaa_Parks_#** (vector) - A copy of the **Park feature** with three additional fields: **CalcAreaM2**, **Pop_Access**, and **SQM_Avail**.
 - **CalcAreaM2** - The park area in square meters. Area calculations will use the geoprocessing environment outputCoordinateSystem if one is set. If an outputCoordinateSystem is not specified, the area calculations will use the spatial coordinate system of the **Cost surface raster**.
 - **Pop_Access** - Population within the accessible area of the park.
 - **SQM_Avail** - Park area in square meters divided by the population within the accessible area of the park.
- ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
- ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*
- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - ***NOTE:** ATiLA will report the location of the saved intermediate products in the Tool Details’ Messages section.*
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - ***NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.*

- **NOTE:** ATtILA will report the name and location of the log file in the Tool Details' Messages section.

Syntax

PAAA (Park_feature, {Dissolve_parks}, Cost_surface_raster, Population_raster_or_polygon_feature, {Population_field}, Maximum_travel_distance, Expand_area_served, Output_raster, {Processing_cell_size}, {Snap_raster}, {Select_options})

Parameter	Explanation	Data Type
Park_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Dissolve_parks (Optional)	Specifies whether park features that overlap or share a boundary should be dissolved into a single feature before calculating accessibility and availability.	Boolean
Cost_surface_raster	A raster with weighted values that represent the relative walkability of each pixel. Users can provide their own cost surface raster or build one with the ATtILA Create Walkability Cost Raster Utility. The input must be an integer raster layer.	
Population_raster_or_polygon_feature	The raster or vector polygon dataset that contains population data.	Feature Layer or Raster Layer
Population_field (Optional)	The field in the Population raster or polygon feature layer that contains population data. The behavior of this parameter is dependent on the input for the Population or polygon feature:	Field
Maximum_travel_distance	The maximum accumulated distance along the cost surface raster that is within the user defined walking distance of the	Long

Parameter	Explanation	Data Type
	park feature. The input must be an integer value.	
Expand_area_served	The number of pixels the resulting service area will be expanded to produce the final park service area. This is useful to fill in gaps in service areas and to account for possible underestimation of length when measured from a lower resolution raster. The service area will be expanded in all directions by this value regardless of the corresponding cost surface raster values or maximum travel distance values. This value must be an integer greater than or equal to zero.	Long
Output_raster	The output accessibility and availability raster. It is always a floating point raster.	Raster Dataset
Processing_cell_size (Optional)	The Processing cell size for the Output raster. The default Processing cell size is the cell size of the input Cost surface raster. Optionally, the user may select a different cell size.	Analysis cell size
Snap_raster (Optional)	The raster that the cell alignment of the Output raster will be aligned to. The default Snap raster is the Cost surface raster.	Raster Layer
Select_options	Tool options available to provide additional information:	Multiple Value

Code Samples

Pedestrian Access and Availability example 1 (Python window)

This example returns a raster. The options to retain the intermediate calculation products, and to record the process steps into a log file have also been selected.

```
import arcpy
from arcpy.sa import *

Park_feature = "D:\\pyexamples\\data.gdb\\Parks"
Dissolve_parks = "false"
Cost_surface_raster = "D:\\pyexamples\\data.gdb\\Walkable_Cost_Surface"
Population_raster_or_polygon_feature = "D:\\pyexamples\\data.gdb\\DASY_Population"
Population_field = ""
Maximum_travel_distance = "800"
Expand_area_served = "8"
Output_raster = "D:\\pyexamples\\results.gdb\\ParksEval_PAAA"
Processing_cell_size = "10"
Snap_raster = "D:\\pyexamples\\data.gdb\\Walkable_Cost_Surface"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric  
Calculation'; 'LOGFILE - Record Process Steps Taken During Metric Calculation'"

    Park_feature,
    Dissolve_parks,
    Cost_surface_raster,
    Population_raster_or_polygon_feature,
    Population_field,
    Maximum_travel_distance,
    Expand_area_served,
    Output_raster,
    Processing_cell_size,
    Snap_raster,
    Select_options
)
```

Pedestrian Access and Availability example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\\filename
|YYYY-MM-DD hh:mm:ss| Setting up environment variables
Intermediates are stored in this directory: D:\\pyexamples\\results.gdb
|YYYY-MM-DD hh:mm:ss| Using Walkable_Cost_Surface's extent for geoprocessing steps.
|YYYY-MM-DD hh:mm:ss| Using Walkable_Cost_Surface's spatial reference for geoprocessing steps:
Albers_Conical_Equal_Area
|YYYY-MM-DD hh:mm:ss| Creating temporary copy of Parks. Intermediate: paaa_Parks_0
|YYYY-MM-DD hh:mm:ss| Collecting id values from OBJECTID_1 for each park.
|YYYY-MM-DD hh:mm:ss| Calculating park area in square meters
|YYYY-MM-DD hh:mm:ss| Calculating access and availability for 20 areas.
|YYYY-MM-DD hh:mm:ss| The following steps will be performed for each park:
---
|YYYY-MM-DD hh:mm:ss| |PER UNIT| 1) Select park by its ID value and create a feature layer.
|YYYY-MM-DD hh:mm:ss| |PER UNIT| 2) Create a buffer around the park extending 5% beyond the maximum travel
distance.
|YYYY-MM-DD hh:mm:ss| |PER UNIT| 3) Create Cost Distance raster to the Maximum travel distance with the
buffer extent as the processing extent.
```

```

|YYYY-MM-DD hh:mm:ss| |PER UNIT| 4) Designate the accessibility area by setting all cells to 1 for any
cell in the Cost Distance raster >= 0.
|YYYY-MM-DD hh:mm:ss| |PER UNIT| 5) Expand the accessibility area if indicated by the Expand area served
parameter.
|YYYY-MM-DD hh:mm:ss| |PER UNIT| 6) Determine Population within accessibility area.
|YYYY-MM-DD hh:mm:ss| |PER UNIT| 6a) If Population parameter input is a raster:
|YYYY-MM-DD hh:mm:ss| |PER UNIT| a1) Set the geoprocessing snap raster and processing cell size
environments to the population raster.
|YYYY-MM-DD hh:mm:ss| |PER UNIT| a2) Use Zonal Statistics As Table with the SUM option to calculate
the accessibility area population.
|YYYY-MM-DD hh:mm:ss| |PER UNIT| 6b) If Population parameter input is a polygon feature:
|YYYY-MM-DD hh:mm:ss| |PER UNIT| b1) Convert the accessibility area raster to a polygon.
|YYYY-MM-DD hh:mm:ss| |PER UNIT| b2) Use Tabulate Intersection with this polygon and the Population
polygons to calculate the accessibility area population.
|YYYY-MM-DD hh:mm:ss| |PER UNIT| 7) Determine Availability (Cost distance value to park area divided by
surrounding population)
---
|YYYY-MM-DD hh:mm:ss| Starting calculations per park...
2 out of 20 loops, 10.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
4 out of 20 loops, 20.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
6 out of 20 loops, 30.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
8 out of 20 loops, 40.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
10 out of 20 loops, 50.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
12 out of 20 loops, 60.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
14 out of 20 loops, 70.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
16 out of 20 loops, 80.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
18 out of 20 loops, 90.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
20 out of 20 loops, 100.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
|YYYY-MM-DD hh:mm:ss| Finished calculations for last park.
|YYYY-MM-DD hh:mm:ss| Merging 20 calculated park/population rasters. Output: ParksEval_PAAA.
|YYYY-MM-DD hh:mm:ss| Deleting 20 individual park rasters.
|YYYY-MM-DD hh:mm:ss| Individual park rasters deletion complete
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)

```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Population Density Metrics

Summary

Calculates population count and density (people per km²) for each reporting unit polygon and creates an output table.

In addition to the default metrics, the tool optionally calculates population change over time.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- ***NOTE:** For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- ***NOTE:** Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Reporting unit feature** must be a polygon feature class or shapefile.
- This tool assumes that population is distributed evenly throughout each **Census feature** polygon. The tool apportions population by area weighting. For example, if 50% of a **Census feature** polygon is within a reporting unit, the tool will assign 50% of the value in the polygon's **Population field** to that reporting unit. Caution should be exercised when **Census feature** polygons do not have even population distributions as this could result in an overweighting or underweighting of population when the tool performs the apportionment. Generally, greater accuracy will be achieved if the **Census feature** polygons are smaller than the smallest **Reporting unit feature** polygons.
- When the optional **POPCHG** (population change) is checked, the tool allows the user to select a **Census T2 feature** and a corresponding **Population T2 field**.
 - For best results, the **Census T2 feature** should contain polygons at a similar scale to that of the **Census feature** (e.g. if **Census feature** represents census block groups, then **Census T2 feature** should also represent census block groups rather than census tracts or some other census geography).
 - **Census T2 feature** may be the same feature layer as used for **Census feature**.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - ***NOTE:** For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.

- When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.
- Field names in the **Output table** follow this naming scheme:
 - For default tool settings (e.g. no optional settings selected):
 - **AREAKM2** - The area of the reporting unit in km².
 - **POPCNT** - The estimated total population within the reporting unit derived by area weighting the population values within each **Census feature** polygon that intersects with the reporting unit and summing the area-weighted values.
 - **POPDENS** - The estimated population density in persons per km² within the reporting unit derived by dividing **POPCNT** by **AREAKM2**.
 - When the **POPCHG** option is selected:
 - **POPCNT_T1** - The estimated total population within the reporting unit derived by area weighting the population values within each **Census feature** polygon (Time 1) that intersects with the reporting unit and summing the area-weighted values.
 - **POPDENS_T1** - The estimated population density in persons per km² within the reporting unit derived by dividing **POPCNT_T1** by **AREAKM2**.
 - **POPCNT_T2** - The estimated total population within the reporting unit derived by area weighting the population values within each **Census T2 feature** polygon (Time 2) that intersects with the reporting unit and summing the area-weighted values.
 - **POPDENS_T2** - The estimated population density in persons per km² within the reporting unit derived by dividing **POPCNT_T2** by **AREAKM2**.
 - **POPCHG** - The estimate percent change in population from Time 1 to Time 2.
- In **Additional Options**, the user may elect to ‘Retain Intermediate Layers Generated During Metric Processing’, and/or ‘Record Process Steps Taken During Metric Calculation’.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **pdm_[Reporting unit feature]_#** (vector) - A polygon feature class containing reporting units with a new area field in km² (AREAKM2).
 - **pdm_populationCnt_#** (table) - A table of the area-weighted population counts within each reporting unit.
 - **NOTE:** *To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*

- **NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.
- **NOTE:** If the optional **POPCHG** is selected, some intermediate products will have a “T1” or “T2” designation added to their filenames to indicate whether the products were generated for the population density calculations at Time 1 or Time 2.
- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - **NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
 - **NOTE:** ATtILA will report the name and location of the log file in the Tool Details’ Messages section.

Syntax

PDM (Reporting_unit_feature, Reporting_unit_ID_field, Census_feature, Population_field, Output_table, {POPCHG}, Census_T2_feature, Population_T2_field, {Select_options})

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer

Parameter	Explanation	Data Type
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Census_feature	The vector polygon dataset that contains population data.	Feature Layer
Population_field	The field in the Census feature layer that contains population data.	Field
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved within a file geodatabase.	Table
POPCHG (Optional)	Specifies whether population change over time (POPCHG) metrics will be included in the output table.	Boolean
Census_T2_feature	The optional vector polygon dataset (Census time 2) that contains population data for the comparison date. It may be the same feature layer as Census feature.	Feature Layer
Population_T2_field	The field in the Census feature layer that contains population data for the comparison date.	Field
Select_options (Optional)	Tool options available to provide additional information:	Multiple Value

Code Samples

Population Density Metrics example 1 (Python window)

This example returns a table with the population density of the reporting unit zone in capita per square kilometer along with the reporting unit's total population count and square kilometer area calculation. The options to retain the intermediate calculation products, and to record the process steps into a log

file have also been selected.

```
import arcpy

Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Census_feature = "D:\\pyexamples\\data.gdb\\Census2010"
Population_field = "TOTALPOP"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_PDM"
POPCHG = ""
Census_T2_feature = ""
Population_T2_field = ""
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric  
Calculation';'LOGFILE - Record Process Steps Taken During Metric Calculation'"

    Reporting_unit_feature,  
    Reporting_unit_ID_field,  
    Census_feature,  
    Population_field,  
    Output_table,  
    POPCHG,  
    Census_T2_feature,  
    Population_T2_field,  
    Select_options  
)
```

Population Density Metrics example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Creating temporary copy of Watersheds. Intermediate: pdm_Watersheds_0
|YYYY-MM-DD hh:mm:ss| Creating output table
|YYYY-MM-DD hh:mm:ss| Calculating population density
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
|YYYY-MM-DD hh:mm:ss| Calculation complete
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Population In Floodplain Metrics

Available with Spatial Analyst license.

Summary

Calculates percentage of total population living in floodplain areas for each reporting unit polygon and creates an output table.

Counts for the total population in the reporting unit and within the floodplain area are also reported.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- ***NOTE:** For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- ***NOTE:** Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.
- As the **Reporting unit feature** is a vector dataset, ArcGIS will perform a vector to raster conversion during processing.
 - To better control the vector to raster conversion, the tool defaults the **Snap raster** and the **Processing cell size** to that of the **Land cover grid**. These may be changed from within the tool.
 - If the extent of the **Reporting unit feature** is smaller than that of the **Land cover grid**, the user may wish to check the **Reduce land cover grid to smallest recommended size** option. This creates a temporary land cover raster with the processing extent equal to that of the **Reporting unit feature** to avoid unnecessary processing.
- If the **Reporting unit feature** has overlapping features, the zonal analysis will be performed for each individual feature.
- If a feature in the **Reporting unit feature** zone data does not overlap any cell centers in the secondary class rasters (e.g., **Land cover grid**), those zones will not get converted to the internal zone raster. As a result, those zones will not be represented in the **Output table**.

- The **Population raster or polygon feature** will allow you to choose either a raster, or a polygon feature.
 - When choosing a raster, this tool assumes the grid cell value to be the population count for the grid cell.
 - When choosing a polygon feature, this tool assumes that population is distributed evenly throughout each feature polygon. The tool apportions population by area weighting. For example, if 50% of a population polygon is within a reporting unit, the tool will assign 50% of the value in the polygon's **Population field** to that reporting unit. Caution should be exercised when population polygons do not have even population distributions as this could result in an overweighting or underweighting of population when the tool performs the apportionment. Generally, greater accuracy will be achieved if the population polygons are smaller than the smallest **Reporting unit feature** polygons.
 - ***NOTE:** The **Population field** dropdown is only available when the **Population raster or polygon feature** is a polygon feature.*
 - ***NOTE:** The **Population field** will default to 'Value' when the **Population raster or polygon feature** is an integer grid.*
 - ***NOTE:** The **Population field** will be empty when the **Population raster or polygon feature** is a non-integer grid.*
- The **Floodplain raster or polygon feature** will allow you to choose either a raster, or a polygon feature.
 - When choosing a raster, floodplains are defined as any non-zero grid cell. The simplest example of this is a binary raster where floodplains are coded as 1 and everything else is coded as 0, but any value or combination of values can be used to designate floodplain areas as long as non-floodplain areas are coded as 0.
 - When choosing a polygon feature, floodplains are defined as any polygon feature. This tool does not make use of any attribute or combination of attributes to select floodplain polygons. All polygons in the floodplain input layer are considered floodplain areas. If polygons exist within the layer that do not represent floodplains, they need to be edited out before using the layer as input for this tool.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - ***NOTE:** For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.
- Field names in the **Output table** are as follows:
 - **RU_POP** - The population for the reporting unit.
 - **FP_POP_C** - The population in a floodplain for the reporting unit.

- **FP_POP_P** - The percentage of the reporting unit population in a floodplain.
- In **Additional Options**, the user may elect to ‘Retain Intermediate Layers Generated During Metric Processing’, and/or ‘Record Process Steps Taken During Metric Calculation’.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **pifm_[Population polygon feature]_Work#** (vector) - A copy of the **Population polygon feature** with the **Population** field retained and a **tmpClass** field added to assist in the population calculations via a Tabulate Intersection operation. This feature class is only produced when a polygon population feature is selected.
 - **pifm_populationCnt_RU#** (table) - A table of the population count within each reporting unit.
 - if the **Population raster or polygon features** is a raster, this table is produced using a Zonal Statistics as Table operation with the **Reporting unit feature** as the in_zone_data, the **Reporting unit ID field** as the zone_field, the **Population raster** as the in_value_raster, and ‘Sum’ as the statistics_type.
 - if the **Population raster or polygon features** is a polygon feature, this table is produced using a Tabulate Intersection operation using **Reporting unit feature** as the in_zone_features, **Reporting unit ID field** as the zone_field, **pifm_[Population polygon feature]_Work#** as the in_class_features, **tmpClass** as the class-fields, and **Population field** as the sum_fields.
 - **pifm_populationFP** (raster) - A raster produced by using SetNull to set non-floodplain areas (VALUE = 0) in the **Floodplain raster** to NoData and replacing the floodplain areas with values from the **Population raster**. This raster is only produced when both the inputs for **Population raster or polygon feature** and **Floodplain raster or polygon feature** are rasters.
 - **pifm_[Floodplain raster feature]_Poly#** (vector) - A polygon feature class produced by setting 0-value areas in the input floodplain raster to NULL and then performing a raster to polygon conversion. A **tmpClass** field is added to assist in the population calculations via a Tabulate Intersection operation. This feature class is only produced when a raster floodplain feature is selected.
 - **pifm_[Floodplain polygon feature]_Work#** (vector) - A copy of the **Floodplain polygon feature** with a **tmpClass** field added to assist in the population calculations via a Tabulate Intersection operation. This feature class is only produced when both population and floodplain input features are polygons.
 - **pifm_[Floodplain raster or polygon feature]_Identity#** (vector) - A feature class of the floodplain areas by reporting unit. This feature class is produced when at least one input is a polygon feature.

- if the Population input is a raster and the Floodplain input is a polygon feature: This feature class is a copy of the **Floodplain polygon feature** with the overlying **Reporting unit feature**'s ID attached via Identity.
 - if the Population input is a polygon feature and the Floodplain input is a raster: This feature class is a copy of the intermediate **pifm_[Floodplain raster feature]_Poly#** with the overlying **Reporting unit feature**'s ID attached via Identity.
 - if the Population input is a polygon feature and the Floodplain input is a polygon feature: This feature is a copy of the intermediate **pifm_[Floodplain polygon feature]_Work#** with the overlying **Reporting unit feature**'s ID attached via Identity.
- **pifm_populationCnt_FP#** (table) - A table of the population count in floodplain areas within each reporting unit. Depending on the inputs used for the Population and Floodplain parameters, the table is produced in one of four ways:
1. Raster Population and Raster Floodplain: this table is produced using a Zonal Statistics as Table operation with the **Reporting unit feature** as the in_zone_data, the **Reporting unit ID field** as the zone_field, the intermediate **pifm_populationFP** as the in_value_raster, and 'Sum' as the statistics_type.
 2. Raster Population and Polygon Floodplain: this table is produced using a Zonal Statistics as Table operation with the intermediate **pifm_[Floodplain polygon feature]_Identity#** as the in_zone_data, the **Reporting unit ID field** as the zone_field, the **Population raster** as the in_value_raster, and 'Sum' as the statistics_type.
 3. Polygon Population and Raster Floodplain: this table is produced using a Tabulate Intersection operation using **pifm_[Floodplain raster]_Identity#** as the in_zone_features, **Reporting unit ID field** as the zone_field, **pifm_[Population polygon feature]_Work#** as the in_class_features, **tmpClass** as the class-fields, and **Population field** as the sum_fields.
 4. Polygon Population and Polygon Floodplain: this table is produced using a Tabulate Intersection operation using **pifm_[Floodplain polygon feature]_Identity#** as the in_zone_features, **Reporting unit ID field** as the zone_field, **pifm_[Population polygon feature]_Work#** as the in_class_features, **tmpClass** as the class-fields, and **Population field** as the sum_fields.
- ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
 - ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*
- Intermediate products are saved to one of the following locations:

- When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - **NOTE:** *ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.*
- Record Process Steps Taken During Metric Calculation
- Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** *In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.*
 - **NOTE:** *ATtILA will report the name and location of the log file in the Tool Details’ Messages section.*

Syntax

PIFM (Reporting_unit_feature, Reporting_unit_ID_field, Population_raster_or_polygon_feature, {Population_field}, Floodplain_raster_or_polygon_feature, Output_table, {Select_options})

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Population_raster_or_polygon_feature	The raster or vector polygon dataset that contains population data.	Feature Layer or Raster Layer

Parameter	Explanation	Data Type
Population_field (Optional)	The field in the Population raster or polygon feature layer that contains population data. The behavior of this parameter is dependent on the input for the Population or polygon feature:	Field
Floodplain_raster_or_polygon_feature	The integer raster or vector polygon dataset that defines the floodplains.	Feature Layer or Raster Layer
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved within a file geodatabase.	Table
Select_options (Optional)	Tool options available to provide additional information:	Multiple Value

Code Samples

Population in Floodplain Metrics example 1 (Python window)

This example returns a table with the population count within the reporting unit, the population count in any floodplain area within the reporting unit, and the percentage of the reporting unit's population that is in a floodplain.

```
import arcpy
from arcpy.sa import *

Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Population_raster_or_polygon_feature = "D:\\pyexamples\\data.gdb\\Census2010"
Population_field = "TOTALPOP"
Floodplain_raster_or_polygon_feature = "D:\\pyexamples\\data.gdb\\Floodplain_Raster"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_PIFM"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric Calculation'; 'LOGFILE - Record Process Steps Taken During Metric Calculation'"

Reporting_unit_feature,
Reporting_unit_ID_field,
Population_raster_or_polygon_feature,
Population_field,
Floodplain_raster_or_polygon_feature,
Output_table,
```



```
Select_options
)
```

Population in Floodplain Metrics example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Calculating population within each reporting unit
|YYYY-MM-DD hh:mm:ss| Converting floodplain raster to a polygon feature
|YYYY-MM-DD hh:mm:ss| Assigning reporting unit IDs to floodplain features
|YYYY-MM-DD hh:mm:ss| Calculating population within floodplain areas for each reporting unit
|YYYY-MM-DD hh:mm:ss| Calculating the percent of the population that is within a floodplain
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
|YYYY-MM-DD hh:mm:ss| Calculation complete
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Population Land Cover Views

Available with Spatial Analyst license.

Summary

Estimates the percentage of the population in each reporting unit that have potential views of selected land cover classes within a given distance from their location and creates an output table. A land cover class is visible if it is found within the neighborhood defined by the View radius and in a group of cells larger than or equal to the Minimum visible patch size. The Minimum visible patch size and View radius are provided by the user. The neighborhood is circular in shape.

Additional fields for total population count within the reporting unit, the total population count in the reporting unit with potential views, the total population count in the reporting unit without potential views, and the percentage of the total population in the reporting unit without potential views are also output.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.
- As the **Reporting unit feature** is a vector dataset, ArcGIS will perform a vector to raster conversion during processing.

- To better control the vector to raster conversion, the tool defaults the **Snap raster** and the **Processing cell size** to that of the **Land cover grid**. These may be changed from within the tool.
- If the extent of the **Reporting unit feature** is smaller than that of the **Land cover grid**, the user may wish to check the **Reduce land cover grid to smallest recommended size** option. This creates a temporary land cover raster with the processing extent equal to that of the **Reporting unit feature** to avoid unnecessary processing.
- If the **Reporting unit feature** has overlapping features, the zonal analysis will be performed for each individual feature.
- If a feature in the **Reporting unit feature** zone data does not overlap any cell centers in the secondary class rasters (e.g., **Land cover grid**), those zones will not get converted to the internal zone raster. As a result, those zones will not be represented in the **Output table**.
- The **Land cover classification scheme** must correspond to the **Land cover grid**.
 - Schemes for common land cover datasets are included with ATtILA. Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD 2001 ALL vs. NLCD 2001 LAND). Schemes designated as “ALL” include all land cover classes in reporting unit area calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded. More information about each of the classification schemes supplied with ATtILA may be found in Supplied Land Cover Classification Schemas.
 - In addition to the common land cover schemes, the tool permits a user-defined land cover classification scheme to be used by specifying a **Land cover classification file**. Refer to Land Cover Classification for more information on how ATtILA may be customized for user-specific needs.
 - ***NOTE:** When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. For example, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. See Values Element in ATtILA’s LCC XML Document and Values Window in the Land Cover Classification Editor for more information on excluded values.*
- The **Population raster** assumes the grid cell value to be the population count for the grid cell.
- The **View radius** defines the circular neighborhood to determine visibility of a land cover class. It is a straight-line distance only and is not a viewshed analysis parameter.
 - ***NOTE:** Only a single cell of the land cover class needs to be in the **View radius** of the focal cell for the focal cell to be categorized as ‘Potential View Area’.*
- Field names in the **Output table** are as follows:
 - **RU_POP** - The population for the reporting unit.
 - **[class Id]_PV_C** - The population count within the reporting unit with potential views of the selected class (e.g. “for_PV_C” is the name of the field for the population count within the reporting unit with potential views of forest land cover).

- **[class Id]_PV_P** - The proportion of the population within the reporting unit with potential views of the selected class (e.g. “for_PV_P” is the name of the field for the proportion of the population with potential views of forest land cover within the reporting unit).
- **[class Id]_MV_C** - The population count within the reporting unit with zero or minimal views of the selected class (e.g. “for_PV_C” is the name of the field for the population count within the reporting unit with zero or minimal views of forest land cover).
- **[class Id]_MV_P** - The proportion of the population within the reporting unit with zero or minimal views of the selected class (e.g. “for_MV_P” is the name of the field for the proportion of the population with zero or minimal views of forest land cover within the reporting unit).
- **NOTE:** *Output field names can be altered by editing the **Land cover classification file** with the Land Cover Classification Editor (Refer to Classes Element in ATtILA's LCC XML Document for general information, or the individual **Land cover classification file** for details).*
- **NOTE:** *The output field name for each class is shown as the second item next to the class's check box in **Report metrics for these classes**.*
- **NOTE:** *Only the **[class Id]_PV_C** field name is presented in the **Report metrics for these classes**.*
- In **Additional Options**, the user may elect to ‘Retain Intermediate Layers Generated During Metric Processing’, and/or ‘Record Process Steps Taken During Metric Calculation’.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **plcv_[class Id]_ViewPatch_#** (raster) - Rasters created from the **Land cover grid** for each class selected in the **Report metrics for these classes** parameter of the tool with reclassified values corresponding to the following scheme: 1 = [class Id], 0 = Other. Patches of [class Id] below the **Minimum visible patch size** have been eliminated (i.e., set to 0).
 - **plcv_[class Id]_ViewArea_#** (raster) - Rasters created by performing a focal statistics sum on the **plcv_[class Id]_ViewPatch_#** raster using a circular neighborhood of **View radius**. The resulting rasters were then reclassified into a single-value raster where 1 = Potential View Area of [class Id]. All other areas were set to NoData.
 - **NOTE:** *Only a single cell of [class Id] needs to be in the **View radius** of the focal cell for the focal cell to be categorized as ‘Potential View Area’.*
 - **plcv_[class Id]_ViewAreaPoly_#** (vector) - A polygon feature class produced by performing a raster to polygon conversion of **plcv_[class Id]_ViewArea_#**.
 - **plcv_[class Id]_ViewPopulation_#** (raster) - A copy of **Population raster** extracted for the **plcv_[class Id]_ViewAreaPoly_#** area.

- **plcv_[class Id]_ViewPopulationByRU_#** (table) - A table of the population count in potential view areas within each reporting unit. The table is produced by performing a Zonal Statistics as Table operation with the **Reporting unit feature** as the **in_zone_data**, the **Reporting unit ID field** as the **zone_field**, the intermediate **plcv_[class Id]_ViewPopulation_#** as the **in_value_raster**, and 'Sum' as the **statistics_type**.
- ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
- ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*
- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named "attilaScratchWorkspace" is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - ***NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details' Messages section.*
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - ***NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.*
 - ***NOTE:** ATtILA will report the name and location of the log file in the Tool Details' Messages section.*

Syntax

PLCV (Reporting_unit_feature, Reporting_unit_ID_field, Land_cover_grid, Land_cover_classification_scheme, Land_cover_classification_file, Report_metrics_for_these_classes, View_radius, Minimum_visible_patch_size, Population_raster, Output_table, {Processing_cell_size}, {Snap_raster}, Select_options)

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Land_cover_grid	The raster dataset representing land cover classes to be summarized within each Reporting unit feature. The grid input must be an integer raster layer.	Raster Layer
Land_cover_classification_scheme	The land cover classification schemes included in ATtILA and a User Defined option. The default schemes correspond to common input land cover datasets. Two schemes are supplied for each dataset included in ATtILA:	String
Land_cover_classification_file	The full pathname to the user-defined .xml file for custom or non-standard land cover classification schemes. Pathname is automatically filled when a default scheme is selected.	File
Report_metrics_for_these_classes	A list of the land cover classes and metric combinations for	Multiple Value

Parameter	Explanation	Data Type
	processing. Check the box to calculate metrics for each land cover class and/or combination class desired within the reporting units.	
View_radius	The radius of the view neighborhood, measured in number of grid cells. The neighborhood is circular in shape. The input must be an integer value.	Long
Minimum_visible_patch_size	The minimum patch size, measured in number of grid cells, necessary to be considered a visible patch. Connectivity is defined within the immediate eight-cell neighborhood. Cells of the same class to the right, left, above, below, or diagonal to each other are considered as part of the same patch. The input must be an integer value.	Long
Population_raster	The raster dataset that contains population data. The grid input can be either an integer or floating-point raster layer.	Raster Layer
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved within a file geodatabase.	Table
Select_options (Optional)	Tool options available to provide additional information:	Multiple Value

Code Samples

Population Land Cover Views example 1 (Python window)

This example returns a table with the percentage of the population in each reporting unit that have potential views of forest land cover within 50 meters from their location. The forest class is visible if it is in a group of cells larger than or equal to one cell. Additional fields for total population count within the reporting unit, the total population count in the reporting unit with potential views, the total population count in the reporting unit without potential views, and the percentage of the total population in the reporting unit without potential views are also output. The options to retain the intermediate calculation products, and to record the process steps into a log file have also been selected.

NOTE: *ATtILA tools that utilize the **Report metrics for these classes** parameter will clear all class selections when the tool is opened. This results in setting any variable set for that parameter to **None** when calling the tool from a script. Because of this, the standard scripting syntax (e.g. `arcpy.ATtILA.LCP(variable 1, variable 2, variable 3, etc.)` will fail. To overcome this idiosyncrasy, use the scripting example provided below. When using this technique, all variables are passed in as strings.*

NOTE: *By running scripts which launch an ATtILA tool via 'metric.run' within ArcGIS Pro's Python Window, all geoprocessing functions run by that tool can be viewed in the History Pane.*

```
import arcpy
from arcpy.sa import *

from ATtILA2 import metric

toolPath = "D:\\Destination Folder\\ToolboxSource\\ATtILA2\\scripts\\PopulationLandCoverViews_ArcGIS.py"
Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Blockgroups"
Reporting_unit_ID_field = "bgrp"
Land_cover_grid = "D:\\pyexamples\\data.gdb\\MULC_2017"
Land_cover_classification_scheme = "MULC Blended Levels ALL"
Land_cover_classification_file = "D:\\Destination Folder\\ToolboxSource\\LandCoverClassifications\\MULC
    Blended Levels ALL.xml"
Report_metrics_for_these_classes = "for - [for_PV_C] Trees/Forest, and Woody Wetlands"
View_radius = "50"
Minimum_visible_patch_size = "1"
Population_raster = "D:\\pyexamples\\data.gdb\\DASY_Population"
Output_table = "D:\\pyexamples\\results.gdb\\Blockgroups_PLCV"
Processing_cell_size = "1"
Snap_raster = "D:\\pyexamples\\data.gdb\\MULC_2017"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric
    Calculation';'LOGFILE - Record Process Steps Taken During Metric Calculation'"

metric.runPopulationLandCoverViews(
    toolPath,
    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Land_cover_grid,
    Land_cover_classification_scheme,
    Land_cover_classification_file,
    Report_metrics_for_these_classes,
    View_radius,
    Minimum_visible_patch_size,
    Population_raster,
    Output_table,
    Processing_cell_size,
    Snap_raster,
    Select_options
)
```


Population Land Cover Views example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Constructing the ATtILA metric output table: Blockgroups_PLCV
|YYYY-MM-DD hh:mm:ss| Calculating population within each reporting unit
|YYYY-MM-DD hh:mm:ss| Reclassifying selected FOR land cover class to 1. All other values = 0...
|YYYY-MM-DD hh:mm:ss| Performing focal SUM on patches of FOR using 50 cell radius neighborhood...
|YYYY-MM-DD hh:mm:ss| Reclassifying focal SUM results into a single-value raster where 1 = potential view
area...
|YYYY-MM-DD hh:mm:ss| Save intermediate grid complete: plcv_FOR_ViewPatch_1
|YYYY-MM-DD hh:mm:ss| Save intermediate grid complete: plcv_FOR_ViewArea_1
|YYYY-MM-DD hh:mm:ss| Converting view raster to a polygon feature
|YYYY-MM-DD hh:mm:ss| Setting geoprocessing environmental parameters for snap raster and cell size to
match DASY_Population_Fresno
|YYYY-MM-DD hh:mm:ss| Extracting population pixels within the potential view area...
|YYYY-MM-DD hh:mm:ss| Save intermediate grid complete: plcv_FOR_ViewPopulation_0
|YYYY-MM-DD hh:mm:ss| Calculating population within minimal-view areas for each reporting unit
|YYYY-MM-DD hh:mm:ss| Restoring snap raster geoprocessing environmental parameter to MULC_2017
|YYYY-MM-DD hh:mm:ss| Restoring cell size geoprocessing environmental parameter to 1
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
|YYYY-MM-DD hh:mm:ss| Calculation complete
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Population Within Zone Metrics

Available with Spatial Analyst license.

Summary

Calculates percentage of total population living within or near a zone for each reporting unit polygon and creates an output table.

Counts for the total population in the reporting unit and within the zone are also reported.

Usage

- **NOTE:** For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.
- **NOTE:** Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.
- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- This tool processes all zone features in the **Zone raster or vector feature** regardless of selections set. The ability to limit calculations to only selected features is not supported in this release.
- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the populations for each polygon are combined and metrics are reported as a single record in the **Output table**.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - **NOTE:** For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.
- When the optional **Group by zone** selection is checked, the tool allows the user to select a **Zone ID field**.

- The **Zone raster or vector feature** will allow you to choose either an integer raster, or a vector polygon, polyline, or point feature layer.
 - When choosing a raster, this tool assumes the grid cell value to be the population count for the grid cell.
 - When choosing a polygon feature, this tool assumes that population is distributed evenly throughout each feature polygon. The tool apportions population by area weighting. For example, if 50% of a population polygon is within a reporting unit, the tool will assign 50% of the value in the polygon's **Population field** to that reporting unit. Caution should be exercised when population polygons do not have even population distributions as this could result in an overweighting or underweighting of population when the tool performs the apportionment. Generally, greater accuracy will be achieved if the population polygons are smaller than the smallest **Reporting unit feature** polygons.
 - ***NOTE:** The **Population field** dropdown is only available when the **Population raster or polygon feature** is a polygon feature.*
 - ***NOTE:** The **Population field** will default to 'Value' when the **Population raster or polygon feature** is an integer grid.*
 - ***NOTE:** The **Population field** will be empty when the **Population raster or polygon feature** is a non-integer grid.*
- Field names in the **Output table** are as follows:
 - **RU_POP** - The population for the reporting unit.
 - **ZN_POP_C** - The population in a zone within the reporting unit.
 - **ZN_POP_P** - The percentage of the reporting unit population in a zone.
- In **Additional Options**, the user may elect to 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **pwzm_[Population raster or polygon feature]_Work_#** (vector) - A copy of the **Population raster or polygon feature** with the **Population field** retained and a **tmpClass** field added to assist in the population calculations via a Tabulate Intersection operation. This feature class is only produced when a polygon population feature is selected.
 - **pwzm_[Zone raster or vector feature]_Buffer_#** (vector) - **Zone raster or vector feature** buffered to the **Buffer distance**. The attribute table will contain fields from **Zone raster or vector feature**, the buffer distance (**BUFF_DIST**), and a field named, 'ORIG_FID'. This feature class is only produced when the **Zone raster or vector feature** is a vector feature and a **Buffer distance** greater than zero is specified.

- **pwzm_populationCnt_#** (table) - A table with the population count within each reporting unit.
 - If the **Population raster or polygon feature** is a polygon feature, the table is generated by performing a Tabulate Intersection operation using the **Reporting unit feature** as the Input Zone Feature, the **Reporting unit ID field** as the Zone Field, and the **pwzm_[Population raster or polygon feature]_Work_#** as the Input Class Features.
 - If the **Population raster or polygon feature** is a raster, the table is generated by performing a Zonal Statistics as Table operation using the **Reporting unit feature** as the Input raster or feature zone data, the **Reporting unit ID field** as the Zone field, and the **Population raster or polygon feature** as the Input value raster. The statistics type is “Sum” and the Ignore NoData in calculations is checked.
 - The population count for the reporting unit is calculated first before the zone population count and will have the lower of the two incremental filename values (i.e., #) of the produced intermediate **pwzm_populationCnt_#** files.
- **pwzm_[Zone raster or vector feature]_Poly_#** (vector) - A polygon feature class produced by setting 0-value areas in the input zone raster to NULL and then performing a raster to polygon conversion. A **tmpClass** field is added to assist in the population calculations. This feature class is only produced when the **Zone raster or vector feature** is a raster and the **Population raster or polygon feature** is a polygon feature or if the **Group by zone** option has been selected and the **Population raster or polygon feature** is a raster.
- **pwzm_[Zone raster or vector feature]_Dissolve_#** (vector) - A polygon feature class produced by dissolving **Zone raster or vector feature**. This feature class is only produced if the **Zone raster or vector feature** is a vector feature or in the special circumstance when the **Zone raster or vector feature** is a raster, the **Population raster or polygon feature** is a raster, and the **Group by zone** option has been selected.
 - If the **Group by zone** option has been selected, the **Zone raster or vector feature** layer is dissolved on the **Zone ID field** unless the **Zone raster or vector feature** is a raster. In that case, the dissolve is performed on the gridcode field in the intermediate **pwzm_[Zone raster or vector feature]_Poly_#** layer.
 - If the **Group by zone** option has not been selected, all features in the **Zone raster or vector feature** layer are dissolved into a single multipart feature.
 - If **Zone raster or vector feature** was buffered, the intermediate **pwzm_[Zone raster or vector feature]_Buffer_#** layer is used as the input for the dissolve operation.
- **pwzm_[Zone raster or vector feature]_Identity_#** (vector) - A feature class of the zone areas by reporting unit. The identity operation will be performed using the **Reporting unit feature** and the intermediate **pwzm_[Zone raster or vector feature]_Dissolve_#** layer.
- ****pwzm_[Zone raster or vector feature]_Poly_#_Identity_#**** (vector) - A feature class of the zone areas by reporting unit. This feature class is produced when the **Zone raster or vector feature** is a raster and the **Population raster or polygon feature** is a

vector. The identity operation will be performed using the **Reporting unit feature** and the intermediate **pwzm_[Zone raster or vector feature]_Poly_#** layer.

- **pwzm_[Zone raster or vector feature]_IdentityDissolve_#** (vector) - A polygon feature class produced by dissolving the intermediate **pwzm_[Zone raster or vector feature]_Identity_#** layer by the **Zone ID field** and **Reporting unit ID field**. This feature is only produced if the **Group by zone** option is selected.
- ****pwzm_[Zone raster or vector feature]_Poly_#_IdentityDissolve_#**** (vector) - A polygon feature class produced by dissolving the intermediate ****pwzm_[Zone raster or vector feature]_Poly_#_Identity_#**** layer by the **Zone ID field** and **Reporting unit ID field**. This feature is only produced if the **Group by zone** option is selected.
- **pwzm_populationZN_#** (raster) - A population raster where grid cells within zone areas contain population values from the **Population raster or polygon feature** raster. Areas outside of designated zone areas are set to NoData. This intermediate product is only produced when the **Population raster or polygon feature** and **Zone raster or vector feature** are both rasters. The cell size and cell boundaries match the **Population raster or polygon feature** raster.
- **pwzm_populationCnt_#+1** (table) - A table with the population count within the zonal areas in each reporting unit.
 1. Raster Population and Raster Zone: this table is produced using a Zonal Statistics as Table operation with the **Reporting unit feature** as the in_zone_data, the **Reporting unit ID field** as the zone_field, the intermediate **pwzm_populationZN_#** as the in_value_raster, and 'Sum' as the statistics_type.
 2. Raster Population and Buffered or Non-Buffered Vector Zone: this table is produced using a Zonal Statistics as Table operation using **pwzm_[Zone raster or vector feature]_Identity_#** as the in_zone_data, the **Reporting unit ID field** as the zone_field, the **Population raster or polygon feature** as the in_value_raster, and 'Sum' as the statistics_type.
 3. Polygon Population and Raster Zone: this table is produced using a Tabulate Intersection operation using ****pwzm_[Zone raster or vector feature]_Poly_#_Identity_#**** as the in_zone_features, **Reporting unit ID field** as the zone_field, **pwzm_[Population raster or polygon feature]_Work_#** as the in_class_features, **tmpClass** as the class-fields, and **Population field** as the sum_fields.
 4. Polygon Population and Buffered or Non-Buffered Vector Zone: this table is produced using a Tabulate Intersection operation using **pwzm_[Zone raster or vector feature]_Identity_#** as the in_zone_features, **Reporting unit ID field** as the zone_field, **pwzm_[Population raster or polygon feature]_Work_#** as the in_class_features, **tmpClass** as the class-fields, and **Population field** as the sum_fields.
- **NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the *CreateScratchName ArcPy* function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.

- **NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.
- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - **NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
 - **NOTE:** ATtILA will report the name and location of the log file in the Tool Details’ Messages section.

Syntax

PAAA (Park_feature, {Dissolve_parks}, Cost_surface_raster, Population_raster_or_polygon_feature, {Population_field}, Maximum_travel_distance, Expand_area_served, Output_raster, {Processing_cell_size}, {Snap_raster}, {Select_options})

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that	Field

Parameter	Explanation	Data Type
	contains the unique ID for each reporting unit. It may be an integer or a string data type.	
Population_raster_or_polygon_feature	The raster or vector polygon dataset that contains population data.	Feature Layer or Raster Layer
Population_field (Optional)	The field in the Population raster or polygon feature layer that contains population data. The behavior of this parameter is dependent on the input for the Population or polygon feature:	Field
Zone_raster_or_vector_feature	The integer raster or vector dataset that defines the zone.	Feature Layer or Raster Layer
Buffer_distance	An additional distance around the zone vector features that population counts are returned for. A valid entry for this parameter is dependent on the input for the Zone raster or vector feature:	Linear Unit
Group_by_zone (Optional)	Specifies whether population counts are returned for each unique intersection of reporting unit and one or more zone features.	Boolean
Zone_ID_field	The field in the Zone raster or vector feature layer that contains the unique ID for each zone.	Field
Output_table	The output reporting unit metrics table to be created. It is recommended that	Table

Parameter	Explanation	Data Type
	the Output table be saved within a file geodatabase.	
Select_options (Optional)	Tool options available to provide additional information:	Multiple Value

Code Samples

Population Within Zone Metrics example 1 (Python window)

This example returns a table with the population count in a reporting unit that is within 2000 meters of a daycare facility, the total population count within the reporting unit, and the percentage of the reporting unit's population that is in the 2000 meter buffer area. The options to retain the intermediate calculation products, and to record the process steps into a log file have also been selected.

```
import arcpy
from arcpy.sa import *

Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Blockgroups"
Reporting_unit_ID_field = "GEOID10"
Population_raster_or_polygon_feature = "D:\\pyexamples\\data.gdb\\Census2010"
Population_field = "TOTALPOP"
Zone_raster_or_polygon_feature = "D:\\pyexamples\\data.gdb\\Daycares"
Buffer_distance = "2000 Meters"
Group_by_zone = ""
Zone_ID_field = ""
Output_table = "D:\\pyexamples\\results.gdb\\Blockgroups_PWZM"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric Calculation'; 'LOGFILE - Record Process Steps Taken During Metric Calculation'"

arcpy.PolygonWithinZones_3(
    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Population_raster_or_polygon_feature,
    Population_field,
    Zone_raster_or_polygon_feature,
    Buffer_distance,
    Group_by_zone,
    Zone_ID_field,
    Output_table,
    Select_options
)
```

Population Within Zone Metrics example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\\filename
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Adding 2000 Meters buffer to Daycares. Intermediate: pwzm_Daycares_Buffer_0
|YYYY-MM-DD hh:mm:ss| Calculating population within each reporting unit. Intermediate:
pwzm_populationCnt_0
```



```
|YYYY-MM-DD hh:mm:ss| Making a working copy of Census2010. Intermediate: pwzm_Census2010_Work_0
|YYYY-MM-DD hh:mm:ss| Making a working copy of Daycares. Intermediate: pwzm_Daycares_Work_0
|YYYY-MM-DD hh:mm:ss| Dissolving pwzm_Daycares_Work_1. Intermediate: pwzm_Daycares_Dissolve_0
|YYYY-MM-DD hh:mm:ss| Assigning reporting unit IDs to Daycares. Intermediate: pwzm_Daycares_Identity_0
|YYYY-MM-DD hh:mm:ss| Calculating population within zone areas for each reporting unit. Intermediate:
pwzm_populationCnt_0
|YYYY-MM-DD hh:mm:ss| Calculating the percent of the population that is within a zone for each reporting
unit
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
|YYYY-MM-DD hh:mm:ss| Calculation complete
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Road Density Metrics

Summary

Calculates metrics based on input vector line dataset(s) for each reporting unit polygon and creates an output table.

Default metrics include total length (km) and line density (km/km²) by reporting unit. Also included in the default metrics is percent impervious surface within each reporting unit, the calculation of which assumes that the input vector line dataset represents roads.

In addition to the default metrics, the tool allows for several optional metrics to be calculated:

- Length and density metrics by Road feature class within each reporting unit. Classes are designated based on values in a specified field in the Road feature layer.
- Metrics that involve spatial relationships between the Road feature and a second linear feature (Stream feature). The user may select to calculate the following:
 - Metrics that measure the frequency of intersections between the Road feature lines and the Stream feature lines in each reporting unit. Measurements by Road feature class are also available.
 - Metrics that measure the length of the Road feature lines within a specified buffer distance from Stream feature lines within each reporting unit. Measurements by Road feature class are also available.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- This tool is designed to use vector line datasets that represent roads and streams, but any vector line dataset may be used. Caution should be exercised when interpreting results, particularly when using line datasets that do not represent roads and/or streams.
- The **Reporting unit feature** must be a polygon feature class or shapefile.
- The optional **Road class field** may be used to create metrics for different classes of **Road feature** lines. The **Road class field** may be any field that differentiates category of **Road feature** lines and does not necessarily need to represent road class.

- Values in the optional **Road class field** have restrictions. They may not contain spaces or special characters such as hyphens, parentheses, brackets, and symbols such as #, \$, %, and &. Essentially, the values in this field are acceptable if they consist only of alphanumeric characters or an underscore. The values in these fields are used to create class-specific metric fields for the three metrics (RDKM, RDDENS, and RDTIA) in the **Output table**. If any of the values in the **Road class field** violate these character restrictions, consider using a different class field or creating a new one that corresponds to the restrictions.
- If a **Road class field** is specified, the tool creates the new field names by appending the **Road class field** characters to the metric field names. Since field name length is dependent on the output table type (i.e., dBase, or geodatabase) care should be used so that the length of the final output field name does not exceed the maximum allowable field name limit for the desired output table. When the limit is exceeded, the **Road class field** characters may replace the characters at the end of these field names.
 - Example: Two road classes exist in the **Road feature** and a dBASE output table type is specified (maximum field length = 10 characters). The **Road class field** designates these as “A31” and “B45100”. When the tool is run with default settings (e.g. no options are selected), six output fields are created in the **Output table**, which are named as follows (**Road class field** characters in bold for emphasis):
 - **A31**: RDKMA**A31** (7 characters), RDDENSA**A31** (9 characters), RDTIAA**A31** (8 characters)
 - **B45100**: RDKMB**B45100** (10 characters), RDDEB**B45100** (10 characters), RDTIB**B45100** (10 characters)
 - ***NOTE**: If the **Road class field** characters replace too many of the characters in the metric field names, duplicate field names will be created.*
- When the optional **STXRD** (stream-road crossing) and/or **RNS** (roads near streams) are checked, the tool allows the user to select a **Stream feature**. When **RNS** is checked, the user is also allowed to enter a **Buffer distance** from the **Stream feature** lines.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - ***NOTE**: For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.
- Field names in the **Output table** follow this naming scheme:
 - For default tool settings (e.g. no optional settings selected):
 - **AREAKM2** - The area of the reporting unit in km².
 - **RDKM** - The total length of **Road feature** lines in km within the reporting unit.
 - **RDDENS** - The density of **Road feature** lines in km/km² within the reporting unit.

- **RDTIA** - An estimate of percent impervious area within the reporting unit. This metric uses road density as the independent variable in a linear regression model to estimate percent impervious surface (see May, et al., 1997). Due to the nature of the regression equation used, values below 1.8 km/km² are assigned a value of 0 for the percent impervious metric, while values above 11 km/km² are considered invalid and are reported as -1.
- When the **STXRD** option is selected:
 - **STRMKM** - The total length of **Stream feature** lines in km within the reporting unit.
 - **STRMDENS** - The density of **Stream feature** lines in km/km² within the reporting unit.
 - **XCNT** - The count of intersections (crossings) between the **Road feature** lines and the **Stream feature** lines within the reporting unit.
 - **STXRD** - The density of stream-road crossings per stream km within the reporting unit.
- When the **RNS** option is selected:
 - **RNS[buffer distance]** - The proportion of the total length of **Road feature** lines within the **Buffer distance** to the total length of **Stream feature** lines by reporting unit.
- When the option to report metrics by **Road class field** is selected, separate fields are generated for each metric/class combination. The metric/class field naming convention is described above.
- In **Additional Options**, the user may elect to ‘Retain Intermediate Layers Generated During Metric Processing’, and/or ‘Record Process Steps Taken During Metric Calculation’.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **rdm_[Reporting unit feature]#** (vector) - A polygon feature class containing reporting units with a new area field in km².
 - **rdm_[Road feature]#** (vector) - A line feature class containing road features. This intermediate vector is only generated if the geometry coordinates for the input Road feature include M or Z values. These values are stripped off in the intermediate file.
 - **rdm_RdsByRU#** (vector) - A line feature class with Road features intersected with reporting units.
 - **rdm_[Stream feature]#** (vector) - A line feature class containing stream features. This intermediate vector is only generated if the geometry coordinates for the input **Stream feature** include M or Z values. These values are stripped off in the intermediate file. This feature class is only produced when the optional **STXRD** selection is checked.
 - **rdm_StrByRU#** (vector) - A line feature class with Stream features intersected with reporting units. This feature class is only produced when the optional **STXRD** or **RNS** selection is checked.

- **rdm_PtsOfXing#** (vector) - A point feature class depicting locations where Road feature lines intersect Stream feature lines. This feature class is only produced when the optional **STXRD** selection is checked.
- **rdm_RdsXStrMP#** (vector) - A multipoint feature class in which all stream-road crossing points from the **rdm_PtsOfXing#** feature class are assembled by reporting unit. This feature class is only produced when the optional **STXRD** selection is checked.
- **rdm_RdsXStrTbl#** (table) - A table of stream lengths and stream-road crossing density within each reporting unit. This table is only produced when the optional **STXRD** selection is checked.
- **rdm_StrBuffers#** (vector) - A polygon feature class containing Stream features from the **rdm_StrByRU#** intermediate feature class buffered by the selected buffer distance. This feature class is only produced when the optional **RNS** selection is checked.
- **rdm_RdsNrStrms#** (vector) - A line feature class containing Road features contained within the **rdm_StrBuffers#** intermediate feature class. This feature class is only produced when the optional **RNS** selection is checked.
- ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
- ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*
- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - ***NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.*
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.

- Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
 - **NOTE:** ATiLA will report the name and location of the log file in the Tool Details' Messages section.

Syntax

RDM (Reporting_unit_feature, Reporting_unit_ID_field, Road_feature, Output_table, Road_class_field, STXRD, RNS, Stream_feature, Buffer_distance, Select_options)

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Road_feature	The vector line dataset representing roads or other linear features to be measured.	Feature Layer
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved within a file geodatabase.	Table
Road_class_field (Optional)	The field in the Road feature layer that distinguishes classes of linear features in order to calculate separate metrics for each class. It may be an integer or a string data type. The values in this field must not contain spaces or	Field

Parameter	Explanation	Data Type
	special characters. As class code values are appended to output field names, care should be taken not to exceed the limit for field name size set by output table type.	
STXRD (Optional)	Specifies whether stream-road crossing (STXRD) metrics will be included in the output table.	Boolean
RNS (Optional)	Specifies whether the roads near streams (RNS) metric will be included in the output table.	Boolean
Stream_feature	The vector line dataset representing streams that is used to calculate the STXRD and/or RNS metrics. This dataset is required when the STXRD and/or RNS options are checked.	Feature Layer
Buffer_distance	The distance around the Stream features in which buffer zones are created. The value must be an integer. If the distance linear units are not specified or are entered as Unknown, the linear unit of the input features' spatial reference is used. This parameter is required when the RNS option is checked.	Linear unit
Select_options	Tool options available to provide additional information:	Multiple Value

Code Samples

Road Density Metrics example 1 (Python window)

This example returns a table with the following fields: the area of the reporting unit in square kilometers, the length of roads within the reporting unit in kilometers, the density of roads within the reporting unit in km/km², the length of streams within the reporting unit in kilometers, the density of streams within the reporting unit in km/km², the count of road and stream crossings, the average number of road and stream crossings per kilometer of stream, and the amount of roads in kilometers within 50 meters of a stream. The options to retain the intermediate calculation products, and to record the process steps into a log file have also been selected.

```
import arcpy

Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Road_feature = "D:\\pyexamples\\data.gdb\\Roads"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_RDM"
Road_class_field = ""
STXRD = "true"
RNS = "true"
Stream_feature = "D:\\pyexamples\\data.gdb\\Stream_Lines"
Buffer_distance = "50 Meters"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric Calculation'; 'LOGFILE - Record Process Steps Taken During Metric Calculation'"

    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Road_feature,
    Output_table,
    Road_class_field,
    STXRD,
    RNS,
    Stream_feature,
    Buffer_distance,
    Select_options
)
```

Road Density Metrics example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Creating temporary copy of Watersheds
|YYYY-MM-DD hh:mm:ss| Calculating reporting unit area
|YYYY-MM-DD hh:mm:ss| Calculating road density
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
|YYYY-MM-DD hh:mm:ss| Compiling calculated values into output table
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
```


Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
|YYYY-MM-DD hh:mm:ss| Calculating Stream and Road Crossings (STXRD)
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
|YYYY-MM-DD hh:mm:ss| Compiling calculated values into output table
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
|YYYY-MM-DD hh:mm:ss| Calculating Roads Near Streams (RNS)
|YYYY-MM-DD hh:mm:ss| Buffering stream features...
|YYYY-MM-DD hh:mm:ss| Intersecting road features with stream buffers...
|YYYY-MM-DD hh:mm:ss| Assigning reporting unit feature ID values to road segments...
|YYYY-MM-DD hh:mm:ss| Dissolving intersection result and calculating values...
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
|YYYY-MM-DD hh:mm:ss| Compiling calculated values into output table
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed

Tags

There are no tags for this item.

Credits

May, C.W., Horner, R.R., Karr, J.R., Mar B.W., Welch, E.B. 1997. *Effects of urbanization on small streams in the Puget Sound Lowland Ecoregion*. Watershed Protection Techniques. 2:4. pp. 483–493.

Use Limitations

There are no access and use limitations for this item.

Riparian Characteristics

Riparian zones, i.e., the areas adjacent to streams and associated waterbodies, provide critical habitat areas and play an important role in determining or preserving surface water quality. While all of the land area within a watershed may contribute pollutants via runoff, the effect of more distant (from stream) areas may be mitigated along the transport path. However, there is little or no mitigation of nutrient or soil runoff from stream side areas. Thus these buffer areas along streams may have a greater impact upon water quality. Conversely, conservation of riparian buffers may help to preserve good water quality.

The Riparian Characteristics toolset consists of the follow metric tools:

- **Floodplain Land Cover Proportions** Calculates the percentage of floodplain area within a reporting unit classified as selected land cover classes and creates an output table.
- **Riparian Land Cover Proportions** is similar to the Land Cover Proportions metric in the Landscape Characteristics toolset except that the results are calculated only for the buffer area around streams instead of over the whole reporting unit. Input features may be lines or polygons or both. Adjacency is determined by a buffer distance specified by the user. Additionally, reporting unit boundaries can be selected to act as an absolute barrier when creating buffer areas or ignored. If boundaries are enforced, buffer areas for features outside of the reporting unit will stop at the reporting unit boundary, otherwise buffer areas for features outside of the reporting unit are allowed inside the reporting unit.
- **Sample Point Land Cover Proportions** Calculates the percentages of selected land cover types within the area adjacent to input point features in each reporting unit and creates an output table. Adjacency is determined by a buffer distance specified by the user. Additionally, reporting unit boundaries can be selected to act as an absolute barrier when creating buffer areas or ignored. If boundaries are enforced, buffer areas for features outside of the reporting unit will stop at the reporting unit boundary, otherwise buffer areas for features outside of the reporting unit are allowed inside the reporting unit.
- **Stream Density Metrics** Calculates density metrics based on input vector line dataset(s) for each reporting unit polygon and creates an output table. Default metrics include area of the reporting unit (km²), total length (km) and line density (km/km²) within the reporting unit. In addition to the default metrics, length and density metrics can be calculated by stream order. Stream order is designated based on values in a specified field in the Stream feature layer.

Tool Floodplain Land Cover Proportions

Available with Spatial Analyst license.

Summary

Calculates the percentage of floodplain area within a reporting unit classified as selected land cover classes and creates an output table.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.
- As the **Reporting unit feature** is a vector dataset, ArcGIS will perform a vector to raster conversion during processing.
 - To better control the vector to raster conversion, the tool defaults the **Snap raster** and the **Processing cell size** to that of the **Land cover grid**. These may be changed from within the tool.
 - If the extent of the **Reporting unit feature** is smaller than that of the **Land cover grid**, the user may wish to check the **Reduce land cover grid to smallest recommended size** option. This creates a temporary land cover raster with the processing extent equal to that of the **Reporting unit feature** to avoid unnecessary processing.

- If the **Reporting unit feature** has overlapping features, the zonal analysis will be performed for each individual feature.
- If a feature in the **Reporting unit feature** zone data does not overlap any cell centers in the secondary class rasters (e.g., **Land cover grid**), those zones will not get converted to the internal zone raster. As a result, those zones will not be represented in the **Output table**.
- The **Land cover classification scheme** must correspond to the **Land cover grid**.
 - Schemes for common land cover datasets are included with ATtILA. Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD 2001 ALL vs. NLCD 2001 LAND). Schemes designated as “ALL” include all land cover classes in reporting unit area calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded. More information about each of the classification schemes supplied with ATtILA may be found in Supplied Land Cover Classification Schemas.
 - In addition to the common land cover schemes, the tool permits a user-defined land cover classification scheme to be used by specifying a **Land cover classification file**. Refer to Land Cover Classification for more information on how ATtILA may be customized for user-specific needs.
 - ***NOTE:** When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. For example, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. See Values Element in ATtILA’s LCC XML Document and Values Window in the Land Cover Classification Editor for more information on excluded values.*
- The **Floodplain raster or polygon feature** will allow you to choose either a raster, or a polygon feature.
 - When choosing a raster, floodplains are defined as any non-zero grid cell. The simplest example of this is a binary raster where floodplains are coded as 1 and everything else is coded as 0, but any value or combination of values can be used to designate floodplain areas as long as non-floodplain areas are coded as 0.
 - When choosing a polygon feature, floodplains are defined as any polygon feature. This tool does not make use of any attribute or combination of attributes to select floodplain polygons. All polygons in the floodplain input layer are considered floodplain areas. If polygons exist within the layer that do not represent floodplains, they need to be edited out before using the layer as input for this tool.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - ***NOTE:** For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.

- Field names in the **Output table** follow this naming scheme:
 - **[class flcpField]** or **f[class Id]** - The percent of the non-excluded floodplain area within a reporting unit occupied by the land cover class.
 - **[class flcpField]** is used if an flcpField attribute is provided in the class's element in the selected **Land cover classification file**.
 - (e.g., **NINDEX_Flood** for <class Id="NI" Name="All Natural Land Use" filter="" flcpField="NINDEX_Flood">)
 - **f[class Id]** - is used if an flcpField attribute is not provided.
 - (e.g., **fNI** for <class Id="NI" Name="All Natural Land Use" filter="">)
 - **NOTE:** *Output field names can be altered by editing the **Land cover classification file** with the Land Cover Classification Editor (Refer to Classes Element in ATtILA's LCC XML Document for general information, or the individual **Land cover classification file** for details).*
 - **NOTE:** *The output field name for each class is shown as the second item next to the class's check box in **Report metrics for these classes**.*
- In **Additional Options**, the user may elect to 'Add Quality Assurance Fields', 'Add Area Fields', 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Add Area Fields
 - Choosing **AREAFIELDS** adds fields to record the raster area measurement of the selected land cover classes occurring in floodplains within each reporting unit. The field naming conventions is dependent on whether an flcpField is provided in the **Land cover classification file**:
 - **[class flcpField]_A** or **f[class Id]_A** - The area of the rasterized floodplain area, by reporting unit, occupied by the land use class.
 - Add Quality Assurance Fields
 - Choosing **QAFIELDS** adds a number of fields to the **Output table** to facilitate quality assurance checking.
 - **FLCP_OVER** - The percent overlap between the floodplain areas in each reporting unit and the **Land cover grid**.
 - **FLCP_TOTA** - The total raster floodplain area for each reporting unit (zone).
 - **FLCP_EFFA** - The effective raster area within a reporting unit's (zone) floodplain area after excluded classes have been removed.
 - **FLCP_EXCA** - The raster area of the excluded classes within a reporting unit's (zone) floodplain area based on exclusions in the **Land cover classification scheme**.
 - **fTOTA** - The percentage of the reporting unit classified as floodplain.
 - **fEFFA** - The percentage of the effective area in the reporting unit found in the floodplain area.

- Retain Intermediate Layers Generated During Metric Calculation

- Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:

- **flcp_Landcover#** (raster) - Copy of the **Land cover grid** where non-floodplain areas (floodplain raster value = 0) are set to NULL. Produced when the **Floodplain raster or polygon feature** is a raster.
- **flcp#** (vector) - Intersection of the **Reporting unit feature** and the **Floodplain raster or polygon feature**. Produced when the **Floodplain raster or polygon feature** is a polygon feature class.
- **flcp_FldplnByRU#** (vector) - Created by performing a Dissolve operation on **flcp#** using the **Reporting unit feature Id field** as the dissolve field. The output contains multipart features. Produced when the **Floodplain raster or polygon feature** is a polygon feature class.
- **flcp_TabArea#** (table) - A table of areas corresponding to the values in the **Land cover grid** occurring in floodplains within each reporting unit.
 - If the **Floodplain raster or polygon feature** is a polygon feature, the table is generated by performing a Tabulate Area operation using the **flcp_FldplnByRU#** and the copied **Reporting unit feature Id field** as the zone identifier and the **Land cover grid** as the raster to be summarized.
 - If the **Floodplain raster or polygon feature** is a raster, the table is generated by performing a Tabulate Area operation using the **Reporting unit feature** and its **Reporting unit feature Id field** as the zone identifier and the **flcp_Landcover#** as the raster to be summarized.
- **flcp_TabAreaFP#** (table) - A table of floodplain (VALUE_1) and non-floodplain (VALUE_0) areas within each reporting unit. Produced when the **Floodplain_raster_or_polygon_feature** is a raster and QAFIELDS is selected in **Additional options**.
- ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
- ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*

- Record Process Steps Taken During Metric Calculation

- Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.

- Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
- Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.
 - **NOTE:** ATtILA will report the name and location of the log file in the Tool Details' Messages section.

Syntax

FLCP (Reporting_unit_feature, Reporting_unit_ID_field, Land_cover_grid, Land_cover_classification_scheme, Land_cover_classification_file, Report_metrics_for_these_classes, Floodplain_raster_or_polygon_feature, Output_table, {Processing_cell_size}, {Snap_raster}, Select_options, {Reduce_land_cover_grid_to_smallest_recommended_size})

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Land_cover_grid	The raster dataset representing land cover classes to be summarized within each Reporting unit feature. The grid input must be an integer raster layer.	Raster Layer
Land_cover_classification_scheme	The land cover classification schemes included in ATtILA and a User Defined option. The default schemes correspond to common input land cover datasets. Two schemes are supplied for each dataset included in	String

Parameter	Explanation	Data Type
ATtILA:		
Land_cover_classification_file	The full pathname to the user-defined .xml file for custom or non-standard land cover classification schemes. Pathname is automatically filled when a default scheme is selected.	File
Report_metrics_for_these_classes	A list of the land cover classes and metric combinations for processing. Check the box to calculate metrics for each land cover class and/or combination class desired within the reporting units.	Multiple Value
Floodplain_raster_or_polygon_feature	The integer raster or vector polygon dataset that defines the floodplains.	Feature Layer or Raster Layer
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved within a file geodatabase.	Table
Processing_cell_size (Optional)	The Processing cell size for the zonal operation. The default Processing cell size is the cell size of the input raster land cover data. Optionally, the user may select a different cell size.	Analysis Cell Size
Snap_raster (Optional)	The raster that the cell alignment of the Land cover grid and rasterized Reporting unit feature layer will be matched to during processing. The default Snap raster is the Land cover grid.	Raster Layer

Parameter	Explanation	Data Type
Select_options	Tool options available to provide additional information:	Multiple Value
Reduce_land_cover_grid_to_smallest_recommended_size (Optional)	Specifies whether the Land cover grid extent will be clipped to that of the Reporting unit feature layer during processing.	Boolean

Code Samples

Floodplain Land Cover Proportions example 1 (Python window)

This example returns a table with the following fields: the proportion of the floodplain in the reporting unit zone comprised of agriculture land cover, and the proportion of the floodplain in the reporting unit zone comprised of developed land cover. The options to retain the intermediate calculation products, to record the process steps into a log file, to add the square area values for the selected land cover classes in the floodplain, and to generate the quality assurance fields have also been selected.

NOTE: ATtILA tools that utilize the **Report metrics for these classes** parameter will clear all class selections when the tool is opened. This results in setting any variable set for that parameter to **None** when calling the tool from a script. Because of this, the standard scripting syntax (e.g. `arcpy.ATtILA.LCP(variable 1, variable 2, variable 3, etc.)` will fail. To overcome this idiosyncrasy, use the scripting example provided below. When using this technique, all variables are passed in as strings.

NOTE: By running scripts which launch an ATtILA tool via 'metric.run' within ArcGIS Pro's Python Window, all geoprocessing functions run by that tool can be viewed in the History Pane.

```
import arcpy
from arcpy.sa import *

from ATtILA2 import metric

toolPath = "D:\\Destination
Folder\\ToolboxSource\\ATtILA2\\scripts\\FloodplainLandCoverProportions_ArcGIS.py"
Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Land_cover_grid = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Land_cover_classification_scheme = "NLCD LAND"
Land_cover_classification_file = "D:\\Destination Folder\\ToolboxSource\\LandCoverClassifications\\NLCD
LAND.xml"
Report_metrics_for_these_classes = "agr - [fagr] Agriculture';'dev - [fdev] Developed"
Floodplain_raster_or_polygon_feature = "D:\\pyexamples\\data.gdb\\Floodplain_Raster"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_FLCF"
Processing_cell_size = "30"
Snap_raster = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Select_options = "'AREAFIELDS - Add Area Fields for All Land Cover Classes';'INTERMEDIATES - Retain
Intermediate Layers Generated During Metric Calculation';'LOGFILE - Record Process Steps Taken
During Metric Calculation';'QAFIELDS - Add Quality Assurance Fields'"
Reduce_land_cover_grid_to_smallest_recommended_size = ""

metric.runFloodplainLandCoverProportions(
    toolPath,
```

```

Reporting_unit_feature,
Reporting_unit_ID_field,
Land_cover_grid,
Land_cover_classification_scheme,
Land_cover_classification_file,
Report_metrics_for_these_classes,
Floodplain_raster_or_polygon_feature,
Output_table,
Processing_cell_size,
Snap_raster,
Select_options,
Reduce_land_cover_grid_to_smallest_recommended_size
)

```

Floodplain Land Cover Proportions example 2 with raster floodplain input (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```

Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Excluded values found in the land cover grid. Calculating effective areas for each
reporting unit
|YYYY-MM-DD hh:mm:ss| Generating land cover in floodplain grid
|YYYY-MM-DD hh:mm:ss| Save intermediate grid complete: flcp_Landcover_0
|YYYY-MM-DD hh:mm:ss| Tabulating the area of the floodplains within each reporting unit
|YYYY-MM-DD hh:mm:ss| Constructing the ATtILA metric output table: Watersheds_FLCP
|YYYY-MM-DD hh:mm:ss| Generating a zonal tabulate area table
|YYYY-MM-DD hh:mm:ss| Processing the tabulate area table and computing metric values
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)

```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Riparian Land Cover Proportions

Available with Spatial Analyst license.

Summary

Calculates the percentages of selected land cover types within the area adjacent to input features in each reporting unit and creates an output table.

Input features may be lines or polygons or both.

Adjacency is determined by a buffer distance specified by the user.

Additionally, reporting unit boundaries can be selected to act as an absolute barrier when creating buffer areas or ignored. If boundaries are enforced, buffer areas for features outside of the reporting unit will stop at the reporting unit boundary, otherwise buffer areas for features outside of the reporting unit are allowed inside the reporting unit.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- This tool is designed to use a vector line or polygon dataset that represents streams, but any vector line or polygon dataset may be used. Caution should be exercised when interpreting results, particularly when using datasets that do not represent streams.
- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.

- **NOTE:** *ArcGIS software has been improved over the years to analyze larger and larger datasets. However, limits do exist and large datasets can still cause problems. The Riparian Land Cover Proportions calculator can fail if an input **Reporting unit feature** contains a large number of reporting units and/or the reporting units cover a significantly large geographical area. Often this type of failure will manifest itself in a Failed to execute (Intersect) error. See Failed to execute (Intersect) for possible workarounds.*
- The tool extracts the **Reporting unit feature** polygons that overlay the **Stream features** buffers using an Intersect and Query. This results in an intermediate polygon feature class containing multipart features of all buffer areas in each reporting unit. The tool then performs a vector to raster conversion of the multipart features to create zones for each reporting unit.
 - To better control the vector to raster conversion, the tool defaults the **Snap raster** and the **Processing cell size** to that of the **Land cover grid**. These may be changed from within the tool.
 - If a large size disparity exists between the extents of the **Reporting unit feature** and the **Land cover grid**, the user may wish to set the Extent in Environment Settings > Processing Extent to the smaller of the two to avoid unnecessary processing.
- If the **Reporting unit feature** has overlapping features, the zonal analysis will be performed for each individual feature.
- If a feature in the **Reporting unit feature** zone data does not overlap any cell centers in the secondary class rasters (e.g., **Land cover grid**), those zones will not get converted to the internal zone raster. As a result, those zones will not be represented in the **Output table**.
- The **Land cover classification scheme** must correspond to the **Land cover grid**.
 - Schemes for common land cover datasets are included with ATtILA. Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD 2001 ALL vs. NLCD 2001 LAND). Schemes designated as “ALL” include all land cover classes in the metric calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded. More information about each of the classification schemes supplied with ATtILA may be found in Supplied Land Cover Classification Schemas.
 - In addition to the common land cover schemes, the tool permits a user-defined land cover classification scheme to be used by specifying a **Land cover classification file**. Refer to Land Cover Classification for more information on how ATtILA may be customized for user-specific needs.
 - **NOTE:** *When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. For example, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. See Values Element in ATtILA’s LCC XML Document and Values Window in the Land Cover Classification Editor for more information on excluded values.*
- **Stream features** may be one or more vector line or polygon datasets. The **Buffer distance** specifies the distance around the input features for which buffer zones are created.
 - When **Stream features** is a line dataset, the buffer occurs on both sides of each line.

- When **Stream features** is a polygon dataset, the tool treats the polygon perimeters as lines. The buffer occurs on both sides of each polygon perimeter.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - **NOTE:** *For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.
- Field names in the **Output table** follow this naming scheme:
 - **[class rlcField]** or **r[class Id][buffer distance]** - The percent of the total buffered area in the reporting unit occupied by the land cover class.
 - **[class rlcField]** is used if an rlcField attribute is provided in the class's element in the selected **Land cover classification file**.
 - (e.g., **RiparianForest** for <class Id="for" Name="Forest and Woody Wetland" filter="" rlcField="RiparianForest">)
 - **r[class Id][buffer distance]** - is used if an rlcField attribute is not provided.
 - (e.g., **rfor45** for <class Id="for" Name="Forest and Woody Wetland" filter=""> and a 45 meter buffer distance is used)
 - **NOTE:** *Output field names can be altered by editing the **Land cover classification file** with the Land Cover Classification Editor (Refer to Classes Element in ATtILA's LCC XML Document for general information, or the individual **Land cover classification file** for details).*
 - **NOTE:** *The output field name for each class is shown as the second item next to the class's check box in **Report metrics for these classes**.*
- In **Additional Options**, the user may elect to 'Add Quality Assurance Fields', 'Add Area Fields', 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Add Area Fields
 - Choosing **AREAFIELDS** adds fields to record the raster area measurement of the selected land cover classes within the buffer area in each reporting unit. The field naming conventions is dependent on whether an rlcField is provided in the **Land cover classification file**:
 - **[class rlcField]_A** or **r[class Id][buffer distance]_A** - The area of the rasterized Stream features buffer, by reporting unit, occupied by the land cover class.
 - Add Quality Assurance Fields
 - Choosing **QAFIELDS** adds a number of fields to the **Output table** to facilitate quality assurance checking.

- **RLCP_OVER** - The percent overlap between the Stream features buffer areas in each reporting unit and the Land cover grid.
 - **RLCP_TOTA** - The total raster zonal area for all Stream features buffers in each reporting unit (zone) after the buffer areas have been rasterized.
 - **RLCP_EFFA** - The effective raster area of all Stream features buffers in each reporting unit (zone) after excluded classes have been removed.
 - **RLCP_EXCA** - The raster area of the excluded classes within the Stream features buffers in each reporting unit (zone) based on exclusions in the **Land cover classification scheme**.
 - **rTOTA[buffer distance]** - The percentage of the reporting unit comprised of the Stream features buffer area.
 - **rEFFA[buffer distance]** - The percentage of the effective area in the reporting unit found in the Stream features buffer area.
- Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **rlcp_Dissolve_[Buffer distance]_#** (vector) - Copy of the **Reporting unit feature** dissolved on the **Reporting unit ID field**. This feature class is only produced if duplicate values are found in the **Reporting unit ID field**.
 - **rlcp_TabAreaRU#** (table) - A table of areas corresponding to the values in the **Land cover grid** occurring in each reporting unit. This table is only produced when **Add Quality Assurance Fields** is selected in **Additional Options**.
 - Intermediates when **Enforce reporting unit boundaries** = TRUE
 - **rlcp_[Input stream feature name]** (vector) - A copy of the input stream feature with any Z-Values and M-Values removed from its vertices. This feature class is only produced if the input stream feature has Z or M values.
 - **rlcp_[Input stream feature name]_intersect#** (vector) - Copy of an input stream feature intersected with **Reporting unit feature**. This step assigns the Reporting Unit's ID value to the input features and eliminates segments outside of the reporting unit boundaries. The **Reporting unit ID field** is also renamed to **new[Reporting unit ID field]**.
 - **rlcp_[Input stream feature name]_buffer#** (vector) - Copy of **rlcp_[Input stream feature name]_intersect#** buffered to the **Buffer distance**. The attribute table will retain the **new[Reporting unit ID field]** field.
 - **rlcp_[Input stream feature name]_2ndintersect#** (vector) - Copy of **rlcp_[Input stream feature name]_buffer#** intersected with **Reporting unit feature**. This step assigns the Reporting Unit's ID value to the buffer areas and eliminates segments outside of the reporting unit boundaries. The attribute table will now contain both the original **Reporting unit ID field** and the **new[Reporting unit ID**

field]. If only one input in **Stream features** is provided, the features with matching values in **Reporting unit ID field** and the **new[Reporting unit ID field]** are dissolved by reporting unit. Features without matching values are eliminated. This dissolved feature class is the final output, **rlcp_Buffer[buffer distance]#**, if only one input in **Stream features** is provided.

- **rlcp_[Input stream feature name]_finalOutput#** (vector) - Dissolved features of **rlcp_[Input stream feature name]_2ndintersect#** where the values in the **Reporting unit ID field** equaled those in the **new[Reporting unit ID field]**. This results in buffer areas that fall outside of the input feature's reporting unit being eliminated. This feature class is only produced if more than one **Stream features** themes are supplied.
- **rlcp_mergeOutput#** (vector) - This feature class is produced by merging all **rlcp_[Input stream feature name]_finalOutput#** layers together. This feature class is only produced if more than one **Stream features** themes are supplied. If no polygon features are provided in the **Stream features**, this layer is then dissolved by the **Reporting unit ID field** to produce the final output, **rlcp_Buffer[buffer distance]#**.
- **rlcp_Buffer45Meters0_preErase#** (vector) - This feature class is produced by dissolving the **rlcp_mergeOutput#** by reporting unit ID. It is only produced if more than one **Stream features** themes are supplied and at least one is a polygon layer.
- **rlcp_erasePolygons#** (vector) - This feature class is produced by merging all polygon input stream features into one layer. When present, the final output, **rlcp_Buffer[buffer distance]#**, is produced by erasing areas in **rlcp_Buffer45Meters0_preErase#** with it. This feature class is only produced if more than one **Stream features** themes are supplied and at least one is a polygon layer.
- Intermediates when **Enforce reporting unit boundaries = FALSE**:
 - **rlcp_[Input stream feature name]_Buffer#** (vector) - Copy of features in the **[Input stream feature name]** buffered to the **Buffer distance**. The attribute table will contain fields from the **Sample point features**, the buffer distance (**BUFF_DIST**), and a field named, 'ORIG_FID'. To eliminate unnecessary features before buffering, only features in the **[Input stream feature name]** that are within the **Buffer distance** of features in **Reporting unit feature** are retained.
 - **rlcp_Merge#** (vector) - This feature class is produced by merging all **rlcp_[Input stream feature name]_Buffer#** layers together. The merge operation will also be performed even if there is only one input. In that instance, the merge will create a copy of the buffer theme without any unnecessary fields. It is essentially a FeatureClassToFeatureClass operation with fieldMappings.
 - **rlcp_Intersect#** (vector) - A polygon feature class with **rlcp_Merge#** intersected with **Reporting unit feature**. This step assigns the Reporting Unit's ID value to the buffer zones and eliminates buffer areas outside of the reporting unit boundaries.
 - **rlcp_ErasePolygons#** (vector) - This feature class is produced by merging all polygon **Stream features**. It is only produced if more than one **Stream features** themes are polygon layers.

- **rlcp_Erase#** (vector) - This feature class is produced by erasing the sections of **rlcp_Intersect#** that occur within the interior of any polygon input stream feature. It is only produced if **Stream features** contains at least one polygon layer.
- **rlcp_Buffer[buffer distance]#** (vector) - A polygon feature class containing all Stream features buffered by the selected buffer distance within each reporting unit merged by **Reporting unit ID field**.
- **rlcp_TabArea#** (table) - A table of areas corresponding to the values in the **Land cover grid** occurring in the buffer area within each reporting unit.
- ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
- ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*
- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - ***NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.*
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - ***NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.*

- **NOTE:** ATtILA will report the name and location of the log file in the Tool Details' Messages section.

Syntax

RLCP (Reporting_unit_feature, Reporting_unit_ID_field, Land_cover_grid, Land_cover_classification_scheme, Land_cover_classification_file, Report_metrics_for_these_classes, Stream_features, Buffer_distance, Output_table, {Processing_cell_size}, {Snap_raster}, Select_options)

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Land_cover_grid	The raster dataset representing land cover classes to be summarized within each Reporting unit feature. The grid input must be an integer raster layer.	Raster Layer
Land_cover_classification_scheme	The land cover classification schemes included in ATtILA and a User Defined option. The default schemes correspond to common input land cover datasets. Two schemes are supplied for each dataset included in ATtILA:	String
Land_cover_classification_file	The full pathname to the user-defined .xml file for custom or non-standard land cover classification schemes. Pathname is automatically filled when a default scheme is selected.	File

Parameter	Explanation	Data Type
Report_metrics_for_these_classes	A list of the land cover classes and metric combinations for processing. Check the box to calculate metrics for each land cover class and/or combination class desired within the reporting units.	Multiple Value
Stream_features	The vector line and/or polygon dataset(s) that provide the basis for the buffer zones. Land cover metrics are calculated for the area within the buffer zones.	Feature layer
Buffer_distance	The distance around the Stream features which comprises the buffer zones. The value must be an integer. If the distance linear units are not specified or are entered as Unknown, the linear unit of the input features' spatial reference is used.	Linear unit
Enforce_reporting_unit_boundaries	Specifies whether a Reporting unit boundary acts as an absolute barrier when creating buffer areas.	Boolean
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved within a file geodatabase.	Table
Processing_cell_size (Optional)	The Processing cell size for the zonal operation. The default Processing cell size is the cell size of the input raster land cover data. Optionally, the user may select a different cell size.	Analysis cell size

Parameter	Explanation	Data Type
Snap_raster (Optional)	The raster that the cell alignment of the Land cover grid and rasterized Reporting unit feature layer will be matched to during processing. The default Snap raster is the Land cover grid.	Raster Layer
Select_options	Tool options available to provide additional information:	Multiple Value

Code Samples

Riparian Land Cover Proportions example 1 (Python window)

This example returns a table with the following fields: the proportion of the buffer area adjacent to streams in the reporting unit zone comprised of agriculture land cover, and the proportion of the buffer area adjacent to streams in the reporting unit zone comprised of forest land cover. The buffer distance was set at 45 meters and the buffer area could not cross the reporting unit boundary. The options to retain the intermediate calculation products, to record the process steps into a log file, and to generate the quality assurance fields have also been selected.

NOTE: *ATtILA tools that utilize the **Report metrics for these classes** parameter will clear all class selections when the tool is opened. This results in setting any variable set for that parameter to **None** when calling the tool from a script. Because of this, the standard scripting syntax (e.g. `arcpy.ATtILA.LCP(variable 1, variable 2, variable 3, etc.)` will fail. To overcome this idiosyncrasy, use the scripting example provided below. When using this technique, all variables are passed in as strings.*

NOTE: *By running scripts which launch an ATtILA tool via 'metric.run' within ArcGIS Pro's Python Window, all geoprocessing functions run by that tool can be viewed in the History Pane.*

```
import arcpy
from arcpy.sa import *

from ATtILA2 import metric

toolPath = "D:\\Destination
Folder\\ToolboxSource\\ATtILA2\\scripts\\RiparianLandCoverProportions_ArcGIS.py"
Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Land_cover_grid = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Land_cover_classification_scheme = "NLCD LAND"
Land_cover_classification_file = "D:\\Destination Folder\\ToolboxSource\\LandCoverClassifications\\NLCD
LAND.xml"
Report_metrics_for_these_classes = "agrc - [ragrc] Cultivated Crop';'for - [rfor] Forest"
Stream_features = "Stream_Lines;Stream_Polygons"
Buffer_distance = "45 Meters"
Enforce_reporting_unit_boundaries = "true"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_RLCP"
Processing_cell_size = "30"
Snap_raster = "D:\\pyexamples\\data.gdb\\NLCD_2016"
```

```
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric
Calculation';'LOGFILE - Record Process Steps Taken During Metric Calculation';'QAFIELDS - Add
Quality Assurance Fields'"

metric.runRiparianLandCoverProportions(
    toolPath,
    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Land_cover_grid,
    Land_cover_classification_scheme,
    Land_cover_classification_file,
    Report_metrics_for_these_classes,
    Stream_features,
    Buffer_distance,
    Enforce_reporting_unit_boundaries,
    Output_table,
    Processing_cell_size,
    Snap_raster,
    Select_options
)
```

Riparian Land Cover Proportions example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Excluded values found in the land cover grid. Calculating effective areas for each
reporting unit...
|YYYY-MM-DD hh:mm:ss| Intersecting Stream_Lines and reporting units: rlcp_Stream_Lines_intersect_0
|YYYY-MM-DD hh:mm:ss| Buffering intersected features: rlcp_Stream_Lines_buffer_0
|YYYY-MM-DD hh:mm:ss| Repairing buffer areas for input linear features...
|YYYY-MM-DD hh:mm:ss| Intersecting buffer features and reporting units: rlcp_Stream_Lines_2ndintersect_0
|YYYY-MM-DD hh:mm:ss| Trimming buffer zones to reporting unit boundaries
|YYYY-MM-DD hh:mm:ss| Dissolving second intersection by reporting unit: rlcp_Stream_Lines_final_0
|YYYY-MM-DD hh:mm:ss| Intersecting Stream_Polygons and reporting units: rlcp_Stream_Polygons_intersect_0
|YYYY-MM-DD hh:mm:ss| Buffering intersected features: rlcp_Stream_Polygons_buffer_0
|YYYY-MM-DD hh:mm:ss| Repairing buffer areas for input areal features...
|YYYY-MM-DD hh:mm:ss| Intersecting buffer features and reporting units:
rlcp_Stream_Polygons_2ndintersect_0
|YYYY-MM-DD hh:mm:ss| Trimming buffer zones to reporting unit boundaries
|YYYY-MM-DD hh:mm:ss| Dissolving second intersection by reporting unit: rlcp_Stream_Polygons_final_0
|YYYY-MM-DD hh:mm:ss| Merging buffer zones from all input Stream features: rlcp_merge_0
|YYYY-MM-DD hh:mm:ss| Dissolving merged buffer zones: rlcp_Buffer45Meters_0
|YYYY-MM-DD hh:mm:ss| Removing interior waterbody areas from dissolve result...
|YYYY-MM-DD hh:mm:ss| Constructing the ATtILA metric output table: Watersheds_RLCP
|YYYY-MM-DD hh:mm:ss| Generating a zonal tabulate area table
|YYYY-MM-DD hh:mm:ss| Processing the tabulate area table and computing metric values
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Sample Point Land Cover Proportions

Available with Spatial Analyst license.

Summary

Calculates the percentages of selected land cover types within the area adjacent to input point features in each reporting unit and creates an output table.

Adjacency is determined by a buffer distance specified by the user.

Additionally, reporting unit boundaries can be selected to act as an absolute barrier when creating buffer areas or ignored. If boundaries are enforced, buffer areas for features outside of the reporting unit will stop at the reporting unit boundary, otherwise buffer areas for features outside of the reporting unit are allowed inside the reporting unit.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *It is highly recommended that the **Land cover grid** has an attribute table along with computed raster statistics before using it as an input to ATtILA. If the **Land cover grid** data source lacks these components, ATtILA will generate them automatically using the Build Raster Attribute Table tool and the Calculate Statistics tool using their default settings. This will alter the input data. If this is acceptable, no additional steps are necessary. If the default settings are not acceptable, it will be necessary for the user to build these components themselves before running the tool.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Reporting unit feature** is a zone dataset.
 - A zone is defined as all areas in the input that have the same value. The areas do not have to be contiguous. The term “value” in this definition refers to the unique values in the **Reporting unit ID field**. Therefore, all polygons with the same reporting unit ID are treated as a single zone.
 - When more than one polygon has the same reporting unit ID, the areas for each polygon are combined and metrics are reported as a single record in the **Output table**.
- The tool extracts the **Reporting unit feature** polygons that overlay the **Sample point features** buffers using an Intersect and Query. This results in an intermediate polygon feature class containing multipart features of all buffer areas in each reporting unit. The tool then performs a vector to raster conversion of the multipart features to create zones for each reporting unit.

- To better control the vector to raster conversion, the tool defaults the **Snap raster** and the **Processing cell size** to that of the **Land cover grid**. These may be changed from within the tool.
- If a large size disparity exists between the extents of the **Reporting unit feature** and the **Land cover grid**, the user may wish to set the Extent in Environment Settings > Processing Extent to the smaller of the two to avoid unnecessary processing.
- If the **Reporting unit feature** has overlapping features, the zonal analysis will be performed for each individual feature.
- If a feature in the **Reporting unit feature** zone data does not overlap any cell centers in the secondary class rasters (e.g., **Land cover grid**), those zones will not get converted to the internal zone raster. As a result, those zones will not be represented in the **Output table**.
- The **Land cover classification scheme** must correspond to the **Land cover grid**.
 - Schemes for common land cover datasets are included with ATtILA. Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD 2001 ALL vs. NLCD 2001 LAND). Schemes designated as “ALL” include all land cover classes in the metric calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded. More information about each of the classification schemes supplied with ATtILA may be found in Supplied Land Cover Classification Schemas.
 - In addition to the common land cover schemes, the tool permits a user-defined land cover classification scheme to be used by specifying a **Land cover classification file**. Refer to Land Cover Classification for more information on how ATtILA may be customized for user-specific needs.
 - **NOTE:** *When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. For example, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. See Values Element in ATtILA’s LCC XML Document and Values Window in the Land Cover Classification Editor for more information on excluded values.*
- If **Enforce reporting unit boundaries** is selected, **Sample point features** must contain an attribute that identifies the reporting unit in which each point lies. This attribute is entered in the **Reporting Unit Link Field** parameter. If no such attribute exists in the **Sample point features** attribute table, it is recommended that the Identity tool be used. The Identity tool is located in the ArcToolbox under Analysis Tools > Overlay.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - **NOTE:** *For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.

- Field names in the **Output table** follow this naming scheme:
 - **[class splcpField]** or **s[class Id][buffer distance]** - The percent of the total buffered area in the reporting unit occupied by the land cover class.
 - **[class splcpField]** is used if an splcpField attribute is provided in the class's element in the selected **Land cover classification file**.
 - (e.g., **NearAgr** for <class Id="agr" Name="Agriculture" filter="" splcpField="NearAgr">)
 - **s[class Id][buffer distance]** - is used if an splcpField attribute is not provided.
 - (e.g., **sagr100** for <class Id="agr" Name="Agriculture" filter=""> and a 100-meter buffer distance used)
 - **NOTE:** *Output field names can be altered by editing the **Land cover classification file** with the Land Cover Classification Editor (Refer to Classes Element in ATtILA's LCC XML Document for general information, or the individual **Land cover classification file** for details).*
 - **NOTE:** *The output field name for each class is shown as the second item next to the class's check box in **Report metrics for these classes**.*
- In **Additional Options**, the user may elect to 'Add Quality Assurance Fields', 'Add Area Fields', 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Add Area Fields
 - Choosing **AREAFIELDS** adds fields to record the raster area measurement of the selected land cover classes occurring in a buffer area within each reporting unit. The field naming conventions is dependent on whether an splcpField is provided in the **Land cover classification file**:
 - **[class splcpField]_A** or **s[class Id][buffer distance]_A** - The area of the rasterized Sample point features buffer, by reporting unit, occupied by the land cover class.
 - Add Quality Assurance Fields
 - Choosing **QAFIELDS** adds a number of fields to the **Output table** to facilitate quality assurance checking.
 - **SPLCP_OVER** - The percent overlap between the Sample point features buffer areas in each reporting unit and the Land cover grid.
 - **SPLCP_TOTA** - The total raster zonal area for all Sample point features buffers in each reporting unit (zone) after the buffer areas have been rasterized.
 - **SPLCP_EFFA** - The effective raster area of all Sample point features buffers in each reporting unit (zone) after excluded classes have been removed.
 - **SPLCP_EXCA** - The raster area of the excluded classes within the Sample point features buffers in each reporting unit (zone) based on exclusions in the **Land cover classification scheme**.

- **sTOTA[buffer distance]** - The percentage of the reporting unit comprised of the Sample point features buffer area.
- **sEFFA[buffer distance]** - The percentage of the effective area in the reporting unit found in the Sample point features buffer area.
- Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **splcp_TabAreaRU#** (table) - A table of areas corresponding to the values in the **Land cover grid** occurring in each reporting unit. This table is only produced when **Add Quality Assurance Fields** is selected in **Additional Options**.
 - **splcp_[Sample point features]_Buffer#** (vector) - **Sample point features** buffered to the **Buffer distance**. The attribute table will contain fields from the **Sample point features**, the buffer distance (**BUFF_DIST**), and a field named, 'ORIG_FID'. This feature class is only produced when the optional **Enforce reporting unit boundaries** selection is NOT checked.
 - **splcp_Merge#** (vector) - Output from a Merge operation with a single input: **splcp_[Sample point features]_Buffer#**. The merge will create a copy of the buffer theme without any unnecessary fields. It is essentially a FeatureClassToFeatureClass operation with fieldMappings. This feature class is only produced when the optional **Enforce reporting unit boundaries** selection is NOT checked.
 - **splcp_Intersect#** (vector) - A polygon feature class with **splcp_Merge#** intersected with **Reporting unit feature**. This step assigns the Reporting Unit's ID value to the buffer zones and eliminates buffer areas outside of the reporting unit boundaries. This feature class is only produced when the optional **Enforce reporting unit boundaries** selection is NOT checked.
 - **splcp_Buffer[buffer distance]#** (vector) - A polygon feature class containing the buffer area for all Sample point features within a reporting unit merged by **Reporting unit ID field**.
 - **splcp_TabArea#** (table) - A table of areas corresponding to the values in the **Land cover grid** occurring in the buffer area within each reporting unit.
 - ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
 - ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*

- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - **NOTE:** *ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.*
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** *In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.*
 - **NOTE:** *ATtILA will report the name and location of the log file in the Tool Details’ Messages section.*

Syntax

SPLCP (Reporting_unit_feature, Reporting_unit_ID_field, Land_cover_grid, Land_cover_classification_scheme, Land_cover_classification_file, Report_metrics_for_these_classes, Sample_point_features, RU_Link_Field, Buffer_distance, Output_table, {Processing_cell_size}, {Snap_raster}, Select_options)

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field

Parameter	Explanation	Data Type
Land_cover_grid	The raster dataset representing land cover classes to be summarized within each Reporting unit feature. The grid input must be an integer raster layer.	Raster Layer
Land_cover_classification_scheme	The land cover classification schemes included in ATtILA and a User Defined option. The default schemes correspond to common input land cover datasets. Two schemes are supplied for each dataset included in ATtILA:	String
Land_cover_classification_file	The full pathname to the user-defined .xml file for custom or non-standard land cover classification schemes. Pathname is automatically filled when a default scheme is selected.	File
Report_metrics_for_these_classes	A list of the land cover classes and metric combinations for processing. Check the box to calculate metrics for each land cover class and/or combination class desired within the reporting units.	Multiple Value
Sample_point_features	The vector point dataset that provides the basis for the buffer zones. Land cover metrics are calculated for the area within the buffer zones.	Feature layer
RU_Link_Field	The field in the Sample point features layer that contains the unique ID for each reporting unit.	Field

Parameter	Explanation	Data Type
	It may be an integer or a string data type.	
Buffer_distance	The distance around the Sample point features which comprises the buffer zones.	
	The value must be an integer.	Linear unit
	If the distance linear units are not specified or are entered as Unknown, the linear unit of the input features' spatial reference is used.	
Enforce_reporting_unit_boundaries	Specifies whether a Reporting unit boundary acts as an absolute barrier when creating buffer areas.	Boolean
Output_table	The output reporting unit metrics table to be created.	
	It is recommended that the Output table be saved within a file geodatabase.	Table
Processing_cell_size (Optional)	The Processing cell size for the zonal operation.	
	The default Processing cell size is the cell size of the input raster land cover data. Optionally, the user may select a different cell size.	Analysis cell size
Snap_raster (Optional)	The raster that the cell alignment of the Land cover grid and rasterized Reporting unit feature layer will be matched to during processing.	Raster Layer
	The default Snap raster is the Land cover grid.	
Select_options	Tool options available to provide additional information:	Multiple Value

Code Samples

Sample Point Land Cover Proportions example 1 (Python window)

This example returns a table with the proportion of the area within 500 meters of a sample point within the reporting unit zone comprised of the agriculture and the developed land cover classes. The options to retain the intermediate calculation products, and to record the process steps into a log file have also been selected.

NOTE: *ATtILA tools that utilize the **Report metrics for these classes** parameter will clear all class selections when the tool is opened. This results in setting any variable set for that parameter to **None** when calling the tool from a script. Because of this, the standard scripting syntax (e.g. `arcpy.ATtILA.LCP(variable 1, variable 2, variable 3, etc.)` will fail. To overcome this idiosyncrasy, use the scripting example provided below. When using this technique, all variables are passed in as strings.*

NOTE: *By running scripts which launch an ATtILA tool via 'metric.run' within ArcGIS Pro's Python Window, all geoprocessing functions run by that tool can be viewed in the History Pane.*

```
import arcpy
from arcpy.sa import *

from ATtILA2 import metric

toolPath = "D:\\Destination
Folder\\ToolboxSource\\ATtILA2\\scripts\\SamplePointLandCoverProportions_ArcGIS.py"
Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Land_cover_grid = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Land_cover_classification_scheme = "NLCD LAND"
Land_cover_classification_file = "D:\\Destination Folder\\ToolboxSource\\LandCoverClassifications\\NLCD
LAND.xml"
Report_metrics_for_these_classes = "agr - [sagr] Agriculture';'dev - [sdev] Developed"
Sample_point_features = "D:\\pyexamples\\data.gdb\\SampleSites"
Reporting_unit_link_field = "HUC_12"
Buffer_distance = "500 Meters"
Enforce_reporting_unit_boundaries = "true"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_SPLCP"
Processing_cell_size = "30"
Snap_raster = "D:\\pyexamples\\data.gdb\\NLCD_2016"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric
Calculation';'LOGFILE - Record Process Steps Taken During Metric Calculation'"

metric.runSamplePointLandCoverProportions(
    toolPath,
    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Land_cover_grid,
    Land_cover_classification_scheme,
    Land_cover_classification_file,
    Report_metrics_for_these_classes,
    Sample_point_features,
    Reporting_unit_link_field,
    Buffer_distance,
    Enforce_reporting_unit_boundaries,
    Output_table,
    Processing_cell_size,
    Snap_raster,
    Select_options
)
```

Sample Point Land Cover Proportions example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Buffering input features...
|YYYY-MM-DD hh:mm:ss| Clipping buffered features to each reporting unit: splcp_Buffer500Meters_0
1 out of 4 loops, 25.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
2 out of 4 loops, 50.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
3 out of 4 loops, 75.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
4 out of 4 loops, 100.0% complete, hh:mm:ss elapsed, est. hh:mm:ss remaining, average time per loop:
hh:mm:ss
|YYYY-MM-DD hh:mm:ss| Constructing the ATtILA metric output table: Watersheds_SPLCP
|YYYY-MM-DD hh:mm:ss| Generating a zonal tabulate area table
|YYYY-MM-DD hh:mm:ss| Processing the tabulate area table and computing metric values
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Tool Stream Density Metrics

Summary

Calculates density metrics based on input vector line dataset(s) for each reporting unit polygon and creates an output table. Default metrics include area of the reporting unit (km²), total length (km) and line density (km/km²) within the reporting unit.

In addition to the default metrics, length and density metrics can be calculated by stream order. Stream order is designated based on values in a specified field in the Stream feature layer.

Usage

- This tool processes all polygons in the **Reporting unit feature** regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- This tool is designed to use a vector line dataset that represent streams, but any vector line dataset may be used. Caution should be exercised when interpreting results, particularly when using line datasets that do not represent streams.
- The **Reporting unit feature** must be a polygon feature class or shapefile.
- The optional **Stream order field** may be used to create density metrics for different classes of **Stream feature** lines. The **Stream order field** may be any field that differentiates classes of **Stream feature** lines and does not necessarily need to represent stream order.
- Values in the **Stream order field** have restrictions. They may may not contain spaces or special characters such as hyphens, parentheses, brackets, and symbols such as #, \$, %, and &. Essentially, the values in this field are acceptable if they consist only of alphanumeric characters or an underscore. The values in these fields are used to create class-specific metric fields for the two metrics (aliases: STRMKM and STRMDENS) in the **Output table**. If any of the values in the **Stream order field** violate these character restrictions, consider using a different class field or creating a new one that corresponds to the restrictions.
- If a **Stream order field** is specified, the tool creates the new field names by appending the **Stream order field** characters to the metric field names. Since field name length is dependent on the output table type (i.e., dBase, or geodatabase) care should be used so that the length of the final output field name does not exceed the maximum allowable field name limit for the desired output table. When the limit is exceeded, the **Stream order field** characters may replace the characters at the end of these field names.

- Example 1: The selected **Stream order field** uses an integer to designate the stream order, and contains only second- and third-order streams (field values “2” or “3”). When the tool is run, four output fields are created in the **Output table** which are named as follows (**Stream order field** characters in bold for emphasis):
 - **2**: STRMKM2 (7 characters), STRMDENS2 (9 characters)
 - **3**: STRMKM3 (7 characters), STRMDENS3 (9 characters)
- Example 2: The selected **Stream order field** uses alphanumeric characters to designate stream feature types (field values “C6” or “F558”). When the tool is run, four output fields are created in the **Output table** which are named as follows (**Stream order field** characters in bold for emphasis):
 - **C6**: STRMKMC6 (10 characters), STRMDENSC6 (10 characters)
 - **F558**: STRMKMF558 (10 characters), STRMDEF558 (10 characters)
- **NOTE**: If the **Stream order field** characters replace too many of the characters in the metric field names, duplicate field names will be created.
- **NOTE**: Do not use ‘ORDER’ as the **Stream order field** name as it is considered a reserved keyword in several database management systems, and its use may cause the Stream Density Metrics tool to fail.
- Final output is written to the location specified in the **Output table** parameter. The **Output table** may be saved as a File Geodatabase Table, or a dBASE Table.
 - **NOTE**: For most consistent results, It is highly recommended that tool output be saved to a file geodatabase.
 - When saving as a File Geodatabase Table, no extension is necessary for the **Output table** name. The output location must be a file geodatabase.
 - When saving as a dBASE Table, include the .dbf extension in the **Output table** name. dBASE tables may not be saved in a file geodatabase.
- Field names in the **Output table** follow this naming scheme:
 - For default tool settings (e.g. no optional settings selected):
 - **AREAKM2** - The area of the reporting unit in km².
 - **STRMKM** - The total length of **Stream feature** lines in km within the reporting unit.
 - **STRMDENS** - The density of **Stream feature** lines in km/km² within the reporting unit.
 - When the option to report metrics by **Stream order field** is selected, separate fields are generated for each metric/order combination. The metric/order field naming convention is described above.
- In **Additional Options**, the user may elect to ‘Retain Intermediate Layers Generated During Metric Processing’, and/or ‘Record Process Steps Taken During Metric Calculation’.
 - Retain Intermediate Layers Generated During Metric Calculation

- Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **sdm_[Reporting unit feature]#** (vector) - A polygon feature class containing reporting units with a new area field in km².
 - **sdm_StrByRU#** (vector) - A line feature class with Stream features intersected with reporting units.
 - ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
 - ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*
- Intermediate products are saved to one of the following locations:
 - When the **Output table** is saved as a File Geodatabase Table, intermediate products are placed in the same file geodatabase.
 - When the **Output table** is saved as a dBASE Table, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output table**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - ***NOTE:** ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.*
- Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - ***NOTE:** In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.*

- **NOTE:** ATtILA will report the name and location of the log file in the Tool Details' Messages section.

Syntax

SDM (Reporting_unit_feature, Reporting_unit_ID_field, Stream_feature, Output_table, {Stream_order_field}, {Select_options})

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
Stream_feature	The vector line dataset representing streams.	Feature Layer
Output_table	The output reporting unit metrics table to be created. It is recommended that the Output table be saved either within a file geodatabase.	Table
Stream_order_field (Optional)	The field in the Stream feature layer that contains the stream order for each feature. It may be an integer or a string data type. The values in this field must not contain spaces or special characters. As order code values are appended to output field names, care should be taken not to exceed the limit for field name size set by output table type.	Field
Select_options (Optional)	Tool options available to provide additional information:	Multiple Value

Code Samples

Stream Density Metrics example 1 (Python window)

This example returns a table with the following fields: the area of the reporting unit in square kilometers, the length of streams by stream order within the reporting unit in kilometers, and the density of streams by stream order within the reporting unit in km/km². The options to retain the intermediate calculation products, and to record the process steps into a log file have also been selected.

```
import arcpy

Reporting_unit_feature = "D:\\pyexamples\\data.gdb\\Watersheds"
Reporting_unit_ID_field = "HUC_12"
Road_feature = "D:\\pyexamples\\data.gdb\\Stream_Lines"
Output_table = "D:\\pyexamples\\results.gdb\\Watersheds_SDM"
Stream_order_field = "StreamOrde"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric Calculation'; 'LOGFILE - Record Process Steps Taken During Metric Calculation'"

    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Road_feature,
    Output_table,
    Stream_order_field,
    Select_options
)
```

Stream Density Metrics example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\Output table_YYYYMMDD_hh-mm-ss.txt
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Creating temporary copy of Watersheds
|YYYY-MM-DD hh:mm:ss| Calculating reporting unit area
|YYYY-MM-DD hh:mm:ss| Calculating feature density
Start Time: Friday, November 1, 2024 6:44:03 AM
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
|YYYY-MM-DD hh:mm:ss| Compiling calculated values into output table
Summarizing the ATtILA metric output table to log file
Summary complete
Log file closed
```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Utilities

The Utilities toolset consists of the following tools:

- **Create or Modify Land Cover Classification (.xml)** Launches a user-friendly interface to create or edit a Land Cover Classification XML file for use in ATtILA.
- **Create Walkability Cost Raster** Creates a cost-surface raster from user-supplied walkable and impassible feature datasets. The output from this tool can be used as input for the cost surface raster in the *Pedestrian Access and Availability* tool in the *People in the Landscape* toolset.
- **Identify Overlapping Polygons** Determines whether an input polygon feature layer contains overlapping polygons, and optionally creates new non-overlapping feature layers.
- **Process NHD for EnviroAtlas Analyses** Processes the National Hydrography Dataset's (NHD) Flowline, Waterbody, and Area features to produce the linear and areal stream features used in producing EnviroAtlas metrics such as the *Percent Forest and Woody Wetlands in Stream Buffer metric* (*Fact Sheet*). This tool allows you to process either a single or multiple geodatabases, as well as a single or multiple folders of shapefiles.
- **Process Roads for EnviroAtlas Analyses** Processes streets data from ESRI StreetMap and NAVTEQ geodatabases to optionally produce road feature classes used in the EnviroAtlas community-based metrics: *Walkable Roads*, *Roads for Intersection Density Analysis*, *Busy Roads - Interstates, Arterials, and Connectors*, and **All Roads - Pedestrian only and ferry routes removed**.
- **Select Zonal Statistics** Calculates select zonal statistics within reporting unit polygons and creates an output table.

Utility Create or Modify Land Cover Classification

Summary

Create custom Land Cover Classification (LCC) files or modify existing files.

The Land Cover Classification Editor will be launched separately from ArcGIS. You may close ArcGIS without disrupting the editor. A LCC file defines the classification scheme and is a required input for some tools.

Illustration

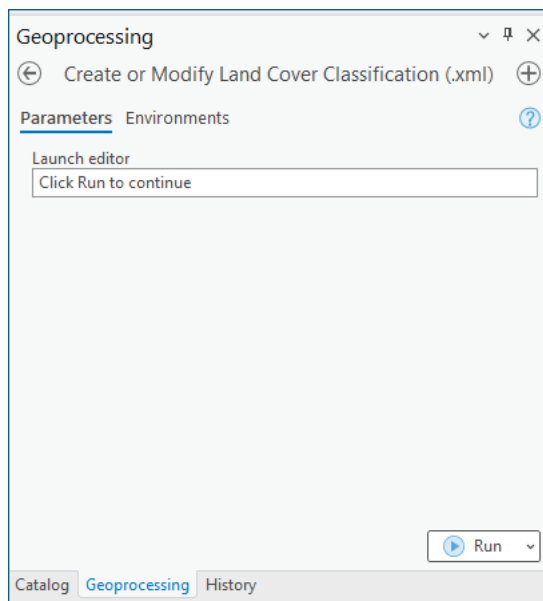


Figure 9. Run Create or Modify Land Cover Classification utility

Usage

- Allows users to modify an existing LCC XML file to suit their specific needs. For example, the user may define a new grouped class by combining multiple individual classes (e.g., deciduous, conifer, and mixed forest into a single forest class).
- Allows users to create a new LCC XML file for a land cover grid other than the standard datasets already included in ATtILA (e.g. a local Land cover grid with a unique classification scheme).

Syntax

LaunchLccEditor ()

This tool has no parameters.

Credits

Use limitations

There are no access and use limitations for this item.

Utility Create Walkability Cost Raster

Available with Spatial Analyst license.

Summary

Generates a cost surface raster suitable for the Pedestrian Access and Availability tool. The output raster will consist of three values representing walkable areas, walking-resistant areas, and impassable areas.

Usage

- Multiple polyline and/or polygon feature layers can be supplied for **Walkable features**. Features from each layer used in the **Output cost raster** creation can be limited to just selected features.
- Multiple polyline and/or polygon feature layers can be supplied for **Impassable features**. Features from each layer used in the **Output cost raster** creation can be limited to just selected features.
- **NOTE:** *For the most accurate results and shortest processing times, it is highly recommended that all input layers be in the same spatial reference.*
- Values entered for the **Maximum travel distance**, **Walk value**, and **Base value** should be in the same linear units as the **Output cost raster**.
 - **NOTE:** *Linear units will be obtained from the geoprocessing environment output coordinate system. If no specific geoprocessing environment output coordinate system is set (i.e., environment output coordinate system equals "NONE", this tool will temporarily set it to match the geocoordinate system of the **Snap raster**.*
- The Impassable value for the **Output cost raster** will be auto-calculated as (**Maximum walking distance** / **Cost raster cell size**) rounded up to the nearest integer.
- The default values of 800 for **Maximum travel distance**, 1 for **Walk value**, and 10 for **Base value** assume the **Output cost raster** has meters for linear units and a 10-meter **Cost raster cell size**. Given these parameters and a calculated impassable value of 800, a pedestrian could travel across 80 continuous **Walk value** cells, and 10 continuous **Base value** cells.
- Final output is written to the location specified in the **Output cost raster** parameter. The **Output cost raster** may be saved in a file folder or a geodatabase.
 - **NOTE:** *For most consistent results, it is highly recommended that tool output be saved to a file geodatabase.*
 - When saving in a File Geodatabase, no extension is permitted for the **Output cost raster** name.
 - When saving in a file folder, the following extensions are allowed for the **Output cost raster** name:
 - .img for ERDAS IMAGINE
 - .tif for TIFF
 - No extension for Esri Grid

- **NOTE:** *Filename lengths and allowable characters are dictated by the output raster format and its file destination. Please see, 'Output raster formats and names', in ArcGIS Pro's Online Help for more detailed information.*
- Where **Walkable features** and **Impassable features** occur in the same **Output cost raster** grid cell, the **Walkable features** take priority and the **Walk value** is assigned to that cell. It is assumed that **Walkable features** can cross an otherwise impassable area via a bridge, a sidewalk, or other such structure.
- Areas in between walkable cells and impassable cells are assigned the **Base value**.
- It is recommended that the extent and cell size of the **Snap raster** match the desired cell size and extent of the Pedestrian Access and Availability tool's **Output raster**.
 - **NOTE:** *If a raster is to be used as the Population or polygon feature for the Pedestrian Access and Availability tool, it is recommended that the **Snap raster** and **Processing cell size** be set to that population raster.*
 - **NOTE:** *If no specific geoprocessing environment extent is set (i.e., environment extent equals "NONE", "MAXOF", or "MINOF"), this tool will temporarily set it to match the geocoordinate system of the **Snap raster**. It will be reset upon tool completion.*
- In **Additional Options**, the user may elect to 'Retain Intermediate Layers Generated During Metric Processing', and/or 'Record Process Steps Taken During Metric Calculation'.
 - Retain Intermediate Layers Generated During Metric Calculation
 - Choosing **INTERMEDIATES** retains several of the key intermediate products (e.g., rasters, feature classes, tables) used to generate the final tool output. By examining the intermediate products, the user can gain a better understanding how the metric is calculated. The following intermediate products are listed in the general order they are generated:
 - **cwcr_Walk_Line_Merge_#** (vector) - A vector feature class created by merging all line features found in **Walkable features**. The attribute table consists of required ArcGIS fields (e.g., OBJECTID, Shape, Shape_Length) and the first field from the first **Walkable features** line layer. This latter field is not used in any future calculations. It contains values for features from the layer it was copied from, but NULL values for the rest of the merged features. This feature class is only produced if more than one line feature layers are in **Walkable features**.
 - **cwcr_Walk_Raster_Line_#** (raster) - Raster created from all line feature layers in **Walkable features**. If more than one line feature layers are present, they are merged into a single feature class, **cwcr_Walk_Line_Merge_#**, before rasterization. This raster is only produced if one or more line feature layers are provided in **Walkable features**.
 - **cwcr_Walk_Poly_Merge_#** (vector) - A vector feature class created by merging all polygon features found in **Walkable features**. The attribute table consists of required ArcGIS fields (e.g., OBJECTID, Shape, Shape_Area) and the first field from the first **Walkable features** polygon layer. This latter field is not used in any future calculations. It contains values for features from the layer it was copied from, but NULL values for the rest of the merged features. This feature class is only produced if more than one polygon feature layers are in **Walkable features**.

- **cwcr_Walk_Raster_Polygon_#** (raster) - Raster created from all polygon feature layers in **Walkable features**. If more than one polygon feature layers are present, they are merged into a single feature class, **cwcr_Walk_Poly_Merge_#**, before rasterization. This raster is only produced if one or more polygon feature layers are provided in **Walkable features**.
- **cwcr_Walk_Raster_#** (raster) - Raster created by combining **cwcr_Walk_Raster_Line_#** and **cwcr_Walk_Raster_Polygon_#**.
- **cwcr_Impass_Line_Merge_#** (vector) - A vector feature class created by merging all line features found in **Impassable features**. The attribute table consists of required ArcGIS fields (e.g., OBJECTID, Shape, Shape_Length) and the first field from the first **Impassable features** line layer. This latter field is not used in any future calculations. It contains values for features from the layer it was copied from, but NULL values for the rest of the merged features. This feature class is only produced if more than one line feature layers are in **Impassable features**.
- **cwcr_Impass_Raster_Line_#** (raster) - Raster created from all line feature layers in **Impassable features**. If more than one line feature layers are present, they are merged into a single feature class, **cwcr_Impass_Line_Merge_#**, before rasterization. This raster is only produced if one or more polygon feature layers are provided in **Impassable features**.
- **cwcr_Impass_Poly_Merge_#** (vector) - A vector feature class created by merging all polygon features found in **Impassable features**. The attribute table consists of required ArcGIS fields (e.g., OBJECTID, Shape, Shape_Area) and the first field from the first **Impassable features** polygon layer. This latter field is not used in any future calculations. It contains values for features from the layer it was copied from, but NULL values for the rest of the merged features. This feature class is only produced if more than one polygon feature layers are in **Impassable features**.
- **cwcr_Impass_Raster_Polygon_#** (raster) - Raster created from all polygon feature layers in **Impassable features**. If more than one polygon feature layers are present, they are merged into a single feature class, **cwcr_Impass_Poly_Merge_#**, before rasterization. This raster is only produced if one or more polygon feature layers are provided in **Impassable features**.
- **cwcr_Impass_Raster_#** (raster) - Raster created by combining **cwcr_Impass_Raster_Line_#** and **cwcr_Impass_Raster_Polygon_#**.
- ***NOTE:** To ensure unique filenames, intermediate filenames are typically generated using the CreateScratchName ArcPy function. This function provides a unique name for the intermediate product by appending a number to a filename prefix. The number starts at 0 and is incremented until a unique filename for the output workspace is found. The generated number is represented in this manual by the hashtag (#) symbol.*
- ***NOTE:** The appended number may not be the same for all intermediate products saved during a tool run. Running a tool with different input options can result in different intermediate products being produced. If a tool is run more than once using the same output workspace, different number suffixes may be generated depending on the files already present in the output workspace.*
- Intermediate products are saved to one of the following locations:

- When the **Output cost raster** is saved as a File Geodatabase Raster, intermediate products are placed in the same file geodatabase.
 - When the **Output cost raster** is saved in a file folder, a file geodatabase named “attilaScratchWorkspace” is automatically generated in the same output location specified for the **Output cost raster**. Intermediate products are placed in the attilaScratchWorkspace file geodatabase.
 - **NOTE:** *ATtILA will report the location of the saved intermediate products in the Tool Details’ Messages section.*
- Record Process Steps Taken During Metric Calculation
- Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** *In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.*
 - **NOTE:** *ATtILA will report the name and location of the log file in the Tool Details’ Messages section.*

Syntax

CWCR (Walkable_features, {Impassable_features}, Maximum_travel_distance, Walk_value, Base_value, Output_cost_raster, Cost_raster_cell_size, Snap_raster, {Select_options})

Parameter	Explanation	Data Type
Walkable_features	The vector polyline and/or polygon Feature Layers that define walkable routes. If selected features exist in the input feature layers, the user can elect to use only those features for generating the Output cost raster.	Multiple Value
Impassable_features (Optional)	The vector polyline and/or polygon Feature Layers that define impassable features.	Multiple Value

Parameter	Explanation	Data Type
	If selected features exist in the input feature layers, the user can elect to use only those features for generating the Output cost raster.	
Maximum_travel_distance	The distance where further travel is not possible. This value is used along with the Cost raster cell size to define the Output cost raster's impassable value in the Output cost raster. Impassable value = (Maximum walking distance / Cost raster cell size) rounded up to the nearest integer. If the output Cost surface raster is to be used in the Pedestrian Access and Availability tool, the value entered should equal the input for that tool's Maximum travel distance parameter. The value units must be a positive integer value, and in the linear units of the Output cost raster.	Long
Walk_value	The cost of traveling through a walkable cell. The value defines the cost of least resistance. Value units must be a positive integer value, and in the linear units of the Output cost raster.	Long
Base_value	The cost of traveling through a walking-resistant cell. The value defines the cost of difficult, but passable, resistance.	Long

Parameter	Explanation	Data Type
	Value units must be a positive integer value, and in the linear units of the Output cost raster.	
Output_cost_raster	The output cost surface raster. It is always an integer raster.	Raster Dataset
Cost_raster_cell_size	The cell size of the output raster that will be created. Units use the linear unit of the geoprocessing outputCoordinateSystem environment setting. If no outputCoordinateSystem is set, the Snap raster's linear unit is used.	Double
Snap_raster	The raster that the cell alignment of the Output cost raster will be set to. The Snap raster's linear unit will be used for the Cost raster cell size if no outputCoordinateSystem environment is set.	Raster Layer
Select_options	Tool options available to provide additional information:	Multiple Value

Code Samples

Create Walkability Cost Raster example 1 (Python window)

This example generates a cost surface raster where the cell values represent that cell's resistance to movement (1 = easy, 10 = resistant, and 80 = impassable). One line feature layer was entered as input for walkable pathways, while barriers to movement were represented by two line and one polygon feature layers. The options to retain the intermediate calculation products, and to record the process steps into a log file have also been selected.

```
import arcpy
from arcpy.sa import *

Walkable_features = "Roads_Walkable"
Impassable_features = "Water_Lines;Roads_NonWalkable;Water_Polygons"
Maximum_travel_distance = "800"
Walk_value = "1"
```

```

Base_value = "10"
Output_raster = "D:\\pyexamples\\results.gdb\\WalkCostSurface"
Density_raster_cell_size = "10"
Snap_raster = "D:\\pyexamples\\data.gdb\\DASY_Population"
Select_options = "'INTERMEDIATES - Retain Intermediate Layers Generated During Metric
Calculation';'LOGFILE - Record Process Steps Taken During Metric Calculation'"

```

```

Walkable_features,
Impassable_features,
Maximum_travel_distance,
Walk_value,
Base_value,
Output_raster,
Density_raster_cell_size,
Snap_raster,
Select_options
)

```

Create Walkability Cost Raster example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```

Start Time: Day, Month DD, YYYY hh:mm:ss AM/PM
Created log file: workspace path\\filename
Started: YYYY-MM-DD hh:mm:ss Setting up environment variables
Intermediates are stored in this directory: workspace path
|YYYY-MM-DD hh:mm:ss| Using DASY_Population's extent for geoprocessing steps.
|YYYY-MM-DD hh:mm:ss| Using DASY_Population's spatial reference for geoprocessing steps:
Albers_Conical_Equal_Area
|YYYY-MM-DD hh:mm:ss| Processing Walkability features
|YYYY-MM-DD hh:mm:ss| Converting Roads_Walkable to raster. Intermediate: cwcr_Walk_Raster_Line_0
|YYYY-MM-DD hh:mm:ss| Setting converted raster cell values to 1 where features exist. Everywhere else will
be set to 10. Intermediate: cwcr_Walk_Raster_0
|YYYY-MM-DD hh:mm:ss| Impass Value = 80. Calculated as (Maximum walking distance / Cost raster cell size)
rounded up to the nearest integer
|YYYY-MM-DD hh:mm:ss| Processing Impassable features
|YYYY-MM-DD hh:mm:ss| Merging 2 line features from input features. Intermediate: cwcr_Impass_Line_Merge_0
|YYYY-MM-DD hh:mm:ss| Converting cwcr_Impass_Line_Merge_0 to raster. Intermediate:
cwcr_Impass_Raster_Line_0
|YYYY-MM-DD hh:mm:ss| Converting Water_Polygons to raster. Intermediate: cwcr_Impass_Raster_Polygon_0
|YYYY-MM-DD hh:mm:ss| Combining converted rasters and setting output cell values to 80 where features
exist. Everywhere else will be set to 10. Intermediate: cwcr_Impass_Raster_0
|YYYY-MM-DD hh:mm:ss| Combining the Walkable raster with the Impassable raster for final output
|YYYY-MM-DD hh:mm:ss| Finalizing WalkCostSurface by adding labels
Log file closed
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)

```

Tags

There are no tags for this item.

Credits

There are no credits for this item.

Use Limitations

There are no access and use limitations for this item.

Utility Identify Overlapping Polygons

Summary

Determines whether an input polygon feature layer contains overlapping polygons, and creates new non-overlapping feature layers.

Usage

- **NOTE:** Processing time for this tool varies depending on the number of polygons in the **Input polygon features** layer. To minimize processing time, consider creating a new **Input polygon features** layer that does not include unwanted polygons. This tool may create multiple polygon layers which will consume disk space in the **Output workspace**. If the **Output workspace** resides on a disk with limited disk space, the tool may fail during processing. To prevent this, consider changing the **Output workspace** to a location with sufficient available disk space, or free up disk space on the disk in which the current **Output workspace** resides to accommodate the new polygon layers.
- This tool processes all polygons in the **Input polygon features** layer regardless of selections set. The ability to limit discovery of overlaps to only selected polygons is not supported in this release.
- When overlapping polygons are discovered in the **Input polygon features** layer, the tool creates new polygon layers. The number of new layers is based on the maximum number of overlaps for any individual polygon. For example, when one or more polygons in the **Input polygon features** layer are overlapped by, at most, one other polygon, the tool produces two new non-overlapping polygon layers. When one or more polygons are overlapped by two other polygons, the tool produces three new non-overlapping polygons layers, etc.
- When the **Check for overlaps only** option is enabled, no new layers are created. Therefore, the **Output workspace** is grayed out.

Syntax

IOP (Input_polygon_features, Output_workspace, Check_for_overlaps_only)

Parameter	Explanation	Data Type
Input_polygon_features	The vector polygon dataset upon which the overlapping polygon check will be performed.	Feature Layer
Output_workspace	The output location in which new polygon layers will be saved if overlapping polygons are present in the Input polygon features. If the Output workspace is a file folder, new layers will be saved as shapefiles. If the Output workspace is a geodatabase, new layers will be saved as	Workspace

Parameter	Explanation	Data Type
	geodatabase feature classes.	
Check_for_overlaps_only (Optional)	Specifies whether the process will result in new non-overlapping polygon layers.	Boolean

Credits

Use limitations

There are no access and use limitations for this item.

Utility Process NHD for EnviroAtlas Analyses

Summary

Process the National Hydrography Dataset's (NHD) Flowline, Waterbody, and Area features to produce the linear and areal stream features used in producing EnviroAtlas metrics such as the Percent Forest and Woody Wetlands in Stream Buffer metric ([Fact Sheet](#)).

The tool allows you to process either a single or multiple geodatabases, as well as a single or multiple folders of shapefiles.

NOTE: The NHD High Resolution (NHD), NHDPlus High Resolution (NHDPlus HR) datasets are available from <https://www.usgs.gov/national-hydrography/access-national-hydrography-products>.

NOTE: The NHDPlus Version 2 datasets are available at https://nhdplus.com/NHDPlus/NHDPlusV2_home.php.

Usage

- **WARNING:** *This tool will ignore the geoprocessing's 'Allow geoprocessing tools to overwrite existing datasets' setting and overwrite generated outputs.*
- **NOTE:** *Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- A list of one or more NHD workspaces to process will be generated depending on the selection in the **Process single or multiple files** and its enabled secondary parameters:
 - Single geodatabase
 - The NHD workspace list will consist of the name of the selected geodatabase file in the **Single geodatabase file** parameter.
 - The geodatabase must have a Hydrography feature dataset containing the feature classes, NHDFlowline, NHDWaterbody, and NHDArea.
 - Multiple geodatabases
 - The NHD workspace list will consist of each name of the geodatabases in the **Geodatabase folder** where a Hydrography feature dataset with the feature classes, NHDFlowline, NHDWaterbody, and NHDArea are found. The search is limited to geodatabases in the **Geodatabase folder** that match the expression in the **Geodatabase file filter**.
 - Single shapefile
 - The NHD workspace list will consist of the name of the folder selected in **Single shapefile folder**. The files, NHDFlowline.shp, NHDWaterbody.shp, and NHDArea.shp can exist either in that folder or one of its subfolders.
 - **WARNING:** *If more than one subfolder exists with the NHD files in the **Single shapefile folder**, all of the subfolders will be processed, but the outputs will be overwritten with each successive iteration.*
 - Multiple shapefiles
 - The NHD workspace list will consist of the name of each child subfolder in the **Multiple shapefiles folder** where the files, NHDFlowline.shp, NHDWaterbody.shp, and

NHDArea.shp are found either in the child subfolder or within one of their subfolders. The search is limited to the child subfolders in the **Multiple shapefiles folder** that match the expression in the **Shapefile folder filter**.

- Three generated layers are produced for each entry in the NHD workspace list:
 - [NHD Workspace Name]_strmLinesNAP - NHDFlowline layer features with ftypes equal to StreamRiver or Connector. NAP stands for 'No Artificial Paths'.
 - [NHD Workspace Name]_strmLines - NHDFlowline layer features with ftypes equal to StreamRiver, Connector, or Artificial Paths.
 - [NHD Workspace Name]_strmAreal - StreamsRivers, Rapids, Lock Chambers from the NHDArea layer. Washes from the NHDArea layer that are within 1 meter of the derived strmLine layer. Reservoir, LakePond, and Ice Mass features from the NHDWaterbody layer that are within 1 meter of the derived strmLines or within 1 meter of the derived strmLine end nodes.
- For each entry in the NHD workspace list, the following operations are performed:
 - Create layer for NHDFlowline.
 - Select from NHDFlowline layer features with ftypes equal to StreamRiver or Connector.
 - Copy selected features to output feature class. The suffix, "_strmLineNAP", will be appended to the filename.
 - Add NHDFlowline features with ArtificialPath ftype to the selected StreamRiver and Connector features.
 - Copy selected NHDFlowline features to output feature class. The suffix, "_strmLine", will be appended to the filename.
 - Find line ends for selected NHDFlowline features.
 - Use NEAR to calculate distances between line ends.
 - Select nodes where the NEAR_DIST value $\neq$ 0. These are the terminal nodes.
 - Copy terminal nodes to intermediate output feature class. The suffix, "_strmEnds", will be appended to the filename.
 - Create layer for NHDArea.
 - Select from NHDArea layer features where the ftype = Wash.
 - Reduce the selected Washes to just those that are within 1 Meter of selected NHDflowlines.
 - Add NHDArea features with ftypes equal to StreamRiver, Rapids, or Lock Chamber features to the selected Washes.
 - Create layer for NHDWaterbody.
 - Select from NHDWaterbody layer features with ftype equal to Reservoir, LakePond, or Ice Mass.
 - Reduce the selected NHDWaterbody features to just those that are within 1 Meter of selected NHDflowlines.
 - Merge the selected NHDArea features with the selected NHDWaterbody features.
 - Add field, "tmp_diss", to the merged feature class attribute table and set its value to 1.
 - Dissolve the merged feature class using the "tmp_diss" field.
 - Perform a second dissolve on the dissolved feature class using the "tmp_diss" field.
 - Create layer for the merged and dissolved areal feature class.
 - Select from the merged and dissolved areal feature class, features that are within 1 Meter of derived terminal nodes.
 - Add to those features any additional merged and dissolved areal features that intersect the selected NHDFlowlines.
 - Copy selected merged and dissolved areal features to output feature class. The suffix, "_strmAreal", will be appended to the filename.

- In **Additional Options**, the user may elect to Retain Intermediate Layers Generated During Metric Processing.
 - Choosing to Retain Intermediate Layers saves the intermediate products to one of the following locations:
 - When **Output workspace** is a geodatabase, intermediate products are placed in the same file geodatabase.
 - When **Output workspace** is a file folder, intermediate products are placed in the same file folder.
- Generated intermediate layers produced are:
 - [NHD Workspace Name]_strmEnds - Terminal nodes (i.e., nodes touching only one line segment) for each stream in [NHD Workspace Name]_strmLine. All of the attributes of the adjacent stream segment are available.

Syntax

PNHD (Single_or_multiple_files, Geodatabase_file, Geodatabase_folder, Geodatabase_file_filter, Single_shapefile_folder, Multiple_shapefiles_folder, Shapefile_folder_filter, Output_workspace, {Select_options})

Parameter	Explanation	Data Type
Single_or_multiple_files	Type and quantity of input NHD feature datasets to process: single or multiple, shapefiles or geodatabases.	String
Geodatabase_file	The path to the NHD geodatabase. The geodatabase must have a Hydrography feature dataset containing the feature classes, NHDFlowline, NHDWaterbody, and NHDArea.	Workspace
Geodatabase_folder	The path to the folder containing multiple NHD geodatabases. Geodatabases within this folder that contain a Hydrography feature dataset with the feature classes: NHDFlowline, NHDWaterbody, and NHDArea, will be identified as NHD workspaces and their names will be used as	Workspace

Parameter	Explanation	Data Type
	output filename bases. The optional wildcard expression in the Geodatabase file filter parameter can be used to limit the geodatabases searched in the Geodatabase folder.	
Geodatabase_file_filter	Wildcard expression used to limit searched geodatabases within the Geodatabase folder for NHD workspaces. Use a single asterisk character (*) to search all geodatabases.	String
Single_shapefile_folder	The path to the NHD shapefile folder. The folder name will be used as the output filename base. The files, NHDFlowline.shp, NHDWaterbody.shp, and NHDArea.shp, can exist in this folder or one of its subfolders.	Workspace
Multiple_shapefiles_folder	The path to the folder containing one or more NHD shapefile folders. Folders within this folder that contain the files: NHDFlowline.shp, NHDWaterbody.shp, and NHDArea.shp, directly or in one of their subfolders, will be identified as NHD workspaces and their names will be used as output filename bases. The optional wildcard expression in the Shapefile folder filter parameter can be used to limit the subfolders searched in the Multiple shapefiles folder.	Workspace

Parameter	Explanation	Data Type
Shapefile_folder_filter	Wildcard expression used to limit searched subfolders within the Multiple shapefiles folder for NHD workspaces. Use a single asterisk character (*) to search all subfolders.	String
Output_workspace	The path to the Output workspace. If the Output workspace is a file folder, new layers will be saved as shapefiles. If the Output workspace is a geodatabase, new layers will be saved as geodatabase feature classes.	Workspace
Select_options (Optional)	Tool options available to provide additional information:	Multiple Value

Credits

Use limitations

There are no access and use limitations for this item.

Utility Process Roads for EnviroAtlas Analyses

Summary

Processes road data from **ESRI StreetMap**, **NAVTEQ 2019**, and **NAVTEQ 2011** structured geodatabases to produce road feature classes used in the EnviroAtlas community-based metrics:

- Walkable Roads
- Roads for Intersection Density Analysis
- Busy Roads - Interstates, Arterials, and Connectors
- All Roads - Pedestrian only and ferry routes removed

Usage

- ***NOTE:** Enabling the Parallel Processing Factor environment can improve computation time for several ATtILA tools, but we have detected, on rare occasions, doing so can lead to spurious results. As a precaution, ATtILA will disable the Parallel Processing Factor environment for this tool. We hope to return this choice to the user once the impact on operational speed and accuracy of swapping geoprocessing tools with their pairwise counterparts has been assessed.*
- The **Version** must correspond to the selected **Roads geodatabase** type.
- The **Roads geodatabase** must be an **ESRI StreetMap**, **NAVTEQ 2019**, or **NAVTEQ 2011** structured geodatabase.
- For **ESRI StreetMap** the following feature classes must be present:
 - .gdb\Streets
 - .gdb\MapLandArea
- For **NAVTEQ 2019** the following feature classes must be present:
 - .gdb\RoutingApplication\Streets
 - .gdb\MapFacilityArea
 - .gdb\MapLanduseArea
 - .gdb\link
- For **NAVTEQ 2011** the following feature classes must be present:
 - .gdb\Streets
 - .gdb\LandUseA
 - .gdb\LandUseB

- **Intersection Density Roads** option requires all input feature classes to be in a projected coordinate system. Its final processing step, merge divided roads, relies on linear distance units, which will create unexpected results in an unprojected coordinate system. Please see, Merge Divided Roads (Cartography), in ArcGIS Pro's Online Help for more information.
- In **Output workspace**, select the geodatabase or folder where the standard tool outputs are to be stored.
 - Standard outputs:
 - **[Filename_prefix]_RdsWalkable** - Polyline feature class of walkable roads using procedures and selections as described below.
 - **[Filename_prefix]_RdsIntDens** - Polyline feature class roads suitable for the Intersection Density tool using procedures and selections as described below.
 - **[Filename_prefix]_RdsIAC** - Polyline feature class of interstates, arterials, and collectors (i.e., busy roads) using procedures and selections as described below.
 - **[Filename_prefix]_AllRds** - Polyline feature class where pedestrian only and ferry routes have been removed using procedures and selections as described below.
 - **WARNING:** *This tool will ignore the geoprocessing's 'Allow geoprocessing tools to overwrite existing datasets' setting and overwrite generated outputs.*
 - **NOTE:** *The filename prefix for the standard outputs is obtained from the **Filename prefix** parameter.*
 - **NOTE:** *If the Output workspace is a file folder, new layers will be saved as shapefiles. If the Output workspace is a geodatabase, new layers will be saved as geodatabase feature classes.*
- In **Additional Options**, the user may elect to 'Record Process Steps Taken During Metric Calculation'.
 - Record Process Steps Taken During Metric Calculation
 - Choosing **LOGFILE** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
 - Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
 - Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** *In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.*
 - **NOTE:** *ATtILA will report the name and location of the log file in the Tool Details' Messages section.*

- Processing steps for **Walkable roads**:

- Process {**ESRI StreetMap**}.gdb\Streets

- select features where FuncClass < 3 or SpeedCat < 4 Or FerryType <> 'H' Or RestrictPedestrians = 'Y' Or Ramp = 'Y', Or ContrAcc = 'Y' Or Tollway = 'Y'
 - INVERT selection
 - copy remaining selected features to [prefix]_RdsWalkable

- Process {**NAVTEQ 2019**}.gdb\RoutingApplication\Streets

- select features where FuncClass < 3 or SpeedCat < 4 Or FerryType <> 'H' Or RestrictPedestrians = 'Y' Or Ramp = 'Y', Or ContrAcc = 'Y' Or Tollway = 'Y'
 - INVERT selection
 - copy remaining selected features to [prefix]_RdsWalkable

- Process {**NAVTEQ 2011**}.gdb\Streets

- select features where FUNC_CLASS IN ('1', '2') Or SPEED_CAT IN ('1', '2', '3') Or FERRY_TYPE <> 'H' Or AR_PEDEST = 'N' Or RAMP = 'Y', Or CONTRACC = 'Y' Or TOLLWAY = 'Y'
 - INVERT selection
 - copy remaining selected features to [prefix]_RdsWalkable

- Processing steps for **Intersection density roads**:

- ***NOTE:** Input feature classes for this option need to be in a projected coordinate system. The final merge divided roads step relies on linear distance units, which will create unexpected results in an unprojected coordinate system. Please see, [Merge Divided Roads \(Cartography\)](#), in ArcGIS Pro's Online Help for more information.*

- Process {**ESRI StreetMap**}.gdb\Streets

- select features where FuncClass < 3 or SpeedCat < 4 Or FerryType <> 'H' Or RestrictPedestrians = 'Y' Or Ramp = 'Y', Or ContrAcc = 'Y' Or Tollway = 'Y'
 - INVERT selection
 - remove from the selection features where SpeedCat = 8
 - assign ESRI_StreetMap.gdbcodes to road segments
 - remove roads with no street names from these land use type areas: AIRPORT, AMUSEMENT PARK, BEACH, CEMETERY, HOSPITAL, INDUSTRIAL COMPLEX, MILITARY BASE, RAILYARD, SHOPPING CENTRE, or GOLF COURSE
 - add a MergeClass field
 - set MergeClass field value to an initial value of 1
 - replace MergeClass field value to 0 where DIR_TRAVEL = 'B'

- convert any multipart roads to singlepart
 - merge divided roads to [prefix]_RdsIntDens using the MergeClass field and a merge distance of '30 Meters'
 - **NOTE:** *Only roads with the same value in the mergeField and within the mergeDistance will be merged. Roads with a MergeClass value equal to zero are locked and will not be merged. All non-merged roads are retained.*
- Process {NAVTEQ 2019}.gdb\RoutingApplication\Streets
 - select features where FuncClass < 3 or SpeedCat < 4 Or FerryType <> 'H' Or RestrictPedestrians = 'Y' Or Ramp = 'Y', Or ContrAcc = 'Y' Or Tollway = 'Y'
 - INVERT selection
 - remove from the selection features where SpeedCat = 8
 - assign Navteq_2019.gdbcodes to road segments
 - assign Navteq_2019.gdbcodes to road segments
 - remove roads with no street names from these land use type areas: AIRPORT, AMUSEMENT PARK, BEACH, CEMETERY, HOSPITAL, INDUSTRIAL COMPLEX, MILITARY BASE, RAILYARD, SHOPPING CENTRE, or GOLF COURSE
 - add a MergeClass field
 - set MergeClass field value to an initial value of 1
 - replace MergeClass field value to 0 where DIR_TRAVEL = 'B'
 - convert any multipart roads to singlepart
 - merge divided roads to [prefix]_RdsIntDens using the MergeClass field and a merge distance of '30 Meters'
 - Process {NAVTEQ 2011}.gdb\Streets
 - select features where FUNC_CLASS IN ('1', '2') Or SPEED_CAT IN ('1', '2', '3') Or FERRY_TYPE <> 'H' Or AR_PEDEST = 'N' Or RAMP = 'Y', Or CONTRACC = 'Y' Or TOLLWAY = 'Y'
 - INVERT selection
 - remove from the selection features where "SPEED_CAT IN ('8')
 - assign Navteq_2011.gdbcodes to road segments
 - assign Navteq_2011.gdbcodes to road segments
 - remove roads with no street names from these land use type areas: AIRPORT, AMUSEMENT PARK, BEACH, CEMETERY, HOSPITAL, INDUSTRIAL COMPLEX, MILITARY BASE, RAILYARD, SHOPPING CENTRE, or GOLF COURSE
 - add a MergeClass field

- set MergeClass field value to an initial value of 1
 - replace MergeClass field value to 0 where DIR_TRAVEL = 'B'
 - convert any multipart roads to singlepart
 - merge divided roads to [prefix]_RdsIntDens using the MergeClass field and a merge distance of '30 Meters'
- Processing steps for **Interstates, arterials, and collectors**:
 - Process {ESRI StreetMap}.gdb\Streets
 - select features where FuncClass <=4
 - remove from the selection features where FERRY_TYPE <> H
 - copy remaining selected features to [prefix]_RdsIAC
 - add field, LANES, to [prefix]_RdsIAC. Calculating its value as TO_REF_NUM_LANES + FROM_REF_NUM_LANES
 - replace LANES field value to 2 for features where LANES field = 0
 - **NOTE:** The user can locate and change these records with the following query: *TO_REF_NUM_LANES = 0 And FROM_REF_NUM_LANES = 0*
 - Process {NAVTEQ 2019}.gdb\RoutingApplication\Streets
 - select features where FuncClass <=4
 - remove from the selection features where FERRY_TYPE <> H
 - copy remaining selected features to [prefix]_RdsIAC
 - add field, LANES, to [prefix]_RdsIAC. Calculating its value as TO_REF_NUM_LANES + FROM_REF_NUM_LANES
 - replace LANES field value to 2 for features where LANES field = 0
 - Process {NAVTEQ 2011}.gdb\Streets
 - select features where FUNC_CLASS = 1, 2, 3, or 4
 - remove from the selection features where FERRY_TYPE <> H
 - copy remaining selected features to [prefix]_RdsIAC
 - add field, LANES, to [prefix]_RdsIAC. Calculating its value as TO_LANES + FROM_LANES
 - replace LANES field value to 2 for features where LANES field = 0
 - Processing steps for **All roads**:
 - Process {ESRI StreetMap}.gdb\Streets

- select and remove features where RestrictCars = Y And RestrictBuses = Y And RestrictTaxis = Y And RestrictDeliveries = Y And RestrictTrucks = Y And RestrictCarpools = Y And RestrictEmergencies = Y And RestrictMotorcycles = Y And FerryType <> H
- Process {NAVTEQ 2019}.gdb\RoutingApplication\Streets
 - select and remove features where RestrictCars = Y And RestrictBuses = Y And RestrictTaxis = Y And RestrictDeliveries = Y And RestrictTrucks = Y And RestrictCarpools = Y And RestrictEmergencies = Y And RestrictMotorcycles = Y And FerryType <> H
- Process {NAVTEQ 2011}.gdb\Streets
 - select and remove features where AR_AUTO = N And AR_BUS = N And AR_TAXIS = N And AR_DELIV = N And AR_TRUCKS = N And AR_CARPOOL = N And AR_EMERVEH = N And AR_MOTOR = N Or FERRY_TYPE <> H

Syntax

PRFEA (Version, Roads_geodatabase, {Walkable_roads}, {Intersection_density_roads}, {Interstates_arterials_and_collectors}, Output_workspace, Filename_prefix, {Select_options})

Parameter	Explanation	Data Type
Version	The version of the Roads geodatabase selected for processing. Must correspond to the selected Roads geodatabase. For ESRI StreetMap the following feature classes must be present:	String
Roads_geodatabase	The path to the geodatabase. NOTE: Necessary file structure for the selected geodatabase must correspond to the selected Version.	Workspace
Walkable_roads (Optional)	Specifies whether the Walkable Roads feature class will be generated.	Boolean
Intersection_density_roads (Optional)	Specifies whether the Intersection Density Roads feature class will be generated.	Boolean
Interstates,_arterials,_and_collectors (Optional)	Specifies whether the Interstate, Arterial, and	Boolean

Parameter	Explanation	Data Type
	Collector Roads feature class will be generated.	
All_roads (Optional)	Specifies whether an All Roads feature class will be generated where all pedestrian only or ferry routes have been removed.	Boolean
Output_workspace	The path to the Output workspace. If the Output workspace is a file folder, new layers will be saved as shapefiles. If the Output workspace is a geodatabase, new layers will be saved as geodatabase feature classes.	Workspace
Filename_prefix	A character string used as the file name prefix for all output feature classes.	String
Select_options (Optional)	Tool options available to provide additional information:	Multiple Value

Code Samples

Process Roads for EnviroAtlas Analyses example 1 (Python window)

This example creates a Walkable Roads feature class and a Roads for Intersection Density feature class to be used in the Intersection Density tool. The option to generate a log file has also been selected.

```
import arcpy
```

```
Version="NAVTEQ 2019",
Roads_geodatabase=r"D:\pyexamples\NAVTEQ2019.gdb",
Walkable_roads=True,
Intersection_density_roads=False,
Interstates_arterials_and_collectors=True,
Output_workspace=r"D:\pyexamples\results.gdb",
Filename_prefix="Roads",
Select_options="'LOGFILE - Record Process Steps Taken During Metric Calculation'"
```

```
Roads_geodatabase=r"D:\pyexamples\NAVTEQ2019.gdb",
Walkable_roads=True,
Intersection_density_roads=True,
Interstates_arterials_and_collectors=False,
All_roads=False,
```

```
Output_workspace=r"D:\pyexamples\results.gdb",  
Filename_prefix="NAVTEQ_2019",  
Select_options="'LOGFILE - Record Process Steps Taken During Metric Calculation'")
```

Process Roads for EnviroAtlas Analyses example 2 (Tool Details Messages)

This is a sample capture of the geoprocessing messages generated during a typical tool run. It is provided here to assist in the understanding of the tool's processing steps.

```
Created log file: C:\...\Profile\Documents\ArcGIS\Projects\Rds_2019_YYYYMMDD_hh-mm-ss.txt  
Started: Time: YYYY-MM-DD hh:mm:ss Setting up initial environment variables
```

```
Adding output(s) to Map view  
Adding NAVTEQ_2019_RdsWalkable to Map view  
Adding NAVTEQ_2019_RdsIntDens to Map view  
Log file closed  
Succeeded at Day, Month DD, YYYY hh:mm:ss AM/PM (Elapsed Time: ##.## units)
```

Tags

There are no tags for this item.

Credits

Use limitations

There are no access and use limitations for this item.

Utility Select Zonal Statistics

Available with Spatial Analyst license.

Summary

Calculates select zonal statistics within reporting unit polygons and creates an output table.

Usage

- This tool processes all polygons in the **Reporting unit feature** layer regardless of selections set. The ability to limit calculations to only selected reporting unit polygons is not supported in this release.
- The list of selectable statistics is dependent on the **Input value raster** type.
 - For integer grids, supported statistics are: **MIN, MAX, RANGE, MEAN, STD, SUM, VARIETY, MAJORITY, MAJORITY_COUNT, MAJORITY_PERCENT, MINORITY, MINORITY_COUNT, MINORITY_PERCENT, MEDIAN, PCT90.**
 - For floating point grids, supported statistics are: **MIN, MAX, RANGE, MEAN, MEDIAN, STD, SUM, PCT90.**
- Field names in the **Output table** follow this naming scheme:
 - **Field_prefix_[selected statistic]**
 - (e.g., **SLOPE_MEAN** for a **Field_prefix** of “SLOPE” and a selected statistic of “MEAN”)
 - The string combination of **Field_prefix** and **[selected statistic]** must conform to the field naming conventions dictated by the output database system.
 - In general, try to restrict field names to just alphanumeric characters and underscores. Use of spaces and special characters should be avoided, as well as beginning a field name with a number or an underscore. For more guidelines on the naming of fields, search on “Fundamentals of adding and deleting fields” in the ArcGIS help documentation.
 - Field names in a geodatabase tables are restricted to 64 characters in length while field names in a dBase table are restricted to 10 characters in length.
- An **AREA_OVER** field is automatically generated and added to the **Output table** for quality assurance purposes. It measures the percent overlap between the **Reporting unit feature** and the **Input value raster** for each reporting unit.
 - **NOTE:** *The area of the **Reporting unit feature** vector polygon and the zonal area calculated from the **Input value raster** are rarely the same even when there is complete overlap between the two. Because of this, values slightly above or below 100 in the **AREA_OVER** field are to be expected. Values noticeably below 100 indicate a possible problem area where data from the **Input value raster** may not extend throughout the reporting unit feature.*

- In **Additional Options**, the user may elect to ‘Record Process Steps Taken During Metric Calculation’.
- Choosing **LOGFILE - Record Process Steps Taken During Metric Calculation** will generate a text file containing information on system setup, tool inputs, parameter selections, script operations, and any encountered error or warning messages.
- Log files will be named, **[Output table]_[date_time].txt**, with the date represented as YYYYMMDD and time as hh_mm_ss.
- Where the log file is saved is dependent on the Current Workspace environment setting:
 - When the Current Workspace is a geodatabase, the log file will be located in the folder that contains that workspace.
 - When the Current Workspace is a folder, the log file will be located in that folder.
 - **NOTE:** *In ArcGIS Pro, the default value for the Current Workspace environments is the project default geodatabase.*
 - **NOTE:** *ATtILA will report the name and location of the log file in the Tool Details’ Messages section.*

Syntax

SZS (Reporting_unit_feature, Reporting_unit_ID_field, In_value_raster, Statistics_type, Output_table, Field_prefix, {Select_options})

Parameter	Explanation	Data Type
Reporting_unit_feature	The vector polygon dataset that defines the reporting units.	Feature Layer
Reporting_unit_ID_field	The field in the Reporting unit feature layer that contains the unique ID for each reporting unit. It may be an integer or a string data type.	Field
In_value_raster	The raster that contains the values for which a statistic will be calculated. Takes integer or floating point raster type.	Raster Layer
Statistics_type	Specifies the statistic type to be calculated.	Multiple Value
Output_table	The output reporting unit statistics table to be created.	Table

Parameter	Explanation	Data Type
	It is recommended that the Output table be saved within a file geodatabase.	
Field_prefix	A character string used as the field name prefix for all output statistics.	String
Select_options	Tool options available to provide additional information:	Multiple Value

Code Samples

Select Zonal Statistics example 1 (Python window)

This example returns a table with a SLOPE raster's MAX, MEAN, and MEDIAN value for each WATERSHED reporting unit zone. The option to record the process steps into a log file has also been selected.

```
import arcpy
from arcpy.sa import *

Reporting_unit_feature = **D:\\pyexamples\\data.gdb\\Watersheds**
Reporting_unit_ID_field = **HUC12**
Input_value_raster = **D:\\pyexamples\\data.gdb\\Slope**
Statistics_type = **MAX;MEAN;MEDIAN**
Output_table = **D:\\pyexamples\\results.gdb\\Watersheds_SZS**
Field_prefix = **SLOPE**
Select_options = **'LOGFILE - Record Process Steps Taken During Metric Calculation'**

    Reporting_unit_feature,
    Reporting_unit_ID_field,
    Input_value_raster,
    Statistics_type,
    Output_table,
    Field_prefix,
    Select_options
)
```

Credits

Use limitations

There are no access and use limitations for this item.

Land Cover Classification

Many ATtILA tools require a Land Cover Classification (LCC) file as one of their input parameters. The LCC file dictates how a tool processes the input Land cover grid. It:

- Defines the text description of land cover classes based on the raster integer value.
- Determines the inclusion or exclusion of specific land cover classes in metric calculations (e.g. limiting calculations to only terrestrial land cover types via the exclusion of water and ice/snow classes).
- Allows for the grouping of classes (e.g. deciduous, coniferous, and mixed forest classes may be grouped into a single “forest” class).
- Allows for the assignment of coefficients associated with specific land cover classes to calculate coefficient-based metrics (e.g. impervious percent, nitrogen loading, and phosphorous loading).

LCC files utilize XML formatting to define these properties for an associated Land cover grid. For the user’s convenience, ATtILA includes LCC files for some commonly-used Land cover datasets including the National Land Cover Database (NLCD), the Coast Change Analysis Program (C-CAP), and the US EPA EnviroAtlas Meter-Scale Urban Land Cover (MULC). Descriptions of the LCC files that accompany ATtILA are in the [Supplied Land Cover Classification Schemas](#) section of this document.

LCC files may be created or edited to allow users to customize the properties of the land cover classes to meet their objectives.

It is recommended that users create or edit LCC schema XML documents using the [Land Cover Classification Editor](#) (LCC Editor) that accompanies ATtILA to reduce the possibility of introducing errors. See [Land Cover Classification Editor](#) for more information on this tool. No previous experience editing XML files is necessary to edit the XML using the LCC Editor, which provides an easy-to-use interface.

LCC XML documents may also be created or edited using any standard text editor following the syntax guidelines provided in [ATtILA’s LCC XML Document](#) section.

LCC XML Document Storage

The [Supplied Land Cover Classification Schemas](#) that accompany ATtILA are stored in the ToolboxSource > LandCoverClassifications folder in the ATtILA toolbox destination folder (see [Installing ATtILA](#)). LCC XML documents stored here automatically populate the “Land cover classification scheme” dropdown in each ATtILA tool that requires a land cover classification file ([Core and Edge Metrics](#), [Floodplain Land Cover Proportions](#), [Facility Land Cover Views](#), [Land Cover on Slopes Proportions](#), [Land Cover Proportions](#), [Neighborhood Proportions](#), [Population Land Cover Views](#), [Patch Metrics](#), [Riparian Land Cover Proportions](#), and [Sample Point Land Cover Proportions](#)).

Customized LCC XML documents can exist anywhere on a local or network computer. Each tool that requires an LCC XML document includes a parameter for navigating to the local or network path at which the XML file is stored. However, users may find it more convenient to access their customized LCC XML documents from the “Land cover classification scheme” dropdown in the ATtILA tools, particularly for frequently-used customized LCC XML documents. To facilitate this, store the customized XML files in the ToolboxSource > LandCoverClassifications folder as indicated above.

NOTE: *It is recommended that the LCC file contain entries for all values in the input Land cover grid. However, if the LCC file does not define properties for a Land cover grid value, ATtILA will alert the user by providing a warning message indicating that a value occurring in the Land cover grid was not*

defined in the LCC file.

LCC XML Document

ATtILA Land Cover Classification (LCC) schema documents require a precise XML structure that consists of the following:

- [XML declaration statement](#)
- [Root element \(lccSchema\)](#)
- [Metadata element](#)
- [Coefficients element](#)
- [Values element](#)
- [Classes element](#)
- [Comments](#)

Detailed descriptions, applications, and possible restrictions for each element are provided their respective sections in this document. The [Supplied Land Cover Classification Schemas](#) that accompany ATtILA may be useful to review while reading through the following sections. What may initially appear to be complex, hard to decipher documents will become easier to interpret with the background information provided below.

Basic structure of ATtILA's Land Cover Classification schema XML document: all required elements.

```
<?xml version='1.0' encoding='utf-8'?>
<lccSchema xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="lcc" xsi:schemaLocation="lcc
  XSD_FILE_PATH">
  <metadata>
    <name></name>
    <description></description>
  </metadata>
  <coefficients>
    <coefficient Id="" Name="" fieldName="" method="" />
  </coefficients>
  <values>
    <value Id="" Name="" excluded="">
      <coefficient Id="" value="" />
    </value>
  </values>
  <classes>
    <class Id="" Name="" filter="">
      <value Id="" />
    </class>
  </classes>
</lccSchema>
```

It is recommended that users create or edit LCC schema XML documents using the [Land Cover Classification Editor](#) that accompanies ATtILA to reduce the possibility of introducing errors. However, LCC XML documents may also be created or edited using any standard text editor and the syntax guidelines provided in each section.

Regardless of how a user chooses to create their LCC XML documents, it is strongly recommended that they first become acquainted with the details on LCC XML formatting. This will give the user a better understanding of how ATtILA interacts with the land cover grid and will provide a background for understanding how ATtILA may be customized for user-specific needs.

XML Fundamentals

An XML document is composed of text content marked up with tags describing the data. These tags look similar to HTML mark-up tags, but unlike HTML, the tags are customized to define what an item is (i.e., a book, a person, a telephone number, etc.) and particular attributes of each defined item (i.e. the book's title, a person's first and last name, whether the telephone number is for a home, business, or mobile device, etc.). In order for an application to correctly parse an XML document into its various pieces, the XML document must be “well-formed”; in other words, it must follow certain rules.

When using the Land Cover Classification Editor in ATtILA to modify or create an LCC XML document, the resulting output document will be well-formed with regard to the ATtILA application. When editing the LCC XML document with a text editor, these general XML rules must be observed:

- Each start-tag (<**tag**>) must have a matching end-tag. The end-tag can be either a non-empty element, (</**tag**>), or an empty element (/>).
- Attribute values must be quoted.
- Attribute names must be unique, and XML is case-sensitive: “**name**” is different from “**Name**” which is different from “**NAME**”.
- Comments and processing commands cannot appear inside tags.

When editing an LCC XML document with a text editor, it is important to check the document for formatting errors and, more importantly, for any violations to the rules and restrictions ATtILA places on document elements and attributes. The XML Validation section of this document provides instructions for performing these checks.

LCC XML Document XML Declaration

*Basic structure of ATtILA's Land Cover Classification schema XML document: **XML declaration**.*

```
<?xml version='1.0' encoding='utf-8'?>
```

The first line of the LCC XML document is an XML declaration statement. It is not required, but if it is included, it must be the first line of the document. No other character or blank space can precede the declaration statement. The declaration statement allows an XML parser to obtain a basic understanding about how the text document is encoded.

LCC XML Document Root Element

*Basic structure of ATtILA's Land Cover Classification schema XML document: **Root element (lccSchema)**.*

```
<lccSchema xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="lcc" xsi:schemaLocation="lcc
XSD_FILE_PATH">
.
.
.
</lccSchema>
```

The <lccSchema> tag is the root element of the LCC XML document. All other elements in the LCC XML document are contained within this element. XML documents are limited to one root element.

The XML root element can also contain information specific to the location and name of an XML Schema Document (XSD). The XSD document is used for validating the form and contents of the XML file. Information about the XSD document is optional.

The XSD information, if included, appears in the following form:

```
<lccSchema xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="lcc" xsi:schemaLocation="lcc
XSD_FILE_PATH">
```

The XSD file prepared for validating ATtILA's XML documents is named LCCSchema.xsd. It is located in the ToolboxSource > LandCoverClassifications folder in the ATtILA toolbox destination folder (see Installing ATtILA).

You can include the following line as your root element if you place your custom XML document in the same folder as the XSD file:

```
<lccSchema xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="lcc" xsi:schemaLocation="lcc
LCCSchema.xsd">
```

LCC XML Document Metadata Element

Basic structure of ATtILA's Land Cover Classification schema XML document: **Metadata element**.

```
<metadata>
  <name></name>
  <description></description>
</metadata>
```

The Metadata element contains the name of the coding schema along with a brief description. ATtILA requires the Metadata element to be present when it parses the the XML document, and it cannot be empty. The Metadata element must contain the following two elements:

- **name** - text, land cover coding schema name
- **description** - text, description of land cover coding schema

Example Metadata element:

```
<metadata>
  <name>NLCD</name>
  <description>National Land Cover Database</description>
</metadata>
```

The information contained in the Metadata element is for informational purposes only. It is displayed in the Land Cover Classification Editor's Metadata window, and only there. The entries for the name and description attributes can be as long or as short as the user desires.

XML Note

- XML is case-sensitive and care should be used when editing the LCC XML documents using a text editor. Wherever possible, element and attribute names for the document have been defined using lowercase characters. The only current exceptions are the attributes “**Id**” and “**Name**” in the *Coefficients*, *Values*, and *Classes* elements. If editing an LCC XML document outside of the Land Cover Classification Editor, these attributes should be capitalized wherever they appear.

- You may notice that in the Metadata element, the <name> element is lowercase. The distinction here is that “name” in the Metadata element pertains to an **element**, and that the capitalization rule only applies to an element's attributes.

LCC XML Document Coefficients Element

*Basic structure of ATtILA's Land Cover Classification schema XML document: **Coefficients element**.*

```
<coefficients>
  <coefficient Id="" Name="" fieldName="" method="" />
</coefficients>
```

The Coefficients element is used to identify and to parameterize the coefficient-based metrics available in the Land Cover Coefficient Calculator tool. This element is required by ATtILA to parse the LCC XML document, but it can be empty (i.e., it does not have to contain any <coefficient> elements).

NOTE: *If no <coefficient> element is in the XML document when the user attempts to use the Land Cover Coefficient Calculator tool, the tool dialog will not populate the “Report metrics for these coefficients” input box when a “Land cover classification scheme” is selected. Despite the missing information, the user is still able to click the OK button to begin the calculation run, but the run will result in an error.*

If any <coefficient> element is provided, it needs to contain the following attributes:

- **Id** - text, unique identifier
- **Name** - text, description of coefficient metric
- **fieldName** - text, name of field to be created for output
- **method** - text, either “P” or “A”

Example Coefficients element with three <coefficient> elements:

```
<coefficients>
  <coefficient Id="IMPERVIOUS" Name="Percent Cover Total Impervious Area" fieldName="PCTIA" method="P" />
  <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover" fieldName="N_Load" method="A" />
  <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover" fieldName="P_Load" method="A" />
</coefficients>
```

Each coefficient represented by a <coefficient> element in the LCC XML document appears in the “Report metrics for these coefficients” parameter in the Land Cover Coefficient Calculator tool dialog and is individually selectable (see below). The element’s attributes are listed next to the check box as follows: **Id** - [fieldName] **Name**

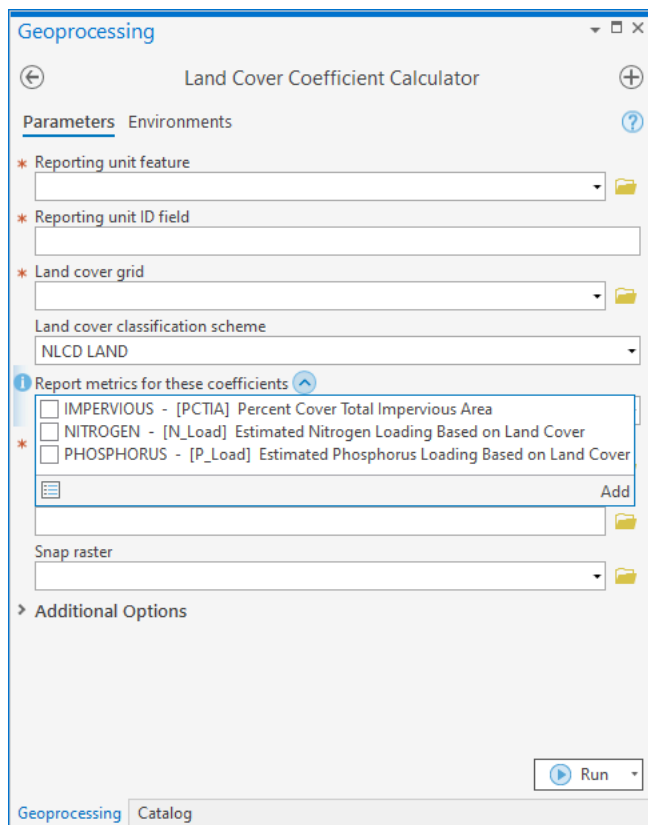


Figure 10. ATtILA Land Cover Coefficient Calculator metric selection dropdown menu

Id Attribute

Id is a text string of any length that supplies the short name for a coefficient. In this version of ATtILA, the text string is restricted to ‘NITROGEN’, ‘PHOSPHORUS’, or ‘IMPERVIOUS’. In a future release, the user will be able to use any string for the **Id** attribute. See the **method** attribute discussion below for more details.

Name Attribute

Name is a text string of any length used to supply a more detailed description of the coefficient and/or provide additional details regarding the metrics that result from tool calculations using the coefficient. It can also be an empty string (e.g., Name=“”).

fieldName Attribute

fieldName is a text string used to customize the name of the coefficient in the output table field. The length of the text string is dependent on the output table type selected in the Land Cover Coefficient Calculator tool as follows:

- dBase table - field names are restricted to 10 characters in length.
- geodatabase table - field names are restricted to 64 characters in length.

ATtILA will truncate field names that are longer than those allowed, and check to see if the truncated field name is already in the output table. If the truncated field name already exists, the text string is truncated further to allow a numerical value to be appended to the field name to create a unique field name. When this occurs, a warning message is added to the tool’s View Details box and the Geoprocessing > History window informing the user of the changes to the fieldName’s text string in the output table.

In addition to size limitations, the **fieldName** attribute string must conform to the field naming conventions dictated by the different database systems (dBASE, or geodatabase). In general, try to restrict field names to just alphanumeric characters and underscores. Use of spaces and special characters should be avoided, as well as beginning a field name with a number or an underscore. Also, avoid using field names that contain words that are considered reserved keywords, such as date, day, month, order, table, text, user, when, where, year, and zone. For more guidelines on the naming of fields, search on “Fundamentals of adding and deleting fields” in the ArcGIS help documentation.

method Attribute

CAUTION: In this version of ATtILA, the **method** attribute is dummy attribute, and exists only as a placeholder.

In future releases, the **method** attribute will be used to pass an appropriate calculation routine for the coefficient to the Land Cover Coefficient Calculator tool. When implemented, the method element will be restricted to two choices:

- “A” - the tool will use a “Per Unit Area” calculation routine to calculate metrics associated with the coefficient.
- “P” - the tool will use a “Percent Area” calculation routine to calculate metrics associated with the coefficient.

More information on the different calculation routines can be found on the Land Cover Coefficient Calculator page.

XML Note

- XML is case-sensitive and care should be used when editing the LCC XML documents using a text editor. Wherever possible, element and attribute names for the document have been defined using lowercase characters. The only current exceptions are the attributes “**Id**” and “**Name**” in the Coefficients, Values, and Classes elements. If editing an LCC XML document outside of the Land Cover Classification Editor, these attributes should be capitalized wherever they appear.

- You may notice that in the Metadata element, the <name> element is lowercase. The distinction here is that “name” in the Metadata element pertains to an **element**, and that the capitalization rule only applies to an element’s attributes.

LCC XML Document Values Element

*Basic structure of ATtILA's Land Cover Classification schema XML document: **Values element***

```
<values>
  <value Id="" Name="" excluded="">
    <coefficient Id="" value="" />
  </value>
</values>
```

The Values element defines the set of values that can exist in a land cover raster. This element is required by ATtILA to parse the XML document, but it can be empty (i.e., it does not have to contain any <value> elements). Value elements are only necessary in two cases:

1. if the user wants to exclude the area occupied by a given grid value when determining a reporting unit's effective area (e.g., basing metric calculations on the land area in a reporting unit vs. the total area of the reporting unit), or
2. if the user is planning to calculate any metrics with the [Land Cover Coefficient Calculator](#). In this case <value> elements should be added where coefficient weights are known (e.g., the nitrogen loading coefficient for a particular agriculture type).

NOTE: *Although values are not necessary in the XML document, if one is using the LCC Editor to construct an LCC XML document, you will be required to have value elements defined in order to assign values to a class element (see [Classes Element](#)).*

If any <value> element is provided, it must have the **Id** attribute. The attributes, **Name** and **excluded**, are allowed, but are not necessary. Properties for the <value> element attributes are:

- **Id** - integer
- **Name** - text (optional, can be the empty string, "")
- **excluded** - boolean (optional, "true" or "false" or "1" or "0")

Example Values element with two <value> elements:

```
<values>
  <value Id="11" Name="Open Water" excluded="true" />
  <value Id="41" Name="Deciduous Forest" />
</values>
```

Id Attribute

Id is an integer value representing a grid code that may be found in a land cover raster. Values included in the LCC XML document can span the range of expected values of a particular land cover dataset (e.g., NLCD, C-CAP, MULC), but all of the values provided in the document do not have to exist in the actual land cover layer. This is often the case when a national-level land cover coding schema is selected by the user, but the input land cover raster has been clipped to a regional study area.

A potentially more important issue is when all the values located in a land cover raster are not accounted for in the LCC XML document. ATtILA will examine all of the values provided in the LCC XML document from both the Values section of the document and those provided in the [Classes Element](#) section and compare them to those found in the land cover raster. Any values in the grid not found in the LCC XML document will be reported to the user with a warning message in the tool's View Details box and the Geoprocessing > History window. The user can then determine if the reported

values were accidentally omitted from the LCC XML document or incorrectly recorded. A report of missing values may also indicate that the wrong LCC Schema was selected for the input raster layer, or that the wrong raster layer was input for the selected LCC Schema.

Name Attribute (optional)

Name is a text string of any length used to supply a more detailed description of the grid code and/or provide other useful information regarding the value. The **Name** attribute may be an empty string (e.g., Name="").

excluded Attribute (optional)

The attribute, **excluded**, is used to identify grid codes whose area is to be excluded from the reporting unit's effective area calculation. Effective area can be thought of as the area of interest within a reporting unit that the user wishes to use for percentage based metric calculations. For example, the user may be interested in basing their metric calculations on just the land area in a reporting unit versus the overall total area of the reporting unit. To make the effective area equal to that of the land area, the user would set the **excluded** attribute of any water related grid value to "true".

The default setting for the **excluded** attribute is excluded="false" (i.e., if no **excluded** attribute is provided in the <value> element, the area associated with that grid value is not excluded from the reporting unit's effective area calculation). When all **excluded** attributes are tagged as 'false', the effective area is equal to the total overall area of the reporting unit.

CAUTION: any value tagged as **excluded** = "true", should not be used in any <class> element definition within the LCC XML document. The <class> element is discussed in the [Classes Element](#) section.

Value Elements with Coefficient Elements

A Value element can also contain one or more Coefficient elements. These Coefficient elements are optional unless the user is planning on generating metrics using the [Land Cover Coefficient Calculator](#) tool. See the caution note below for more specifics on when <coefficient> elements must be supplied. If any <coefficient> element is provided, it needs to contain the following attributes:

- **Id** - text (Either ‘NITROGEN’, ‘PHOSPHORUS’, or ‘IMPERVIOUS’).
- **value** - decimal

Example Values element with one <value> element with two <coefficient> elements:

```
<coefficients>
  <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover" apField="A"
    fieldName="N_Load" />
  <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover" apField=""
    fieldName="P_Load" />
</coefficients>

<values>
  <value Id="41" Name="Deciduous Forest">
    <coefficient Id="NITROGEN" value="2.447" />
    <coefficient Id="PHOSPHORUS" value="0.089" />
  </value>
</values>
```

Coefficient Element: Id Attribute

Id is a text string used to reference a <coefficient> element within the LCC XML document’s [Coefficients Element](#) section. This text string needs to match the **Id** attribute for that <coefficient> element exactly. At present, the text string for this attribute is restricted to equal either ‘NITROGEN’, ‘PHOSPHORUS’, or ‘IMPERVIOUS’. In future releases of ATtILA, we plan to allow the user to use any string for the this element’s **Id** attribute as long as it matches an **Id** attribute found in a corresponding <coefficient> element. See the discussion on the attribute, **method**, in [Coefficients Element](#) for more details on this.

Coefficient Element: value Attribute

The attribute **value** is a decimal number representing the coefficient weighting factor assigned to a land cover/land use type. Values for NITROGEN and PHOSPHOROUS should be given in kg per hectare per year. Values for IMPERVIOUS should represent percentages.

CAUTION: For any coefficient-based metric the user selects to calculate when running the [Land Cover Coefficient Calculator](#) tool, that coefficient <element> must be present in all of the <value> elements provided in the <values> element. Not all values in the input land cover raster need to be accounted for in the LCC XML document when using the [Land Cover Coefficient Calculator](#) tool, but if a value is included, its <value> element must contain the corresponding <coefficient> element for the desired metric.

***XML Note** - XML is case-sensitive and care should be used when editing the LCC XML documents using a text editor. Wherever possible, element and attribute names for the document have been defined using lowercase characters. The only current exceptions are the attributes “Id” and “Name” in the Coefficients, Values, and Classes elements. If editing an LCC XML document outside of the [Land Cover Classification Editor](#), these attributes should be capitalized wherever they appear.*

- *You may notice that in the Metadata element, the `<name>` element is lowercase. The distinction here is that “name” in the Metadata element pertains to an **element**, and that the capitalization rule only applies to an element’s attributes.*

LCC XML Document Classes Element

Basic structure of *ATtILA's Land Cover Classification schema XML document*: *Classes element*.

```
<classes>
  <class Id="" Name="" filter="">
    <value Id="" />
  </class>
</classes>
```

The <classes> element contains values from a land cover raster grouped into one or more <class> elements. The <classes> element is required by ATtILA to parse the XML document, and it must contain at least one <class> element. Any provided <class> element needs to contain an **Id** attribute. The attributes, *Name*, *filter*, and *xxxField* (where *xxxx* equals a tool name abbreviation), are allowed, but are not necessary. Properties for the <class> element attributes are:

- **Id** - text, unique identifier
- *Name* - text (optional, can be the empty string, "")
- *filter* - text, a string of one or more tool name abbreviations (*caem*, *flcp*, *flcv*, *lcosp*, *lcp*, *np*, *plcv*, *pm*, *rlcp*, or *splcp*) separated by a ";" (optional, can be the empty string, "")
- *xxxField* - text, where *xxxx* equals a tool name abbreviation (*caem*, *flcp*, *flcv*, *lcosp*, *lcp*, *rlcp*, or *splcp*). A separate *xxxField* attribute can exist for each tool (optional, can be the empty string, "")

A <class> element can contain either <value> elements or additional <class> elements but not both types. A <class> element contained within another <class> element is known as a child class. The containing <class> element is the parent class. A <value> element within a <class> element has only one attribute:

- **Id** - integer, corresponds to a land cover/land use grid value

Example Classes element with one <class> element:

```
<classes>
  <class Id="nat" Name="All natural land use" filter="" lcpField="NINDEX">
    <value Id="41" />
    <value Id="42" />
    <value Id="43" />
    <value Id="51" />
    <value Id="52" />
  </class>
</classes>
```

Example Classes element with one parent <class> element and two child classes:

```
<classes>
  <class Id="nat" Name="All natural land use" lcpField="NINDEX">
    <class Id="for" Name="Forest">
      <value Id="41" />
      <value Id="42" />
      <value Id="43" />
    </class>
    <class Id="shrb" Name="Shrubland">
      <value Id="51" />
    </class>
  </class>
</classes>
```



```

        <value Id="52" />
    </class>
</class>
</classes>

```

NOTE: When a <class> element (parent class) has additional <class> elements nested within it (child classes), the <value> elements from all child classes are combined to define the parent class. For example, if a class, “nat”, contains two classes, “for” and “shrb”, and “for” is comprised of values 41, 42, and 43 and “shrb” is defined as values 51 and 52, “nat” would be defined as having values 41, 42, 43, 51, and 52.

NOTE: Empty <class> elements are ignored by ATiLA.

NOTE: if a value is marked as excluded = “true” in the Values Element section, but is included in a class definition, ATiLA will ignore the area of that excluded value when summing up the area of the selected class for metric calculations. If all values assigned to the class are tagged as excluded, the output metric is calculated as 0%.

Each <class> element in the LCC XML document will appear as a selectable metric in the “Report metrics for these classes” or the “Generate output for these classes” input parameter area for tools that utilize the LCC XML document (i.e., caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp). An example for lcp is shown below using the supplied land cover classification schema, NLCD LAND.

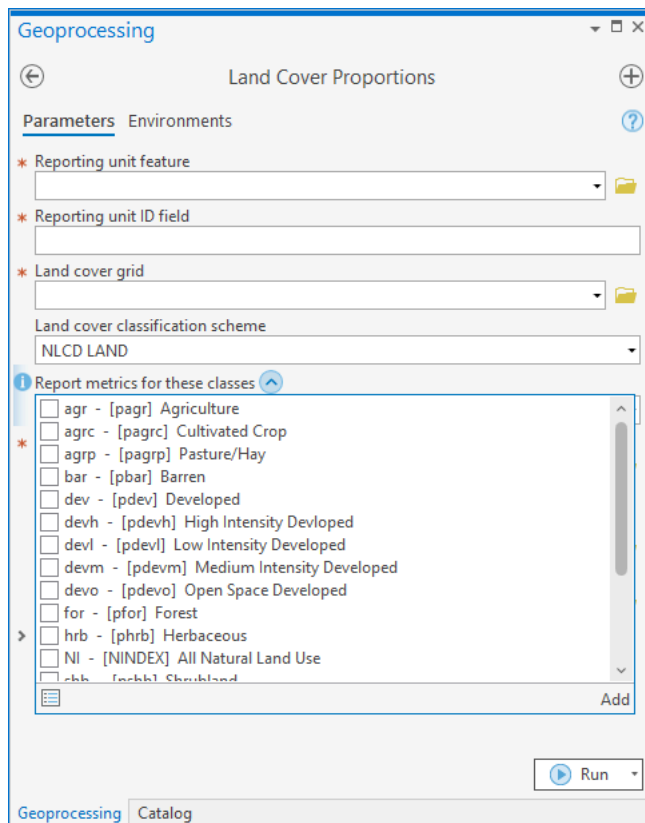


Figure 11. Land Cover Proportions dialog’s Report metrics for these classes selection dropdown menu

In the above example, the <class> element’s attributes are listed next to its check box in the following order and format:

- **Id** - [xxxx**Field** or auto-generated field name] **Name**

Id Attribute

Id is a text string used to identify the defined set of raster grid values as a group (i.e., class) for analysis. In addition, the class **Id** attribute is used as a base to auto-generate output metric field names when no **xxxxField** class attribute is provided (see below) or if the **xxxxField** attribute is an empty string. Each ATtILA tool has a predefined prefix and/or suffix which it will use in conjunction with the **Id** attribute string to auto-generate field names. See the help section for each individual tool ([caem](#), [flcp](#), [fleuv](#), [lcosp](#), [lcp](#), [rlcp](#), and [splcp](#)) to learn more.

xxxxField Attribute (optional)

The attribute, **xxxxField**, where xxxx equals a tool name abbreviation, is a text string used as the default output field name. Only the tools [caem](#), [flcp](#), [fleuv](#), [lcosp](#), [lcp](#), [rlcp](#), and [splcp](#) can make use of the **xxxxField** attribute. If no **xxxxField** attribute is provided or if the **xxxxField** attribute contains an empty string, ATtILA will auto-generate an output field name using the class **Id** attribute as a base (see above). A separate **xxxxField** attribute can exist in the <class> element for each applicable tool (e.g., <class Id="nat" Name="All natural land use" filter="" lcpField="NINDEX" rlcpField="RNatural">).

CAUTION: The **Id** attribute, if used for output field naming, or the **xxxxField** attribute, if provided, must be unique among the different <class> elements or identical field names can be generated between classes. If duplicate field names are present, only one is added to the output table. During metric runs, all previous values calculated for that field will be overwritten by the last metric processed with the duplicate field name.

CAUTION: Field name size is limited by the Output table type: dBASE table field names can be 10 characters in length, and File Geodatabase table field names can be 64 characters in length.

- For **xxxxField** attribute strings: ATtILA will truncate strings that are longer than those allowed, and check to see if the truncated field name is already in the output table. If the truncated field name already exists, the text string will be truncated further to allow a numerical value to be appended to the field name to create a unique entity. A warning message will then be added to the tool's View Details box and the Geoprocessing > History window informing the user of what the provided **xxxxField** attribute text string was changed to for the output table.
- For auto-generated field names: If the auto-generated field name is longer than what is allowed, ATtILA will truncate the field name base (i.e., the class **Id** attribute string), keeping the metric's predefined prefix and/or suffix intact, to shorten the field name. If the truncated field name already exists in the output table, the base string will be truncated further to allow a numerical value to be appended to it to create a unique entity. A warning message will then be added to the tool's View Details box and the Geoprocessing > History window informing the user of what the auto-generated field name was changed to before adding it to the output table.

CAUTION: In addition to size limitations, the **Id** attribute, if used for output field naming, or the **xxxxField** attribute, if provided, must conform to the field naming conventions dictated by the different database systems (dBASE, or geodatabase). In general, try to restrict field names to just alphanumeric characters and underscores. Use of spaces and special characters should be avoided, as well as beginning a field name with a number or an underscore. Also, avoid using field names that contain words that are considered reserved keywords, such as date, day, month, order, table, text, user, when, where, year, and zone. For more guidelines on the naming of fields, search on "Fundamentals of adding and deleting fields" in the ArcGIS help documentation.

CAUTION: If an invalid field name is provided in the **xxxxField** attribute, such as a string of spaces, the metric run will fail during execution, and an "Invalid field name" error message will appear in the tool's View Details box and the Geoprocessing > History window.

Name Attribute (optional)

Name, is a text string of any length used to supply a more detailed description of the class and/or provide other useful information regarding it. The **Name** attribute may be an empty string (e.g., Name=""). The **Name** attribute is not used in metric calculations or output field naming routines.

filter Attribute (optional)

The **filter** attribute is a string of one or more tool name abbreviations (caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, or splcp) separated by a semi-colon. It is used to exclude a class from a tool's list of selectable classes in the "Report metrics for these classes" or the "Generate output for these classes" input parameter area. This can be done to reduce clutter within a tool's GUI (e.g., dropping all classes except those related to agriculture for the Land Cover on Slopes Proportions tool) or to prevent classes incongruous for the analysis from being selected (e.g., <class Id="wetl" Name="All wetland land cover" filter="lcosp"> will eliminate the wetl class from the Land Cover on Slopes Proportions tool).

Value element: Id Attribute

The <value> element attribute, **Id**, is an integer value representing a grid code that may be found in a land cover raster. Groupings of <value> elements within a <class> element is dictated solely by the research needs of the user.

XML Note

- XML is case-sensitive and care should be used when editing the LCC XML documents using a text editor. Wherever possible, element and attribute names for the document have been defined using lowercase characters. The only current exceptions are the attributes "Id" and "Name" in the Coefficients, Values, and Classes elements. If editing an LCC XML document outside of the Land Cover Classification Editor, these attributes should be capitalized wherever they appear.

- You may notice that in the Metadata element, the <name> element is lowercase. The distinction here is that "name" in the Metadata element pertains to an **element**, and that the capitalization rule only applies to an element's attributes.*

LCC XML Document Comments

XML documents can contain commented characters for notes and brief descriptions of sections. Comments are delimited by <!-- and end with the first occurrence of -->.

The Supplied Land Cover Classification schemas that come with ATtILA come with a standardized set of comments (see below). They provide information to the user regarding the contents of the following elements: <coefficients>, <values>, and <classes>. These standardized comments are also inserted when LCC XML documents are created with the Land Cover Classification Editor. Although comments are supplied by the program, they are unnecessary for the functioning of the XML document, and can be deleted. Any LCC XML document created by the user outside of the Land Cover Classification Editor can omit comments.

```
<!--
* The coefficients node contains coefficients to be assigned to values.

* REQUIRED ATTRIBUTES
* Id - text, unique identifier
* Name - text, word or phrase describing coefficient
* fieldName - text, name of field to be created for output
* - must conform to the field naming conventions dictated by the output database system
* apMethod - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
-->
```

```
<!--
* The values node defines the full set of values that can exist in a land cover raster.
* REQUIRED ATTRIBUTES
* Id - integer, raster code
*
* OPTIONAL ATTRIBUTES
* Name - text, word or phrase describing value
* excluded - boolean, "true" or "false" or "1" or "0"
* - used to exclude values from effective area calculations
* - excluded=false is the default

* A value element can optionally contain one or more coefficient elements

* REQUIRED COEFFICIENT ATTRIBUTES
* Id - text, must match an Id attribute from a coefficients node element
* value - decimal, weighting/calculation factor
-->
```

```
<!--
* The classes node contains values from a land cover raster grouped into one or more classes.

* REQUIRED ATTRIBUTES
* Id - text, unique identifier, also used for automated generation of output field name
* - must conform to the field naming conventions dictated by the output database system

* OPTIONAL ATTRIBUTES
* Name - text, word or phrase describing class
* filter - text, a string of one or more tool name abbreviations separated by a ";"
* - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp
* - used to exclude the class from the selectable classes in the tool's GUI
* xxxxField - text, overrides ATtILA-generated field name for output
```

- * - where xxxx equals a tool name abbreviation
- * - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
- * - a separate xxxxField attribute can exist for each tool
- * - must conform to the field naming conventions dictated by the output database system

- * A class can contain either values or classes but not both types.
- * Value elements contain only an Id attribute which refers to a value in a raster.
- * Values tagged as excluded="true" in the values node should not be included in any class.

--\>

LCC XML Document XML Validation

Validation ensures that any self-edited LCC XML document is both properly formatted as an XML document, and that all elements necessary for ATtILA are present and correctly populated.

Validation is not necessary for any LCC XML documents created with the Land Cover Classification Editor or for any of the Supplied Land Cover Classification Schemas that came with ATtILA . If you have used the Land Cover Classification Editor to create your LCC XML document, you may skip this section.

Validation is only recommended for LCC XML documents constructed outside of the Land Cover Classification Editor. If you have created a self-edited LCC XML file, please continue reading.

To perform validations, it is necessary to have either an XML editor/validator installed on your computer or by having access to the Internet and to one of several websites that provide validation service. If you do not currently have access to an XML program, an internet search using the phrase, “validation of XML”, will locate several options.

While many programs/websites are available, the procedure for conducting syntax checks and validations are similar between them. We will describe the steps used within the Notepad++ v7.8.9 free source code editor. Notepad++ is available for download from <https://notepad-plus-plus.org>.

To use Notepad++ for validation, the XML plugin must be installed. To install the plugin, select ‘Plugins Admin...’ from the Plugins dropdown menu.

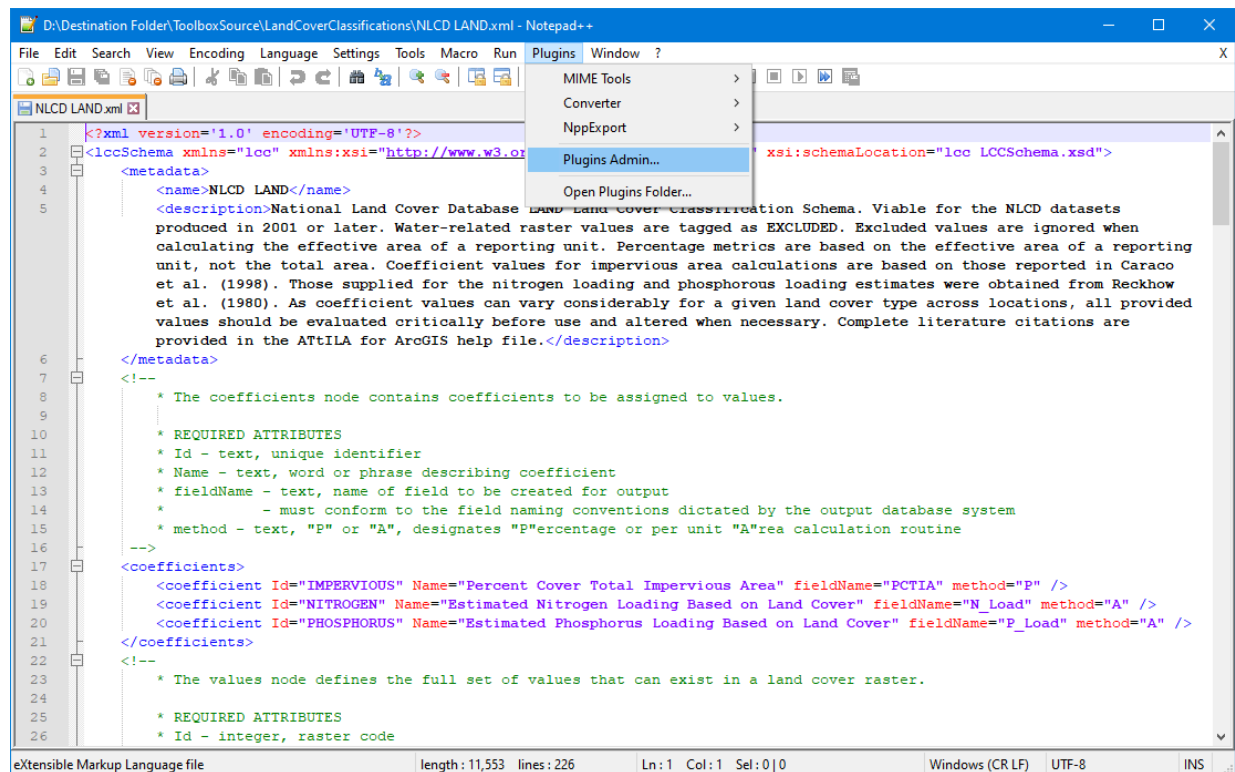


Figure 12. Screenshot of the Notepad++ application window with the Plugins Admin selection highlighted

Once the Plugin Manager dialog box opens, select the Available tab and then click on XML Tools. Select Install. Notepad++ will typically ask to close and restart to finish the installation. If not, after installation is complete, close the Plugin Manager dialog.

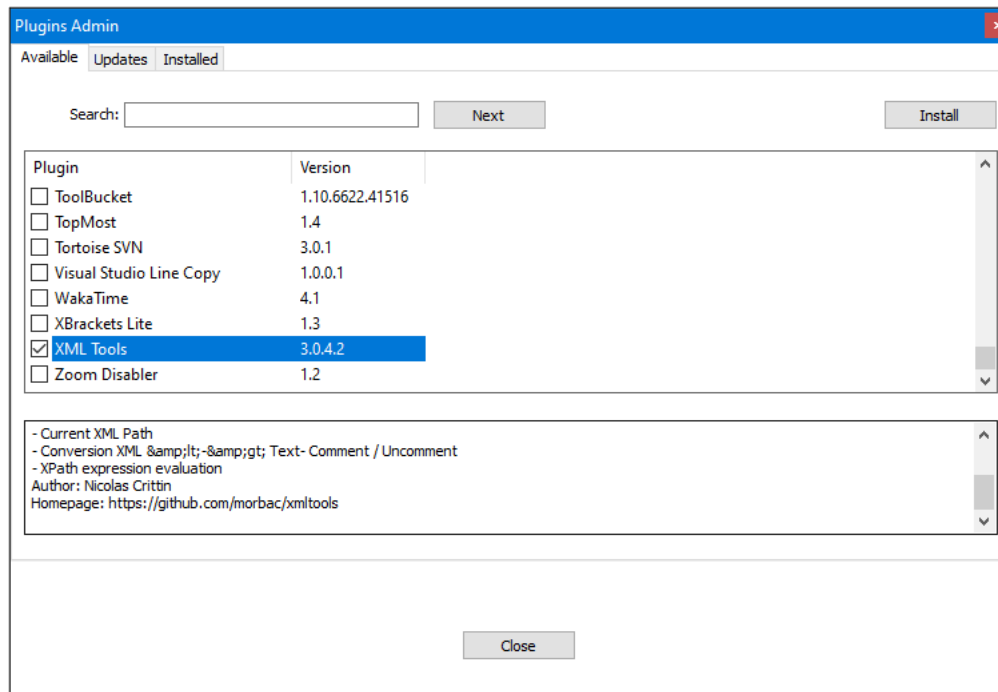


Figure 13. Screenshot of the Plugins Admin manager dialog in Notepad++

The XML Tools item will now be available from the Plugins dropdown menu.

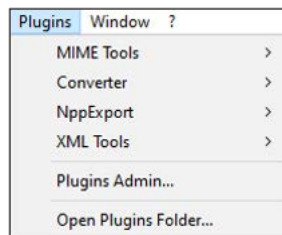


Figure 14. Example of Plugins Menu for Notepad++ with XML Tools added

XML Validation - Step One

Validation is typically a two step process. The first step ensures that the XML document follows standard formatting rules. Some standard rules include:

- Each start-tag (**<tag>**) must have a matching end-tag. The end-tag can be either a non-empty element, (**</tag>**), or an empty element (**/>**).
- Attribute values must be quoted.
- Attribute names must be unique, but XML is case-sensitive so “name” is different from “Name” which is different from “NAME”.
- Comments and processing commands cannot appear inside tags.

Checking of XML syntax rules is rather straightforward. Once your XML LCC documented is loaded into Notepad++, select “Check XML syntax now” from the XML Tools plugin menu.

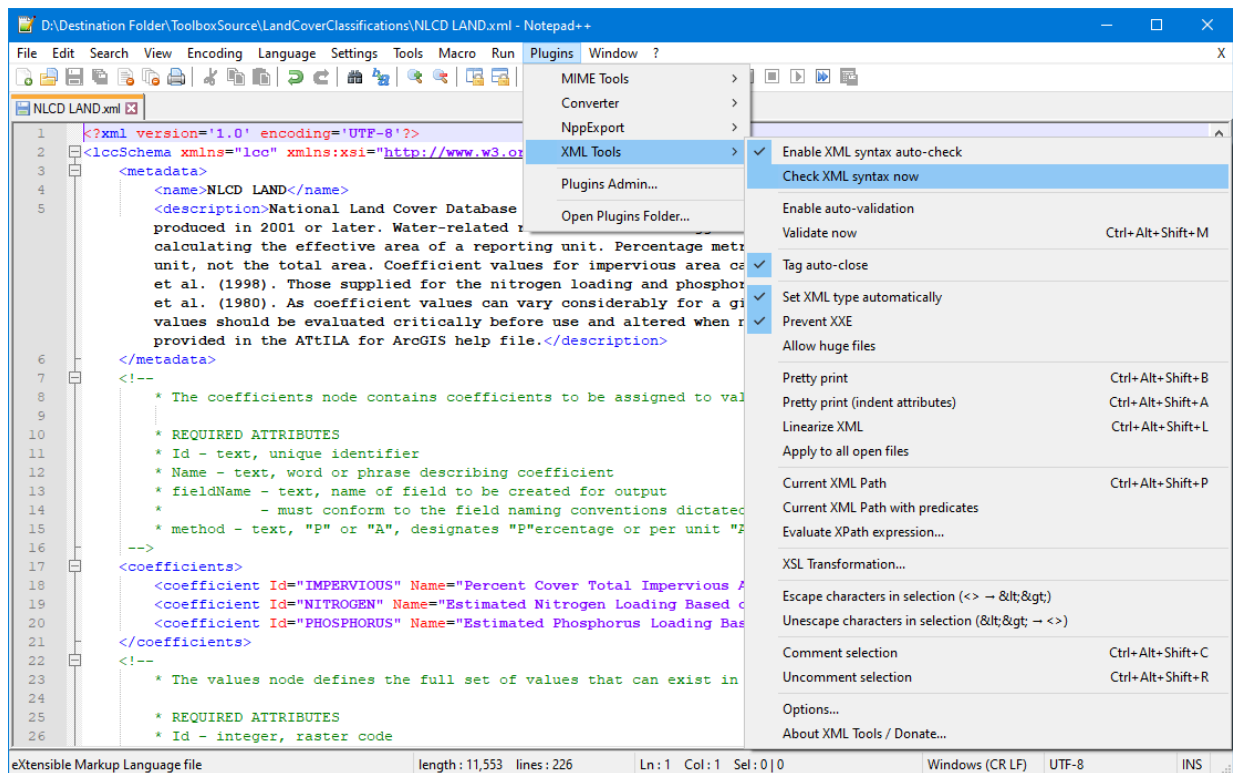


Figure 15. Screenshot of XML Tools menu options

If all is correct, a message similar to the one below should appear indicating that no errors were detected, and you can proceed to step two.

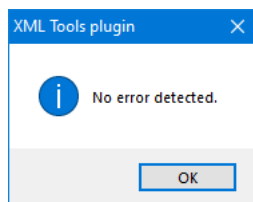


Figure 16. Screenshot of XML Tools' No syntax error detected notice

If an error does exist, an error dialog will appear, and the error must be located and corrected before continuing. A couple examples of error messages are provided below.

Example: End tag error.

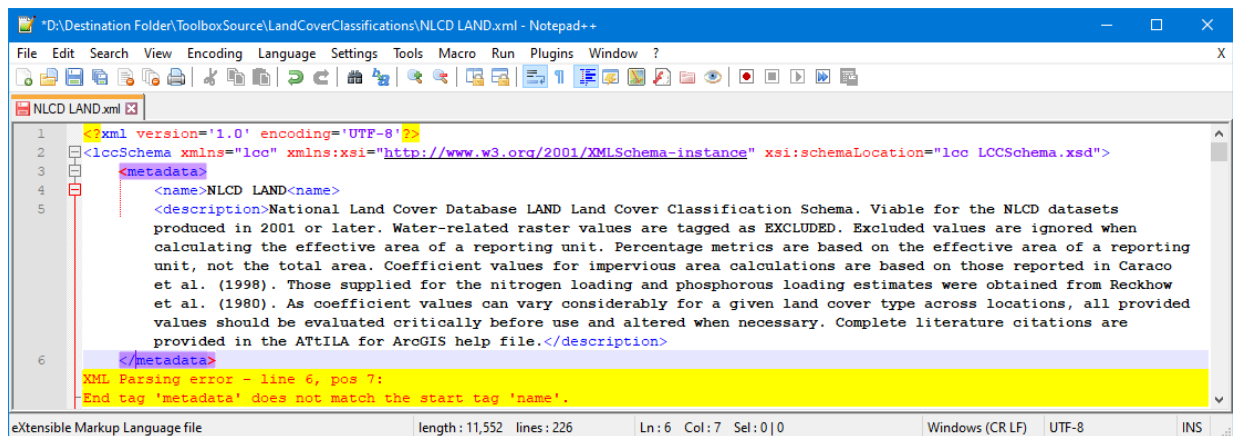


Figure 17. Screenshot of an XML end tag validation error

On line four, in the metadata element, the end-tag for the <name> attribute is missing a backslash. Each start-tag (e.g., <name>) must have a matching end-tag. The end-tag can be either a non-empty element, (e.g., </name>), or an empty element (</>).

Example: Mismatched tags error

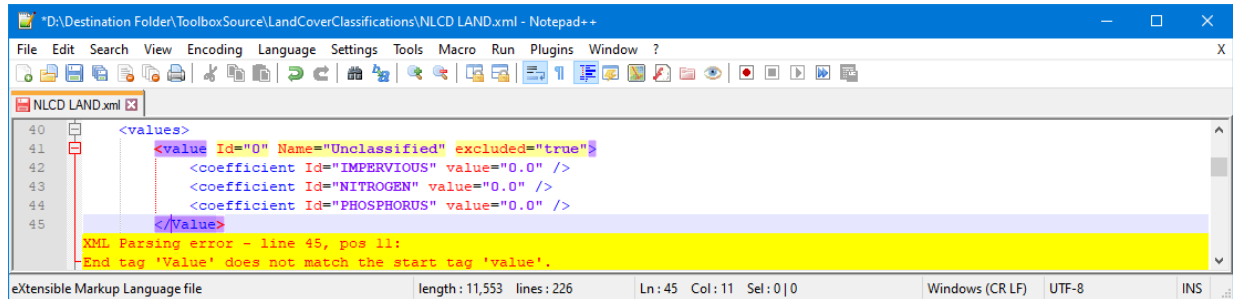


Figure 18. Screenshot of an XML mismatched tags validation error

On line 45, the end-tag is </Value> while the start tag on line 41 was <value>. This causes an error because XML is case-sensitive. This error can be corrected by changing the end-tag from </Value> to </value>.

Example: Unquoted attribute error

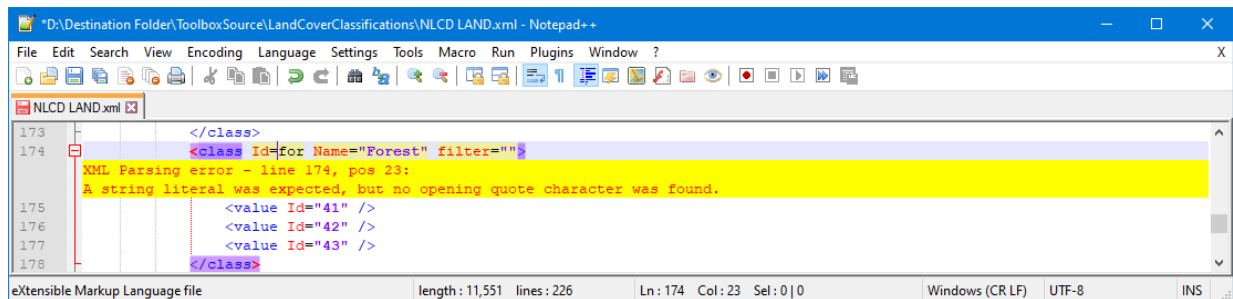


Figure 19. Screenshot of XML unquoted attribute validation error

On line 174, the value for the Id attribute is missing quotation marks. Attribute values must be quoted. This error can be corrected by changing the attribute assignment from, Id=for, to Id="for".

Example: Misquoted attribute error

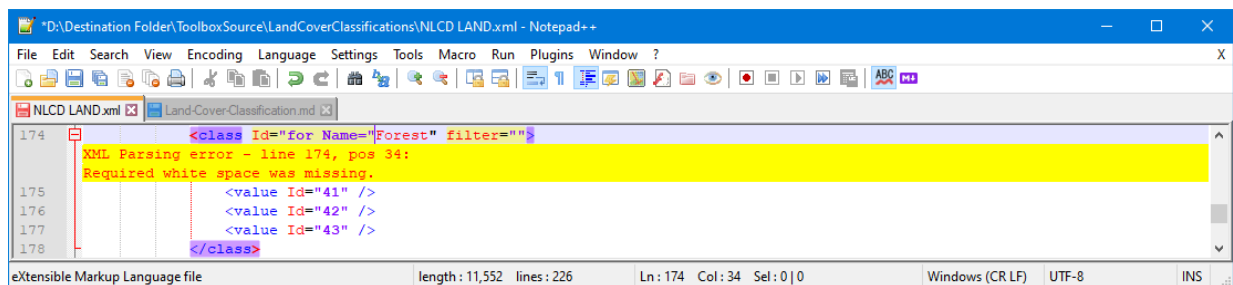


Figure 20. Screenshot of XML misquoted attribute validation error

On line 174, the error is caused by not closing the quotations for the Id attribute value. The value assignment of 'for' is missing its end-quote. Correct this error by changing the attribute assignment from, Id="for, to Id="for".

Example: Missing bracket error

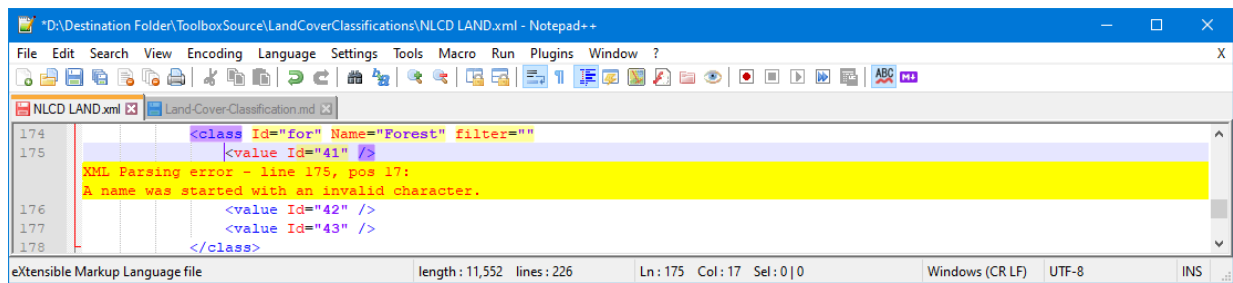


Figure 21. Screenshot of XML missing bracket validation error

On line 175, an XML Parsing error occurred. This error may be more difficult to troubleshoot than previous examples. It is caused by the class start-tag on line 174 missing its closing sharp bracket. The correct start-tag phrasing for this example is: `<class Id="for" Name="Forest" filter="">`.

Using the information from the error message will often quickly lead to the problem area in the text, but in certain cases, such as in our last example error message, locating and correcting the cause of the error requires a careful examination of the entire XML document. If the errors are not readily apparent, forgo manually editing your LCC XML document, and perform your final edits in the Land Cover Classification Editor.

Once all syntax errors have been corrected, you can proceed to step two.

XML Validation - Step Two

The second step of the validation process is ensuring that all XML elements necessary for ATtILA are present and correctly populated. ATtILA requires several customized elements and attributes to also be in the XML document (see ATtILA's LCC XML Document). If these items are missing or their content rules are not followed, ATtILA may fail to operate or, worse yet, produce incorrect results.

To assist with this step of the validation, an XML Schema Definition (XSD) file has been provided for the ATtILA user. An XSD is a type of file that is used to formally describe the elements of an XML document, and to constrain and verify their contents. The XSD provided for ATtILA specifically controls which elements and attributes are permitted in the LCC XML document, the contents of those items, their order, and whether or not the items are required or optional.

ATtILA's XSD file is located in the ToolboxSource > LandCoverClassifications folder in the ATtILA toolbox destination folder (see Installing ATtILA). It is named, "LCCSchema.xsd". A copy of its contents can be found here.

You perform the validation step in much the same way as checking the XML's syntax, but you select "Validate now" from the XML Tools plugin menu instead.

If you have placed a reference to the XSD file in the root element of the LCC XML document, the XML editor should be able to locate the XSD file and perform the validation checks. If the XSD file is not found in the referenced location or if there is an error in the root element, the following error message may appear:

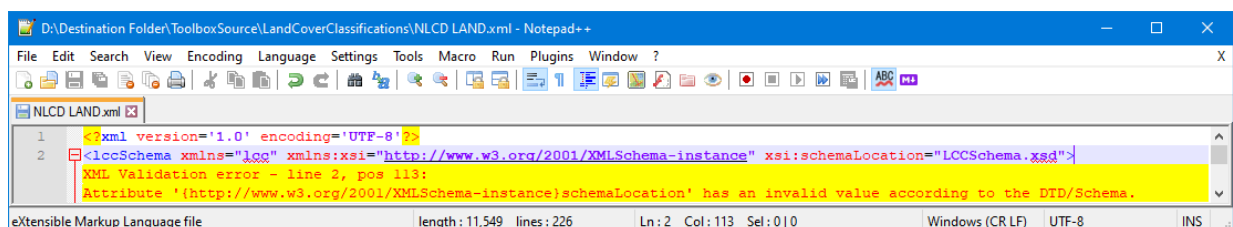


Figure 22. XSD file not found error message

If the LCC XML document does not contain a reference to an XSD file in its root element, Notepad++ will open a “Select file...” dialog allowing you to locate the XSD file yourself.

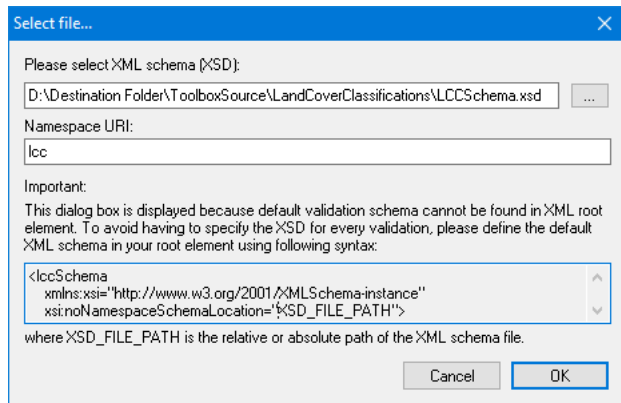


Figure 23. XSD file selection dialog box

Again, the ATtILA XSD file is located in the ToolboxSource > LandCoverClassifications folder in the ATtILA toolbox destination folder (see [Installing ATtILA](#)). It is named, “LCCSchema.xsd”, and a copy of its contents can be found [here](#).

If all goes well, a “No error detected” statement will appear.

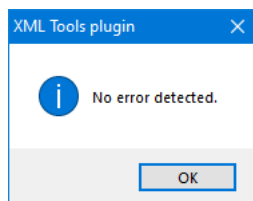


Figure 24. No error detected dialog box

If any problems are encountered, Notepad++ will alert you with an Information dialog describing the problem. The descriptions tend to be detailed, but, depending on your validation application, the exact location of the troublesome element in the LCC XML document may or may not be difficult to locate (see example errors below).

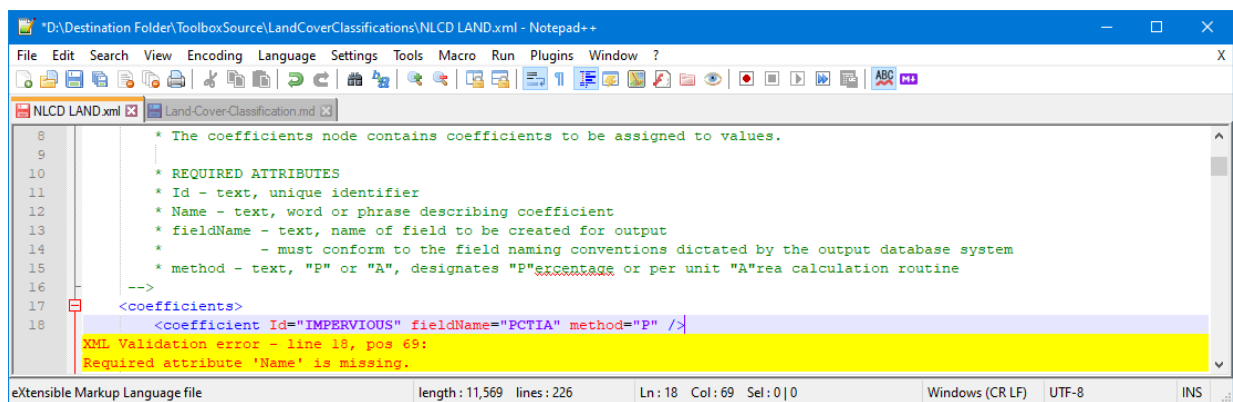


Figure 25. Example missing attribute XSD validation error screenshot

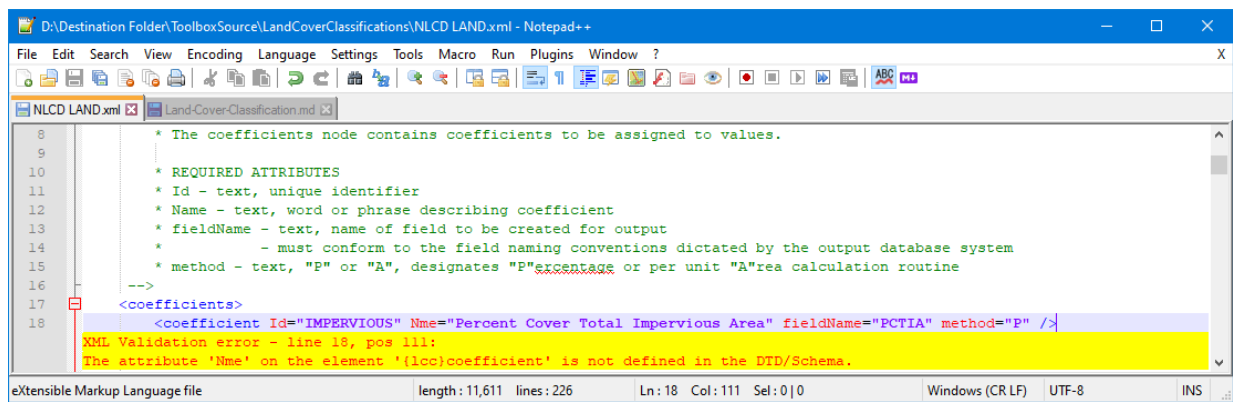


Figure 26. Example undefined attribute XSD validation error screenshot

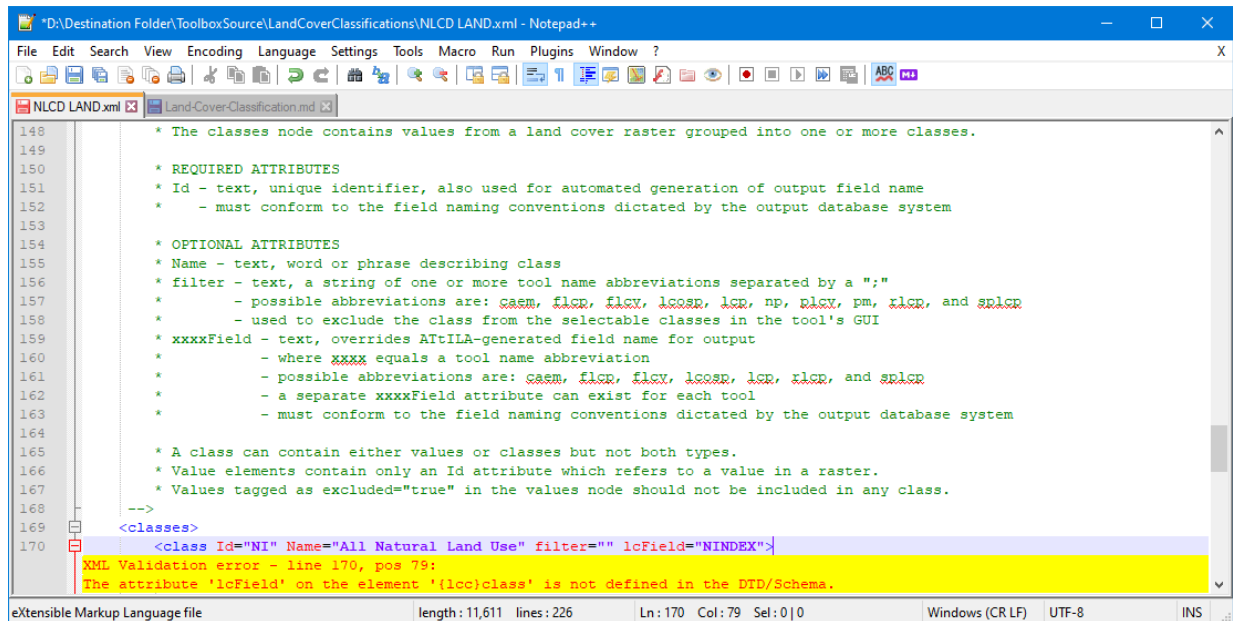


Figure 27. Example undefined attribute XSD validation error screenshot

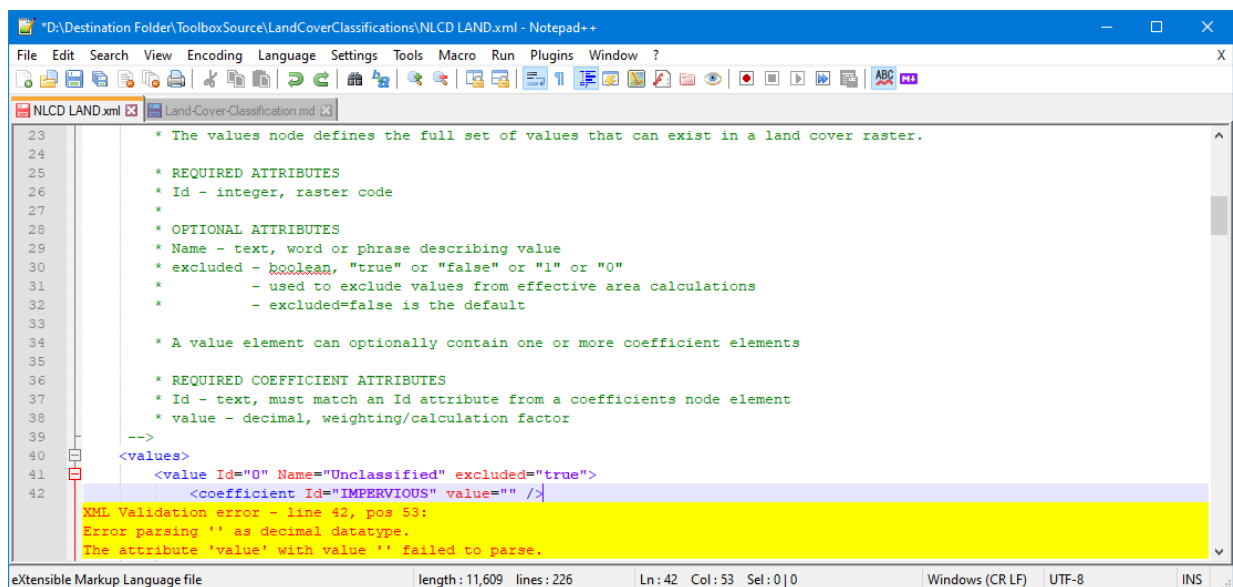


Figure 28. Example invalid data content attribute XSD validation error screenshot

Once validation steps one and two are performed without error reports, your LCC XML document should be ready for use with ATtILA. If any problems do occur during tool execution, please reread the section, ATtILA's LCC XML Document, and amend your file as necessary. As mentioned above, if the

errors are not readily apparent, forgo manually editing your LCC XML document, and perform your final edits in the Land Cover Classification Editor.

Editor

The Land Cover Classification Editor provides a user-friendly interface to create or edit a Land Cover Classification XML file for use in ATtILA.

To start editing, click the Create or Modify Land Cover Classification (.xml) tool in the ATtILA Toolbox.

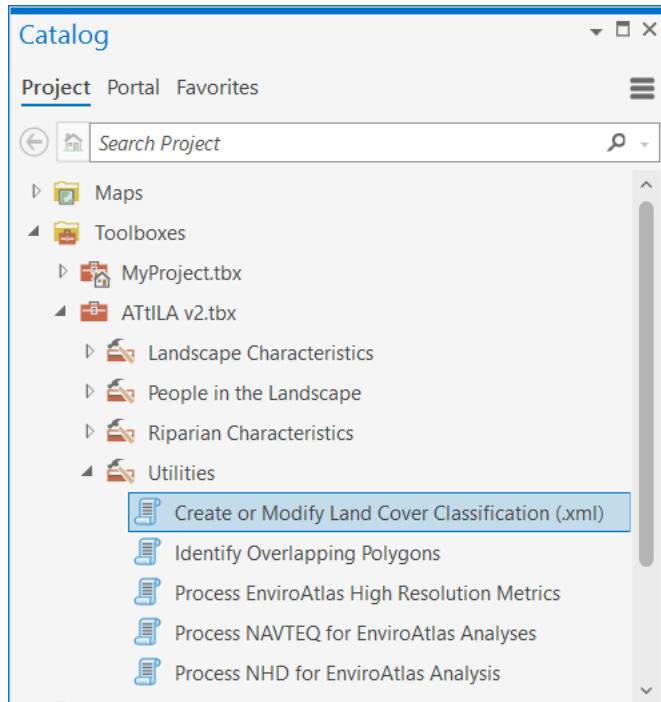


Figure 29. Create or Modify Land Cover Classification utility selection

The tool will open with a single parameter, Launch editor. Simply click “Run” to open the LCC Editor graphical user interface.

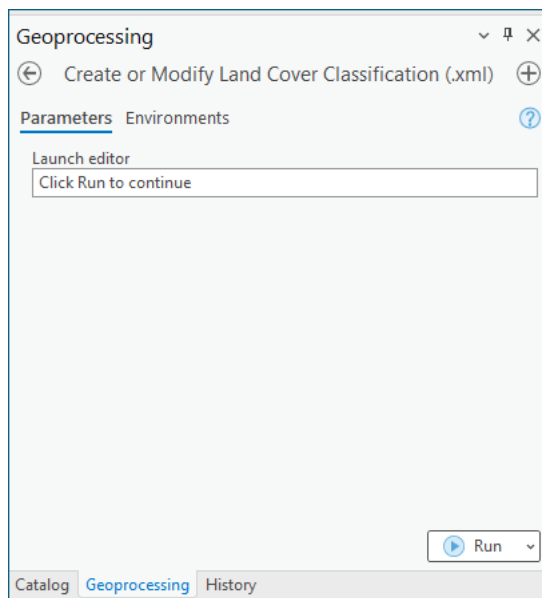


Figure 30. Run Create or Modify Land Cover Classification utility

NOTE: The Land Cover Classification Editor can also be opened directly by launching its executable file, *LCCEditor.exe*, located in the *ToolboxSource > ATiLA2 > scripts > bin* folder in the *ATiLA* toolbox destination folder (see [Installing ATiLA](#)).

Land Cover Classification Editor Interface

The LCC Editor consists of four windows (VALUES, COEFFICIENTS, METADATA, and CLASSES, with buttons to edit information within the windows. In addition, the Editor contains menu dropdowns with commands to manipulate the LCC XML file and the appearance of the LCC Editor, and an icon toolbar below the menu dropdowns that also controls the Editor's appearance.

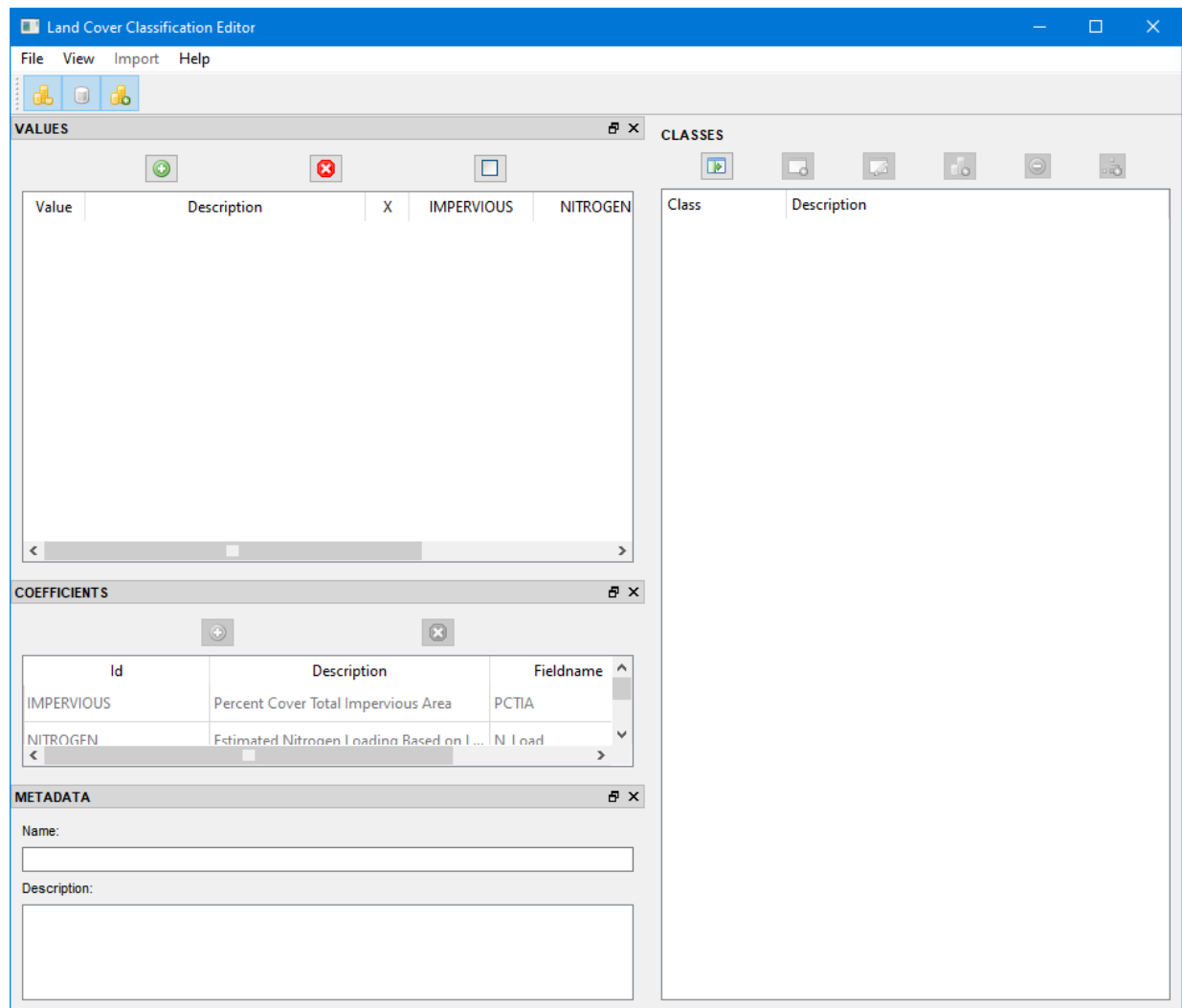


Figure 31. Create or Modify Land Cover Classification graphical user interface

Menu Items

File

The File menu contains file utilities necessary for manipulation of the XML files:

- **New** - Clears the Editor for a new file or for a clean start.
- **Open** - Opens an existing XML file.
- **Open Recent** - Allows for quick navigation to the five most recent XML files viewed.
- **Restore AutoSave** - Restores an auto-saved version of the current work. AutoSave occurs every five seconds; select to restore the last saved version of the file.

- **Save** - Saves the existing work in the LCC editor to a new XML file if none exists, or to the opened file name.
- **Save As** - Saves the existing work in the LCC editor with a new file name.
- **Quit** - Closes the Editor

View

The View menu allows the user to turn on or off the three dockable windows on the left side of the tool: Values, Coefficients, and Metadata.

- **Show/Hide Values Window Dock** - Toggles the Values Window on or off.
- **Show/Hide Coefficients Window Dock** - Toggles the Coefficients Window on or off.
- **Show/Hide Metadata Window Dock** - Toggles the Metadata Window on or off.

The Icon Toolbar provides the same functionality.

The windows may also be viewed as tabs by clicking on the title bar of a window, holding the mouse button and dragging the window onto another window, then releasing the mouse button. To untab, click the title bar of the window and drag to an open location (the other windows will “move” out of the way).

Double-clicking on the title bar of a window will pop it out from the editor.

Import

Reserved for future use.

Help

The Help menu provides access to the ATtILA help document and provides information about the editor.

- **LCCEditor Help** - Opens the ATtILA help document.
- **About LCCEditor** - Provides system information about the editor.

Icon Toolbar



Figure 32. Land Cover Classification editor toolbar

The Icon Toolbar provides the same functionality as View.

-  **Show/Hide Values** - Toggles the VALUES Window on or off.
-  **Show/Hide Metadata** - Toggles the METADATA Window on or off.
-  **Show/Hide Coefficients** - Toggles the COEFFICIENTS Window on or off.

Entering and Editing Data

Information is entered into the VALUES Window, the METADATA Window and the CLASSES Window. In this version of ATtILA, the COEFFICIENTS Window is uneditable. Future versions will allow for editing within the COEFFICIENTS Window.

The general workflow is as follows with additional detail in the specific table sections:

1. Raster values, their descriptions, and their properties (exclusion and coefficient values) are entered in the Values table.
2. Classes are created in the Classes tree. Classes may be comprised of a single land cover type, multiple land cover types grouped together, or groups of groups.
3. Items from the Values table are added to the appropriate categories(s) in the Classes tree.
4. Metadata containing the name and description of the Land Cover Classification schema are added to the Metadata section.

Editor Values Window

The Values Window consists of three buttons and a table:

1.  **Add Value** - Enter new values into the table
2.  **Remove Value** - Remove values from the table
3.  **Deselect Exclusion** - Remove all exclusions set

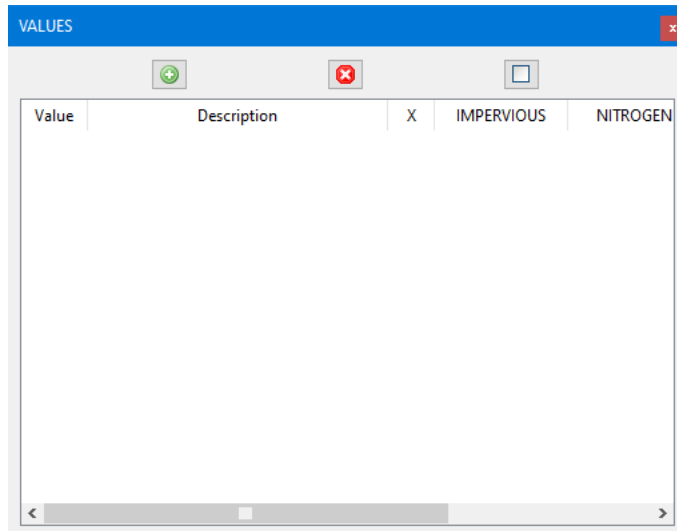


Figure 33. LCC Editor VALUES window with editing buttons

Add Value

Values are entered into the Values Table by pressing the Add Value button. Here, the user may then enter the following:

- **Value** (required) - The integer value of the land cover class in the Land cover grid.
- **Description** (required) - A text description of the land cover value.
- **X** - A checkbox that determines whether the land cover value will be excluded from ATtILA metric calculations. This is frequently checked when the user elects to exclude open water and perennial snow/ice values out of calculations, but may be used on any value.
- **IMPERVIOUS** - Numerical coefficient that estimates the percent of impervious surfaces that is generally found within the land cover type. This coefficient is frequently used with developed land cover types. *NOTE: The coefficient should be entered as a decimal value rather than a percentage.*
- **NITROGEN** - Numerical coefficient that estimates the amount of nitrogen output (stream loading) in kilograms per hectare per year that is generally found within the land cover type. This coefficient is frequently used with developed and agricultural land cover types.
- **PHOSPHORUS** - Numerical coefficient that estimates the amount of phosphorus output (stream loading) in kilograms per hectare per year that is generally found within the land cover type. This coefficient is frequently used with developed and agricultural land cover types.

The Add Row button may be clicked to add one or more additional values, if needed. Once the desired number of values have been added, they should be reviewed for accuracy. Edits may be made directly in the fields, or entire values may be deleted by clicking on the value to be deleted, then clicking the Remove Selected Row button. Once satisfied with the entries, click the OK button to add the new values to the Values table.

***NOTE:** Once in the Values table, the Value cannot be modified. However, the Description, e[X]clusion or Coefficients may be modified from the table.*

*Value	*Description	X	IMPERVIOUS	NITROGEN
		☐	0.0	NITROGEN

* This is a required field.

Buttons: Add Row, Remove Selected Row, OK, Cancel

Figure 34. LCC Editor Add Value Dialog

Remove Value

To delete a single value from the Values table, click on the value to be deleted in the table, then click the Remove Value button. To remove multiple values, click the Remove Value button *without* selecting a value. The Remove Value dialog will appear. Select multiple values by holding the Shift key while selecting the values with the mouse, and then click OK to delete.

Current List of Values

- 11
- 12
- 21
- 22
- 23
- 24

Buttons: OK, Cancel

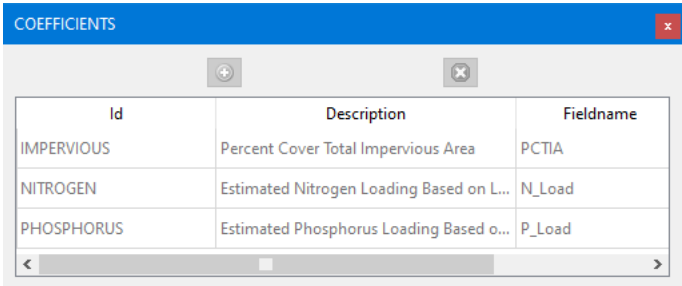
Figure 35. LCC Editor Remove Value Dialog

Deselect Exclusion

To remove all exclusions set on the values in the Values table, click the Deselect Exclusions button. To reset exclusions, simply click the X checkbox for each value to be excluded.

Editor Coefficients Window

The Coefficients Window lists the three coefficients that are utilized in this version of ATtILA. These are not editable. Future versions of ATtILA will provide users the opportunity to add coefficients.

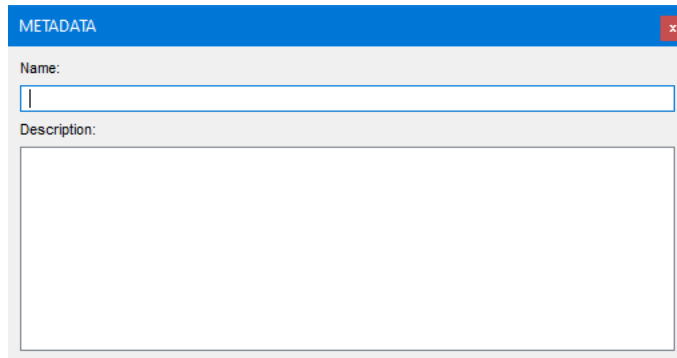


Id	Description	Fieldname
IMPERVIOUS	Percent Cover Total Impervious Area	PCTIA
NITROGEN	Estimated Nitrogen Loading Based on L...	N_Load
PHOSPHORUS	Estimated Phosphorus Loading Based o...	P_Load

Figure 36. LCC Editor COEFFICIENTS window

Editor Metadata Window

The Land Cover Classification name and description are entered in these fields. To minimize errors, it is recommended that the Name be consistent with the name of the LCC XML file.



The screenshot shows a window titled "METADATA" with a blue header bar. Below the header, there are two input fields. The first is labeled "Name:" and has a single-line text box. The second is labeled "Description:" and has a multi-line text box.

Figure 37. LCC Editor METADATA window

Editor Classes Window

The Classes Window provides an expandable and collapsible view of the classes, class groups, and the land cover values that belong to the classes. This window is used to start, edit, and view the Classes tree with the following buttons:

1.  **Start Tree/Add Sibling Class** - *Start a Classes tree or add a sibling class to the Classes tree*
2.  **Add Child Class** - *Add a child class to the Classes tree*
3.  **Edit Class** - *Edit a class*
4.  **Insert Values** - *Insert values into the Classes tree*
5.  **Remove Class/Value** - *Remove class or value from Classes tree*
6.  **Expand** - *Expand the entire Classes tree*
7.  **Collapse** - *Collapse the entire Classes tree*

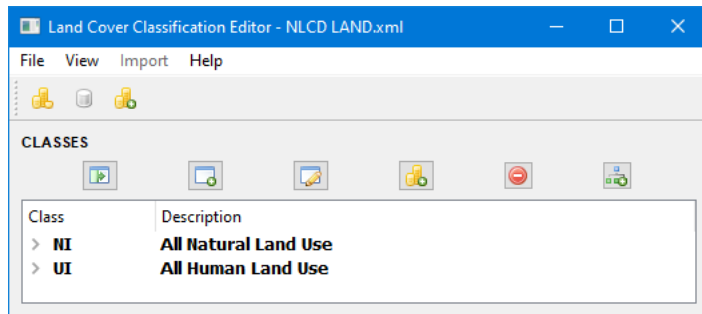


Figure 38. LCC Editor CLASSES window

Entering a Class

The first step to creating a land classification scheme is to start the Classes tree. Click the Start Tree/Add Sibling Class button. If there are no entries in the Classes Window, this will be the only button active. The Add Sibling Dialog opens. Here, the user may enter the following:

- **Class** (required) - The alphanumeric value to assign to the class. It is recommended that this be a recognizable short (three to five characters) abbreviation of the class description.
- **Description** (required) - A text description of the class.
- **caemField** - A text string used as the default output field name in the Core and Edge Metrics output table. If no attribute is provided, ATtILA will auto-generate an output field name using the **Class** entry as a base.
- **flcpField** - A text string used as the default output field name in the Floodplain Land Cover Proportions output table. If no attribute is provided, ATtILA will auto-generate an output field name using the **Class** entry as a base.
- **flcvField** - A text string used as the default output field name in the Facility Land Cover Views output table. If no attribute is provided, ATtILA will auto-generate an output field name using the **Class** entry as a base.
- **lcospField** - A text string used as the default output field name in the Land Cover on Slope Proportions output table. If no attribute is provided, ATtILA will auto-generate an output field name using the **Class** entry as a base.

- **lcpField** - A text string used as the default output field name in the Land Cover Proportions output table. If no attribute is provided, ATtILA will auto-generate an output field name using the **Class** entry as a base.
- **rlcpField** - A text string used as the default output field name in the Riparian Land Cover Proportions output table. If no attribute is provided, ATtILA will auto-generate an output field name using the **Class** entry as a base.
- **splcpField** - A text string used as the default output field name in the Sample Point Land Cover Proportions output table. If no attribute is provided, ATtILA will auto-generate an output field name using the **Class** entry as a base.

Once a class is entered into the Classes tree, a new class may be added by selecting a class and clicking either the Start Tree/Add Sibling Class or Add Child Class buttons. A “sibling” is a class at the same level as the selected class, while a “child” is a subclass that has membership within the upper-level class. The fields for entry are the same as described above. Once the entry is complete, click OK to accept the class. The class will now be visible in the Classes tree.

Figure 39. LCC Editor Add Sibling Class Dialog

Figure 40. LCC Editor Add Child Class Dialog

Editing a Class

To edit a class, click the Edit Class button. The Edit Class Dialog pop-up will appear, allowing modifications in each of the fields. Press OK to accept the modifications.

Figure 41. LCC Editor Edit Class Dialog

Inserting Values into the Classes Tree

The Classes tree is populated with items from the Values table in one of three ways:

- Drag a value from the Values table - Click and hold the mouse cursor on the desired value and drag it to the appropriate class in the Classes Table.
- Insert a value using the Insert Values button - Select the class in the Classes Table in which the value is to be inserted. Click Insert Values, which will display the Add Value to Class Tree Dialog. Select the value or values to be inserted and click OK.
- Right click on the class and select Insert Value - Right-click on the class in which the value is to be inserted. Click Insert Values, which will display the Add Value to Class Tree Dialog. Select the value or values to be inserted and click OK.

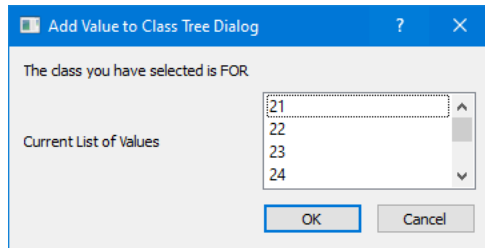


Figure 42. LCC Editor Add Value to Class Dialog

Removing a Class or a Value from the Classes Tree

Classes or values may be removed from the classes tree in one of two ways:

- Remove Class/Value button - Select the class or value to be removed and click the Remove Class/Value button.
- Right click on the class and select Remove - Right-click on the class or value to be removed and click Remove.

CAUTION: This action will erase all child classes and values in the tree below the selected class. This action cannot be undone.

NOTE: An alternative way to exclude classes from ATiLA metric calculations is to utilize the exclusion checkboxes in the Values table. This removes all occurrences of the selected values from the Classes tree and therefore should be used with caution. The values may be restored in the Classes tree by unchecking the checkboxes.

Expanding or Collapsing the Class Tree

To expand or collapse the Classes tree, click the Expand/Collapse Tree button. This will expand or collapse the entire tree. To expand or collapse an individual group, click the plus or minus sign next to the class name or double-click the class description.

1.  **Expand** - Expand all classes in the Classes tree
2.  **Collapse** - Collapse all classes in the Classes tree

Editor Shortcut Keys

Action	Ctrl	Function	Alt
Menu Items:			
New	Ctrl+N		
Open	Ctrl+O		
Restore AutoSave	Ctrl+R		
Save	Ctrl+S		
SaveAs		F12	
Quit	Ctrl+Q		
Help		F1	
Buttons:			
ValueTable			
Add Value	Ctrl+A		
Remove Values(s)	Ctrl+D		
Remove Exclusion	Ctrl+I		
Class Tree			
Start Tree/Add Sibling			Alt+S
Add Child			Alt+C
Edit Class			Alt+E
Insert Value			Alt+I
Remove Value			Alt+D
Expend/Collapse Tree			Alt+T

Supplied Land Cover Classification Schemas

An attempt has been made to standardize the land cover classification compiled metric fields among the C-CAP, MULC, NLCD (1992 Beta), NLCD, and WorldCover data sets. Slightly different land cover classification schemes were used for each of these, although in some cases the only real difference is in the nomenclature. For example, NLCD (1992 Beta) describes three urban classes as low density residential, high density residential, and commercial/industrial. Later NLCD outputs and C-CAP describe the three urban classes as low intensity developed, medium intensity developed, and high intensity developed. In the standard land cover classifications provided here, the latter terminology was adopted, primarily to aid the user in making comparisons among these data sets. The Developed metric for each consists of these three urban descriptions plus urban grasslands.

In a few cases, it was not possible to complete standardization. The NLCD (1992 Beta) data contains separate descriptions of natural and man-made barren lands, while the later NLCD and C-CAP data do not make this distinction. Therefore, the user must be cautious in making comparisons or drawing conclusions based on the barren metric.

Supplied schemes may be designated as either “ALL” or “LAND” (e.g. NLCD ALL vs. NLCD LAND). Schemes designated as “ALL” include all land cover classes in reporting unit area calculations, while those designated as “LAND” include only terrestrial land cover classes, with non-terrestrial land cover classes such as water and snow/ice excluded.

When a classification scheme with excluded land cover classes is selected, the areas of the excluded classes are disregarded in metric calculations. This means, when selecting a “LAND” classification scheme, the tool will process individual land cover classes and calculate metrics based on the total terrestrial area they occupy within the reporting unit, rather than the percent of the total area within the reporting unit. Thus, when using a “LAND” classification scheme, a calculated output value of 70% forest in say the Land Cover Proportions tool means that 70% of the land area in the reporting unit is covered by forest.

Five sets of land cover classification schemas supplied with ATtILA:

- **C-CAP ALL**
- **C-CAP LAND**

- **MULC Blended Levels ALL**
- **MULC Blended Levels LAND**

- **NLCD (1992 Beta) ALL**
- **NLCD (1992 Beta) LAND**

- **NLCD ALL**
- **NLCD LAND**

- **WorldCover 2020 ALL**
- **WorldCover 2020 LAND**

They are stored in the ToolboxSource > LandCoverClassifications folder in the ATtILA toolbox destination folder (see [Installing ATtILA](#)).

LCC C CAP ALL

Coastal Change Analysis Program ALL

No raster values except those representing NODATA are excluded from the calculation of reporting unit effective area (i.e., effective area equals the total raster area within a reporting unit).

Coefficient values for impervious area calculations are based on those reported in Caraco et al. (1998). Those supplied for the nitrogen loading and phosphorous loading estimates were obtained from Reckhow et al. (1980). As coefficient values can vary considerably for a given land cover type across locations, all provided values should be evaluated critically before use and altered when necessary.

For more information about the C-CAP dataset go to:
<https://coast.noaa.gov/digitalcoast/data/ccapregional.html>

C-CAP ALL - Value Table

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class(
0	Background	yes				
1	Unclassified					UC
2	High Intensity Developed		5.5	1.2	0.9	UI, dev, devh
3	Medium Intensity Developed		5.5	1.2	0.6	UI, dev, devm
4	Low Intensity Developed		5.5	1.2	0.4	UI, dev, devl
5	Open Spaces Developed				0.1	UI, dev, devo
6	Cultivated Crop		5.0	0.9		UI, agr, agrc
7	Pasture/Hay		5.0	0.9		UI, agr, agrp
8	Grassland		0.3	0.06	0.02	NI, hrb
9	Deciduous Forest		2.5	0.25	0.02	NI, for
10	Evergreen Forest		2.5	0.25	0.02	NI, for
11	Mixed Forest		2.5	0.25	0.02	NI, for
12	Scrub/Shrub		0.4	0.04	0.02	NI, shb
13	Palustrine Forested Wetland		2.5	0.25	0.02	NI, wtl, wtlw

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class(es)
14	Palustrine Scrub/Shrub Wetland		0.4	0.04	0.02	NI, wtl, wtlw
15	Palustrine Emergent Wetland		0.3	0.06	0.02	NI, wtl, wtle
16	Estuarine Forested Wetland		2.5	0.25	0.02	NI, wtl, wtlw
17	Estuarine Scrub/Shrub Wetland		0.4	0.04	0.02	NI, wtl, wtlw
18	Estuarine Emergent Wetland		0.3	0.06	0.02	NI, wtl, wtle
19	Unconsolidated Shore					NI, bar
20	Barren Land					NI, bar
21	Open Water					NI, wtr
22	Palustrine Aquatic Bed					NI, wtr
23	Estuarine Aquatic Bed					NI, wtr
24	Tundra		0.3	0.06	0.02	NI, tun
25	Perennial Ice/Snow					NI, wtr

C-CAP ALL - Class Table

Class Name	abv	Value
All Natural Land Use	NI	8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25
Herbaceous	hrb	8
Forest	for	9,10,11
Shrubland	shb	12
Wetland	wtl	13,14,15,16,17,18
Woody Wetland	wtlw	13,14,16,17

Class Name	abbv	Value
Emergent Wetland	wtle	15,18
Barren	bar	19,20
Water	wtr	21,22,23,25
Tundra	tun	24
All Human Land Use	UI	2,3,4,5,6,7
Developed	dev	2,3,4,5
High Intensity Developed	devh	2
Medium Intensity Developed	devm	3
Low Intensity Developed	devl	4
Open Space Developed	devo	5
Agriculture	agr	6,7
Cultivated Crop	agrc	6
Pasture/Hay	agrp	7
Unclassified	UC	1

C-CAP ALL XML

```
<?xml version='1.0' encoding='UTF-8'?>
<lccSchema xmlns="lcc" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="lcc
  LCCSchema.xsd">
  <metadata>
    <name>C-CAP ALL</name>
    <description>C-CAP ALL Land Cover Classification Schema. No raster values except those
      representing NODATA are excluded from the calculation of reporting unit effective area (i.e.,
      effective area equals the total raster area within a reporting unit). Coefficient values for
      impervious area calculations are based on those reported in Caraco et al. (1998). Those supplied
      for the nitrogen loading and phosphorous loading estimates were obtained from Reckhow et al.
      (1980). As coefficient values can vary considerably for a given land cover type across locations,
      all provided values should be evaluated critically before use and altered when necessary.
      Complete literature citations are provided in the ATtILA online help
      (https://github.com/USEPA/ATtILA2/wiki).</description>
  </metadata>
  <!--
    * The coefficients node contains coefficients to be assigned to values.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier
    * Name - text, word or phrase describing coefficient
    * fieldName - text, name of field to be created for output
    * - must conform to the field naming conventions dictated by the output database system
    * method - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
```

```

-->
<coefficients>
  <coefficient Id="IMPERVIOUS" Name="Percent Cover Total Impervious Area" fieldName="PCTIA"
    method="P" />
  <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover"
    fieldName="N_Load" method="A" />
  <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover"
    fieldName="P_Load" method="A" />
</coefficients>
<!--
  * The values node defines the full set of values that can exist in a land cover raster.

  * REQUIRED ATTRIBUTES
  * Id - integer, raster code
  *
  * OPTIONAL ATTRIBUTES
  * Name - text, word or phrase describing value
  * excluded - boolean, "true" or "false" or "1" or "0"
  *           - used to exclude values from effective area calculations
  *           - excluded=false is the default

  * A value element can optionally contain one or more coefficient elements

  * REQUIRED COEFFICIENT ATTRIBUTES
  * Id - text, must match an Id attribute from a coefficients node element
  * value - decimal, weighting/calculation factor
-->
<values>
  <value Id="0" Name="Background" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="1" Name="Unclassified">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="2" Name="High Intensity Developed">
    <coefficient Id="IMPERVIOUS" value="0.9" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
  </value>
  <value Id="3" Name="Medium Intensity Developed">
    <coefficient Id="IMPERVIOUS" value="0.6" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
  </value>
  <value Id="4" Name="Low Intensity Developed">
    <coefficient Id="IMPERVIOUS" value="0.4" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
  </value>
  <value Id="5" Name="Open Space Developed">
    <coefficient Id="IMPERVIOUS" value="0.1" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="6" Name="Cultivated Crop">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="5.0" />
    <coefficient Id="PHOSPHORUS" value="0.9" />
  </value>
  <value Id="7" Name="Pasture/Hay">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="5.0" />
    <coefficient Id="PHOSPHORUS" value="0.9" />
  </value>

```

```
</value>
<value Id="8" Name="Grassland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="9" Name="Deciduous Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="10" Name="Evergreen Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="11" Name="Mixed Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="12" Name="Scrub/Shrub">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.4" />
  <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="13" Name="Palustrine Forested Wetland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="14" Name="Palustrine Scrub/Shrub Wetland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.4" />
  <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="15" Name="Palustrine Emergent Wetland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="16" Name="Estuarine Forested Wetland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="17" Name="Estuarine Scrub/Shrub Wetland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.4" />
  <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="18" Name="Estuarine Emergent Wetland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="19" Name="Unconsolidated Shore">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="20" Name="Barren Land">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
```



```

<value Id="21" Name="Open Water">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="22" Name="Palustrine Aquatic Bed">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="23" Name="Estuarine Aquatic Bed">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="24" Name="Tundra">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="25" Name="Perennial Ice/Snow">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
</values>
<!--
  * The classes node contains values from a Land cover raster grouped into one or more classes.

  * REQUIRED ATTRIBUTES
  * Id - text, unique identifier, also used for automated generation of output field name
  *   - must conform to the field naming conventions dictated by the output database system

  * OPTIONAL ATTRIBUTES
  * Name - text, word or phrase describing class
  * filter - text, a string of one or more tool name abbreviations separated by a ";"
  *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp
  *   - used to exclude the class from the selectable classes in the tool's GUI
  * xxxxField - text, overrides ATtILA-generated field name for output
  *   - where xxxx equals a tool name abbreviation
  *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
  *   - a separate xxxxField attribute can exist for each tool
  *   - must conform to the field naming conventions dictated by the output database system

  * A class can contain either values or classes but not both types.
  * Value elements contain only an Id attribute which refers to a value in a raster.
  * Values tagged as excluded="true" in the values node should not be included in any class.
-->
<classes>
  <class Id="NI" Name="All Natural Land Use" filter="" lcpField="NINDEX">
    <class Id="hrb" Name="Herbaceous" filter="">
      <value Id="8" />
    </class>
    <class Id="for" Name="Forest" filter="">
      <value Id="9" />
      <value Id="10" />
      <value Id="11" />
    </class>
    <class Id="shb" Name="Shrubland" filter="">
      <value Id="12" />
    </class>
    <class Id="wtl" Name="Wetland" filter="">
      <class Id="wtlw" Name="Woody Wetland" filter="">
        <value Id="13" />
        <value Id="14" />
        <value Id="16" />
      </class>
    </class>
  </class>

```

```

        <value Id="17" />
    </class>
    <class Id="wtle" Name="Emergent Wetland" filter="">
        <value Id="15" />
        <value Id="18" />
    </class>
</class>
<class Id="bar" Name="Barren" filter="">
    <value Id="19" />
    <value Id="20" />
</class>
<class Id="wtr" Name="Water" filter="">
    <value Id="21" />
    <value Id="22" />
    <value Id="23" />
    <value Id="25" />
</class>
<class Id="tun" Name="Tundra" filter="">
    <value Id="24" />
</class>
</class>
<class Id="UI" Name="All Human Land Use" filter="" lcpField="UINDEX">
    <class Id="dev" Name="Developed" filter="">
        <class Id="devh" Name="High Intensity Developed" filter="">
            <value Id="2" />
        </class>
        <class Id="devm" Name="Medium Intensity Developed" filter="">
            <value Id="3" />
        </class>
        <class Id="devl" Name="Low Intensity Developed" filter="">
            <value Id="4" />
        </class>
        <class Id="devo" Name="Open Space Developed" filter="">
            <value Id="5" />
        </class>
    </class>
    <class Id="agr" Name="Agriculture" filter="">
        <class Id="agrc" Name="Cultivated Crop" filter="">
            <value Id="6" />
        </class>
        <class Id="agrp" Name="Pasture/Hay" filter="">
            <value Id="7" />
        </class>
    </class>
</class>
<class Id="UC" Name="Unclassified" filter="">
    <value Id="1" />
</class>
</classes>
</lccSchema>

```

LCC C CAP LAND

Coastal Change Analysis Program LAND

Water-related raster values are tagged as EXCLUDED. Excluded values are ignored when calculating the effective area of a reporting unit. Percentage metrics are based on the effective area of a reporting unit, not the total area.

Coefficient values for impervious area calculations are based on those reported in Caraco et al. (1998). Those supplied for the nitrogen loading and phosphorous loading estimates were obtained from Reckhow et al. (1980). As coefficient values can vary considerably for a given land cover type across locations, all provided values should be evaluated critically before use and altered when necessary.

For more information about the C-CAP dataset go to:
<https://coast.noaa.gov/digitalcoast/data/ccapregional.html>

C-CAP LAND - Value Table

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class(c
0	Background	yes				
1	Unclassified	yes				
2	High Intensity Developed		5.5	1.2	0.9	UI, dev, devh
3	Medium Intensity Developed		5.5	1.2	0.6	UI, dev, devm
4	Low Intensity Developed		5.5	1.2	0.4	UI, dev, devl
5	Open Spaces Developed				0.1	UI, dev, devo
6	Cultivated Crop		5.0	0.9		UI, agr, agrc
7	Pasture/Hay		5.0	0.9		UI, agr, agrp
8	Grassland		0.3	0.06	0.02	NI, hrb
9	Deciduous Forest		2.5	0.25	0.02	NI, for
10	Evergreen Forest		2.5	0.25	0.02	NI, for
11	Mixed Forest		2.5	0.25	0.02	NI, for
12	Scrub/Shrub		0.4	0.04	0.02	NI, shb
13	Palustrine Forested Wetland		2.5	0.25	0.02	NI, wtl, wtlw

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class(es)
14	Palustrine Scrub/Shrub Wetland		0.4	0.04	0.02	NI, wtl, wtlw
15	Palustrine Emergent Wetland		0.3	0.06	0.02	NI, wtl, wtle
16	Estuarine Forested Wetland		2.5	0.25	0.02	NI, wtl, wtlw
17	Estuarine Scrub/Shrub Wetland		0.4	0.04	0.02	NI, wtl, wtlw
18	Estuarine Emergent Wetland		0.3	0.06	0.02	NI, wtl, wtle
19	Unconsolidated Shore					NI, bar
20	Barren Land					NI, bar
21	Open Water	yes				
22	Palustrine Aquatic Bed	yes				
23	Estuarine Aquatic Bed	yes				
24	Tundra		0.3	0.06	0.02	NI, tun
25	Perennial Ice/Snow	yes				

C-CAP LAND - Class Table

Class Name	abv	Value
All NIural Land Use	NI	8,9,10,11,12,13,14,15,16,17,18,19,20,24
Herbaceous	hrb	8
Forest	for	9,10,11
Shrubland	shb	12
Wetland	wtl	13,14,15,16,17,18
Woody Wetland	wtlw	13,14,16,17
Emergent Wetland	wtle	15,18
Barren	bar	19,20

Class Name	abbrev	Value
Tundra	tun	24
All Human Land Use	UI	2,3,4,5,6,7
Developed	dev	2,3,4,5
High Intensity Developed	devh	2
Medium Intensity Developed	devm	3
Low Intensity Developed	devl	4
Open Space Developed	devo	5
Agriculture	agr	6,7
Cultivated Crop	agrc	6
Pasture/Hay	agrp	7
Unclassified	UC	1

C-CAP LAND XML

```
<?xml version='1.0' encoding='UTF-8'?>
<lccSchema xmlns="lcc" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="lcc
  LCCSchema.xsd">
  <metadata>
    <name>C-CAP LAND</name>
    <description>C-CAP LAND Land Cover Classification Schema. Water-related raster values are tagged
      as EXCLUDED. Excluded values are ignored when calculating the effective area of a reporting unit.
      Percentage metrics are based on the effective area of a reporting unit, not the total area.
      Coefficient values for impervious area calculations are based on those reported in Caraco et al.
      (1998). Those supplied for the nitrogen loading and phosphorous loading estimates were obtained
      from Reckhow et al. (1980). As coefficient values can vary considerably for a given land cover
      type across locations, all provided values should be evaluated critically before use and altered
      when necessary. Complete literature citations are provided in the ATtILA online help
      (https://github.com/USEPA/ATtILA2/wiki).</description>
  </metadata>
  <!--
    * The coefficients node contains coefficients to be assigned to values.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier
    * Name - text, word or phrase describing coefficient
    * fieldName - text, name of field to be created for output
    *      - must conform to the field naming conventions dictated by the output database system
    * method - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
  -->
  <coefficients>
    <coefficient Id="IMPERVIOUS" Name="Percent Cover Total Impervious Area" fieldName="PCTIA"
      method="P" />
    <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover"
      fieldName="N_Load" method="A" />
    <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover"
      fieldName="P_Load" method="A" />
  </coefficients>
  <!--
    * The values node defines the full set of values that can exist in a land cover raster.

    * REQUIRED ATTRIBUTES
    * Id - integer, raster code
    *
```

** OPTIONAL ATTRIBUTES*

** Name - text, word or phrase describing value*

** excluded - boolean, "true" or "false" or "1" or "0"*

** - used to exclude values from effective area calculations*

** - excluded=false is the default*

** A value element can optionally contain one or more coefficient elements*

** REQUIRED COEFFICIENT ATTRIBUTES*

** Id - text, must match an Id attribute from a coefficients node element*

** value - decimal, weighting/calculation factor*

-->

```
<values>
  <value Id="0" Name="Background" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="1" Name="Unclassified" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="2" Name="High Intensity Developed">
    <coefficient Id="IMPERVIOUS" value="0.9" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
  </value>
  <value Id="3" Name="Medium Intensity Developed">
    <coefficient Id="IMPERVIOUS" value="0.6" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
  </value>
  <value Id="4" Name="Low Intensity Developed">
    <coefficient Id="IMPERVIOUS" value="0.4" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
  </value>
  <value Id="5" Name="Open Space Developed">
    <coefficient Id="IMPERVIOUS" value="0.1" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="6" Name="Cultivated Crop">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="5.0" />
    <coefficient Id="PHOSPHORUS" value="0.9" />
  </value>
  <value Id="7" Name="Pasture/Hay">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="5.0" />
    <coefficient Id="PHOSPHORUS" value="0.9" />
  </value>
  <value Id="8" Name="Grassland">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="0.3" />
    <coefficient Id="PHOSPHORUS" value="0.06" />
  </value>
  <value Id="9" Name="Deciduous Forest">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="2.5" />
    <coefficient Id="PHOSPHORUS" value="0.25" />
  </value>
  <value Id="10" Name="Evergreen Forest">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="2.5" />
```

```

    <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="11" Name="Mixed Forest">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="2.5" />
    <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="12" Name="Scrub/Shrub">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="0.4" />
    <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="13" Name="Palustrine Forested Wetland">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="2.5" />
    <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="14" Name="Palustrine Scrub/Shrub Wetland">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="0.4" />
    <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="15" Name="Palustrine Emergent Wetland">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="0.3" />
    <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="16" Name="Estuarine Forested Wetland">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="2.5" />
    <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="17" Name="Estuarine Scrub/Shrub Wetland">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="0.4" />
    <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="18" Name="Estuarine Emergent Wetland">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="0.3" />
    <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="19" Name="Unconsolidated Shore">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="20" Name="Barren Land">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="21" Name="Open Water" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="22" Name="Palustrine Aquatic Bed" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="23" Name="Estuarine Aquatic Bed" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />

```

```

</value>
<value Id="24" Name="Tundra">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="25" Name="Perennial Ice/Snow" excluded="true">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
</values>
<!--
  * The classes node contains values from a land cover raster grouped into one or more classes.

  * REQUIRED ATTRIBUTES
  * Id - text, unique identifier, also used for automated generation of output field name
  *     - must conform to the field naming conventions dictated by the output database system

  * OPTIONAL ATTRIBUTES
  * Name - text, word or phrase describing class
  * filter - text, a string of one or more tool name abbreviations separated by a ";"
  *         - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp
  *         - used to exclude the class from the selectable classes in the tool's GUI
  * xxxxField - text, overrides ATtILA-generated field name for output
  *             - where xxxx equals a tool name abbreviation
  *             - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
  *             - a separate xxxxField attribute can exist for each tool
  *             - must conform to the field naming conventions dictated by the output database system

  * A class can contain either values or classes but not both types.
  * Value elements contain only an Id attribute which refers to a value in a raster.
  * Values tagged as excluded="true" in the values node should not be included in any class.
-->
<classes>
  <class Id="NI" Name="All Natural Land Use" filter="" lcpField="NINDEX">
    <class Id="hrb" Name="Herbaceous" filter="">
      <value Id="8" />
    </class>
    <class Id="for" Name="Forest" filter="">
      <value Id="9" />
      <value Id="10" />
      <value Id="11" />
    </class>
    <class Id="shb" Name="Shrubland" filter="">
      <value Id="12" />
    </class>
    <class Id="wtl" Name="Wetland" filter="">
      <class Id="wtlw" Name="Woody Wetland" filter="">
        <value Id="13" />
        <value Id="14" />
        <value Id="16" />
        <value Id="17" />
      </class>
      <class Id="wtle" Name="Emergent Wetland" filter="">
        <value Id="15" />
        <value Id="18" />
      </class>
    </class>
    <class Id="bar" Name="Barren" filter="">
      <value Id="19" />
      <value Id="20" />
    </class>
    <class Id="tun" Name="Tundra" filter="">
      <value Id="24" />
    </class>
  </class>

```



```
</class>
<class Id="UI" Name="All Human Land Use" filter="" lcpField="UINDEX">
  <class Id="dev" Name="Developed" filter="">
    <class Id="devh" Name="High Intensity Developed" filter="">
      <value Id="2" />
    </class>
    <class Id="devm" Name="Medium Intensity Developed" filter="">
      <value Id="3" />
    </class>
    <class Id="devl" Name="Low Intensity Developed" filter="">
      <value Id="4" />
    </class>
    <class Id="devo" Name="Open Space Developed" filter="">
      <value Id="5" />
    </class>
  </class>
  <class Id="agr" Name="Agriculture" filter="">
    <class Id="agrc" Name="Cultivated Crop" filter="">
      <value Id="6" />
    </class>
    <class Id="agrp" Name="Pasture/Hay" filter="">
      <value Id="7" />
    </class>
  </class>
</class>
<class Id="UC" Name="Unclassified" filter="" />
</classes>
</lccSchema>
```

LCC MULC Blended Levels ALL

US EPA EnviroAtlas Meter-Scale Urban Land Cover ALL

No raster values except those representing NODATA are excluded from the calculation of reporting unit effective area (i.e., effective area equals the total raster area within a reporting unit).

No coefficient values for impervious area, nitrogen loading, or phosphorous loading calculations are provided.

For more information about the MULC Blended Levels dataset go to: <https://www.mdpi.com/2072-4292/12/12/1909>

For US EPA EnviroAtlas community specific MULC, you can access fact sheets from <https://enviroatlas.epa.gov/enviroatlas/DataFactSheets/pdf/Supplemental/Landcover.pdf>

MULC Blended Levels ALL - Value Table

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class(es)
10	Water					
20	Impervious Surface					imp
30	Soil and Barren					bar
40	Trees and Forest					green, veg, for, mfor
52	Shrubs					green, veg, shb
70	Grass and Herbaceous					green, veg, grs
80	Row Crops					green, ag, agrc
82	Orchards					green, ag, agor, veg, mfor
91	Woody Wetlands					green, veg, for, mfor, wet, wetw
92	Emergent Wetlands					green, veg, wet, wete

MULC Blended Levels ALL - Class Table

Class Name	abbv	Value
All Vegetative Areas	green	40,52,70,80,82,91,92
Agriculture	ag	80,82
Row Crops	agrc	80
Orchards	agor	82
Natural Vegetation	veg	40,52,70,82,91,92
Trees/Forest, and Woody Wetlands	for	40,91
Trees/Forest, Orchards, and Woody Wetlands	mfor	40,82,91
Grass and Herbaceous	grs	70
Shrub	shb	52
Wetlands	wet	91, 92
Woody Wetlands	wetw	91
Emergent Wetlands	wete	92
Soil and Barren	bar	30
Impervious Surface	imp	20
Water	wtr	10

MULC Blended Levels ALL XML

```
<?xml version='1.0' encoding='UTF-8'?>
<lccSchema xmlns="lcc" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="lcc
LCCSchema.xsd">
  <metadata>
    <name>EnviroAtlas Meter-Scale Urban Land Cover LAND</name>
    <description>The EnviroAtlas One Meter-scale Urban Land Cover (MULC) Data were generated
individually for each EnviroAtlas community. Source imagery varies by community. Land cover
classes mapped also vary by community and include the following: water, impervious surfaces, soil
and barren land, trees, shrub, grass and herbaceous, agriculture, orchards, woody wetlands, and
emergent wetlands. No raster values except those representing NODATA are excluded from the
calculation of reporting unit effective area (i.e., effective area equals the total raster area
within a reporting unit). Percentage metrics are based on the effective area of a reporting unit,
not necessarily its total area.</description>
  </metadata>
  <!--
    * The coefficients node contains coefficients to be assigned to values.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier
    * Name - text, word or phrase describing coefficient
    * fieldName - text, name of field to be created for output
    *      - must conform to the field naming conventions dictated by the output database system
    * method - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
  -->
  <coefficients>
    <coefficient Id="IMPERVIOUS" Name="Percent Cover Total Impervious Area" fieldName="PCTIA"
method="P" />
    <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover"
fieldName="N_Load" method="A" />
  </coefficients>
</lccSchema>
```

```

    <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover"
      fieldName="P_Load" method="A" />
  </coefficients>
<!--
  * The values node defines the full set of values that can exist in a land cover raster.

  * REQUIRED ATTRIBUTES
  * Id - integer, raster code
  *
  * OPTIONAL ATTRIBUTES
  * Name - text, word or phrase describing value
  * excluded - boolean, "true" or "false" or "1" or "0"
  *           - used to exclude values from effective area calculations
  *           - excluded=false is the default

  * A value element can optionally contain one or more coefficient elements

  * REQUIRED COEFFICIENT ATTRIBUTES
  * Id - text, must match an Id attribute from a coefficients node element
  * value - decimal, weighting/calculation factor
-->
<values>
  <value Id="10" Name="Water">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="20" Name="Impervious Surface">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="30" Name="Soil and Barren">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="40" Name="Trees and Forest">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="52" Name="Shrubs">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="70" Name="Grass and Herbaceous">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="80" Name="Row Crops">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="82" Name="Orchards">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="91" Name="Woody Wetlands">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>

```

```

</value>
<value Id="92" Name="Emergent Wetlands">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
</values>
<!--
  * The classes node contains values from a Land cover raster grouped into one or more classes.

  * REQUIRED ATTRIBUTES
  * Id - text, unique identifier, also used for automated generation of output field name
  *   - must conform to the field naming conventions dictated by the output database system

  * OPTIONAL ATTRIBUTES
  * Name - text, word or phrase describing class
  * filter - text, a string of one or more tool name abbreviations separated by a ";"
  *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp
  *   - used to exclude the class from the selectable classes in the tool's GUI
  * xxxxField - text, overrides ATtILA-generated field name for output
  *   - where xxxx equals a tool name abbreviation
  *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
  *   - a separate xxxxField attribute can exist for each tool
  *   - must conform to the field naming conventions dictated by the output database system

  * A class can contain either values or classes but not both types.
  * Value elements contain only an Id attribute which refers to a value in a raster.
  * Values tagged as excluded="true" in the values node should not be included in any class.
-->
<classes>
  <class Id="green" Name="All Vegetative Areas" filter="">
    <class Id="ag" Name="Agriculture" filter="">
      <class Id="agrc" Name="Row Crops" filter="">
        <value Id="80" />
      </class>
      <class Id="agor" Name="Orchards" filter="">
        <value Id="82" />
      </class>
    </class>
    <class Id="veg" Name="Natural Vegetation" filter="">
      <class Id="for" Name="Trees/Forest, and Woody Wetlands" filter="">
        <value Id="40" />
        <value Id="91" />
      </class>
      <class Id="mfor" Name="Trees/Forest, Orchards, and Woody Wetlands" filter="">
        <value Id="40" />
        <value Id="82" />
        <value Id="91" />
      </class>
      <class Id="grs" Name="Grass and Herbaceous" filter="">
        <value Id="70" />
      </class>
      <class Id="shb" Name="Shrub" filter="">
        <value Id="52" />
      </class>
      <class Id="wet" Name="Wetlands" filter="">
        <class Id="wetw" Name="Woody Wetlands" filter="">
          <value Id="91" />
        </class>
        <class Id="wete" Name="Emergent Wetlands" filter="">
          <value Id="92" />
        </class>
      </class>
    </class>
  </class>
  <class Id="bar" Name="Soil and Barren" filter="">

```

```
        <value Id="30" />
    </class>
    <class Id="imp" Name="Impervious Surface" filter="">
        <value Id="20" />
    </class>
    <class Id="wtr" Name="Water" filter="">
        <value Id="10" />
    </class>
</classes>
</lccSchema>
```

LCC MULC Blended Levels LAND

US EPA EnviroAtlas Meter-Scale Urban Land Cover LAND

Water-related raster values are tagged as EXCLUDED. Excluded values are ignored when calculating the effective area of a reporting unit. Percentage metrics are based on the effective area of a reporting unit, not the total area.

No coefficient values for impervious area, nitrogen loading, or phosphorous loading calculations are provided.

For more information about the MULC Blended Levels dataset go to: <https://www.mdpi.com/2072-4292/12/12/1909>

For US EPA EnviroAtlas community specific MULC, you can access fact sheets from <https://enviroatlas.epa.gov/enviroatlas/DataFactSheets/pdf/Supplemental/Landcover.pdf>

MULC Blended Levels LAND - Value Table

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class(es)
10	Water	yes				
20	Impervious Surface					imp
30	Soil and Barren					bar
40	Trees and Forest					green, veg, for, mfor
52	Shrubs					green, veg, shb
70	Grass and Herbaceous					green, veg, grs
80	Row Crops					green, ag, agrc
82	Orchards					green, ag, agor, veg, mfor
91	Woody Wetlands					green, veg, for, mfor, wet, wetw
92	Emergent Wetlands					green, veg, wet, wete

MULC Blended Levels LAND - Class Table

Class Name	abbv	Value
All Vegetative Areas	green	40,52,70,80,82,91,92
Agriculture	ag	80,82
Row Crops	agrc	80
Orchards	agor	82
Natural Vegetation	veg	40,52,70,82,91,92
Trees/Forest, and Woody Wetlands	for	40,91
Trees/Forest, Orchards, and Woody Wetlands	mfor	40,82,91
Grass and Herbaceous	grs	70
Shrub	shb	52
Wetlands	wet	91, 92
Woody Wetlands	wetw	91
Emergent Wetlands	wete	92
Soil and Barren	bar	30
Impervious Surface	imp	20

MULC Blended Levels LAND XML

```
<?xml version='1.0' encoding='UTF-8'?>
<lccSchema xmlns="lcc" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="lcc
  LCCSchema.xsd">
  <metadata>
    <name>EnviroAtlas Meter-Scale Urban Land Cover LAND</name>
    <description>The EnviroAtlas One Meter-scale Urban Land Cover (MULC) Data were generated
    individually for each EnviroAtlas community. Source imagery varies by community. Land cover
    classes mapped also vary by community and include the following: water, impervious surfaces, soil
    and barren land, trees, shrub, grass and herbaceous, agriculture, orchards, woody wetlands, and
    emergent wetlands. Water-related raster values are tagged as EXCLUDED. Excluded values are
    ignored when calculating the effective area of a reporting unit. Percentage metrics are based on
    the effective area of a reporting unit, not its total area.</description>
  </metadata>
  <!--
    * The coefficients node contains coefficients to be assigned to values.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier
    * Name - text, word or phrase describing coefficient
    * fieldName - text, name of field to be created for output
    *      - must conform to the field naming conventions dictated by the output database system
    * method - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
  -->
  <coefficients>
    <coefficient Id="IMPERVIOUS" Name="Percent Cover Total Impervious Area" fieldName="PCTIA"
      method="P" />
    <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover"
      fieldName="N_Load" method="A" />
    <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover"
      fieldName="P_Load" method="A" />
  </coefficients>
```



```

<!--
    * The values node defines the full set of values that can exist in a land cover raster.

    * REQUIRED ATTRIBUTES
    * Id - integer, raster code
    *
    * OPTIONAL ATTRIBUTES
    * Name - text, word or phrase describing value
    * excluded - boolean, "true" or "false" or "1" or "0"
    *
        - used to exclude values from effective area calculations
    *
        - excluded=false is the default

    * A value element can optionally contain one or more coefficient elements

    * REQUIRED COEFFICIENT ATTRIBUTES
    * Id - text, must match an Id attribute from a coefficients node element
    * value - decimal, weighting/calculation factor
-->
<values>
  <value Id="10" Name="Water" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="20" Name="Impervious Surface">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="30" Name="Soil and Barren">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="40" Name="Trees and Forest">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="52" Name="Shrubs">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="70" Name="Grass and Herbaceous">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="80" Name="Row Crops">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="82" Name="Orchards">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="91" Name="Woody Wetlands">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="92" Name="Emergent Wetlands">
    <coefficient Id="IMPERVIOUS" value="0.0" />

```

```

        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
</values>
<!--
    * The classes node contains values from a Land cover raster grouped into one or more classes.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier, also used for automated generation of output field name
    *     - must conform to the field naming conventions dictated by the output database system

    * OPTIONAL ATTRIBUTES
    * Name - text, word or phrase describing class
    * filter - text, a string of one or more tool name abbreviations separated by a ";"
    *         - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp
    *         - used to exclude the class from the selectable classes in the tool's GUI
    * xxxxField - text, overrides ATtILA-generated field name for output
    *             - where xxxx equals a tool name abbreviation
    *             - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
    *             - a separate xxxxField attribute can exist for each tool
    *             - must conform to the field naming conventions dictated by the output database system

    * A class can contain either values or classes but not both types.
    * Value elements contain only an Id attribute which refers to a value in a raster.
    * Values tagged as excluded="true" in the values node should not be included in any class.
-->
<classes>
    <class Id="green" Name="All Vegetative Areas" filter="">
        <class Id="ag" Name="Agriculture" filter="">
            <class Id="agrc" Name="Row Crops" filter="">
                <value Id="80" />
            </class>
            <class Id="agor" Name="Orchards" filter="">
                <value Id="82" />
            </class>
        </class>
        <class Id="veg" Name="Natural Vegetation" filter="">
            <class Id="for" Name="Trees/Forest, and Woody Wetlands" filter="">
                <value Id="40" />
                <value Id="91" />
            </class>
            <class Id="mfor" Name="Trees/Forest, Orchards, and Woody Wetlands" filter="">
                <value Id="40" />
                <value Id="82" />
                <value Id="91" />
            </class>
            <class Id="grs" Name="Grass and Herbaceous" filter="">
                <value Id="70" />
            </class>
            <class Id="shb" Name="Shrub" filter="">
                <value Id="52" />
            </class>
            <class Id="wet" Name="Wetlands" filter="">
                <class Id="wetw" Name="Woody Wetlands" filter="">
                    <value Id="91" />
                </class>
                <class Id="wete" Name="Emergent Wetlands" filter="">
                    <value Id="92" />
                </class>
            </class>
        </class>
    </class>
    <class Id="bar" Name="Soil and Barren" filter="">
        <value Id="30" />
    </class>
    <class Id="imp" Name="Impervious Surface" filter="">

```

```
        <value Id="20" />
    </class>
</classes>
</lccSchema>
```

LCC NLCD 1992 Beta ALL

National Land Cover Database (1992 Beta) ALL

No raster values except those representing NODATA are excluded from the calculation of reporting unit effective area (i.e., effective area equals the total raster area within a reporting unit).

Coefficient values for impervious area calculations are based on those reported in Caraco et al. (1998). Those supplied for the nitrogen loading and phosphorous loading estimates were obtained from Reckhow et al. (1980). As coefficient values can vary considerably for a given land cover type across locations, all provided values should be evaluated critically before use and altered when necessary.

Information for the 1992 version of the NLCD has been archived, but for general information about the National Land Cover Dataset go to: <http://www.mrlc.gov>

NLCD (1992 Beta) ALL - Value Table

Value	Description	Excluded	Nitrogen	Phosphorus
11	Open Water			
12	Perennial Ice/Snow			
21	Low Intensity Developed (Residential)		5.5	1.2
22	Medium Intensity Developed (Residential)		5.5	1.2
23	High Intensity Developed (Commercial/Industrial/Transportation)		5.5	1.2
31	Barren Land (Rock/Sand/Clay)			
32	Quarry/Strip Mine/Gravel Pit			
33	Transitional			
41	Deciduous Forest		2.5	0.25
42	Evergreen Forest		2.5	0.25
43	Mixed Forest		2.5	0.25
51	Shrubland		0.4	0.04
61	Orchard/Vineyard/Other Cultivated Crop		6.0	0.8
71	Grassland		0.3	0.06
81	Pasture/Hay		5.0	0.9
82	Row Crop		8.5	2.3
83	Small Grain		6.0	0.8

Value	Description	Excluded	Nitrogen	Phosphorus
84	Fallow			
85	Open Space Developed (Urban Grass/Recreational Grass)			
91	Woody Wetland			
92	Emergent Wetland			

NLCD (1992 Beta) ALL - Class Table

Class Name	abbv	Value
All Natural Land Use	NI	11,12,31,41,42,43,51,71,91,92
Water	wtr	11,12
Barren	bar	31
Forest	for	41,42,43
Shrubland	shb	51
Herbaceous	hrb	71
Wetland	wtl	91,92
Woody Wetland	wtlw	91
Emergent Wetland	wtle	92
All Human Land Use	UI	21,22,23,32,33,61,81,82,83,84,85
Developed	dev	21,22,23,85
Man-Made Barren	mmb	32,33
Agriculture	agr	61,81,82,83,84
Pasture/Hay	agrp	81
Cultivated Crop	agrc	61,82,83,84

NLCD (1992 Beta) ALL XML

```

<?xml version='1.0' encoding='UTF-8'?>
<lccSchema xmlns="lcc" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="lcc
  lccSchema.xsd">
  <metadata>
    <name>NLCD (1992 Beta) ALL</name>

```

```

<description>National Land Cover Database (1992 Beta) ALL Land Cover Classification Schema. Only
viable for the NLCD dataset produced in 1992. No raster values except those representing NODATA
are excluded from the calculation of reporting unit effective area (i.e., effective area equals
the total raster area within a reporting unit). Coefficient values for impervious area
calculations are based on those reported in Caraco et al. (1998). Those supplied for the nitrogen
loading and phosphorous loading estimates were obtained from Reckhow et al. (1980). As
coefficient values can vary considerably for a given land cover type across locations, all
provided values should be evaluated critically before use and altered when necessary. Complete
literature citations are provided in the ATTILA online help
(https://github.com/USEPA/ATTILA2/wiki).</description>
</metadata>
<!--
    * The coefficients node contains coefficients to be assigned to values.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier
    * Name - text, word or phrase describing coefficient
    * fieldName - text, name of field to be created for output
    *           - must conform to the field naming conventions dictated by the output database system
    * method - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
-->
<coefficients>
  <coefficient Id="IMPERVIOUS" Name="Percent Cover Total Impervious Area" fieldName="PCTIA"
    method="P" />
  <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover"
    fieldName="N_Load" method="A" />
  <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover"
    fieldName="P_Load" method="A" />
</coefficients>
<!--
    * The values node defines the full set of values that can exist in a land cover raster.

    * REQUIRED ATTRIBUTES
    * Id - integer, raster code
    *
    * OPTIONAL ATTRIBUTES
    * Name - text, word or phrase describing value
    * excluded - boolean, "true" or "false" or "1" or "0"
    *           - used to exclude values from effective area calculations
    *           - excluded=false is the default

    * A value element can optionally contain one or more coefficient elements

    * REQUIRED COEFFICIENT ATTRIBUTES
    * Id - text, must match an Id attribute from a coefficients node element
    * value - decimal, weighting/calculation factor
-->
<values>
  <value Id="11" Name="Open Water">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="12" Name="Perennial Ice/Snow">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="21" Name="Low Intensity Developed (Residential)">
    <coefficient Id="IMPERVIOUS" value="0.4" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
  </value>
  <value Id="22" Name="Medium Intensity Developed (Residential)">
    <coefficient Id="IMPERVIOUS" value="0.6" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
  </value>

```

```
<value Id="23" Name="High Intensity Developed (Commercial/Industrial/Transportation)">
  <coefficient Id="IMPERVIOUS" value="0.9" />
  <coefficient Id="NITROGEN" value="5.5" />
  <coefficient Id="PHOSPHORUS" value="1.2" />
</value>
<value Id="31" Name="Barren Land (Rock/Sand/Clay)">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="32" Name="Quarry/Strip Mine/Gravel Pit">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="33" Name="Transitional">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="41" Name="Deciduous Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="42" Name="Evergreen Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="43" Name="Mixed Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="51" Name="Shrubland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.4" />
  <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="61" Name="Orchard/Vineyard/Other Cultivated Crop">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="6.0" />
  <coefficient Id="PHOSPHORUS" value="0.8" />
</value>
<value Id="71" Name="Grassland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="81" Name="Pature/Hay">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="5.0" />
  <coefficient Id="PHOSPHORUS" value="0.9" />
</value>
<value Id="82" Name="Row Crop">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="8.5" />
  <coefficient Id="PHOSPHORUS" value="2.3" />
</value>
<value Id="83" Name="Small Grain">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="6.0" />
  <coefficient Id="PHOSPHORUS" value="0.8" />
</value>
<value Id="84" Name="Fallow">
```

```

        <coefficient Id="IMPERVIOUS" value="0.0" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="85" Name="Open Space Developed (Urban Grass/Recreational Grass)">
        <coefficient Id="IMPERVIOUS" value="0.1" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="91" Name="Woody Wetland">
        <coefficient Id="IMPERVIOUS" value="0.02" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="92" Name="Emergent Wetland">
        <coefficient Id="IMPERVIOUS" value="0.02" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
</values>
<!--
    * The classes node contains values from a Land cover raster grouped into one or more classes.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier, also used for automated generation of output field name
    *     - must conform to the field naming conventions dictated by the output database system

    * OPTIONAL ATTRIBUTES
    * Name - text, word or phrase describing class
    * filter - text, a string of one or more tool name abbreviations separated by a ";"
    *         - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp
    *         - used to exclude the class from the selectable classes in the tool's GUI
    * xxxxField - text, overrides ATtILA-generated field name for output
    *             - where xxxx equals a tool name abbreviation
    *             - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
    *             - a separate xxxxField attribute can exist for each tool
    *             - must conform to the field naming conventions dictated by the output database system

    * A class can contain either values or classes but not both types.
    * Value elements contain only an Id attribute which refers to a value in a raster.
    * Values tagged as excluded="true" in the values node should not be included in any class.
-->
<classes>
    <class Id="NI" Name="All Natural Land Use" filter="" lcpField="NINDEX">
        <class Id="wtr" Name="Water" filter="">
            <value Id="11" />
            <value Id="12" />
        </class>
        <class Id="bar" Name="Barren" filter="">
            <value Id="31" />
        </class>
        <class Id="for" Name="Forest" filter="">
            <value Id="41" />
            <value Id="42" />
            <value Id="43" />
        </class>
        <class Id="shb" Name="Shrubland" filter="">
            <value Id="51" />
        </class>
        <class Id="hrb" Name="Herbaceous" filter="">
            <value Id="71" />
        </class>
        <class Id="wtl" Name="Wetland" filter="">
            <class Id="wtlw" Name="Woody Wetland" filter="">
                <value Id="91" />
            </class>

```



```

        <class Id="wtle" Name="Emergent Wetland" filter="">
            <value Id="92" />
        </class>
    </class>
</class>
<class Id="UI" Name="All Human Land Use" filter="" lcpField="UINDEX">
    <class Id="dev" Name="Developed" filter="">
        <class Id="dev1" Name="Low Intensity Developed" filter="">
            <value Id="21" />
        </class>
        <class Id="devm" Name="Medium Intensity Developed" filter="">
            <value Id="22" />
        </class>
        <class Id="devh" Name="High Intensity Developed" filter="">
            <value Id="23" />
        </class>
        <class Id="devo" Name="Open Space Developed" filter="">
            <value Id="85" />
        </class>
    </class>
    <class Id="mmb" Name="Man-Made Barren" filter="">
        <value Id="32" />
        <value Id="33" />
    </class>
    <class Id="agr" Name="Agriculture" filter="">
        <class Id="agrp" Name="Pasture/Hay" filter="">
            <value Id="81" />
        </class>
        <class Id="agrc" Name="Cultivated Crop" filter="">
            <value Id="61" />
            <value Id="82" />
            <value Id="83" />
            <value Id="84" />
        </class>
    </class>
</class>
</classes>
</lccSchema>

```

LCC NLCD 1992 Beta LAND

National Land Cover Database (1992 Beta) LAND

Water-related raster values are tagged as EXCLUDED. Excluded values are ignored when calculating the effective area of a reporting unit. Percentage metrics are based on the effective area of a reporting unit, not the total area.

Coefficient values for impervious area calculations are based on those reported in Caraco et al. (1998). Those supplied for the nitrogen loading and phosphorous loading estimates were obtained from Reckhow et al. (1980). As coefficient values can vary considerably for a given land cover type across locations, all provided values should be evaluated critically before use and altered when necessary.

Information for the 1992 version of the NLCD has been archived, but for general information about the National Land Cover Dataset go to: <http://www.mrlc.gov>

NLCD (1992 Beta) LAND - Value Table

Value	Description	Excluded	Nitrogen	Phosphorus
11	Open Water	yes		
12	Perennial Ice/Snow	yes		
21	Low Intensity Developed (Residential)		5.5	1.2
22	Medium Intensity Developed (Residential)		5.5	1.2
23	High Intensity Developed (Commercial/Industrial/Transportation)		5.5	1.2
31	Barren Land (Rock/Sand/Clay)			
32	Quarry/Strip Mine/Gravel Pit			
33	Transitional			
41	Deciduous Forest		2.5	0.25
42	Evergreen Forest		2.5	0.25
43	Mixed Forest		2.5	0.25
51	Shrubland		0.4	0.04
61	Orchard/Vineyard/Other Cultivated Crop		6.0	0.8
71	Grassland		0.3	0.06
81	Pasture/Hay		5.0	0.9
82	Row Crop		8.5	2.3

Value	Description	Excluded	Nitrogen	Phosphorus
83	Small Grain		6.0	0.8
84	Fallow			
85	Open Space Developed (Urban Grass/Recreational Grass)			
91	Woody Wetland			
92	Emergent Wetland			

NLCD (1992 Beta) LAND - Class Table

Class Name	abbv	Value
All Natural Land Use	NI	31,41,42,43,51,71,91,92
Barren	bar	31
Forest	for	41,42,43
Shrubland	shb	51
Herbaceous	hrb	71
Wetland	wtl	91,92
Woody Wetland	wtlw	91
Emergent Wetland	wtle	92
All Human Land Use	UI	21,22,23,32,33,61,81,82,83,84,85
Developed	dev	21,22,23,85
Man-Made Barren	mmb	32,33
Agriculture	agr	61,81,82,83,84
Pasture/Hay	agrp	81
Cultivated Crop	agrc	61,82,83,84

NLCD (1992 Beta) LAND XML

```

<?xml version='1.0' encoding='UTF-8'?>
<lccSchema xmlns="lcc" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="lcc
  LCCSchema.xsd">
  <metadata>
    <name>NLCD (1992 Beta) LAND</name>

```

```

<description>National Land Cover Database 1992 LAND Land Cover Classification Schema. Only viable
for the NLCD dataset produced in 1992. Water-related raster values are tagged as EXCLUDED.
Excluded values are ignored when calculating the effective area of a reporting unit. Percentage
metrics are based on the effective area of a reporting unit, not the total area. Coefficient
values for impervious area calculations are based on those reported in Caraco et al. (1998).
Those supplied for the nitrogen loading and phosphorous loading estimates were obtained from
Reckhow et al. (1980). As coefficient values can vary considerably for a given land cover type
across locations, all provided values should be evaluated critically before use and altered when
necessary. Complete literature citations are provided in the ATtILA online help
(https://github.com/USEPA/ATtILA2/wiki).</description>
</metadata>
<!--
    * The coefficients node contains coefficients to be assigned to values.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier
    * Name - text, word or phrase describing coefficient
    * fieldName - text, name of field to be created for output
    *           - must conform to the field naming conventions dictated by the output database system
    * method - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
-->
<coefficients>
  <coefficient Id="IMPERVIOUS" Name="Percent Cover Total Impervious Area" fieldName="PCTIA"
    method="P" />
  <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover"
    fieldName="N_Load" method="A" />
  <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover"
    fieldName="P_Load" method="A" />
</coefficients>
<!--
    * The values node defines the full set of values that can exist in a Land cover raster.

    * REQUIRED ATTRIBUTES
    * Id - integer, raster code
    *
    * OPTIONAL ATTRIBUTES
    * Name - text, word or phrase describing value
    * excluded - boolean, "true" or "false" or "1" or "0"
    *           - used to exclude values from effective area calculations
    *           - excluded=false is the default

    * A value element can optionally contain one or more coefficient elements

    * REQUIRED COEFFICIENT ATTRIBUTES
    * Id - text, must match an Id attribute from a coefficients node element
    * value - decimal, weighting/calculation factor
-->
<values>
  <value Id="11" Name="Open Water" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="12" Name="Perennial Ice/Snow" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="21" Name="Low Intensity Developed (Residential)">
    <coefficient Id="IMPERVIOUS" value="0.4" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
  </value>
  <value Id="22" Name="Medium Intensity Developed (Residential)">
    <coefficient Id="IMPERVIOUS" value="0.6" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
  </value>

```

```
<value Id="23" Name="High Intensity Developed (Commercial/Industrial/Transportation)">
  <coefficient Id="IMPERVIOUS" value="0.9" />
  <coefficient Id="NITROGEN" value="5.5" />
  <coefficient Id="PHOSPHORUS" value="1.2" />
</value>
<value Id="31" Name="Barren Land (Rock/Sand/Clay)">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="32" Name="Quarry/Strip Mine/Gravel Pit">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="33" Name="Transitional">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="41" Name="Deciduous Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="42" Name="Evergreen Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="43" Name="Mixed Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="51" Name="Shrubland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.4" />
  <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="61" Name="Orchard/Vineyard/Other Cultivated Crop">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="6.0" />
  <coefficient Id="PHOSPHORUS" value="0.8" />
</value>
<value Id="71" Name="Grassland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="81" Name="Pature/Hay">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="5.0" />
  <coefficient Id="PHOSPHORUS" value="0.9" />
</value>
<value Id="82" Name="Row Crop">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="8.5" />
  <coefficient Id="PHOSPHORUS" value="2.3" />
</value>
<value Id="83" Name="Small Grain">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="6.0" />
  <coefficient Id="PHOSPHORUS" value="0.8" />
</value>
<value Id="84" Name="Fallow">
```

```

        <coefficient Id="IMPERVIOUS" value="0.0" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="85" Name="Open Space Developed (Urban Grass/Recreational Grass)">
        <coefficient Id="IMPERVIOUS" value="0.1" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="91" Name="Woody Wetland">
        <coefficient Id="IMPERVIOUS" value="0.02" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="92" Name="Emergent Wetland">
        <coefficient Id="IMPERVIOUS" value="0.02" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
</values>
<!--
    * The classes node contains values from a Land cover raster grouped into one or more classes.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier, also used for automated generation of output field name
    *   - must conform to the field naming conventions dictated by the output database system

    * OPTIONAL ATTRIBUTES
    * Name - text, word or phrase describing class
    * filter - text, a string of one or more tool name abbreviations separated by a ";"
    *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp
    *   - used to exclude the class from the selectable classes in the tool's GUI
    * xxxxField - text, overrides ATtILA-generated field name for output
    *   - where xxxx equals a tool name abbreviation
    *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
    *   - a separate xxxxField attribute can exist for each tool
    *   - must conform to the field naming conventions dictated by the output database system

    * A class can contain either values or classes but not both types.
    * Value elements contain only an Id attribute which refers to a value in a raster.
    * Values tagged as excluded="true" in the values node should not be included in any class.
-->
<classes>
    <class Id="NI" Name="All Natural Land Use" filter="" lcpField="NINDEX">
        <class Id="bar" Name="Barren" filter="">
            <value Id="31" />
        </class>
        <class Id="for" Name="Forest" filter="">
            <value Id="41" />
            <value Id="42" />
            <value Id="43" />
        </class>
        <class Id="shb" Name="Shrubland" filter="">
            <value Id="51" />
        </class>
        <class Id="hrb" Name="Herbaceous" filter="">
            <value Id="71" />
        </class>
        <class Id="wtl" Name="Wetland" filter="">
            <class Id="wtlw" Name="Woody Wetland" filter="">
                <value Id="91" />
            </class>
            <class Id="wtle" Name="Emergent Wetland" filter="">
                <value Id="92" />
            </class>
        </class>
    </class>

```

```

</class>
<class Id="UI" Name="All Human Land Use" filter="" lcpField="UINDEX">
  <class Id="dev" Name="Developed" filter="">
    <class Id="dev1" Name="Low Intensity Developed" filter="">
      <value Id="21" />
    </class>
    <class Id="devm" Name="Medium Intensity Developed" filter="">
      <value Id="22" />
    </class>
    <class Id="devh" Name="High Intensity Developed" filter="">
      <value Id="23" />
    </class>
    <class Id="devo" Name="Open Space Developed" filter="">
      <value Id="85" />
    </class>
  </class>
  <class Id="mmb" Name="Man-Made Barren" filter="">
    <value Id="32" />
    <value Id="33" />
  </class>
  <class Id="agr" Name="Agriculture" filter="">
    <class Id="agrp" Name="Pasture/Hay" filter="">
      <value Id="81" />
    </class>
    <class Id="agrc" Name="Cultivated Crop" filter="">
      <value Id="61" />
      <value Id="82" />
      <value Id="83" />
      <value Id="84" />
    </class>
  </class>
</class>
</classes>
</lccSchema>

```

LCC NLCD ALL

National Land Cover Database ALL

No raster values except those representing NODATA are excluded from the calculation of reporting unit effective area (i.e., effective area equals the total raster area within a reporting unit).

Coefficient values for impervious area calculations are based on those reported in Caraco et al. (1998). Those supplied for the nitrogen loading and phosphorous loading estimates were obtained from Reckhow et al. (1980). As coefficient values can vary considerably for a given land cover type across locations, all provided values should be evaluated critically before use and altered when necessary.

For more information about the National Land Cover Dataset go to: <http://www.mrlc.gov>

NLCD ALL - Value Table

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class
11	Open Water					NI
12	Perennial Ice/Snow					NI
21	Open Space Developed				0.1	UI, c devo
22	Low Intensity Developed		5.5	1.2	0.4	UI, c devl
23	Medium Intensity Developed		5.5	1.2	0.6	UI, c devn
24	High Intensity Developed		5.5	1.2	0.9	UI, c devh
31	Barren Land (Rock/Sand/Clay)					NI, b
41	Deciduous Forest		2.5	0.25	0.02	NI, f
42	Evergreen Forest		2.5	0.25	0.02	NI, f
43	Mixed Forest		2.5	0.25	0.02	NI, f
51	Dwarf Scrub		0.4	0.04	0.02	NI, t
52	Scrub/Shrub		0.4	0.04	0.02	NI, s
71	Grassland		0.3	0.06	0.02	NI, h
72	Sedge/Herbaceous		0.3	0.06	0.02	NI, t
73	Lichen		0.3	0.06	0.02	NI, t
74	Moss		0.3	0.06	0.02	NI, t
81	Pasture/Hay		5.0	0.9		UI, a agrp

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class
82	Cultivated Crop		5.0	0.9		UI, a agrc
90	Woody Wetland				0.02	NI, v wtlw
95	Emergent Herbaceous Wetland				0.02	NI, v wtle

NLCD ALL - Class Table

Class Name	abbv	Value
All Natural Land Use	NI	11,12,31,41,42,43,51,52,71,72,73,74,90,95
Water	wtr	11,12
Barren	bar	31
Forest	for	41,42,43
Tundra	tun	51,72,73,74
Shrubland	shb	52
Herbaceous	hrb	71
Wetland	wtl	90,95
Woody Wetland	wtlw	90
Emergent Wetland	wtle	95
All Human Land Use	UI	21,22,23,24,81,82
Developed	dev	21,22,23,24
Open Space Developed	devo	21
Low Intensity Developed	devl	22
Medium Intensity Developed	devm	23
High Intensity Developed	devh	24
Agriculture	agr	81,82
Pasture/Hay	agrp	81
Cultivated Crop	agrc	82

NLCD ALL XML

```
<?xml version='1.0' encoding='UTF-8'?>
<lccSchema xmlns="lcc" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="lcc
  LCCSchema.xsd">
  <metadata>
    <name>NLCD ALL</name>
    <description>National Land Cover Database ALL Land Cover Classification Schema. Viable for the
      NLCD datasets produced in 2001 or later. No raster values except those representing NODATA are
      excluded from the calculation of reporting unit effective area (i.e., effective area equals the
      total raster area within a reporting unit). Coefficient values for impervious area calculations
      are based on those reported in Caraco et al. (1998). Those supplied for the nitrogen loading and
      phosphorous loading estimates were obtained from Reckhow et al. (1980). As coefficient values can
      vary considerably for a given land cover type across locations, all provided values should be
      evaluated critically before use and altered when necessary. Complete literature citations are
      provided in the ATtILA online help (https://github.com/USEPA/ATtILA2/wiki).</description>
  </metadata>
  <!--
    * The coefficients node contains coefficients to be assigned to values.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier
    * Name - text, word or phrase describing coefficient
    * fieldName - text, name of field to be created for output
    *      - must conform to the field naming conventions dictated by the output database system
    * method - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
  -->
  <coefficients>
    <coefficient Id="IMPERVIOUS" Name="Percent Cover Total Impervious Area" fieldName="PCTIA"
      method="P" />
    <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover"
      fieldName="N_Load" method="A" />
    <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover"
      fieldName="P_Load" method="A" />
  </coefficients>
  <!--
    * The values node defines the full set of values that can exist in a land cover raster.

    * REQUIRED ATTRIBUTES
    * Id - integer, raster code
    *
    * OPTIONAL ATTRIBUTES
    * Name - text, word or phrase describing value
    * excluded - boolean, "true" or "false" or "1" or "0"
    *      - used to exclude values from effective area calculations
    *      - excluded=false is the default

    * A value element can optionally contain one or more coefficient elements

    * REQUIRED COEFFICIENT ATTRIBUTES
    * Id - text, must match an Id attribute from a coefficients node element
    * value - decimal, weighting/calculation factor
  -->
  <values>
    <value Id="0" Name="Unclassified" excluded="true">
      <coefficient Id="IMPERVIOUS" value="0.0" />
      <coefficient Id="NITROGEN" value="0.0" />
      <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="11" Name="Open Water">
      <coefficient Id="IMPERVIOUS" value="0.0" />
      <coefficient Id="NITROGEN" value="0.0" />
      <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="12" Name="Perennial Ice/Snow">
      <coefficient Id="IMPERVIOUS" value="0.0" />
      <coefficient Id="NITROGEN" value="0.0" />
```

```

    <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="21" Name="Open Space Developed">
    <coefficient Id="IMPERVIOUS" value="0.1" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="22" Name="Low Intensity Developed">
    <coefficient Id="IMPERVIOUS" value="0.4" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
</value>
<value Id="23" Name="Medium Intensity Developed">
    <coefficient Id="IMPERVIOUS" value="0.6" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
</value>
<value Id="24" Name="High Intensity Developed">
    <coefficient Id="IMPERVIOUS" value="0.9" />
    <coefficient Id="NITROGEN" value="5.5" />
    <coefficient Id="PHOSPHORUS" value="1.2" />
</value>
<value Id="31" Name="Barren Land (Rock/Sand/Clay)">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="41" Name="Deciduous Forest">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="2.5" />
    <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="42" Name="Evergreen Forest">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="2.5" />
    <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="43" Name="Mixed Forest">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="2.5" />
    <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="51" Name="Dwarf Scrub">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="0.4" />
    <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="52" Name="Scrub/Shrub">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="0.4" />
    <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="71" Name="Grassland">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="0.3" />
    <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="72" Name="Sedge/Herbaceous">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="0.3" />
    <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="73" Name="Lichen">
    <coefficient Id="IMPERVIOUS" value="0.02" />
    <coefficient Id="NITROGEN" value="0.3" />
    <coefficient Id="PHOSPHORUS" value="0.06" />

```

```

</value>
<value Id="74" Name="Moss">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="81" Name="Pasture/Hay">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="5.0" />
  <coefficient Id="PHOSPHORUS" value="0.9" />
</value>
<value Id="82" Name="Cultivated Crop">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="5.0" />
  <coefficient Id="PHOSPHORUS" value="0.9" />
</value>
<value Id="90" Name="Woody Wetland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="95" Name="Emergent Herbaceous Wetland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
</values>
<!--
  * The classes node contains values from a Land cover raster grouped into one or more classes.

  * REQUIRED ATTRIBUTES
  * Id - text, unique identifier, also used for automated generation of output field name
  *   - must conform to the field naming conventions dictated by the output database system

  * OPTIONAL ATTRIBUTES
  * Name - text, word or phrase describing class
  * filter - text, a string of one or more tool name abbreviations separated by a ";"
  *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp
  *   - used to exclude the class from the selectable classes in the tool's GUI
  * xxxxField - text, overrides ATtILA-generated field name for output
  *   - where xxxx equals a tool name abbreviation
  *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
  *   - a separate xxxxField attribute can exist for each tool
  *   - must conform to the field naming conventions dictated by the output database system

  * A class can contain either values or classes but not both types.
  * Value elements contain only an Id attribute which refers to a value in a raster.
  * Values tagged as excluded="true" in the values node should not be included in any class.
-->
<classes>
  <class Id="NI" Name="All Natural Land Use" filter="" lcpField="NINDEX">
    <class Id="wtr" Name="Water" filter="">
      <value Id="11" />
      <value Id="12" />
    </class>
    <class Id="bar" Name="Barren" filter="">
      <value Id="31" />
    </class>
    <class Id="for" Name="Forest" filter="">
      <value Id="41" />
      <value Id="42" />
      <value Id="43" />
    </class>
    <class Id="tun" Name="Tundra" filter="">
      <value Id="51" />
      <value Id="72" />

```

```

        <value Id="73" />
        <value Id="74" />
    </class>
    <class Id="shb" Name="Shrubland" filter="">
        <value Id="52" />
    </class>
    <class Id="hrb" Name="Herbaceous" filter="">
        <value Id="71" />
    </class>
    <class Id="wtl" Name="Wetland" filter="">
        <class Id="wtlw" Name="Woody Wetland" filter="">
            <value Id="90" />
        </class>
        <class Id="wtle" Name="Emergent Wetland" filter="">
            <value Id="95" />
        </class>
    </class>
</class>
<class Id="UI" Name="All Human Land Use" filter="" lcpField="UINDEX">
    <class Id="dev" Name="Developed" filter="">
        <class Id="devo" Name="Open Space Developed" filter="">
            <value Id="21" />
        </class>
        <class Id="devl" Name="Low Intensity Developed" filter="">
            <value Id="22" />
        </class>
        <class Id="devm" Name="Medium Intensity Developed" filter="">
            <value Id="23" />
        </class>
        <class Id="devh" Name="High Intensity Developed" filter="">
            <value Id="24" />
        </class>
    </class>
    <class Id="agr" Name="Agriculture" filter="">
        <class Id="agrp" Name="Pasture/Hay" filter="">
            <value Id="81" />
        </class>
        <class Id="agrc" Name="Cultivated Crop" filter="">
            <value Id="82" />
        </class>
    </class>
</class>
</classes>
</lccSchema>

```

LCC NLCD LAND

National Land Cover Database LAND

Water-related raster values are tagged as EXCLUDED. Excluded values are ignored when calculating the effective area of a reporting unit. Percentage metrics are based on the effective area of a reporting unit, not the total area.

Coefficient values for impervious area calculations are based on those reported in Caraco et al. (1998). Those supplied for the nitrogen loading and phosphorous loading estimates were obtained from Reckhow et al. (1980). As coefficient values can vary considerably for a given land cover type across locations, all provided values should be evaluated critically before use and altered when necessary.

For more information about the National Land Cover Dataset go to: <http://www.mrlc.gov>

NLCD LAND - Value Table

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class
11	Open Water	yes				
12	Perennial Ice/Snow	yes				
21	Open Space Developed				0.1	UI, c devo
22	Low Intensity Developed		5.5	1.2	0.4	UI, c devl
23	Medium Intensity Developed		5.5	1.2	0.6	UI, c devn
24	High Intensity Developed		5.5	1.2	0.9	UI, c devh
31	Barren Land (Rock/Sand/Clay)					NI, b
41	Deciduous Forest		2.5	0.25	0.02	NI, f
42	Evergreen Forest		2.5	0.25	0.02	NI, f
43	Mixed Forest		2.5	0.25	0.02	NI, f
51	Dwarf Scrub		0.4	0.04	0.02	NI, t
52	Scrub/Shrub		0.4	0.04	0.02	NI, s
71	Grassland		0.3	0.06	0.02	NI, h
72	Sedge/Herbaceous		0.3	0.06	0.02	NI, t
73	Lichen		0.3	0.06	0.02	NI, t
74	Moss		0.3	0.06	0.02	NI, t
81	Pasture/Hay		5.0	0.9		UI, a agrp

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class
82	Cultivated Crop		5.0	0.9		UI, a agrc
90	Woody Wetland				0.02	NI, v wtlw
95	Emergent Herbaceous Wetland				0.02	NI, v wtle

NLCD LAND - Class Table

Class Name	abbv	Value
All Natural Land Use	NI	31,41,42,43,51,52,71,72,73,74,90,95
Barren	bar	31
Forest	for	41,42,43
Tundra	tun	51,72,73,74
Shrubland	shb	52
Herbaceous	hrb	71
Wetland	wtl	90,95
Woody Wetland	wtlw	90
Emergent Wetland	wtle	95
All Human Land Use	UI	21,22,23,24,81,82
Developed	dev	21,22,23,24
Open Space Developed	devo	21
Low Intensity Developed	devl	22
Medium Intensity Developed	devm	23
High Intensity Developed	devh	24
Agriculture	agr	81,82
Pasture/Hay	agrp	81
Cultivated Crop	agrc	82

NLCD LAND XML

```

<?xml version='1.0' encoding='UTF-8'?>
<lccSchema xmlns="lcc" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="lcc
  LCCSchema.xsd">
  <metadata>
    <name>NLCD LAND</name>

```

```

<description>National Land Cover Database LAND Land Cover Classification Schema. Viable for the
NLCD datasets produced in 2001 or later. Water-related raster values are tagged as EXCLUDED.
Excluded values are ignored when calculating the effective area of a reporting unit. Percentage
metrics are based on the effective area of a reporting unit, not the total area. Coefficient
values for impervious area calculations are based on those reported in Caraco et al. (1998).
Those supplied for the nitrogen loading and phosphorous loading estimates were obtained from
Reckhow et al. (1980). As coefficient values can vary considerably for a given land cover type
across locations, all provided values should be evaluated critically before use and altered when
necessary. Complete literature citations are provided in the ATtILA online help
(https://github.com/USEPA/ATtILA2/wiki).</description>
</metadata>
<!--
    * The coefficients node contains coefficients to be assigned to values.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier
    * Name - text, word or phrase describing coefficient
    * fieldName - text, name of field to be created for output
    *           - must conform to the field naming conventions dictated by the output database system
    * method - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
-->
<coefficients>
  <coefficient Id="IMPERVIOUS" Name="Percent Cover Total Impervious Area" fieldName="PCTIA"
    method="P" />
  <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover"
    fieldName="N_Load" method="A" />
  <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover"
    fieldName="P_Load" method="A" />
</coefficients>
<!--
    * The values node defines the full set of values that can exist in a land cover raster.

    * REQUIRED ATTRIBUTES
    * Id - integer, raster code
    *
    * OPTIONAL ATTRIBUTES
    * Name - text, word or phrase describing value
    * excluded - boolean, "true" or "false" or "1" or "0"
    *           - used to exclude values from effective area calculations
    *           - excluded=false is the default

    * A value element can optionally contain one or more coefficient elements

    * REQUIRED COEFFICIENT ATTRIBUTES
    * Id - text, must match an Id attribute from a coefficients node element
    * value - decimal, weighting/calculation factor
-->
<values>
  <value Id="0" Name="Unclassified" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="11" Name="Open Water" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="12" Name="Perennial Ice/Snow" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="21" Name="Open Space Developed">
    <coefficient Id="IMPERVIOUS" value="0.1" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>

```



```
<value Id="22" Name="Low Intensity Developed">
  <coefficient Id="IMPERVIOUS" value="0.4" />
  <coefficient Id="NITROGEN" value="5.5" />
  <coefficient Id="PHOSPHORUS" value="1.2" />
</value>
<value Id="23" Name="Medium Intensity Developed">
  <coefficient Id="IMPERVIOUS" value="0.6" />
  <coefficient Id="NITROGEN" value="5.5" />
  <coefficient Id="PHOSPHORUS" value="1.2" />
</value>
<value Id="24" Name="High Intensity Developed">
  <coefficient Id="IMPERVIOUS" value="0.9" />
  <coefficient Id="NITROGEN" value="5.5" />
  <coefficient Id="PHOSPHORUS" value="1.2" />
</value>
<value Id="31" Name="Barren Land (Rock/Sand/Clay)">
  <coefficient Id="IMPERVIOUS" value="0.0" />
  <coefficient Id="NITROGEN" value="0.0" />
  <coefficient Id="PHOSPHORUS" value="0.0" />
</value>
<value Id="41" Name="Deciduous Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="42" Name="Evergreen Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="43" Name="Mixed Forest">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="2.5" />
  <coefficient Id="PHOSPHORUS" value="0.25" />
</value>
<value Id="51" Name="Dwarf Scrub">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.4" />
  <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="52" Name="Scrub/Shrub">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.4" />
  <coefficient Id="PHOSPHORUS" value="0.04" />
</value>
<value Id="71" Name="Grassland">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="72" Name="Sedge/Herbaceous">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="73" Name="Lichen">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="74" Name="Moss">
  <coefficient Id="IMPERVIOUS" value="0.02" />
  <coefficient Id="NITROGEN" value="0.3" />
  <coefficient Id="PHOSPHORUS" value="0.06" />
</value>
<value Id="81" Name="Pasture/Hay">
```

```

        <coefficient Id="IMPERVIOUS" value="0.0" />
        <coefficient Id="NITROGEN" value="5.0" />
        <coefficient Id="PHOSPHORUS" value="0.9" />
    </value>
    <value Id="82" Name="Cultivated Crop">
        <coefficient Id="IMPERVIOUS" value="0.0" />
        <coefficient Id="NITROGEN" value="5.0" />
        <coefficient Id="PHOSPHORUS" value="0.9" />
    </value>
    <value Id="90" Name="Woody Wetland">
        <coefficient Id="IMPERVIOUS" value="0.02" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="95" Name="Emergent Herbaceous Wetland">
        <coefficient Id="IMPERVIOUS" value="0.02" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
</values>
<!--
    * The classes node contains values from a Land cover raster grouped into one or more classes.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier, also used for automated generation of output field name
    *   - must conform to the field naming conventions dictated by the output database system

    * OPTIONAL ATTRIBUTES
    * Name - text, word or phrase describing class
    * filter - text, a string of one or more tool name abbreviations separated by a ";"
    *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp
    *   - used to exclude the class from the selectable classes in the tool's GUI
    * xxxxField - text, overrides ATtILA-generated field name for output
    *   - where xxxx equals a tool name abbreviation
    *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
    *   - a separate xxxxField attribute can exist for each tool
    *   - must conform to the field naming conventions dictated by the output database system

    * A class can contain either values or classes but not both types.
    * Value elements contain only an Id attribute which refers to a value in a raster.
    * Values tagged as excluded="true" in the values node should not be included in any class.
-->
<classes>
    <class Id="NI" Name="All Natural Land Use" filter="" lcpField="NINDEX">
        <class Id="bar" Name="Barren" filter="">
            <value Id="31" />
        </class>
        <class Id="for" Name="Forest" filter="">
            <value Id="41" />
            <value Id="42" />
            <value Id="43" />
        </class>
        <class Id="tun" Name="Tundra" filter="">
            <value Id="51" />
            <value Id="72" />
            <value Id="73" />
            <value Id="74" />
        </class>
        <class Id="shb" Name="Shrubland" filter="">
            <value Id="52" />
        </class>
        <class Id="hrb" Name="Herbaceous" filter="">
            <value Id="71" />
        </class>
        <class Id="wtl" Name="Wetland" filter="">
            <class Id="wtlw" Name="Woody Wetland" filter="">

```

```

        <value Id="90" />
    </class>
    <class Id="wtle" Name="Emergent Wetland" filter="">
        <value Id="95" />
    </class>
</class>
</class>
<class Id="UI" Name="All Human Land Use" filter="" lcpField="UINDEX">
    <class Id="dev" Name="Developed" filter="">
        <class Id="devo" Name="Open Space Developed" filter="">
            <value Id="21" />
        </class>
        <class Id="devl" Name="Low Intensity Developed" filter="">
            <value Id="22" />
        </class>
        <class Id="devm" Name="Medium Intensity Developed" filter="">
            <value Id="23" />
        </class>
        <class Id="devh" Name="High Intensity Developed" filter="">
            <value Id="24" />
        </class>
    </class>
    <class Id="agr" Name="Agriculture" filter="">
        <class Id="agrp" Name="Pasture/Hay" filter="">
            <value Id="81" />
        </class>
        <class Id="agrc" Name="Cultivated Crop" filter="">
            <value Id="82" />
        </class>
    </class>
</class>
</classes>
</lccSchema>

```

LCC Worldcover 2020 ALL

European Space Agency WorldCover 2020 ALL

No raster values except those representing NODATA are excluded from the calculation of reporting unit effective area (i.e., effective area equals the total raster area within a reporting unit).

No coefficient values for impervious area, nitrogen loading, or phosphorous loading calculations are provided.

For more information about the WorldCover 2020 dataset go to: <https://worldcover2020.esa.int/>

WorldCover 2020 ALL - Value Table

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class(es)
0	NODATA					
10	Tree Cover					NI, veg, for, mfor
20	Shrubland					NI, veg, shb
30	Grassland					NI, veg, grs
40	Cropland					UI, ag, agrc
50	Built-up					UI, imp
60	Bare/Sparse Vegetation					NI, bar
70	Snow and Ice					NI, ice
80	Permanent Water Bodies					
90	Herbaceous Wetland					NI, veg, wet, wete
95	Mangroves					NI, veg, mfor, wet, wetw
100	Moss and Lichen					NI, veg, tun

WorldCover 2020 ALL - Class Table

Class Name	abbv	Value
All Natural Land Use	NI	10,20,30,60,70,90,95,100
Natural Vegetation	veg	10,20,30,90,95,100
Trees/Forest	for	10
Trees/Forest, Orchards, and Woody Wetlands	mfor	10,95
Grass and Herbaceous	grs	30
Shrub	shb	20
Tundra	tun	100
Wetlands	wet	90,95
Woody Wetlands	wetw	95
Emergent Wetlands	wete	90
Soil and Barren	bar	60
Snow and Ice	ice	70
All Human Land Use	UI	40,50
Impervious Surface	imp	50
Agriculture	ag	40
Row Crops	agrc	40

WorldCover 2020 ALL XML

```
<?xml version='1.0' encoding='UTF-8'?>
<lccSchema xmlns="lcc" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="lcc
  LCCSchema.xsd">
  <metadata>
    <name>WorldCover 2020 ALL</name>
    <description>WorldCover 2020 ALL Land Cover Classification Schema. WorldCover is a 10 m resolution
      2020 global land cover product from the European Space Agency (ESA) based on both Sentinel-1 and
      Sentinel-2 data, containing 11 land cover classes. No raster values except those representing
      NODATA are excluded from the calculation of reporting unit effective area (i.e., effective area
      equals the total raster area within a reporting unit). Percentage metrics are based on the
      effective area of a reporting unit, not necessarily its total area.</description>
  </metadata>
  <!--
    * The coefficients node contains coefficients to be assigned to values.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier
    * Name - text, word or phrase describing coefficient
    * fieldName - text, name of field to be created for output
    * - must conform to the field naming conventions dictated by the output database system
    * method - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
  -->
  <coefficients>
    <coefficient Id="IMPERVIOUS" Name="Percent Cover Total Impervious Area" fieldName="PCTIA"
      method="P" />
    <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover"
      fieldName="N_Load" method="A" />
    <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover"
      fieldName="P_Load" method="A" />
  </coefficients>
```

```

<!--
  * The values node defines the full set of values that can exist in a land cover raster.

  * REQUIRED ATTRIBUTES
  * Id - integer, raster code
  *
  * OPTIONAL ATTRIBUTES
  * Name - text, word or phrase describing value
  * excluded - boolean, "true" or "false" or "1" or "0"
  *           - used to exclude values from effective area calculations
  *           - excluded=false is the default

  * A value element can optionally contain one or more coefficient elements

  * REQUIRED COEFFICIENT ATTRIBUTES
  * Id - text, must match an Id attribute from a coefficients node element
  * value - decimal, weighting/calculation factor
-->
<values>
  <value Id="0" Name="NODATA" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="10" Name="Tree Cover">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="20" Name="Shrubland">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="30" Name="Grassland">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="40" Name="Cropland">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="50" Name="Built-up">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="60" Name="Bare/Sparse Vegetation">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="70" Name="Snow and Ice">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="80" Name="Permanent Water Bodies">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="90" Name="Herbaceous Wetland">
    <coefficient Id="IMPERVIOUS" value="0.0" />

```

```

        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="95" Name="Mangroves">
        <coefficient Id="IMPERVIOUS" value="0.0" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="100" Name="Moss and Lichen">
        <coefficient Id="IMPERVIOUS" value="0.0" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
</values>
<!--
    * The classes node contains values from a land cover raster grouped into one or more classes.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier, also used for automated generation of output field name
    *   - must conform to the field naming conventions dictated by the output database system

    * OPTIONAL ATTRIBUTES
    * Name - text, word or phrase describing class
    * filter - text, a string of one or more tool name abbreviations separated by a ";"
    *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp
    *   - used to exclude the class from the selectable classes in the tool's GUI
    * xxxxField - text, overrides ATtILA-generated field name for output
    *   - where xxxx equals a tool name abbreviation
    *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
    *   - a separate xxxxField attribute can exist for each tool
    *   - must conform to the field naming conventions dictated by the output database system

    * A class can contain either values or classes but not both types.
    * Value elements contain only an Id attribute which refers to a value in a raster.
    * Values tagged as excluded="true" in the values node should not be included in any class.
-->
<classes>
    <class Id="NI" Name="All Natural Land Use" filter="" lcpField="NINDEX">
        <class Id="veg" Name="Natural Vegetation" filter="">
            <class Id="for" Name="Trees/Forest" filter="">
                <value Id="10" />
            </class>
            <class Id="mfor" Name="Trees/Forest, Orchards, and Woody Wetlands" filter="">
                <value Id="10" />
                <value Id="95" />
            </class>
            <class Id="grs" Name="Grass and Herbaceous" filter="">
                <value Id="30" />
            </class>
            <class Id="shb" Name="Shrub" filter="">
                <value Id="20" />
            </class>
            <class Id="tun" Name="Tundra" filter="">
                <value Id="100" />
            </class>
            <class Id="wet" Name="Wetlands" filter="">
                <class Id="wetw" Name="Woody Wetlands" filter="">
                    <value Id="95" />
                </class>
                <class Id="wete" Name="Emergent Wetlands" filter="">
                    <value Id="90" />
                </class>
            </class>
            <class Id="bar" Name="Soil and Barren" filter="">
                <value Id="60" />
            </class>
        </class>
    </class>

```

```
</class>
<class Id="ice" Name="Snow and Ice" filter="">
  <value Id="70" />
</class>
<class Id="wtr" Name="Water" filter="">
  <value Id="80" />
</class>
</class>
<class Id="UI" Name="All Human Land Use" filter="" lcpField="UINDEX">
  <class Id="imp" Name="Impervious Surface" filter="">
    <value Id="50" />
  </class>
  <class Id="ag" Name="Agriculture" filter="">
    <class Id="agrc" Name="Row Crops" filter="">
      <value Id="40" />
    </class>
  </class>
</class>
</classes>
</lccSchema>
```


LCC Worldcover 2020 LAND

European Space Agency WorldCover 2020 LAND

Water-related raster values are tagged as EXCLUDED. Excluded values are ignored when calculating the effective area of a reporting unit. Percentage metrics are based on the effective area of a reporting unit, not the total area.

No coefficient values for impervious area, nitrogen loading, or phosphorous loading calculations are provided.

For more information about the WorldCover 2020 dataset go to: <https://worldcover2020.esa.int/>

WorldCover 2020 LAND - Value Table

Value	Description	Excluded	Nitrogen	Phosphorus	Impervious	Class(es)
0	NODATA	yes				
10	Tree Cover					NI, veg, for, mfor
20	Shrubland					NI, veg, shb
30	Grassland					NI, veg, grs
40	Cropland					UI, ag, agrc
50	Built-up					UI, imp
60	Bare/Sparse Vegetation					NI, bar
70	Snow and Ice					NI, ice
80	Permanent Water Bodies	yes				
90	Herbaceous Wetland					NI, veg, wet, wete
95	Mangroves					NI, veg, mfor, wet, wetw
100	Moss and Lichen					NI, veg, tun

WorldCover 2020 LAND - Class Table

Class Name	abbv	Value
All Natural Land Use	NI	10,20,30,60,70,90,95,100
Natural Vegetation	veg	10,20,30,90,95,100
Trees/Forest	for	10
Trees/Forest, Orchards, and Woody Wetlands	mfor	10,95
Grass and Herbaceous	grs	30
Shrub	shb	20
Tundra	tun	100
Wetlands	wet	90,95
Woody Wetlands	wetw	95
Emergent Wetlands	wete	90
Soil and Barren	bar	60
Snow and Ice	ice	70
All Human Land Use	UI	40,50
Impervious Surface	imp	50
Agriculture	ag	40
Row Crops	agrc	40

WorldCover 2020 LAND XML

```
<?xml version='1.0' encoding='UTF-8'?>
<lccSchema xmlns="lcc" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="lcc
  LCCSchema.xsd">
  <metadata>
    <name>WorldCover 2020 LAND</name>
    <description>WorldCover 2020 LAND Land Cover Classification Schema. Water-related raster values
      are tagged as EXCLUDED. Excluded values are ignored when calculating the effective area of a
      reporting unit. Percentage metrics are based on the effective area of a reporting unit, not its
      total area (i.e., class area / effective area * 100 = metric value). WorldCover is a 10 m
      resolution 2020 global land cover product from the European Space Agency (ESA) based on both
      Sentinel-1 and Sentinel-2 data, containing 11 land cover classes.</description>
  </metadata>
  <!--
    * The coefficients node contains coefficients to be assigned to values.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier
    * Name - text, word or phrase describing coefficient
    * fieldName - text, name of field to be created for output
    * - must conform to the field naming conventions dictated by the output database system
    * method - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
  -->
  <coefficients>
    <coefficient Id="IMPERVIOUS" Name="Percent Cover Total Impervious Area" fieldName="PCTIA"
      method="P" />
    <coefficient Id="NITROGEN" Name="Estimated Nitrogen Loading Based on Land Cover"
      fieldName="N_Load" method="A" />
    <coefficient Id="PHOSPHORUS" Name="Estimated Phosphorus Loading Based on Land Cover"
      fieldName="P_Load" method="A" />
  </coefficients>
```

```

<!--
  * The values node defines the full set of values that can exist in a land cover raster.

  * REQUIRED ATTRIBUTES
  * Id - integer, raster code
  *
  * OPTIONAL ATTRIBUTES
  * Name - text, word or phrase describing value
  * excluded - boolean, "true" or "false" or "1" or "0"
  *           - used to exclude values from effective area calculations
  *           - excluded=false is the default

  * A value element can optionally contain one or more coefficient elements

  * REQUIRED COEFFICIENT ATTRIBUTES
  * Id - text, must match an Id attribute from a coefficients node element
  * value - decimal, weighting/calculation factor
-->
<values>
  <value Id="0" Name="NODATA" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="10" Name="Tree Cover">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="20" Name="Shrubland">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="30" Name="Grassland">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="40" Name="Cropland">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="50" Name="Built-up">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="60" Name="Bare/Sparse Vegetation">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="70" Name="Snow and Ice">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="80" Name="Permanent Water Bodies" excluded="true">
    <coefficient Id="IMPERVIOUS" value="0.0" />
    <coefficient Id="NITROGEN" value="0.0" />
    <coefficient Id="PHOSPHORUS" value="0.0" />
  </value>
  <value Id="90" Name="Herbaceous Wetland">
    <coefficient Id="IMPERVIOUS" value="0.0" />

```

```

        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="95" Name="Mangroves">
        <coefficient Id="IMPERVIOUS" value="0.0" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
    <value Id="100" Name="Moss and Lichen">
        <coefficient Id="IMPERVIOUS" value="0.0" />
        <coefficient Id="NITROGEN" value="0.0" />
        <coefficient Id="PHOSPHORUS" value="0.0" />
    </value>
</values>
<!--
    * The classes node contains values from a Land cover raster grouped into one or more classes.

    * REQUIRED ATTRIBUTES
    * Id - text, unique identifier, also used for automated generation of output field name
    *   - must conform to the field naming conventions dictated by the output database system

    * OPTIONAL ATTRIBUTES
    * Name - text, word or phrase describing class
    * filter - text, a string of one or more tool name abbreviations separated by a ";"
    *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and splcp
    *   - used to exclude the class from the selectable classes in the tool's GUI
    * xxxxField - text, overrides ATtILA-generated field name for output
    *   - where xxxx equals a tool name abbreviation
    *   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
    *   - a separate xxxxField attribute can exist for each tool
    *   - must conform to the field naming conventions dictated by the output database system

    * A class can contain either values or classes but not both types.
    * Value elements contain only an Id attribute which refers to a value in a raster.
    * Values tagged as excluded="true" in the values node should not be included in any class.
-->
<classes>
    <class Id="NI" Name="All Natural Land Use" filter="" lcpField="NINDEX">
        <class Id="veg" Name="Natural Vegetation" filter="">
            <class Id="for" Name="Trees/Forest" filter="">
                <value Id="10" />
            </class>
            <class Id="mfor" Name="Trees/Forest, Orchards, and Woody Wetlands" filter="">
                <value Id="10" />
                <value Id="95" />
            </class>
            <class Id="grs" Name="Grass and Herbaceous" filter="">
                <value Id="30" />
            </class>
            <class Id="shb" Name="Shrub" filter="">
                <value Id="20" />
            </class>
            <class Id="tun" Name="Tundra" filter="">
                <value Id="100" />
            </class>
            <class Id="wet" Name="Wetlands" filter="">
                <class Id="wetw" Name="Woody Wetlands" filter="">
                    <value Id="95" />
                </class>
                <class Id="wete" Name="Emergent Wetlands" filter="">
                    <value Id="90" />
                </class>
            </class>
        </class>
    </class>
    <class Id="bar" Name="Soil and Barren" filter="">
        <value Id="60" />
    </class>

```

```
    </class>
    <class Id="ice" Name="Snow and Ice" filter="">
      <value Id="70" />
    </class>
  </class>
</class>
<class Id="UI" Name="All Human Land Use" filter="" lcpField="UINDEX">
  <class Id="imp" Name="Impervious Surface" filter="">
    <value Id="50" />
  </class>
  <class Id="ag" Name="Agriculture" filter="">
    <class Id="agrc" Name="Row Crops" filter="">
      <value Id="40" />
    </class>
  </class>
</class>
</classes>
</lccSchema>
```

References

May, C.W., Horner, R.R., Karr, J.R., Mar B.W., Welch, E.B. 1997. Effects of urbanization on small streams in the Puget Sound Lowland Ecoregion. *Watershed Protection Techniques*. 2:4. pp. 483–493.

Reckhow, K.H., Beaulac, M.N., and Simpson, J.T. 1980. *Modeling Phosphorus Loading and Lake Response Under Uncertainty: A Manual and Compilation of Export Coefficients*. USEPA 440/5-80-011. Washington, DC: Office of Water Regulations and Standards, U.S. Environmental Protection Agency. Washington, DC, USA.

Troubleshooting

When ATtILA encounters a potential problem or an unexpected occurrence during the execution of one of its tools, it will attempt to relay that information to the user via the tool's View Details box and the Geoprocessing > History window. Depending on the situation, ATtILA may provide potential solutions, possible consequences, or some other piece of information deemed useful to the user. In most cases, the message provided in the tool's View Details box and the Geoprocessing > History window will be sufficient to inform the user fully of the situation. In other cases, a more detailed explanation than what can be readily expressed is necessary and a referral to this document is provided.

Please report any additional undocumented errors to: <https://ecomments.epa.gov/enviroatlas/> and include "ATtILA" in the subject line.

***NOTE:** The errors listed below have been encountered over the lifetime of ATtILA's development. While several of them may not still be an issue with the current iteration of ATtILA, they are included here as a precaution.*

Errors encountered:

- **Background Processing error**
- **Excluded Values in Class Definitions**
- **Failed to execute (AddField)**
- **Failed to execute (TabulateArea)**
- **Failed to execute (Intersect)**
- **Failed to save raster dataset**
- **{FIELD} already exists error**
- **{FIELD} does not exist within table error**
- **Invalid extent error**
- **Tool appears to run but does not finish**
- **Unable to Determine Conversion Factor**
- **Undefined Values in LCC XML Document**

Troubleshooting Background Processing Error

When running a tool in which the output and intermediate files are directed to be saved to a file geodatabase, the following Background Processing error may occur:

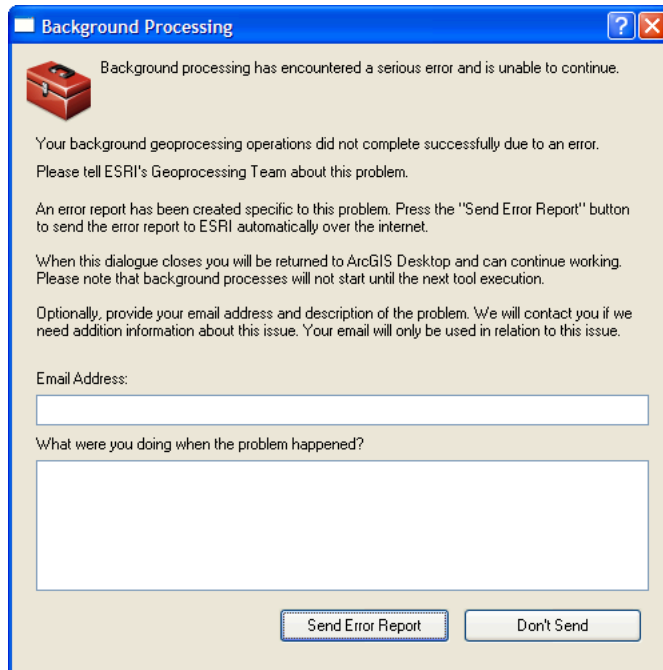


Figure 43. Background Processing error dialog

This error may be due to an incompletely deleted file geodatabase, such as the `attilaScratchWorkspace.gdb`. This occurs periodically when the `attilaScratchWorkspace.gdb` is deleted using ArcCatalog. While the geodatabase may be successfully deleted and no longer appear in the ArcCatalog interface, some folders and files may linger.

To correct this, simply open Windows Explorer, navigate to the disk location where the file geodatabase formerly resided, and delete the lingering geodatabase folder and its contents.

NOTE: ArcGIS Pro will report 'ERROR 999999: Something unexpected caused the tool to fail.' when ATILA attempts to write to a corrupted geodatabase. The solution is the same.

Troubleshooting Excluded Values In Class Definitions

This warning message is triggered when a value is marked as excluded = “true” in the Values Element section of the LCC XML document, but is also included in one or more class definitions in the Classes Element section. The warning message will also provide the class name that contains the value tagged as excluded.

Example: > LCC XML file: excluded value(s) [23] in class DEV definition - Please refer to the ATtILA documentation regarding excluded values.

The value attribute, ***excluded***, (see Values Element in ATtILA's LCC XML Document) is used to identify grid codes whose area is to be excluded from the reporting unit's effective area calculation. Effective area can be thought of as the area of interest within a reporting unit that the user wishes to use for percentage based metric calculations. For example, the user may be interested in basing their metric calculations on just the land area in a reporting unit versus the overall total area of the reporting unit. To make the effective area equal to that of the land area, the user would set the ***excluded*** attribute of any water related grid value to “true”.

Values attributed as excluded = “true” included in any class definition in the Classes Element section of the LCC XML document create an ambiguous situation for ATtILA. When this occurs, ATtILA will ignore the area of that excluded value when summing up the area of the selected class for metric calculations, as well as ignoring its area in the calculation of the effective area.

The following example illustrates the point.

Assume a land cover raster layer contains four land cover types, values 1, 2, 3, and 4 with value 1 attributed as excluded = “true”. Only one class has been defined named “Forest”, and consists of values 1 and 2. For our example, the different land cover types have the following areas within a reporting unit:

value 1 = 100 square meters

value 2 = 200 square meters

value 3 = 300 square meters

value 4 = 400 square meters

If the Land Cover Proportions tool is run, the output for pForest will be calculated as follows:

If value 1 was not attributed as excluded = “true”, then the calculations would be as follows:

But if value 1 is attributed as excluded = “true”, then the calculations will adjust to the following:

NOTE: If all values assigned to the class are tagged as excluded, the output metric is calculated as 0%.

Troubleshooting Failed To Execute AddField

A field name generated by ATtILA was either too large for the output metric table or did not conform to the naming conventions dictated by the different database systems (dBASE, or geodatabase).



Figure 44. Example Add Field failure screenshot

This error can largely be avoided by directing the output table to be placed inside of a geodatabase where the field name size limitation is 64 characters. An output table created as a .dbf table is limited to 10 characters in length.

The error can also be addressed by editing the selected Land Cover Classification scheme XML file (see [Classes Element](#) in the [Land Cover Classification XML Document](#)) and shortening either the xxxxFld Attribute or the Id Attribute if the xxxxFld Attribute is unused.

If field name size is not the problem, restrict field names to just alphanumeric characters and underscores. Use of spaces and special characters should be avoided, as well as beginning a field name with a number or an underscore. Also, avoid using field names that contain words that are considered

reserved keywords, such as date, day, month, order, table, text, user, when, where, year, and zone. For more guidelines on the naming of fields, search on “Fundamentals of adding and deleting fields” in the ArcGIS help documentation.

Troubleshooting Failed To Execute Intersect

This error has been detected when executing the Riparian Land Cover Proportions tool. Please report any additional instances of this error to: <https://ecomments.epa.gov/enviroatlas/> and include “ATtILA” in the subject line.

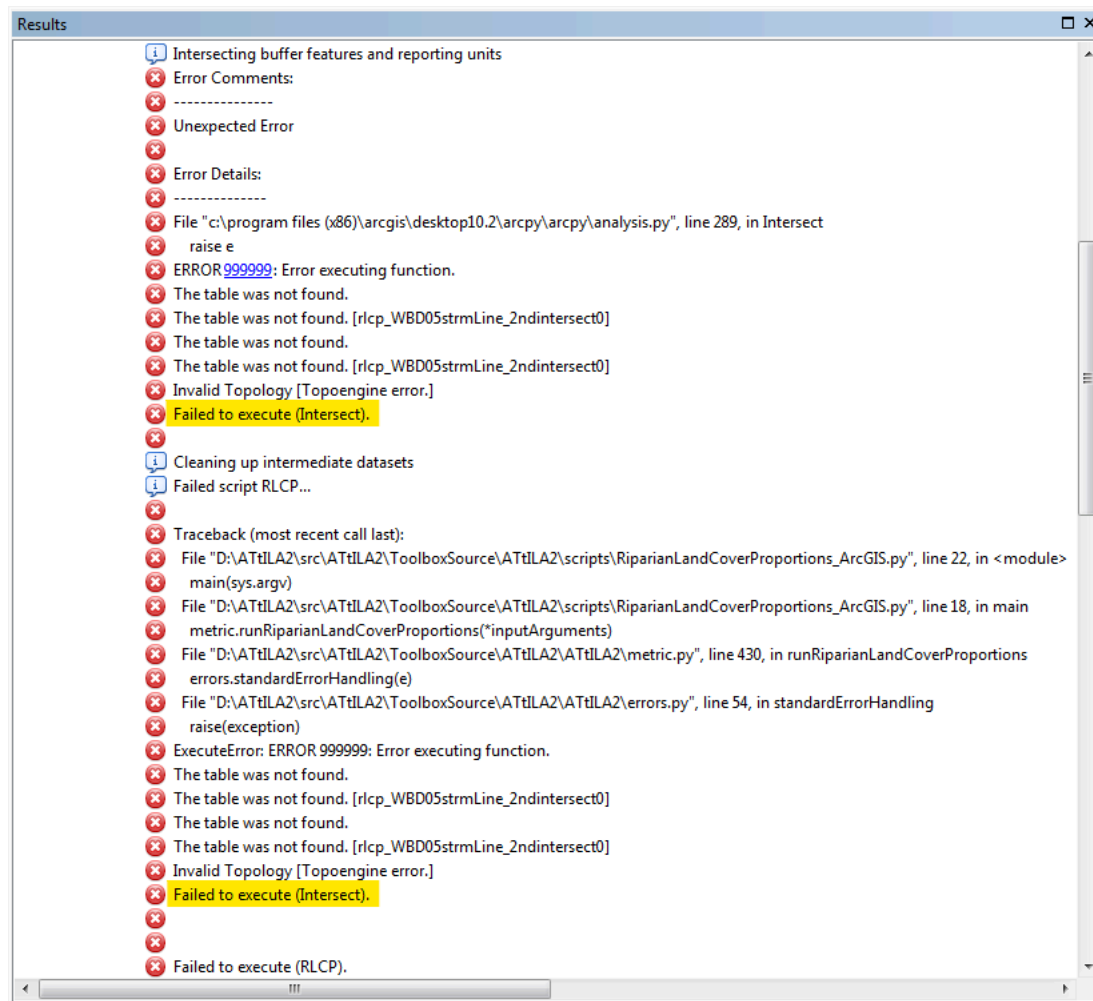


Figure 45. Example Intersect failure screenshot

This problem appears to occur when the input reporting unit feature consists of a large number of reporting units and/or covers a large geographic area. When the ArcGIS software tries to tile the data to process the dataset, the tiling process appears to fail, usually during the reassembly stage. This is an inconsistent error. Sometimes when the metric run is reattempted, using the exact same inputs, the metric run will finish.

If the metric run fails a second or third time, we recommend using the selection tool and selecting approximately half of the input reporting unit features, exporting the selected reporting units into a new dataset, and reattempting the metric run with the new dataset as the input reporting unit theme.

Continue the reduction process until the metric run runs to completion or until a single reporting unit remains.

If the metric run fails with only one reporting unit feature as the input reporting unit theme, it is likely that there is an error in one of the other input datasets within the confines of the recalcitrant reporting unit. Close inspection of the other input datasets in this area may reveal the problem (e.g. poor digitization of input data as in nodes not being used at line intersections).

Troubleshooting Failed To Execute TabulateArea

When running a tool that uses a Land cover grid ([Core and Edge Metrics](#), [Floodplain Land Cover Proportions](#), [Facility Land Cover Views](#), [Land Cover Coefficient Calculator](#), [Land Cover Diversity](#), [Land Cover on Slope Proportions](#), [Land Cover Proportions](#), [Population Land Cover Views](#), [Patch Metrics](#), [Riparian Land Cover Proportions](#), and [Sample Point Land Cover Proportions](#)), an error may occur indicating that TabulateArea failed to execute:

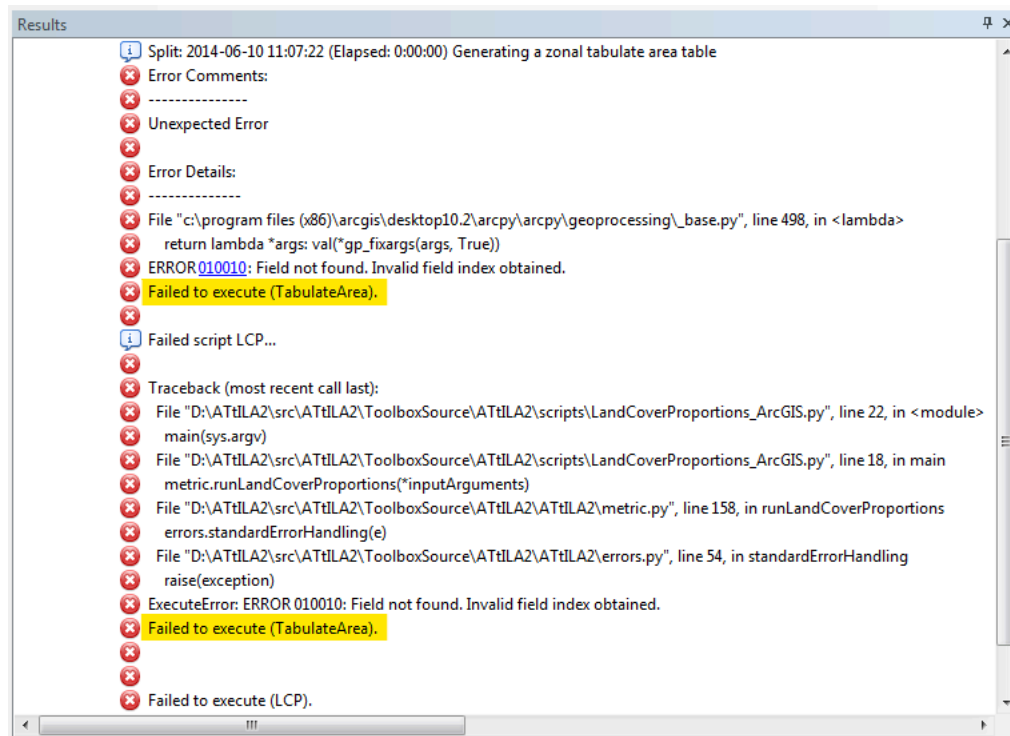


Figure 46. Example TabulateArea failure screenshot

The issue is that raster geoprocessing tools convert other formats to GRID format internally, then perform the analysis, then re-convert the GRID back out to the desired format. The limit is reached at the conversion to the GRID format; GRID's have an upper limit of 2.1 billion cells in their INFO table. The only workaround for this is breaking your data down into smaller chunks before running the tabulate area tool or by selecting a larger processing cell size. The larger processing cell size may allow the process to finish, but the output results will be of a more generalized nature.

Troubleshooting Failed To Save Raster Dataset

This error has been detected when executing the Neighborhood Proportions tool. Please report any additional instances of this error to: <https://ecomments.epa.gov/enviroatlas/> and include “ATtILA” in the subject line.

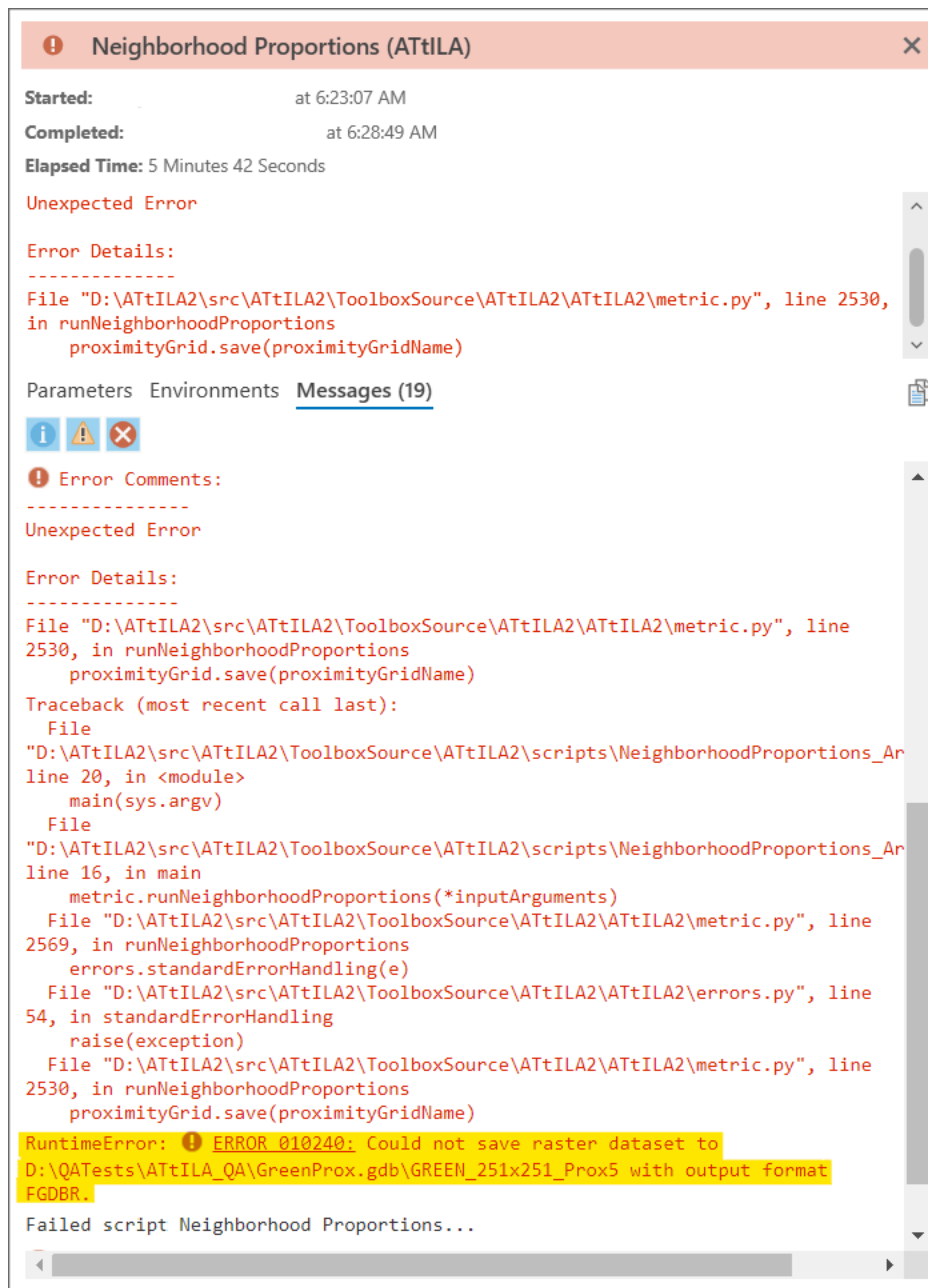


Figure 47. Example Failed to save raster dataset error screenshot

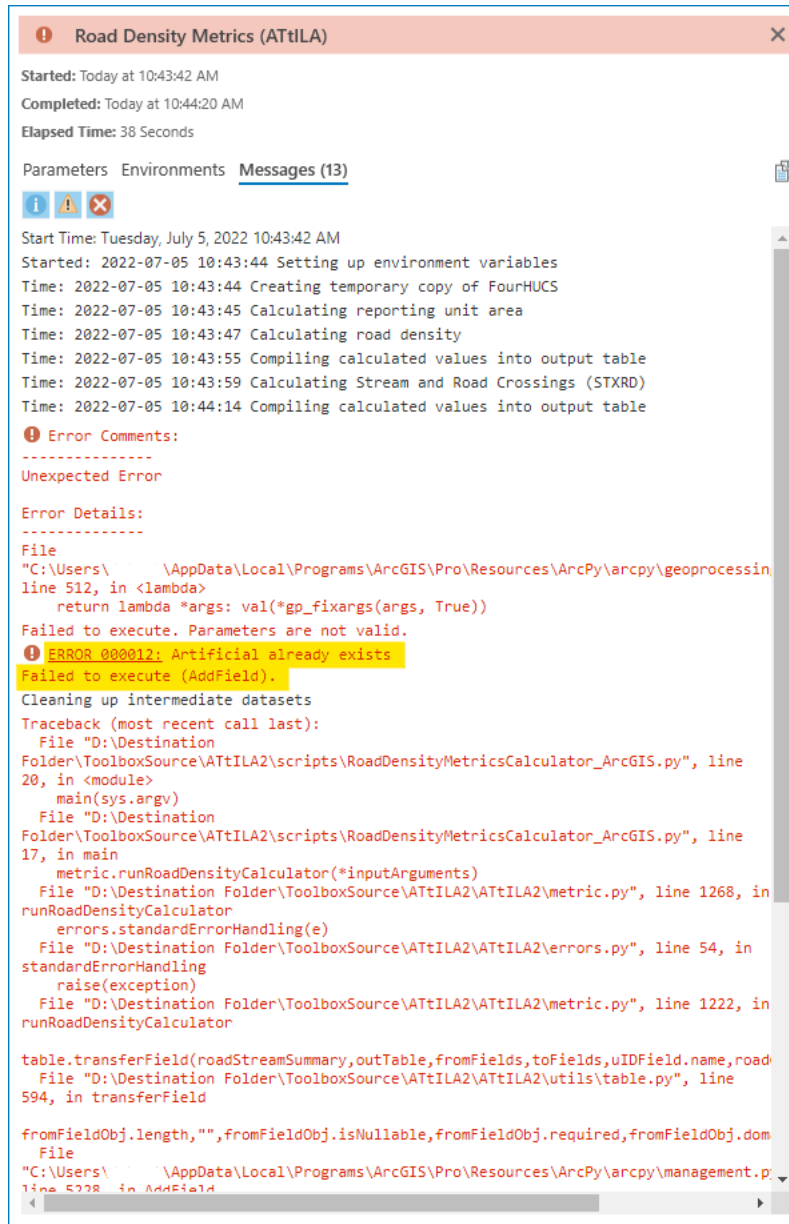
This problem appears to occur when the **Overwrite existing outputs option** is toggled on or off between successive runs of the Neighborhood Proportions tool, but may occur on its first run. Typically the 010240 error is caused when a raster with the same name or format already exists in the output workspace. While ATtILA checks for this condition during tool operation, a residual lock file may exist in the output workspace.

This is an inconsistent error. Sometimes when the metric run is reattempted, using the exact same inputs, the metric run will finish. Another option is to select a new geodatabase as the output workspace.

Finally, another option is to compact the geodatabase. For more information about compacting geodatabases, please see, [‘Compact file geodatabases’](#), in ArcGIS Pro’s Online Help.

Troubleshooting Field Already Exists Error

When running the Road Density Metrics or the Stream Density Metrics tool, an error may occur indicating that a field already exists:



```

Road Density Metrics (ATtILA)
Started: Today at 10:43:42 AM
Completed: Today at 10:44:20 AM
Elapsed Time: 38 Seconds
Parameters Environments Messages (13)
Start Time: Tuesday, July 5, 2022 10:43:42 AM
Started: 2022-07-05 10:43:44 Setting up environment variables
Time: 2022-07-05 10:43:44 Creating temporary copy of FourHUCS
Time: 2022-07-05 10:43:45 Calculating reporting unit area
Time: 2022-07-05 10:43:47 Calculating road density
Time: 2022-07-05 10:43:55 Compiling calculated values into output table
Time: 2022-07-05 10:43:59 Calculating Stream and Road Crossings (STXRD)
Time: 2022-07-05 10:44:14 Compiling calculated values into output table
Error Comments:
-----
Unexpected Error
Error Details:
-----
File
"C:\Users\... \AppData\Local\Programs\ArcGIS\Pro\Resources\ArcPy\arcpy\geoprocessing
line 512, in <lambda>
    return lambda *args: val(*gp_fixargs(args, True))
Failed to execute. Parameters are not valid.
ERROR 000012: Artificial already exists
Failed to execute (AddField).
Cleaning up intermediate datasets
Traceback (most recent call last):
  File "D:\Destination
Folder\ToolboxSource\ATtILA2\scripts\RoadDensityMetricsCalculator_ArcGIS.py", line
20, in <module>
    main(sys.argv)
  File "D:\Destination
Folder\ToolboxSource\ATtILA2\scripts\RoadDensityMetricsCalculator_ArcGIS.py", line
17, in main
    metric.runRoadDensityCalculator(*inputArguments)
  File "D:\Destination Folder\ToolboxSource\ATtILA2\ATtILA2\metric.py", line 1268, in
runRoadDensityCalculator
    errors.standardErrorHandling(e)
  File "D:\Destination Folder\ToolboxSource\ATtILA2\ATtILA2\errors.py", line 54, in
standardErrorHandling
    raise(exception)
  File "D:\Destination Folder\ToolboxSource\ATtILA2\ATtILA2\metric.py", line 1222, in
runRoadDensityCalculator
    table.transferField(roadStreamSummary,outTable,fromFields,toFields,uIDField.name,road
  File "D:\Destination Folder\ToolboxSource\ATtILA2\ATtILA2\utils\table.py", line
594, in transferField
    fromFieldObj.length,"",fromFieldObj.isNullable,fromFieldObj.required,fromFieldObj.dom
  File
"C:\Users\... \AppData\Local\Programs\ArcGIS\Pro\Resources\ArcPy\arcpy\management.p
line 5228, in AddField

```

Figure 48. Example Field already exists error screenshot

This error may be caused by problems with the field used to classify the different road or stream features (Road class field or Stream order field, respectively). Specific rules for the length of the fields and the characters contained within the fields are found in each tool's help section.

Troubleshooting Field Does Not Exist Within Table Error

When running the Road Density Metrics or the Stream Density Metrics tool, an error may occur indicating that a field does not exist:

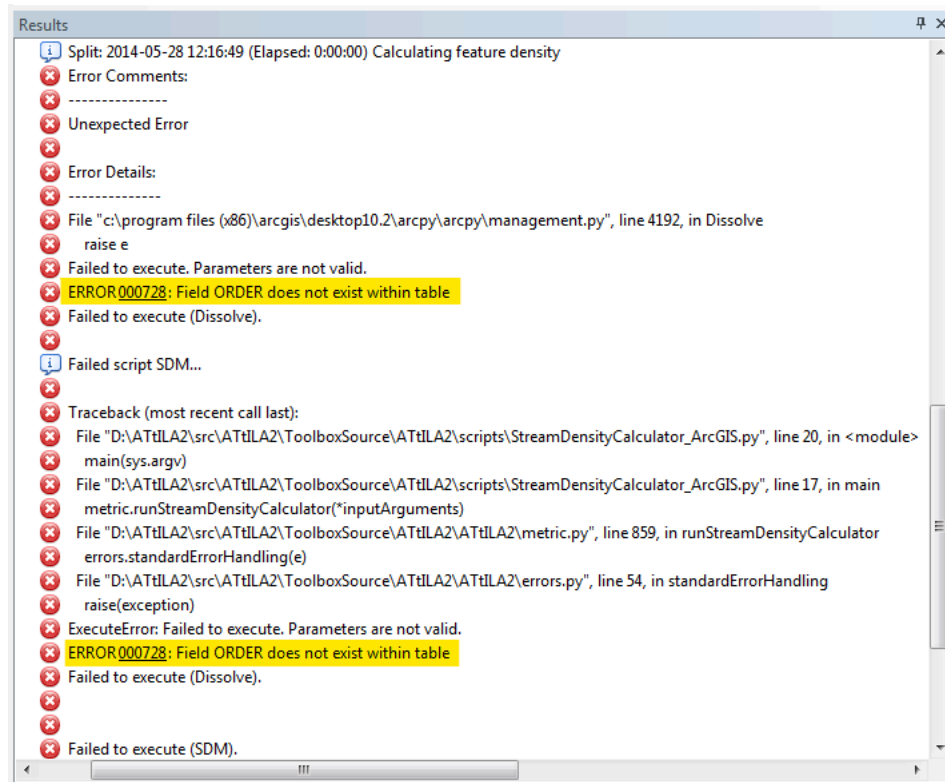


Figure 49. Example field does not exist error screenshot

This error may be caused by problems with the field used to classify the different road or stream features (Road class field or Stream order field, respectively).

If an input feature layer contains a reserved keyword such as ORDER as a field name, that field name is altered slightly by ArcGIS when the input feature layer is copied. ATtILA copies several of the input feature layers while processing selected metrics. This can result in ATtILA being unable to locate the specified field in the copied dataset. For more guidelines on the naming of fields, search on “Fundamentals of adding and deleting fields” in the ArcGIS help documentation.

Troubleshooting Invalid Extent Error

Most ATtILA tools involve processing two or more geospatial inputs. If at least one of these inputs does not overlap with the others, the tool operation will fail. The example below shows the error message reported when non-overlapping inputs for **Reporting unit feature** and **Population raster or polygon features** are used in the Population in Floodplain Metrics tool.

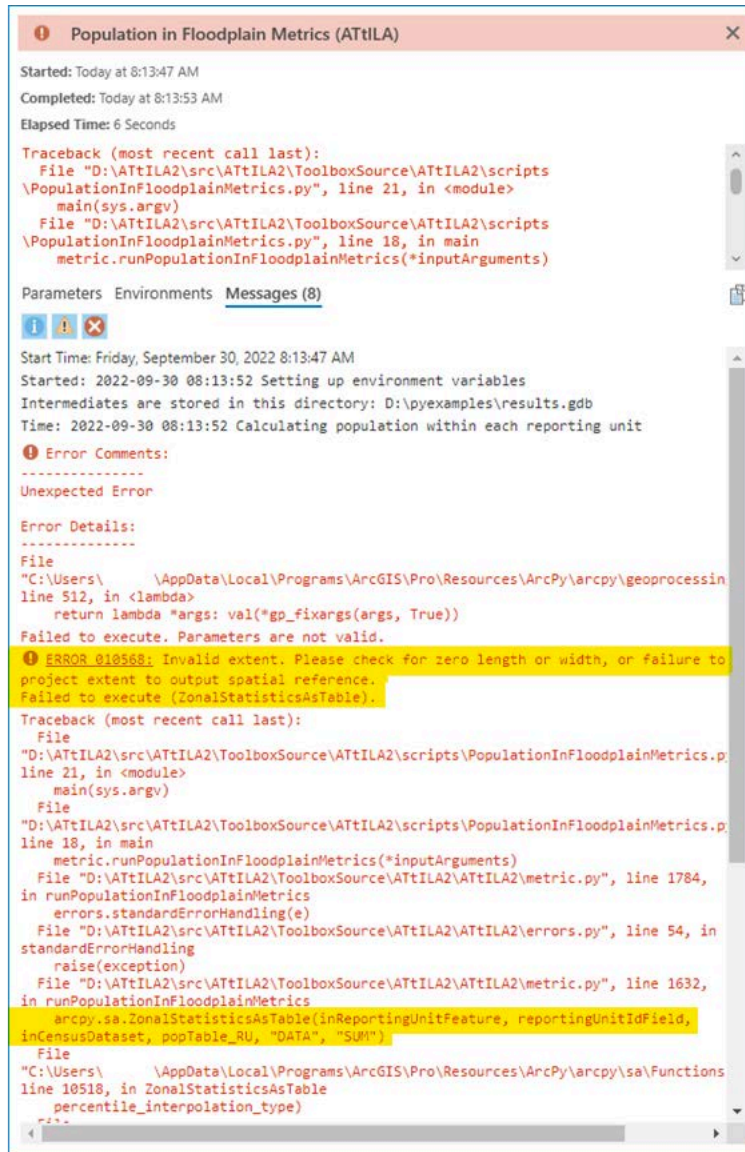


Figure 50. Example Field already exists error screenshot

According to ArcGIS Pro Online Help, when ERROR 010568 is encountered:

“Set the Output Extent environment to the extent you want to process. If this does not resolve the error, make sure your dataset extents overlap. To confirm that your dataset extents overlap, project them into the same spatial reference.”

Troubleshooting Tool Appears To Run But Does Not Finish

If one of ATtILA's tools 'hangs' (i.e., you enter all the necessary tool inputs and click OK, but the tool never finishes):

- Open Windows Task Manager and select the hung ArcGIS application in the Applications Tab - Click End Task
- Reopen the ArcGIS application where the problem occurred
- Right-click on the tool and open it's properties
- Check under the source tab to be sure that the 'Run Python script in process' box is checked

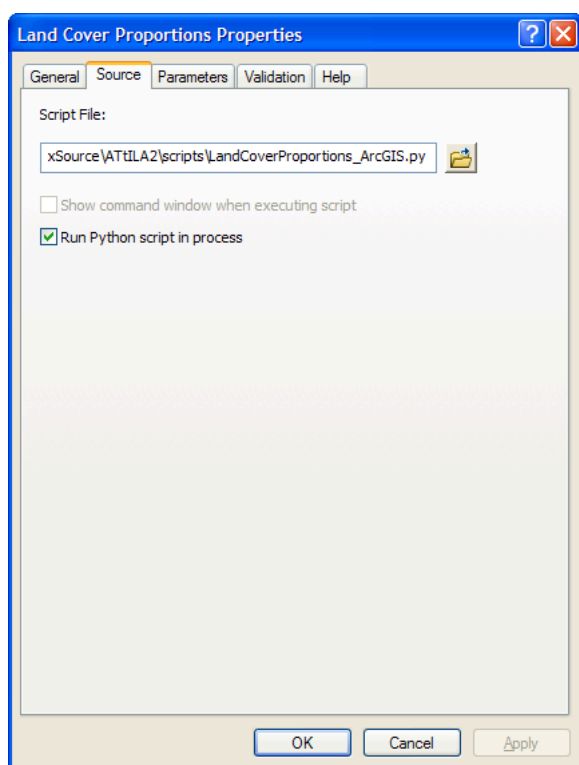


Figure 51. Screenshot of ArcGIS Desktop tool properties dialog box

That should solve the problem.

To relaunch the tool with all it's previously set parameters, checkboxes, and options, do the following:

- Open the 'Not Run' shortcut menu in the Geoprocessing > Results Window
- Right-click on the tool execution that failed
- Select 'Re-Run'

Troubleshooting Unable To Determine Conversion Factor

This error is triggered when ATtILA is unable to determine the conversion factor necessary to convert output linear units to metric units. The following two possible scenario's have been identified when this may occur.

1. Output Coordinates are not set to a system defined with angular units of Decimal Degrees.

Please be sure that the Environment Settings > Output Coordinates are not set to a system defined with angular units of Decimal Degrees. If an Environment Settings > Output Coordinates is not set, please be sure to select an 'Input reporting unit feature' that does not have a spatial reference defined with angular units of decimal degrees.

2. Output Coordinates unit name is not found in the look up tables used by ATtILA for metric conversions

It is also possible for this error to occur if the linear units name defined in the Output Coordinates, if one is set, or the linear units property of the 'Input reporting unit feature', if an Output Coordinates is not set, is not found in ATtILA's look up table for metric conversion factors. Please compare your linear units with those found in the list of Conversion Factors located in the appendix. If the linear units name is not found, it will be necessary to change the Output Coordinates in the Environment Settings to one found in the look table or to project the 'Input reporting unit feature' to a projection with a linear units property known to ATtILA.

Troubleshooting Undefined Values In Lcc Xml Document

For tools in ATtILA that utilize an LCC XML document, (i.e., tools that utilize a land cover raster input) ATtILA will examine all of the values found in that tool's selected land cover raster and compare them to those values supplied in the LCC XML document. The values in the LCC XML document are gathered from both the Values Element section of the document and the Classes Element section. Any values found in the grid but not found in the LCC XML document will be reported to the user with a warning message is added to the tool's View Details box and the Geoprocessing > History window. The user can then determine if the reported values were either accidentally or purposely omitted from the LCC XML document or were possibly incorrectly transcribed. A report of missing values may also indicate that the wrong LCC Schema was selected for the input raster layer, or that the wrong raster layer was input for the selected LCC Schema.

Example: >LCC XML file: grid value(s) [31, 41, 42, 43, 52, 71, 81, 82, 90, 95] undefined in LCC XML file - Please refer to the ATtILA documentation regarding undefined values.

NOTE: Not all values found in the land cover grid need to be included in the LCC XML document. Only land cover values that are of interest to the user (e.g., values for forest or agriculture land cover types) need to be accounted for. If this is the case, and the LCC XML document was constructed to analyze only a subset of land cover classes from the land cover raster, then the warning message from ATtILA from ArcGIS is superfluous and can be ignored.

Appendix

The Appendix consists of the following pages:

- **Change Log** A list of what's changed in the current version of ATtILA.
- **Conversion Factors** Tables used by ATtILA to convert input dataset's distance and area units into meters and square meters.
- **LCCSchema.xsd** The supplied XSD file, located in the ToolboxSource > LandCoverClassifications folder in the ATtILA toolbox destination folder (see [Installing ATtILA](#)), used to help verify that any Land Cover Classification Schema XML file created outside of the LCCEditor application contains the necessary XML elements needed by ATtILA.
- **Tool Output Field Names** Table of output field naming conventions for each tool with examples.

Appendix Change Log

New\Enhanced Features

- Added a new tool, Pedestrian Access and Availability, that estimates the availability of parkland per person based on accessibility of parkland. Accessibility is determined using walkable routes along a cost surface raster and a user-specified maximum travel distance. The output is a raster.
- Added a new tool, Population Within Zone Metrics, that calculates percentage of total population living within or near a zone for each reporting unit polygon and creates an output table. Counts for the total population in the reporting unit and within the zone are also reported.
- Added log file generation as an Optional selection for all tools in the Landscape Characteristics, People in the Landscape, and Riparian Characteristics toolsets.
- Added a new utility, Create Walkability Cost Raster, that will generate a cost surface raster for use in a new tool to evaluate the availability of park space to individuals.
- Added a new utility, Select Zonal Statistics, that will calculate select zonal statistics within reporting unit polygons, creating an output table, and, optionally, a log file.
- Added a new utility, Process Roads for EnviroAtlas Analyses, that processes a street geodatabase to optionally produced three road feature classes used in the EnviroAtlas community-based metrics: Walkable Roads, Roads for Intersection Density Analysis, and Busy Roads - Interstates, Arterials, and Connectors and one road feature class used in the Road Density metrics: All Roads. Supported street geodatabases include ESRI StreetMap, NAVTEQ circa 2019, and NAVTEQ circa 2011.
- Adjusted Patch Metrics to include the median patch size in the output table and to be able to record higher numbers for all output metrics.
- Added 'Launch editor' string parameter to the Create or Modify Land Cover Classification utility with the default setting of, 'Click Run to continue' to make it more clear to the user how to launch the LCC Editor.

Bug Fixes

- Bug Fix. Changed default short field types to either double or long to allow larger numbers in count and sum type fields. Kept float default field type for percentage value fields. This fixed a bug where ATtILA would crash when very large calculated values were unable to be entered into an output table field with a short field type.
- Bug Fix. Placed checks in all tools that use a land cover grid to look for an attribute table and the presence of raster statistics for the land cover grid, and to automatically generate them if they are missing. This feature fixed an issue where the new release of NLCD rasters comes formatted as a tif files that lacks these components.
- Bug Fix. Needed to place a conditional operation in Patch Metrics to check for the existence of an ATtILA generated field index in the Reporting unit feature's attribute table before attempting to delete it. The index could sometimes persist between Patch Metrics runs. The temporary field index is used to speed up patch selections.
- Bug Fix. Fixed issue where the Patch Metrics MDCP option would fail when no land cover class patches existed after processing the input land cover grid to identify connected patches of a minimum size.

- Bug Fix. Fixed issue where Patch Metrics would create the MDCP field when the option was not selected.
- Bug Fix. Needed to go to zoneValue >= 1 in the Land Cover Proportions tool's Per Capita calculations as dividing the amount of land cover in the reporting unit with values below 1 assigns more land cover to the individual than exists in the reporting unit. In addition, if the zoneValue is very, very small (e.g. 0.000009), the perValueCalc value can be so large that it will not fit in the output field and will cause the tool to fail. Reporting units with population counts below 1 now receive a value of -99999 in the Per Capita field.
- Bug Fix. Added a Spatial Analyst needed message in the metadata for the tools that need it. Place validation checks in the respective tool's GUI to prevent them from running if Spatial Analyst is not available.
- Bug Fix. Fixed bug where there is no arcpy.mp.ArcGISProject("CURRENT") for Neighborhood Proportions and Intersection Density when running them on the High Performance Computer.
- Bug Fix. Placed a check in the scripts to see if ATtILA was running via ArcGISPro.exe before restoring environment settings via global variables. This fixed an issue when running multiple ATtILA tools consecutively using the High Performance Computer.
- Bug Fix. Fixed bug where selecting QA Fields in the Additional Options caused the tool to crash for Riparian Land Cover Proportions, Sample Point Land Cover Proportions, and Floodplain Land Cover Proportions.
- Bug Fix. Corrected bug in Facility Land Cover Views tool that occurred when the metric output table was being created. Any field in the additionalFields list needs to have 5 parameters to define it. Integer fields do not need the last parameter, but some ATtILA scripts will error out when it is missing.
- Bug Fix. Removed OID as a field choice for the Reporting unit ID field parameter for all applicable tools. Selecting an OID field as a **Reporting unit ID field** created a fatal error in the Riparian Land Cover Proportions tool. To address this and possibly other problems, it was decided to remove OID as a selectable field type for any field parameter for all tools.
- Bug Fix. Altered the behavior of the Population Land Cover Views tool when determining potential view areas. In earlier versions of the tool, if any cell within a cell's potential viewable area contained a NODATA value, that cell was categorized as 'not visible'. This became problematic when the view radius became large enough to extend beyond the extent of the input land cover grid. As a result, the potential view area for a reporting unit would be underrepresented if they contained cells whose distance from the edge was within the view radius.
- Bug Fix. Corrected bug in the Population Density Metrics tool that caused the population count from the second input population feature class to be incorrectly calculated.

Appendix Conversion Factors

These tables were copied from the Esri document, ‘ArcGIS Pro 2.8 Projected Coordinate Systems Tables’. It is available for download at:

https://pro.arcgis.com/en/pro-app/2.8/help/mapping/properties/pdf/projected_coordinate_systems.pdf

Conversion factors to convert to square meters

Area Unit of Measure Name	Conversion Factor
<i>150_Kilometers</i>	22500000000.0
<i>50_Kilometers</i>	2500000000.0
<i>Centimeter</i>	0.0001
<i>Chain</i>	404.68564224
<i>Chain_Benoit_1895_A</i>	404.68493412895
<i>Chain_Benoit_1895_B</i>	404.684937926029
<i>Chain_Clarke</i>	404.678379705448
<i>Chain_Sears</i>	404.684238955715
<i>Chain_US</i>	404.687260987426
<i>Decimeter</i>	0.01
<i>Fathom</i>	3.34450944
<i>Foot</i>	0.09290304
<i>Foot_1865</i>	0.092903548000694
<i>Foot_Benoit_1895_A</i>	0.092902877440071
<i>Foot_Benoit_1895_B</i>	0.09290287831176
<i>Foot_British_1936</i>	0.092903496651921
<i>Foot_Clarke</i>	0.09290137275148
<i>Foot_Gold_Coast</i>	0.092902863326732
<i>Foot_Indian</i>	0.09290274144751
<i>Foot_Indian_1937</i>	0.092902070738528
<i>Foot_Indian_1962</i>	0.09290279616016
<i>Foot_Indian_1975</i>	0.09290273520025
<i>Foot_Sears</i>	0.092902717850256
<i>Foot_US</i>	0.092903411613275
<i>Inch</i>	0.00064516
<i>Inch_US</i>	0.000645162580648

Area Unit of Measure Name	Conversion Factor
<i>Kilometer</i>	1000000.0
<i>Link</i>	0.040468564224
<i>Link_Benoit_1895_A</i>	0.040468493412895
<i>Link_Benoit_1895_B</i>	0.040468493792603
<i>Link_Clarke</i>	0.040467837970545
<i>Link_Sears</i>	0.040468423895572
<i>Link_US</i>	0.040468726098743
<i>Meter</i>	1.0
<i>Meter_German</i>	1.00002719318486
<i>Mile_US</i>	2589998.47031953
<i>Millimeter</i>	0.000001
<i>Nautical_Mile</i>	3429904.0
<i>Nautical_Mile_UK</i>	3434290.937856
<i>Nautical_Mile_US</i>	3434528.149504
<i>Rod</i>	25.29285264
<i>Rod_US</i>	25.2929538117141
<i>Statute_Mile</i>	2589988.110336
<i>Yard</i>	0.83612736
<i>Yard_Benoit_1895_A</i>	0.83612589696064
<i>Yard_Benoit_1895_B</i>	0.836125904805841
<i>Yard_Clarke</i>	0.836112354763322
<i>Yard_Indian</i>	0.836124673027593
<i>Yard_Indian_1937</i>	0.836118636646753
<i>Yard_Indian_1962</i>	0.83612516544144
<i>Yard_Indian_1975</i>	0.83612461680225
<i>Yard_Sears</i>	0.836124460652307
<i>Yard_Sears_1922_Truncated</i>	0.836123702404
<i>Yard_US</i>	0.836130704519474

Conversion factors to convert to meters

Linear Unit of Measure Name	Conversion Factor
<i>150_Kilometers</i>	150000.0
<i>50_Kilometers</i>	50000.0
<i>Centimeter</i>	0.01
<i>Chain</i>	20.1168
<i>Chain_Benoit_1895_A</i>	20.1167824
<i>Chain_Benoit_1895_B</i>	20.1167824943759
<i>Chain_Clarke</i>	20.11661949
<i>Chain_Sears</i>	20.1167651215526
<i>Chain_US</i>	20.1168402336805
<i>Decimeter</i>	0.1
<i>Fathom</i>	1.8288
<i>Foot</i>	0.3048
<i>Foot_1865</i>	0.3048008333333333
<i>Foot_Benoit_1895_A</i>	0.3047997333333333
<i>Foot_Benoit_1895_B</i>	0.304799734763271
<i>Foot_British_1936</i>	0.3048007491
<i>Foot_Clarke</i>	0.304797265
<i>Foot_Gold_Coast</i>	0.304799710181509
<i>Foot_Indian</i>	0.304799510248147
<i>Foot_Indian_1937</i>	0.30479841
<i>Foot_Indian_1962</i>	0.3047996
<i>Foot_Indian_1975</i>	0.3047995
<i>Foot_Sears</i>	0.304799471538676
<i>Foot_US</i>	0.304800609601219
<i>Inch</i>	0.0254
<i>Inch_US</i>	0.0254000508001016
<i>Kilometer</i>	1000.0
<i>Link</i>	0.201168
<i>Link_Benoit_1895_A</i>	0.201167824
<i>Link_Benoit_1895_B</i>	0.201167824943759
<i>Link_Clarke</i>	0.2011661949

Linear Unit of Measure Name	Conversion Factor
<i>Link_Sears</i>	0.201167651215526
<i>Link_US</i>	0.201168402336805
<i>Meter</i>	1.0
<i>Meter_German</i>	1.0000135965
<i>Mile_US</i>	1609.34721869444
<i>Millimeter</i>	0.001
<i>Nautical_Mile</i>	1852.0
<i>Nautical_Mile_UK</i>	1853.184
<i>Nautical_Mile_US</i>	1853.248
<i>Rod</i>	5.0292
<i>Rod_US</i>	5.02921005842012
<i>Statute_Mile</i>	1609.344
<i>Yard</i>	0.9144
<i>Yard_Benoit_1895_A</i>	0.9143992
<i>Yard_Benoit_1895_B</i>	0.914399204289812
<i>Yard_Clarke</i>	0.914391795
<i>Yard_Indian</i>	0.914398530744441
<i>Yard_Indian_1937</i>	0.91439523
<i>Yard_Indian_1962</i>	0.9143988
<i>Yard_Indian_1975</i>	0.9143985
<i>Yard_Sears</i>	0.914398414616029
<i>Yard_Sears_1922_Truncated</i>	0.914398
<i>Yard_US</i>	0.914401828803658

Appendix LCCSchema XSD

An XML Schema Definition (XSD) file is a type of file used to define the structure and content of an XML file. The supplied file, LCCSchema.xsd, located in the ToolboxSource > LandCoverClassifications folder in the ATtILA toolbox destination folder (see [Installing ATtILA](#)), is provided to the user to help verify that any Land Cover Classification Schema XML file created outside of the LCCEditor application contains the necessary XML elements needed by ATtILA. How to use the XSD to validate a custom XML file is described in the [XML Validation](#) section of this document.

A copy of LCCSchema.xsd is supplied below:

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema attributeFormDefault="unqualified" elementFormDefault="qualified"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  targetNamespace="lcc" xmlns="lcc" xmlns:k="lcc">
  <xs:element name="lccSchema">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="metadata">
          <xs:complexType>
            <xs:sequence>
              <xs:element type="xs:string" name="name"/>
              <xs:element type="xs:string" name="description"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
        <xs:element name="coefficients">
          <xs:annotation>
            <xs:documentation>
              * The coefficients node contains coefficients to be assigned to values.

              * REQUIRED ATTRIBUTES
              * Id - text, unique identifier
              * Name - text, word or phrase describing coefficient
              * fieldName - text, name of field to be created for output
              * - must conform to the field naming conventions dictated by the output database
              system
              * method - text, "P" or "A", designates "P"ercentage or per unit "A"rea calculation routine
            </xs:documentation>
          </xs:annotation>
          <xs:complexType>
            <xs:sequence>
              <xs:element name="coefficient" maxOccurs="unbounded" minOccurs="0">
                <xs:complexType>
                  <xs:attribute name="Id" use="required">
                    <xs:simpleType>
                      <xs:restriction base="xs:string">
                        <xs:enumeration value="NITROGEN"/>
                        <xs:enumeration value="PHOSPHORUS"/>
                        <xs:enumeration value="IMPERVIOUS"/>
                      </xs:restriction>
                    </xs:simpleType>
                  </xs:attribute>
                  <xs:attribute type="xs:string" name="Name" use="required"/>
                  <xs:attribute name="method" use="required">
                    <xs:simpleType>
                      <xs:restriction base="xs:string">
                        <xs:enumeration value="A"/>
                        <xs:enumeration value="P"/>
                      </xs:restriction>
                    </xs:simpleType>
                  </xs:attribute>
                </xs:complexType>
              </xs:element>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

```

        <xs:attribute name="fieldName" use="required">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:minLength value="1"/>
            </xs:restriction>
          </xs:simpleType>
        </xs:attribute>
      </xs:complexType>
    </xs:element>
  </xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="values">
  <xs:annotation>
    <xs:documentation>
      * The values node defines the full set of values that can exist in a land cover raster.

      * REQUIRED ATTRIBUTES
      * Id - integer, raster code
      *
      * OPTIONAL ATTRIBUTES
      * Name - text, word or phrase describing value
      * excluded - boolean, "true" or "false" or "1" or "0"
      *           - used to exclude values from effective area calculations
      *           - excluded=false is the default

      * A value element can optionally contain one or more coefficient elements

      * REQUIRED COEFFICIENT ATTRIBUTES
      * Id - text, must match an Id attribute from a coefficients node element
      * value - decimal, weighting/calculation factor
    </xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element name="value" maxOccurs="unbounded" minOccurs="0">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="coefficient" maxOccurs="unbounded" minOccurs="0">
              <xs:complexType>
                <xs:simpleContent>
                  <xs:extension base="xs:string">
                    <xs:attribute name="Id" use="required"/>
                    <xs:attribute type="xs:decimal" name="value" use="required"/>
                  </xs:extension>
                </xs:simpleContent>
              </xs:complexType>
            </xs:element>
          </xs:sequence>
          <xs:attribute type="xs:int" name="Id" use="required"/>
          <xs:attribute type="xs:string" name="Name" use="optional"/>
          <xs:attribute type="xs:boolean" name="excluded" use="optional"/>
        </xs:complexType>
        <xs:unique name="uniqueCoefficient">
          <xs:selector xpath="coefficient"/>
          <xs:field xpath="@Id"/>
        </xs:unique>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
<xs:element name="classes">
  <xs:annotation>
    <xs:documentation>
      * The classes node contains values from a land cover raster grouped into one or more
      classes.

```

```

* REQUIRED ATTRIBUTES
* Id - text, unique identifier, also used for automated generation of output field name
*   - must conform to the field naming conventions dictated by the output database system

* OPTIONAL ATTRIBUTES
* Name - text, word or phrase describing class
* filter - text, a string of one or more tool name abbreviations separated by a ";"
*   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, np, plcv, pm, rlcp, and
splcp
*   - used to exclude the class from the selectable classes in the tool's GUI
* xxxxField - text, overrides ATtILA-generated field name for output
*   - where xxxx equals a tool name abbreviation
*   - possible abbreviations are: caem, flcp, flcv, lcosp, lcp, rlcp, and splcp
*   - a separate xxxxField attribute can exist for each tool
*   - must conform to the field naming conventions dictated by the output database
system

* A class can contain either values or classes but not both types.
* Value elements contain only an Id attribute which refers to a value in a raster.
* Values tagged as excluded="true" in the values node should not be included in any class.
</xs:documentation>
</xs:annotation>
<xs:complexType>
  <xs:sequence>
    <xs:element name="class" maxOccurs="unbounded" minOccurs="1" type="classType"/>
  </xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
<!--<xs:key name="coefficientKey">
  <xs:selector xpath="k:coefficients/k:coefficient"/>
  <xs:field xpath="@Id"/>
</xs:key>
<xs:keyref name="coefficientKeyRef" refer="k:coefficientKey">
  <xs:selector xpath="k:values/k:value/k:coefficient"/>
  <xs:field xpath="@Id"/>
</xs:keyref-->
</xs:element>
<xs:complexType name="classType">
  <xs:sequence>
    <xs:choice minOccurs="1" maxOccurs="1">
      <xs:element name="class" maxOccurs="unbounded" minOccurs="0" type="classType"/>
      <xs:element name="value" maxOccurs="unbounded" minOccurs="0">
        <xs:complexType>
          <xs:simpleContent>
            <xs:extension base="xs:string">
              <xs:attribute type="xs:int" name="Id" use="required"/>
            </xs:extension>
          </xs:simpleContent>
        </xs:complexType>
      </xs:element>
    </xs:choice>
  </xs:sequence>
  <xs:attribute name="Id" use="required">
    <xs:simpleType>
      <xs:restriction base="xs:string">
        <xs:pattern value="[a-zA-Z0-9_]*"/>
      </xs:restriction>
    </xs:simpleType>
  </xs:attribute>
  <xs:attribute type="xs:string" name="Name" use="optional"/>
  <xs:attribute type="xs:string" name="caemField" use="optional"/>
  <xs:attribute type="xs:string" name="flcpField" use="optional"/>
  <xs:attribute type="xs:string" name="flcvField" use="optional"/>
  <xs:attribute type="xs:string" name="lcpField" use="optional"/>

```



```
<xs:attribute type="xs:string" name="rlcpField" use="optional"/>
<xs:attribute type="xs:string" name="lcospField" use="optional"/>
<xs:attribute type="xs:string" name="splcpField" use="optional"/>
<xs:attribute type="xs:string" name="filter" use="optional"/>
</xs:complexType>
</xs:schema>
```

Appendix Tool Output Field Names

This table is provided as a quick guide to the default output field names generated by ATtILA.

For more information about these fields, please visit the individual tool page.

TOOL	CONVENTION	EXAMPLE
<u>Core and Edge Metrics</u>	[class Id]_E2A[edge width] [class Id]_COR[edge width] [class Id]_EDG[edge width]	agr_E2A3
<u>Facility Land Cover Views</u>	[class Id]_Low[view threshold] [class Id]_High[view threshold] FAC_Count	agr_Low10
<u>Floodplain Land Cover Proportions</u>	f[class Id]	fagr –optional– fagr_A fTOTA fEFA
<u>Intersection Density</u>	None - Output is a raster with a user-defined name	
<u>Land Cover Coefficient Calculator</u>	PCTIA N_Load P_Load	
<u>Land Cover Diversity</u>	H H_Prime C S	
<u>Land Cover on Slope Proportions</u>	[class Id]_SL[slope threshold]	agrSL3 –optional– agrSL3_A
<u>Land Cover Proportions</u>	p[class Id] [class Id]_PC [class Id]_M2	pagr agr_PC agr_M2 –optional– pagr_A
<u>Neighborhood Proportions</u>	None - Output is a raster named: [class Id]_[neighborhood width]_Prox#	AGR_251_Prox
<u>Patch Metrics</u>	[class Id]_PLGP [class Id]_NUM [class Id]_AVG [class Id]_MDN	agr_PLGP

TOOL	CONVENTION	EXAMPLE
	[class Id]_DENS [class Id]_LRG [class Id]_MDCP	
<u>Population Density Metrics</u>	AREAKM2 POPCNT POPCNT_T1 POPCNT_T2 POPDENS POPDENS_T1 POPDENS_T2 POPCHG	_T1 and _T2 are added if POPCHG is selected
<u>Population in Floodplain Metrics</u>	RU_POP FP_POP_C FP_POP_P	
<u>Population Land Cover Views</u>	RU_POP [class Id]_PV_C [class Id]_PV_P [class Id]_MV_C [class Id]_MV_P	agr_PV_C agr_PV_P
<u>Population within Zone Metrics</u>	RU_POP FP_POP_C FP_POP_P	
<u>Riparian Land Cover Proportions</u>	r[class Id][buffer distance]	ragr45 –optional– ragr45_A rTOTA45 rEFA45
<u>Road Density Metrics</u>	AREAKM2 RDKM[road class category] RDDENS[road class category] XCNT[road class category] STXRD[road class category] RNS[buffer distance][road class category]	RDKM –optional– RDKMClass1 RDKMClass2 RNS100Class2
<u>Sample Point Land Cover Proportions</u>	s[class Id][buffer distance]	sagr500 –optional– sagr500_A sTOTA500 sEFA200
<u>Stream Density Metrics</u>	AREAKM2 STRMKM[stream order category] STRMDENS[stream order category]	STRMKM –optional– STRMKMorder1 STRMKMorder2

